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Quantum Mechanics

Non-relativistic Theory

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Contents

CHAPTER I

Fundamental Concepts of Quantum Mechanics

§ 1. The Uncertainty Principle

When classical mechanics and electrodynamics are applied in an attempt to explain atomic phenomena, they give results sharply at variance with experiment. This is already seen most clearly in the contradiction produced by applying ordinary electrodynamics to a model of the atom in which electrons move around the nucleus in classical orbits. Since any accelerated charge radiates, electrons in such orbits would be obliged to emit electromagnetic waves without interruption. Through radiation they would lose energy and eventually fall into the nucleus. Classical electrodynamics would therefore render the atom unstable—a conclusion entirely at odds with reality.

So profound a conflict between theory and experiment shows that a theory applicable to atomic phenomena—phenomena involving particles of very small mass in very small regions of space—requires a fundamental alteration of the basic classical ideas and laws.

A convenient point of departure for identifying these changes is the experimentally observed phenomenon known as electron diffraction[1]. When a uniform beam of electrons passes through a crystal, the transmitted beam displays a succession of intensity maxima and minima entirely analogous to the diffraction pattern observed for electromagnetic waves. Under certain conditions, therefore, the behavior of material particles—electrons—shows features characteristic of wave processes.

The depth of this phenomenon's conflict with ordinary ideas of motion is best shown by the following thought experiment, an idealization of electron diffraction by a crystal. Imagine a barrier impermeable to electrons, with two slits cut in it. If we observe an electron beam[2] passing through one slit while the other is closed, we obtain some intensity distribution on a solid screen behind the slit. Opening the second slit and closing the first gives another distribution. With both slits open at once, ordinary reasoning would lead us to expect simply the superposition of those two distributions: each electron, moving on its own trajectory, passes through one slit without affecting electrons passing through the other. Electron diffraction shows instead that the actual result is a diffraction pattern which, because of interference, is by no means the sum of the patterns from the separate slits. Plainly, this result cannot be reconciled in any way with the idea that electrons move along trajectories.

The mechanics governing atomic phenomena—the so-called *quantum* or *wave mechanics*—must therefore rest on concepts of motion fundamentally different from those of classical mechanics. Quantum mechanics has no concept of a particle trajectory.

[1] In fact, electron diffraction was discovered after quantum mechanics had been created. Our presentation, however, does not follow the historical sequence in which the theory developed; it is arranged instead to show as clearly as possible how the fundamental principles of quantum mechanics are connected with observed phenomena.

[2] The beam is assumed to be so dilute that interactions among its particles play no part.

This fact constitutes the content of the so-called *uncertainty principle*, a fundamental principle of quantum mechanics, due to *Heisenberg* (*W. Heisenberg*, 1927)[3]. Since it rejects the customary ideas of classical mechanics, the uncertainty principle may be said to have negative content. By itself it is naturally quite insufficient as the basis for constructing a new mechanics of particles. Such a theory must of course rest on certain positive propositions, which will be considered below (§ 2). Before those propositions can be formulated, however, the character of the problems posed to quantum mechanics must first be made clear.

For this purpose, let us first consider the special relation between quantum and classical mechanics.

Ordinarily, a more general theory can be formulated as a logically closed system independently of the less general theory that arises as its limiting case. Relativistic mechanics, for example, can be constructed from its own fundamental principles without any reference to Newtonian mechanics. The fundamental propositions of quantum mechanics, on the other hand, cannot in principle be formulated without invoking classical mechanics.

The absence of a definite trajectory for an electron[4] also deprives the electron, considered by itself, of any other dynamical characteristics[5]. It is therefore clear that no logically closed mechanics could be constructed for a system consisting solely of quantum objects. A quantitative description of electron motion requires the presence of physical objects which obey classical mechanics to sufficient accuracy. When an electron interacts with a "classical object," the state of that object is, in general, altered. Both the character and the magnitude of this alteration are governed by the electron's state and can therefore serve as its quantitative characteristic.

In this connection, the "classical object" is usually called an "apparatus," and its interaction with the electron is spoken of as a "measurement." It must be emphasized, however, that this does not mean a process of "measurement" involving a physicist as observer. In quantum mechanics, measurement means any interaction between a classical and a quantum object, occurring apart from and independently of any observer. The recognition of the profound role of the measurement concept in quantum mechanics is due to *Bohr* (*N. Bohr*).

We have defined an apparatus as a physical object which obeys classical mechanics to sufficient accuracy; a body of sufficiently great mass is one example. It should not be supposed, however, that an apparatus must be macroscopic. Under suitable conditions an object known to be microscopic can also play the part of an apparatus, since the meaning of "with sufficient accuracy" depends on the specific problem under consideration. In a Wilson chamber, for instance, one tracks the electron by its mist trail, whose thickness far exceeds atomic dimensions; at such a level of accuracy in locating the trajectory, the

[3]It is noteworthy that the complete mathematical apparatus of quantum mechanics had already been constructed by *W. Heisenberg* and *E. Schrödinger* in 1925–1926—prior to formulation of the uncertainty principle, which lays bare the physical content of that apparatus.

[4]In this and the next section, for brevity, we speak of an electron while meaning any quantum object in general: a particle or system of particles which obeys quantum but not classical mechanics.

[5]Here we mean quantities characterizing the electron's motion, not those characterizing the electron as a particle (charge and mass), which are parameters.

electron is a fully classical object.

Quantum mechanics thus occupies a very distinctive place among physical theories: it contains classical mechanics as its limiting case and, at the same time, requires that limiting case for its own foundation.

We can now formulate the problem addressed by quantum mechanics. A typical problem is to predict the outcome of a repeated measurement from the known outcomes of earlier measurements. We shall also see below that, compared with classical mechanics, quantum mechanics generally restricts the set of values that various physical quantities (energy, for example) may assume, that is, the values that may be found by measuring the quantity in question. The formalism of quantum mechanics must make it possible to determine these allowed values.

In quantum mechanics the measurement process has an essential peculiarity: it always acts upon the electron being measured, and for a given measurement accuracy this action cannot in principle be made arbitrarily weak. The more accurate the measurement, the stronger its action; only in measurements of very low accuracy can the action on the measured object be weak. This property of measurements is logically connected with the fact that the dynamical characteristics of an electron arise only through the measurement itself. It is clear that, if the action of the measurement process on the object could be made arbitrarily weak, the quantity being measured would have a definite value by itself, irrespective of the measurement.

Of the various types of measurement, the principal one is the measurement of an electron's coordinates. Within the domain of applicability of quantum mechanics, the electron's coordinates can always be measured[6] to any accuracy.

Suppose that successive measurements of the coordinates of an electron are made at specified time intervals Δt. As a rule, these results will fail to follow any smooth curve. Quite the opposite: the greater the measurement precision, the more abrupt and erratic the succession of results becomes, consistent with the absence of a path concept for the electron. A more or less smooth path is obtained only when the electron's coordinates are measured with a low degree of accuracy, for example, by the condensation of droplets of vapour in a Wilson chamber.

If instead, while keeping the measurement accuracy unchanged, the intervals Δt between measurements are reduced, adjacent measurements will, of course, give nearby coordinate values. Yet although the results of a series of successive measurements will lie in a small region of space, within that region they will be distributed quite irregularly and will by no means fit any smooth curve. In particular, as Δt tends to zero, the results of nearby measurements do not tend at all to lie on one straight line.

This last observation makes plain that a classical particle velocity does not exist in quantum mechanics: the ratio of the coordinate change between two time instants to the separation Δt tends to no well-defined limit. As we shall see later, however, quantum mechanics does admit a meaningful definition of a particle's velocity at a given instant, one that reduces to the classical velocity upon passing to classical mechanics.

In classical mechanics a particle possesses, at each instant, definite coordinates and

[6] We emphasize once again that by a "measurement made" we mean the interaction of the electron with a classical "apparatus"; this in no way presupposes the presence of an outside observer.

velocity; in quantum mechanics the situation is entirely different. If a measurement has given an electron definite coordinates, it then has no definite velocity at all. Conversely, an electron having a definite velocity cannot have a definite position in space. Indeed, the simultaneous existence of coordinates and velocity at any instant would imply the presence of a definite velocity, which the electron does not have. Thus, in quantum mechanics the electron's coordinates and velocity are quantities that cannot be measured exactly at the same time, that is, cannot simultaneously have definite values. One may say that the electron's coordinates and velocity are quantities which do not exist simultaneously. A quantitative relation governing the possibility of measuring coordinates and velocity inexactly at the same instant will be derived below. In classical mechanics, the state of a physical system is completely described by specifying all its coordinates and velocities at a given instant; from these initial data the equations of motion fully determine the system's behaviour at all future instants. In quantum mechanics such a description is impossible in principle, since the coordinates and their corresponding velocities do not exist simultaneously. Hence the state of a quantum system is described by fewer quantities than in classical mechanics, that is, its description is less detailed than a classical one.

From this follows a very important consequence regarding the character of predictions in quantum mechanics. Whereas a classical description is sufficient to predict with complete precision the future motion of a mechanical system, the less detailed quantum-mechanical description manifestly cannot be adequate for this purpose. Consequently, even when an electron is in a state described as completely as possible, its behaviour at subsequent instants is in principle not unique. Quantum mechanics therefore cannot make strictly definite predictions about the electron's future behaviour. For a specified initial state of an electron, a subsequent measurement may give different results. The task of quantum mechanics is only to determine the probability that one or another result will be obtained in that measurement. In some cases, of course, the probability of a particular measurement result may equal unity, that is, become a certainty, so that the result of the measurement is unique.

Every measurement process in quantum mechanics falls into one of two categories. The first, which includes the majority of measurements, comprises those that yield no determinate result with certainty for any state of the system. The second comprises measurements such that for every possible result there exists a state in which that result is obtained with certainty. It is precisely these latter measurements, which may be called *predictable*, that have the principal role in quantum mechanics. The quantitative characteristics of a state determined by such measurements are what quantum mechanics calls physical quantities. If in some state a measurement gives a unique result with certainty, we shall say that the corresponding physical quantity has a definite value in that state. Below we shall everywhere understand the expression "physical quantity" in precisely the sense stated here.

We shall repeatedly find below that by no means every collection of physical quantities in quantum mechanics can be measured simultaneously, that is, can have definite values simultaneously (we have already discussed one example: the velocity and coordinates of an electron).

Sets of physical quantities with the following property have an important role in quantum mechanics: the quantities can be measured simultaneously and, when they simultaneously have definite values, no other physical quantity (unless it is a function of them) can have a definite value in that state. We shall refer to such sets of physical quantities as *complete sets*.

Every description of an electron's state arises from some measurement. We now state what a complete description of a state means in quantum mechanics. Completely described states arise from the simultaneous measurement of a complete set of physical quantities. From the results of such a measurement one can, in particular, determine the probabilities of the results of any subsequent measurement, independently of everything that happened to the electron before the first measurement.

Below, everywhere except in § 14, by states of a quantum system we shall mean states described in precisely this complete manner.

§ 2. The superposition principle

The radical change in physical ideas about motion that quantum mechanics requires in comparison with classical mechanics naturally calls for an equally radical change in the theory's mathematical formalism. In this connection the first question that arises concerns the means of describing a state in quantum mechanics.

Let q stand for the totality of coordinates of a quantum system, and dq for the product of their differentials (it is called a volume element of the system's *configuration space*); for one particle, dq coincides with the volume element dV of ordinary space.

The mathematical formalism of quantum mechanics rests on the assertion that a system's state may be described by a certain (generally complex) function of the coordinates $\Psi(q)$, and that the squared modulus of this function determines the probability distribution of the coordinate values: $|\Psi|^2\,dq$ is the probability that a measurement made on the system will find coordinate values in the element dq of configuration space. The function Ψ is called the system's *wave function*[7].

Knowledge of the wave function makes it possible in principle to calculate the probabilities of the various outcomes of any measurement in general (not necessarily a coordinate measurement). Every one of these probabilities is given by an expression bilinear in Ψ and Ψ^*. In full generality, such an expression has the form

$$\iint \Psi(q)\Psi^*(q')\varphi(q,q')\,dq\,dq', \tag{2.1}$$

where the function $\varphi(q,q')$ depends on the kind and outcome of the measurement, and the integrations extend over the entire configuration space. The probability $\Psi\Psi^*$ itself for the different coordinate values is also an expression of this type[8].

As time passes, the system's state, and with it the wave function, generally changes. In this sense the wave function may also be regarded as a function of time. If the wave function is known at some initial instant, then, by the very meaning of a complete description of a state, it is thereby determined in principle at every future instant. The

[7] It was first introduced into quantum mechanics by *Schrödinger* (*E. Schrödinger*, 1926).

[8] It is obtained from (2.1) by taking $\varphi(q,q') = \delta(q-q_0)\,\delta(q'-q_0)$, where δ denotes the so-called δ-function defined below, in § 5; q_0 denotes the coordinate value whose probability is sought.

actual time dependence of the wave function is specified by equations to be derived below.

By definition, the sum of the probabilities of all possible values of the system's coordinates must equal unity. The integral of $|\Psi|^2$ over the whole configuration space must therefore equal unity:

$$\int |\Psi|^2 \, dq = 1. \tag{2.2}$$

This equality constitutes what is known as the *normalization condition* on wave functions. When the integral of $|\Psi|^2$ is convergent, by choosing a suitable constant coefficient, the function Ψ can always be normalized, as it is said. We shall see below, however, that the integral of $|\Psi|^2$ may diverge, in which case Ψ cannot be normalized by condition (2.2). Here $|\Psi|^2$ naturally does not yield the absolute coordinate probabilities; the ratio of $|\Psi|^2$ between two distinct points of configuration space, however, does determine the relative probability of the coordinate values.

Since all quantities with direct physical meaning that are calculated using the wave function have the form (2.1), in which Ψ occurs multiplied by Ψ^*, it is clear that a normalized wave function is defined only up to a constant *phase factor* of the form $e^{i\alpha}$, where α is any real number. This ambiguity is fundamental and cannot be removed; it is, however, immaterial, since it has no effect on any physical result.

The positive content of quantum mechanics is founded on a number of assertions about the properties of the wave function, as follows.

Suppose that in the state with wave function $\Psi_1(q)$ a certain measurement leads with certainty to a definite outcome, outcome 1, while in the state $\Psi_2(q)$ it leads to outcome 2. It is then postulated that every linear combination of Ψ_1 and Ψ_2, that is, every function of the form $c_1\Psi_1 + c_2\Psi_2$ (c_1, c_2 being constants), describes a state in which the same measurement yields either outcome 1 or outcome 2. Moreover, if the time dependence of the states is known, being given in one case by the function $\Psi_1(q, t)$ and in the other by $\Psi_2(q, t)$, then any linear combination of them also gives a possible time dependence of a state.

These assertions constitute the so-called *superposition principle for states*, the fundamental positive principle of quantum mechanics. It follows from this principle, in particular, that every equation satisfied by wave functions must be linear in Ψ.

Consider a system made up of two parts, and suppose its state is specified in such a way that each part is described completely[9]. One can then assert that the coordinate probabilities for q_1 in the first part are independent of those for the coordinates q_2 of the second, and therefore that the probability distribution for the whole system must equal the product of the probability distributions for its parts. It follows that the system's wave function $\Psi_{12}(q_1, q_2)$ may be written as a product of the wave functions $\Psi_1(q_1)$ and $\Psi_2(q_2)$ of its parts:

$$\Psi_{12}(q_1, q_2) = \Psi_1(q_1)\Psi_2(q_2). \tag{2.3}$$

If the two parts do not interact with each other, this relation between the wave functions

[9]This, of course, also gives a complete description of the state of the system as a whole. We emphasize, however, that the converse statement is by no means valid: in general, a complete description of the state of the system as a whole does not yet describe the states of its separate parts completely (see also § 14).

of the system and of its parts is preserved at future instants as well:

$$\Psi_{12}(q_1, q_2, t) = \Psi_1(q_1, t)\Psi_2(q_2, t). \tag{2.4}$$

§ 3. Operators

Consider some physical quantity f that characterizes the state of a quantum system. Strictly speaking, in what follows we ought to speak not of a single quantity but of an entire complete set of them at once. However, none of the reasoning is thereby essentially altered, and for the sake of brevity and simplicity we speak below of only a single physical quantity.

Those values which a given physical quantity may assume are termed its *eigenvalues* in quantum mechanics, and their aggregate is referred to as the eigenvalue *spectrum* of that quantity. In classical mechanics, quantities range, in general, over a continuous series of values. In quantum mechanics too there exist physical quantities (for example, coordinates) whose eigenvalues fill a continuous range; in such cases one speaks of a *continuous spectrum* of eigenvalues. Alongside these quantities, however, quantum mechanics also possesses quantities whose eigenvalues constitute a discrete set; one then speaks of a *discrete spectrum*.

For simplicity we shall first assume that the quantity f under consideration has a discrete spectrum; the case of a continuous spectrum is considered in § 5. We denote the eigenvalues of f by f_n, where the index n runs through the values $0, 1, 2, 3, \ldots$ We denote the wave function of the system in the state in which f has the value f_n by Ψ_n. The wave functions Ψ_n bear the name *eigenfunctions* of the physical quantity f. Every such function is taken to be normalized:

$$\int |\Psi_n|^2 \, dq = 1. \tag{3.1}$$

If the system is in some arbitrary state with wave function Ψ, then a measurement of f performed on it will yield as a result one of the eigenvalues f_n. By the superposition principle, the wave function Ψ must be a linear combination of those eigenfunctions Ψ_n that correspond to the values f_n which can be found with non-zero probability in a measurement performed on the system in the state under consideration. In the general case of an arbitrary state, Ψ may accordingly be written as the series

$$\Psi = \sum_n a_n \Psi_n, \tag{3.2}$$

where the summation is over all n, and a_n are certain constant coefficients.

Thus we arrive at the conclusion that every wave function can be, as one says, expanded in the eigenfunctions of any physical quantity. A system of functions in terms of which such an expansion can be carried out is spoken of as a *complete system of functions*.

The expansion (3.2) makes it possible to determine the probabilities of finding (by means of measurements) one or another value f_n of f for a system in the state with wave function Ψ. Indeed, according to what was said in the preceding section, these probabilities must be determined by certain expressions bilinear in Ψ and Ψ^* and must therefore be bilinear in a_n and a_n^*. Moreover, positivity of these expressions is obviously required. As for the probability associated with a given f_n: it must become unity for a system in the state $\Psi = \Psi_n$ and must vanish whenever expansion (3.2) of Ψ lacks a term containing the corresponding Ψ_n. The only essentially positive quantity satisfying this condition is the squared modulus of the coefficient a_n. Thus we arrive at the result that the squared modulus $|a_n|^2$ of every expansion coefficient in (3.2) gives the probability of the respective value f_n of f in the state with wave function Ψ. The sum of the probabilities of all possible values f_n must equal unity; in other words, the relation

$$\sum_n |a_n|^2 = 1. \tag{3.3}$$

must hold.

If Ψ were not normalized, then the relation (3.3) would also not hold. The sum $\sum_n |a_n|^2$ would then have to be determined by some expression bilinear in Ψ and Ψ^* and reducing to unity for normalized Ψ. The only such expression is the integral $\int \Psi\Psi^*\,dq$. Thus the equality

$$\sum_n a_n a_n^* = \int \Psi\Psi^*\,dq. \tag{3.4}$$

must hold.

On the other hand, multiplying the expansion $\Psi^* = \sum_n a_n^* \Psi_n^*$ of the complex conjugate function Ψ^* by Ψ and integrating, we obtain

$$\int \Psi\Psi^*\,dq = \sum_n a_n^* \int \Psi_n^*\Psi\,dq.$$

Comparing this with (3.4), we have

$$\sum_n a_n a_n^* = \sum_n a_n^* \int \Psi_n^*\Psi\,dq,$$

from which we find the following formula determining the coefficients a_n of the expansion of Ψ in the eigenfunctions Ψ_n:

$$a_n = \int \Psi\Psi_n^*\,dq. \tag{3.5}$$

Substituting (3.2) here, we obtain

$$a_n = \sum_m a_m \int \Psi_m\Psi_n^*\,dq,$$

whence it is seen that the eigenfunctions must satisfy the conditions

$$\int \Psi_m\Psi_n^*\,dq = \delta_{nm}, \tag{3.6}$$

where $\delta_{nm} = 1$ for $n = m$ and $\delta_{nm} = 0$ for $n \neq m$. The fact that the integrals of the products $\Psi_m \Psi_n^*$ with $m \neq n$ vanish is spoken of as the mutual *orthogonality* of the functions Ψ_n. The collection of eigenfunctions Ψ_n thus constitutes a complete system of normalized and mutually orthogonal (or, as one says for short, *orthonormal*) functions.

We introduce the concept of the *mean value* $\bar{f}$ of a quantity f in a given state. Following the standard definition, $\bar{f}$ is the sum of every eigenvalue f_n of the quantity in question, weighted by the respective probability $|a_n|^2$:

$$\bar{f} = \sum_n f_n |a_n|^2. \tag{3.7}$$

We write $\bar{f}$ in the form of an expression that involves not the expansion coefficients of Ψ but the function itself. Since (3.7) contains the products $a_n a_n^*$, it is clear that the desired expression must be bilinear in Ψ and Ψ^*. We introduce a certain mathematical *operator*, which we denote by $\hat{f}$[10], and define as follows. Let $(\hat{f}\Psi)$ denote the result of the action of the operator $\hat{f}$ on the function Ψ. We define $\hat{f}$ by requiring that the integral of $(\hat{f}\Psi)$ multiplied by the conjugate Ψ^* yield $\bar{f}$:

$$\bar{f} = \int \Psi^* (\hat{f}\Psi)\, dq. \tag{3.8}$$

It is easy to see that in the general case the operator $\hat{f}$ is some linear integral operator.[11] Indeed, using the expression (3.5) for a_n, the definition (3.7) of the mean value can be cast as

$$\bar{f} = \sum_n f_n a_n a_n^* = \int \Psi^* \left(\sum_n a_n f_n \Psi_n \right) dq.$$

Comparing with (3.8), we see that the result of the action of the operator $\hat{f}$ on the function Ψ has the form

$$(\hat{f}\Psi) = \sum_n a_n f_n \Psi_n. \tag{3.9}$$

Substituting here the expression (3.5) for a_n, we find that $\hat{f}$ is an integral operator of the form

$$(\hat{f}\Psi) = \int K(q, q')\Psi(q')\, dq', \tag{3.10}$$

where the function $K(q, q')$ (the so-called kernel of the operator) is

$$K(q, q') = \sum_n f_n \Psi_n^*(q') \Psi_n(q). \tag{3.11}$$

Thus, to each physical quantity in quantum mechanics there is assigned a definite linear operator.

[10]We shall everywhere denote operators by letters with a hat.

[11]A linear operator is one possessing the properties: $\hat{f}(\Psi_1 + \Psi_2) = \hat{f}\Psi_1 + \hat{f}\Psi_2$, $\hat{f}(a\Psi) = a\hat{f}\Psi$, where Ψ_1, Ψ_2 are arbitrary functions and a is an arbitrary constant.

From (3.9) it is seen that if Ψ is one of the eigenfunctions Ψ_n (so that all a_n except one are zero), then upon the action of the operator $\hat{f}$ on it, this function is simply multiplied by the eigenvalue f_n[12]:

$$\hat{f}\Psi_n = f_n\Psi_n. \tag{3.12}$$

The eigenfunctions of a physical quantity f are therefore solutions of the equation

$$\hat{f}\Psi = f\Psi,$$

in this equation f denotes a constant; the eigenvalues are those particular values of f that yield solutions obeying the necessary conditions. As will be seen later, the explicit form of operators for the various physical quantities can be established on direct physical grounds, whereupon this operator property enables one to obtain eigenfunctions and eigenvalues by solving the equations $\hat{f}\Psi = f\Psi$.

For a real physical quantity, both the eigenvalues themselves and their mean values in any state are real. This fact places a definite constraint on the properties of the associated operators. Setting the expression (3.8) equal to its own complex conjugate gives

$$\int \Psi^*(\hat{f}\Psi)\,dq = \int \Psi(\hat{f}^*\Psi^*)\,dq, \tag{3.13}$$

where $\hat{f}^*$ denotes the operator complex conjugate to $\hat{f}$[13]. For an arbitrary linear operator such a relation does not, in general, hold, so that it represents a certain restriction imposed on the possible form of the operators $\hat{f}$. For an arbitrary operator $\hat{f}$ one can define, as one says, the *transposed* operator $\tilde{\hat{f}}$, defined so that

$$\int \Phi(\hat{f}\Psi)\,dq = \int \Psi(\tilde{\hat{f}}\Phi)\,dq, \tag{3.14}$$

where Ψ, Φ are two different functions. Choosing as Φ the conjugate function Ψ^*, comparison with (3.13) reveals that

$$\tilde{\hat{f}} = \hat{f}^*. \tag{3.15}$$

An operator that satisfies this condition is called *Hermitian*[14]. Operators representing real physical quantities within the mathematical apparatus of quantum mechanics must therefore be Hermitian.

One may also formally consider complex physical quantities, i.e. quantities possessing complex eigenvalues. Suppose f is one such quantity; one may then form the complex-conjugate quantity f^*, whose eigenvalues are the complex conjugates of the eigenvalues of f. The operator corresponding to the quantity f^* will be denoted by $\hat{f}^+$. It is called the *adjoint* operator to $\hat{f}$ and must, in general, be distinguished from the complex-conjugate

[12] Below, wherever it cannot lead to misunderstanding, we shall omit the parentheses in the expression $(\hat{f}\Psi)$, the operator being understood to act on the expression written after it.

[13] By definition, if for the operator $\hat{f}$ we have $\hat{f}\psi = \varphi$, then the complex-conjugate operator $\hat{f}^*$ is the operator for which $\hat{f}^*\psi^* = \varphi^*$.

[14] For a linear integral operator of the form (3.10), Hermiticity requires that the kernel of the operator satisfy $K(q, q') = K^*(q', q)$.

operator $\hat{f}^*$. Indeed, by definition of the operator $\hat{f}^+$, the mean value of f^* in some state Ψ is

$$\overline{f^*} = \int \Psi^* \hat{f}^+ \Psi \, dq.$$

On the other hand, we have

$$(\bar{f})^* = \left[\int \Psi^* \hat{f} \Psi \, dq\right]^* = \int \Psi \hat{f}^* \Psi^* \, dq = \int \Psi^* \tilde{\hat{f}}^* \Psi \, dq.$$

Equating the two expressions, we find that

$$\hat{f}^+ = \tilde{\hat{f}}^*, \tag{3.16}$$

from which it is clear that $\hat{f}^+$ does not, in general, coincide with $\hat{f}^*$. The condition (3.15) can now be written in the form

$$\hat{f} = \hat{f}^+, \tag{3.17}$$

i.e. the operator of a real physical quantity coincides with its adjoint (Hermitian operators are also called *self-adjoint*).

Let us show how one can directly prove the mutual orthogonality of the eigenfunctions of a Hermitian operator corresponding to different eigenvalues. Let f_n, f_m be two different eigenvalues of the real quantity f, and Ψ_n, Ψ_m the corresponding eigenfunctions:

$$\hat{f}\Psi_n = f_n \Psi_n, \qquad \hat{f}\Psi_m = f_m \Psi_m.$$

Multiplying both sides of the first of these equalities by Ψ_m^*, and the complex conjugate of the second equality by Ψ_n, and subtracting these products termwise from each other, we obtain

$$\Psi_m^* \hat{f} \Psi_n - \Psi_n \hat{f}^* \Psi_m^* = (f_n - f_m)\Psi_n \Psi_m^*.$$

Integrate both sides of this equality over dq. Since $\hat{f}^* = \tilde{\hat{f}}$, by virtue of (3.14) the integral of the left-hand side of the equality vanishes, so that we obtain

$$(f_n - f_m) \int \Psi_n \Psi_m^* \, dq = 0,$$

whence, in view of $f_n \neq f_m$, the required orthogonality property of the functions Ψ_n and Ψ_m follows.

We have been speaking here all along of only a single physical quantity f, whereas we ought to speak, as was noted at the beginning of the section, of a complete system of simultaneously measurable physical quantities. Then we would find that to each of these quantities $f, g, \ldots$ there corresponds its own operator $\hat{f}, \hat{g}, \ldots$ The eigenfunctions Ψ_n correspond to states in which all the quantities under consideration have definite values, i.e. correspond to definite sets of eigenvalues $f_n, g_n, \ldots$, and are joint solutions of the system of equations

$$\hat{f}\Psi = f\Psi, \qquad \hat{g}\Psi = g\Psi, \ldots$$

§ 4. Addition and Multiplication of Operators

If $\hat{f}$ and $\hat{g}$ are the operators corresponding to two physical quantities f and g, then $\hat{f} + \hat{g}$ corresponds to their sum $f + g$. In quantum mechanics, however, the meaning of adding different physical quantities differs essentially according as the quantities are or are not measurable simultaneously. If f and g are simultaneously measurable, the operators $\hat{f}$ and $\hat{g}$ have common eigenfunctions. These are also eigenfunctions of $\hat{f} + \hat{g}$, whose eigenvalues are the sums $f_n + g_n$.

If f and g cannot have definite values simultaneously, their sum $f + g$ has a more restricted meaning. All that can be stated is that its mean value in an arbitrary state equals the sum of the separate mean values of the two terms:

$$\overline{f + g} = \bar{f} + \bar{g}. \tag{4.1}$$

The eigenvalues and eigenfunctions of $\hat{f} + \hat{g}$, on the other hand, will in general have no relation to those of f and g. Clearly, when $\hat{f}$ and $\hat{g}$ are Hermitian, so is $\hat{f} + \hat{g}$; its eigenvalues are therefore real and are the eigenvalues of the new quantity $f + g$ defined in this way.

We note the following theorem. Let f_0 and g_0 be the least eigenvalues of f and g, and let $(f + g)_0$ denote the corresponding quantity for $f + g$. Then

$$(f + g)_0 \geq f_0 + g_0. \tag{4.2}$$

The equality sign applies if f and g are simultaneously measurable. The proof follows from the evident fact that a quantity's mean value is always at least its least eigenvalue. In the state in which $(f + g)$ has the value $(f + g)_0$, we have $\overline{(f + g)} = (f + g)_0$; on the other hand, $\overline{(f + g)} = \bar{f} + \bar{g} \geq f_0 + g_0$, which gives (4.2).

Suppose again that f and g are simultaneously measurable. Besides their sum, we may introduce their product as a quantity whose eigenvalues are the products of the eigenvalues of f and g. The corresponding operator is readily seen to act by applying one operator to a function and then the other. Mathematically it is written as the product of $\hat{f}$ and $\hat{g}$. Indeed, if Ψ_n are common eigenfunctions of $\hat{f}$ and $\hat{g}$, then

$$\hat{f}\hat{g}\Psi_n = \hat{f}(\hat{g}\Psi_n) = \hat{f}g_n\Psi_n = g_n\hat{f}\Psi_n = g_n f_n\Psi_n$$

(the symbol $\hat{f}\hat{g}$ denotes the operator which acts on a function Ψ first by applying $\hat{g}$ to Ψ, and then by applying $\hat{f}$ to the function $\hat{g}\Psi$). We could equally well have used $\hat{g}\hat{f}$, whose factors occur in the opposite order. Plainly, the actions of these two operators on the functions Ψ_n give the same result. Since every wave function Ψ is expandable in the functions Ψ_n, it follows that $\hat{f}\hat{g}$ and $\hat{g}\hat{f}$ yield the same result when acting on an arbitrary function. This fact may be written symbolically as $\hat{f}\hat{g} = \hat{g}\hat{f}$, or

$$\hat{f}\hat{g} - \hat{g}\hat{f} = 0. \tag{4.3}$$

Two such operators $\hat{f}$ and $\hat{g}$ are said to *commute* with one another. We have therefore reached an important result: if two quantities f and g can have definite values simultaneously, their operators commute.

The converse theorem can also be proved (see § 11): when $\hat{f}$ and $\hat{g}$ commute, all their eigenfunctions can be chosen in common, which physically means that the corresponding physical quantities are simultaneously measurable. Thus commutation of the operators is a necessary and sufficient condition for simultaneous measurability of physical quantities.

Raising an operator to a power is a special case of multiplying operators. It follows from what has been said that the eigenvalues of $\hat{f}^p$ (where p is an integer) are the eigenvalues of $\hat{f}$ raised to the same pth power. More generally, any function of an operator, $\varphi(\hat{f})$, may be defined as the operator whose eigenvalues are the same function $\varphi(f)$ of the eigenvalues of $\hat{f}$. When $\varphi(f)$ has a Taylor-series expansion, this expansion reduces the action of $\varphi(\hat{f})$ to the actions of the various powers $\hat{f}^p$.

In particular, $\hat{f}^{-1}$ is called the operator *inverse* to $\hat{f}$. Clearly, successive application of $\hat{f}$ and $\hat{f}^{-1}$ to an arbitrary function leaves it unchanged; that is, $\hat{f}\hat{f}^{-1} = \hat{f}^{-1}\hat{f} = 1$. When f and g are not simultaneously measurable, their product has no such direct meaning. This is already apparent because $\hat{f}\hat{g}$ is then not Hermitian and consequently cannot correspond to a real physical quantity. Indeed, by the definition of the transposed operator,

$$\int \Psi \hat{f}\hat{g}\Phi \, dq = \int \Psi \hat{f}(\hat{g}\Phi) \, dq = \int (\hat{g}\Phi)(\tilde{f}\Psi) \, dq.$$

Here $\tilde{f}$ acts only on Ψ, while $\hat{g}$ acts on Φ; the integrand is therefore simply the product of the two functions $\hat{g}\Phi$ and $\tilde{f}\Psi$. Applying the definition of the transposed operator once more gives

$$\int \Psi \hat{f}\hat{g}\Phi \, dq = \int (\tilde{f}\Psi)(\hat{g}\Phi) \, dq = \int \Phi \tilde{g} \tilde{f} \Psi \, dq.$$

We have thus obtained an integral in which Ψ and Φ have exchanged places relative to the original one. In other words, $\tilde{g}\tilde{f}$ is the operator transposed to $\hat{f}\hat{g}$, so we may write

$$\widetilde{fg} = \tilde{g}\tilde{f}, \tag{4.4}$$

that is, the transpose of the product $\hat{f}\hat{g}$ is the product of the transposed factors written in reverse order. Taking the complex conjugate of both sides of (4.4), we find

$$(\hat{f}\hat{g})^+ = \hat{g}^+\hat{f}^+. \tag{4.5}$$

If both $\hat{f}$ and $\hat{g}$ are Hermitian, then $(\hat{f}\hat{g})^+ = \hat{g}\hat{f}$. Hence $\hat{f}\hat{g}$ is Hermitian only when the factors $\hat{f}$ and $\hat{g}$ commute.

From the products $\hat{f}\hat{g}$ and $\hat{g}\hat{f}$ of two noncommuting Hermitian operators one can nevertheless form another Hermitian operator, their *symmetrized product*

$$\frac{1}{2}(\hat{f}\hat{g} + \hat{g}\hat{f}). \tag{4.6}$$

It is also readily verified that the difference $\hat{f}\hat{g} - \hat{g}\hat{f}$ is an "anti-Hermitian" operator (that is, its transpose is equal to the negative of its complex conjugate). Multiplication by i makes it Hermitian; hence

$$i(\hat{f}\hat{g} - \hat{g}\hat{f}) \tag{4.7}$$

is likewise a Hermitian operator.

For brevity, later on we shall sometimes use the notation

$$\{\hat{f}, \hat{g}\} = \hat{f}\hat{g} - \hat{g}\hat{f} \tag{4.8}$$

for the so-called *commutator* of the operators. It is easy to verify the relation

$$\{\hat{f}\hat{g}, \hat{h}\} = \{\hat{f}, \hat{h}\}\hat{g} + \hat{f}\{\hat{g}, \hat{h}\}. \tag{4.9}$$

Notice that $\{\hat{f}, \hat{h}\} = 0$ and $\{\hat{g}, \hat{h}\} = 0$ do not in general imply that $\hat{f}$ and $\hat{g}$ commute with each other.

§ 5. Continuous spectrum

All the relations obtained in § 3, § 4 for the properties of discrete-spectrum eigenfunctions extend without difficulty to a continuous spectrum of eigenvalues.

Let f be a physical quantity possessing a continuous spectrum. We shall denote its eigenvalues simply by the same letter f without an index; the corresponding eigenfunctions will be written Ψ_f. In the same way that an arbitrary wave function Ψ admits the series expansion (3.2) in eigenfunctions of a discrete-spectrum quantity, it can also be expanded — this time as an integral — over the complete system of eigenfunctions of a continuous-spectrum quantity. Such an expansion takes the form

$$\Psi(q) = \int a_f \Psi_f(q)\, df, \tag{5.1}$$

where the integration is taken over the whole domain of values that the quantity f can assume.

More complicated than in the discrete-spectrum case is the question of normalizing the continuous-spectrum eigenfunctions. As we shall see below, the requirement that the integral of the squared modulus equal unity cannot be satisfied here. Instead, let us set ourselves the aim of normalizing the functions Ψ_f in such a way that $|a_f|^2\, df$ represents the probability for the physical quantity under consideration, in the state described by the wave function Ψ, to have a value in the given interval between f and $f + df$. Since the sum of the probabilities of all possible values of f must be equal to unity, we have

$$\int |a_f|^2\, df = 1 \tag{5.2}$$

(analogously to relation (3.3) for the discrete spectrum).

Proceeding exactly as we did in deriving formula (3.5), and using the same arguments, we write, on one hand,

$$\int \Psi\Psi^*\, dq = \int |a_f|^2\, df,$$

and, on other hand,

$$\int \Psi\Psi^*\, dq = \iint a_f^* \Psi_f^* \Psi\, df\, dq.$$

By comparing the two expressions we find the formula that determines the expansion coefficients,

$$a_f = \int \Psi(q)\Psi_f^*(q)\, dq, \tag{5.3}$$

exactly analogous to (3.5).

To derive the normalization condition we now substitute (5.1) into (5.3):

$$a_f = \int a_{f'} \left(\int \Psi_{f'} \Psi_f^* \, dq \right) df'.$$

This relation must hold for arbitrary a_f and must therefore be satisfied identically. For this to be so it is first of all necessary that the coefficient of $a_{f'}$ under the integral sign (i.e. the integral $\int \Psi_{f'} \Psi_f^* \, dq$) vanish for all $f' \neq f$. For $f' = f$ this coefficient must become infinite (otherwise the integral over df' would simply be equal to zero). Thus the integral $\int \Psi_{f'} \Psi_f^* \, dq$ is a function of the difference $f - f'$ which vanishes for non-zero values of the argument and becomes infinite when the argument is zero. We denote this function by $\delta(f' - f)$:

$$\int \Psi_{f'} \Psi_f^* \, dq = \delta(f' - f). \tag{5.4}$$

The manner in which the function $\delta(f' - f)$ becomes infinite at $f' - f = 0$ is determined by the requirement that there should be

$$\int \delta(f' - f) a_{f'} \, df' = a_f.$$

Clearly, for this to hold it is necessary that

$$\int \delta(f' - f) \, df' = 1.$$

A function defined in this way is called the *delta-function*[15]. Let us write out once more the formulas that define it. We have

$$\delta(x) = 0 \quad \text{at } x \neq 0, \qquad \delta(0) = \infty, \tag{5.5}$$

and moreover in such a way that

$$\int_{-\infty}^{+\infty} \delta(x) \, dx = 1. \tag{5.6}$$

As the limits of integration one may write any other pair between which the point $x = 0$ lies. If $f(x)$ is a function continuous at $x = 0$, then

$$\int_{-\infty}^{+\infty} \delta(x) f(x) \, dx = f(0). \tag{5.7}$$

In more general form this formula can be written as

$$\int \delta(x - a) f(x) \, dx = f(a), \tag{5.8}$$

where the region of integration includes the point $x = a$, and $f(x)$ is continuous at $x = a$. It is also obvious that the delta-function is even, i.e.

$$\delta(-x) = \delta(x). \tag{5.9}$$

[15]The delta-function was introduced into theoretical physics by *Dirac* (*P. A. M. Dirac*).

Finally, writing

$$\int_{-\infty}^{\infty} \delta(\alpha x)\, dx = \int_{-\infty}^{\infty} \delta(y) \frac{dy}{|\alpha|} = \frac{1}{|\alpha|},$$

we arrive at the conclusion that

$$\delta(\alpha x) = \frac{1}{|\alpha|}\delta(x), \tag{5.10}$$

where α is any constant. Formula (5.4) expresses the normalization rule for the eigenfunctions of the continuous spectrum; it replaces condition (3.6) of the discrete spectrum. We see that the functions Ψ_f and $\Psi_{f'}$ with $f \neq f'$ remain orthogonal to each other. The integrals of the squares $|\Psi_f|^2$ of the functions of the continuous spectrum, however, diverge.

The functions $\Psi_f(q)$ satisfy one further relation, similar to (5.4). To derive it we substitute (5.3) into (5.1), which gives

$$\Psi(q) = \int \Psi(q') \left(\int \Psi_f^*(q')\Psi_f(q)\, df \right) dq',$$

whence we immediately conclude that one must have

$$\int \Psi_f^*(q')\Psi_f(q)\, df = \delta(q - q'). \tag{5.11}$$

An analogous relation can, of course, be derived for the discrete spectrum, where it takes the form

$$\sum_n \Psi_n^*(q')\Psi_n(q) = \delta(q' - q). \tag{5.12}$$

On comparing the pair of formulas (5.1) and (5.4) with the pair (5.3) and (5.11), we see that, on the one hand, the functions $\Psi_f(q)$ effect the expansion of the function $\Psi(q)$ with the expansion coefficients a_f, while, on the other hand, formula (5.3) can be regarded as a completely analogous expansion of the function $a_f \equiv a(f)$ in the functions $\Psi_f^*(q)$, with $\Psi(q)$ playing the role of the expansion coefficients. The function $a(f)$, like $\Psi(q)$, determines the state of the system completely; it is spoken of as the wave function in the *f-representation* (and the function $\Psi(q)$ as the wave function in the *q-representation*). Just as $|\Psi(q)|^2$ determines the probability for the system to have coordinates in a given interval dq, so $|a(f)|^2$ determines the probability of values of the quantity f in a given interval df. The functions $\Psi_f(q)$, for their part, are, on the one hand, the eigenfunctions of the quantity f in the q-representation, while their complex conjugates $\Psi_f^*(q)$ serve as eigenfunctions of the coordinate q in the f-representation.

Suppose $\varphi(f)$ is a function of the quantity f, with φ and f standing in one-to-one correspondence. Every function $\Psi_f(q)$ may then equally be viewed as an eigenfunction of the quantity φ. In doing so, however, the normalization of these functions must be changed. Indeed, the eigen- functions $\Psi_\varphi(q)$ of the quantity φ must be normalized by the condition

$$\int \Psi_{\varphi(f')}\Psi_{\varphi(f)}^*\, dq = \delta[\varphi(f') - \varphi(f)],$$

whereas the functions Ψ_f are normalized by condition (5.4). The argument of the delta-function vanishes at $f' = f$. For f' close to f we have

$$\varphi(f') - \varphi(f) = \frac{d\varphi(f)}{df}(f' - f).$$

Making use of expression (5.10), one can write[16]

$$\delta[\varphi(f') - \varphi(f)] = \frac{1}{|d\varphi(f)/df|}\delta(f' - f). \tag{5.13}$$

Comparison of (5.13) with (5.4) now shows that the functions Ψ_φ and Ψ_f are related to each other by the relation

$$\Psi_{\varphi(f)} = \frac{1}{\sqrt{|d\varphi(f)/df|}}\Psi_f. \tag{5.14}$$

There exist physical quantities which, over some domain of their values, possess a discrete spectrum, and over another possess a continuous one. For the eigenfunctions of such a quantity, of course, all the same relations that were derived in this and the preceding sections continue to hold. It need only be noted that the complete system of functions is formed by the collection of the eigenfunctions of both spectra together. Therefore the expansion of an arbitrary wave function in the eigenfunctions of such a quantity has the form

$$\Psi(q) = \sum_n a_n\Psi_n(q) + \int a_f\Psi_f(q)\,df, \tag{5.15}$$

where the summation runs over the discrete spectrum, and the integration over the entire continuous spectrum.

An example of a quantity possessing a continuous spectrum is the coordinate q itself. One readily sees that the associated operator reduces to multiplication by q. Indeed, since the square $|\Psi(q)|^2$ governs the probability of different coordinate values, the mean value of the coordinate is

$$\bar{q} = \int q|\Psi|^2\,dq = \int \Psi^* q\Psi\,dq.$$

Comparing this expression with the definition of operators according to (3.8), we see that[17]

$$\hat{q} = q. \tag{5.16}$$

The eigenfunctions of this operator must be determined, according to the general rule, by the equation $q\Psi_{q_0} = q_0\Psi_{q_0}$, where q_0 is used temporarily to denote concrete values of

[16] In general, if $\varphi(x)$ is a single-valued function (its inverse, however, may be many-valued), then the formula

$$\delta[\varphi(x)] = \sum_i \frac{1}{|\varphi'(\alpha_i)|}\delta(x - \alpha_i), \tag{5.13a}$$

holds, where α_i are the roots of the equation $\varphi(x) = 0$.

[17] In what follows we shall agree, for simplicity of notation, to write operators that reduce to multiplication by some quantity everywhere simply as that quantity itself.

the coordinate, in distinction from the variable q. Since this equality can be satisfied either when $\Psi_{q_0} = 0$ or when $q = q_0$, it is clear that the normalized eigenfunctions are[18]

$$\Psi_{q_0} = \delta(q - q_0). \tag{5.17}$$

§ 6. The Limiting Transition

Classical mechanics is contained in quantum mechanics as a limiting case. We must determine how this limiting transition takes place.

In quantum mechanics a wave function describes the electron, governing the various possible values of its coordinate; thus far we know only that this function solves a certain linear partial differential equation. Classical mechanics, by contrast, treats the electron as a material particle moving along a trajectory fixed completely by the equations of motion. A relation analogous in a certain sense to that between quantum and classical mechanics occurs in electrodynamics between wave optics and geometrical optics. Wave optics describes electromagnetic waves by electric- and magnetic-field vectors satisfying a particular system of linear differential equations (Maxwell's equations). Geometrical optics, however, describes the propagation of light along definite paths—rays. This analogy leads one to conclude that the quantum-to-classical limiting transition is analogous to that from wave optics to geometrical optics.

We now recall how this latter transition is carried out mathematically (see II, § 53). Let u be any component of the field in an electromagnetic wave. It may be written $u = ae^{i\varphi}$, with real amplitude a and phase φ (the phase is called the eikonal in geometrical optics). The geometrical-optics limit corresponds to short wavelengths; mathematically, this means that φ changes greatly over short distances. In particular, its absolute magnitude may be regarded as large.

Correspondingly, assume that the classical limiting case in quantum mechanics is represented by wave functions of the form $\Psi = ae^{i\varphi}$, where a varies slowly and φ takes large values. In mechanics, as is known, a particle trajectory can be found from a variational principle according to which the so-called action S of the mechanical system must be a minimum (the principle of least action). In geometrical optics the ray path is instead fixed by Fermat's principle: the "optical path length" of the ray, that is, the difference between its phases at the end and at the beginning of the path, must be a minimum.

This analogy allows us to conclude that, in the classical limiting case, the phase φ of the wave function must be proportional to the mechanical action S of the physical system under consideration: $S = \text{const} \cdot \varphi$. The proportionality factor is called *Planck's constant* and is denoted by $\hbar$[19]. Since φ is dimensionless, it has the dimensions of action, and its value is

$$\hbar = 1{,}055 \cdot 10^{-27}\ \text{erg} \cdot \text{s}.$$

[18]The expansion coefficients of an arbitrary function Ψ in these eigenfunctions are $a_{q_0} = \int \Psi(q)\,\delta(q - q_0)\,dq = \Psi(q_0)$. The probability of values of the coordinate in a given interval dq_0 is $|a_{q_0}|^2\,dq_0 = |\Psi(q_0)|^2\,dq_0$, as it should be.

[19]It was introduced into physics by *Planck* (*M. Planck*, 1900). The constant $\hbar$ used throughout this book is, strictly speaking, Planck's constant h divided by 2π (Dirac's notation).

Thus the wave function of an "almost classical" (or, as it is called, *quasiclassical*) physical system has the form

$$\Psi = a e^{iS/\hbar}. \tag{6.1}$$

Planck's constant has a fundamental part in every quantum phenomenon. Its relative magnitude (in comparison with other quantities having the same dimensions) determines the "degree of quantumness" of a given physical system. Passage from quantum to classical mechanics corresponds to a large phase and can formally be described by taking the limit $\hbar \to 0$ (just as the passage from wave to geometrical optics corresponds to the zero-wavelength limit, $\lambda \to 0$).

We have established the limiting form of the wave function, but must still ask how it is related to classical motion along a trajectory. In general, motion described by a wave function does not at all turn into motion along one fixed trajectory. Its connection with classical motion is this: if the wave function, and hence the probability distribution of the coordinates, is specified at an initial instant, the distribution thereafter "moves" as required by the laws of classical mechanics (see the end of § 17 for details).

To obtain motion along a definite trajectory one must begin with a wave function of a special kind, appreciably different from zero only within a very small region of space (a so-called *wave packet*); the dimensions of this region may be made to tend to zero together with $\hbar$. One can then state that, in the quasiclassical case, the wave packet moves through space along the particle's classical trajectory.

Finally, in the limit, quantum-mechanical operators must reduce simply to multiplication by their corresponding physical quantities.

§ 7. The Wave Function and Measurements

Let us return to the measurement process, whose properties were discussed qualitatively in § 1, and show how those properties are connected with the mathematical apparatus of quantum mechanics.

Consider a system made up of two parts: a classical apparatus and an electron (treated as a quantum object). Measurement consists in bringing these two parts into interaction. The apparatus consequently passes from its initial state into some other state, and from this change we draw conclusions about the state of the electron. States of the apparatus are distinguished by the values of a physical quantity characterizing it (or quantities)—the apparatus "reading." Denote this quantity provisionally by g, and its eigenvalues by g_n. Because the apparatus is classical, these values generally range over a continuum; solely to simplify the notation in the formulas below, however, we shall suppose that the spectrum is discrete. The states of the apparatus are described by quasiclassical wave functions, which we denote by $\Phi_n(\xi)$; the index n corresponds to the reading g_n, and ξ stands collectively for its coordinates. That the apparatus is classical manifests itself in the circumstance that at each instant one can assert with certainty that it occupies one of the known states Φ_n, with some definite value of g. Such a statement would of course be false for a quantum system.

Let $\Phi_0(\xi)$ be the wave function of the apparatus in its initial state (before measurement), and let $\Psi(q)$ be an arbitrary normalized initial wave function of the electron (q denotes its coordinates). These functions describe apparatus and electron independently, so the initial wave function of the complete system is the product

$$\Psi(q)\Phi_0(\xi). \tag{7.1}$$

The apparatus and electron then interact. In principle, the equations of quantum mechanics allow the time variation of the system's wave function to be followed. After the measurement it will plainly no longer be a product of functions of ξ and q. Expanding it in the eigenfunctions Φ_n of the apparatus (which form a complete system of functions), we obtain a sum of the form

$$\sum_n A_n(q)\Phi_n(\xi), \tag{7.2}$$

where the $A_n(q)$ are certain functions of q.

At this point the "classical nature" of the apparatus enters, together with the double role of classical mechanics as both a limiting case and a basis of quantum mechanics. As already noted, because the apparatus is classical, the quantity g (its "reading") has a definite value at every instant. It follows that after measurement the actual state of apparatus plus electron is described not by the whole sum (7.2), but by only the one term corresponding to the apparatus reading g_n:

$$A_n(q)\Phi_n(\xi). \tag{7.3}$$

Consequently, $A_n(q)$ is proportional to the electron's wave function after measurement. It is not itself that wave function, as is already evident from the fact that $A_n(q)$ is not normalized. It contains both information about the properties of the state produced and the probability, determined by the system's initial state, that the apparatus will give its nth reading.

By the linearity of the equations of quantum mechanics, the relation between $A_n(q)$ and the initial electron wave function $\Psi(q)$ is in general given by a linear integral operator,

$$A_n(q) = \int K_n(q, q')\Psi(q')\,dq', \tag{7.4}$$

whose kernel $K_n(q, q')$ characterizes the measurement process in question.

We suppose that the measurement under consideration yields a complete description of the electron's state. In other words (see § 1), in the state produced, the probabilities for all quantities must be independent of the electron's preceding state (before measurement). Mathematically, this means that the form of the functions $A_n(q)$ must be fixed by the measurement process itself and must not depend on the initial electron wave function $\Psi(q)$.

The A_n must therefore have the form

$$A_n(q) = a_n\varphi_n(q), \tag{7.5}$$

where the φ_n are certain functions that we shall take to be normalized, while only the constants a_n depend on the initial state $\Psi(q)$. In the integral relation (7.4), this corresponds to a kernel $K_n(q, q')$ which factors into functions of q and q' separately:

$$K_n(q, q') = \varphi_n(q)\Psi_n^*(q'). \tag{7.6}$$

The linear dependence of the constants a_n on $\Psi(q)$ is then expressed by formulas of the form

$$a_n = \int \Psi(q)\Psi_n^*(q)\,dq, \tag{7.7}$$

where the $\Psi_n(q)$ are certain functions fixed by the measurement process.

The functions $\varphi_n(q)$ are the normalized wave functions of the electron after measurement. We thus see how the theory's mathematical formalism represents the possibility of producing, by a measurement, an electron state described by a definite wave function. When a measurement is made on an electron having a specified wave function $\Psi(q)$, the constants a_n have a simple physical meaning: by the general rules, $|a_n|^2$ is the probability that the measurement gives the nth result. The probabilities of all results sum to unity:

$$\sum_n |a_n|^2 = 1. \tag{7.8}$$

For an arbitrary normalized function $\Psi(q)$, the validity of (7.7) and (7.8) is equivalent (compare § 3) to the statement that $\Psi(q)$ can be expanded in the functions $\Psi_n(q)$. Hence the functions $\Psi_n(q)$ form a complete normalized and mutually orthogonal set.

If the electron's initial wave function coincides with one of the functions $\Psi_n(q)$, the corresponding constant a_n is plainly unity and all the others vanish. Put differently, if the measurement is performed on an electron in the state $\Psi_n(q)$, it will with certainty give a definite (nth) result.

Taken together, these properties of the functions $\Psi_n(q)$ demonstrate that they are eigenfunctions of a certain physical quantity that characterizes the electron (call it f), and the measurement being considered may be called a measurement of that quantity.

It is essential that, in general, the functions $\Psi_n(q)$ do not coincide with the functions $\varphi_n(q)$ (indeed, the latter need not even be mutually orthogonal or form an eigenfunction system of any operator). This fact expresses first of all the non-repeatability of measurement results in quantum mechanics. If the electron was in the state $\Psi_n(q)$, a measurement of f made upon it will reveal the value f_n with certainty. After the measurement, however, the electron is in a different state $\varphi_n(q)$, in which f need no longer have any definite value. Thus, if a second measurement were made immediately after the first, the value found for f would not coincide with the result of the first measurement[20]. To predict (in the sense of calculating the probability of) the result of a repeated measurement when the first result is known, one takes from the first measurement the wave function $\varphi_n(q)$ of the state it produced, and from the second the wave function $\Psi_m(q)$ of the state whose probability is sought. More explicitly, the equations of quantum mechanics determine a wave function $\varphi_n(q,t)$ which equals $\varphi_n(q)$ at the time of the first measurement. If the second measurement is made at time t, the probability of its mth result is the squared modulus of the integral $\int \varphi_n(q,t)\Psi_m^*(q)\,dq$.

We see that measurement in quantum mechanics is "two-faced": its roles with respect to past and future do not coincide. Toward the past it "verifies" the probabilities of the

[20]There is, however, an important exception to the non-repeatability of measurements: coordinate is the one quantity whose measurement is repeatable. Two measurements of an electron's coordinate made at a sufficiently short interval must give nearby values; otherwise the electron would have infinite velocity. The mathematical origin of this exception lies in the commutativity of the coordinate with the electron–apparatus interaction energy operator, which (in the nonrelativistic theory) depends solely on coordinates.

various possible results predicted from the state created by the preceding measurement. Toward the future it creates a new state (see also § 44). A profound irreversibility is therefore inherent in the measurement process itself.

This irreversibility has important consequences of principle. As will be seen later (see the end of § 18), the fundamental equations of quantum mechanics by themselves are symmetric under reversal of the sign of time; in this respect quantum mechanics does not differ from classical mechanics. The irreversibility of measurement, however, makes the two directions of time physically inequivalent in quantum phenomena, and thus produces a distinction between future and past.

CHAPTER II

Energy and Momentum

§ 8. The Hamiltonian

In quantum mechanics the wave function Ψ fully specifies the state of a physical system. This means that prescribing Ψ at some instant not only describes all the properties of the system at that instant, but also determines its behavior at every later instant—of course, only to the degree of completeness that quantum mechanics permits at all. Mathematically, this means that at each instant the value of the time derivative $\partial\Psi/\partial t$ must be determined by the value of Ψ itself at that same instant, and, by the superposition principle, this dependence must be linear. In the most general form we may write

$$i\hbar\frac{\partial\Psi}{\partial t} = \hat{H}\Psi, \tag{8.1}$$

where $\hat{H}$ is a certain linear operator; the factor $i\hbar$ has been introduced here for a reason that will become clear below.

Since the integral $\int \Psi^*\Psi\, dq$ is constant in time, we have

$$\frac{d}{dt}\int |\Psi|^2\, dq = \int \frac{\partial\Psi^*}{\partial t}\Psi\, dq + \int \Psi^*\frac{\partial\Psi}{\partial t}\, dq = 0.$$

Substituting (8.1) here and applying the definition of the transposed operator in the first integral, we obtain (after suppressing the common factor $i/\hbar$)

$$\begin{aligned}\int \Psi\hat{H}^*\Psi^*\, dq - \int \Psi^*\hat{H}\Psi\, dq &= \int \Psi^*\tilde{H}^*\Psi\, dq - \int \Psi^*\hat{H}\Psi\, dq \\ &= \int \Psi^*(\tilde{H}^* - \hat{H})\Psi\, dq = 0.\end{aligned}$$

Because this equality must hold for an arbitrary function Ψ, it follows that identically $\hat{H}^+ = \hat{H}$; the operator $\hat{H}$ is therefore Hermitian. Let us determine the physical quantity to which it corresponds. We use the limiting expression (6.1) for the wave function and write

$$\frac{\partial\Psi}{\partial t} = \frac{i}{\hbar}\frac{\partial S}{\partial t}\Psi$$

(the slowly varying amplitude a need not be differentiated). Comparing this equality with definition (8.1), we see that in the limiting case $\hat{H}$ reduces to multiplication by $-\partial S/\partial t$. Thus this last quantity is the physical quantity into which the Hermitian operator $\hat{H}$ passes.

But $-\partial S/\partial t$ is precisely the Hamilton function H of the mechanical system. Hence $\hat{H}$ is the operator corresponding in quantum mechanics to the Hamilton function. This operator is termed the *Hamiltonian operator*, or simply the *Hamiltonian* of the system.

Once the form of the Hamiltonian is known, equation (8.1) fixes the wave functions of the given physical system. This fundamental equation of quantum mechanics is called the *wave equation*.

§ 9. Time Differentiation of Operators

In quantum mechanics, the concept of the time derivative of a physical quantity cannot be defined with the sense that it carries in classical mechanics. Indeed, the definition of the derivative in classical mechanics is connected with considering the values of the quantity at two close but different instants. But in quantum mechanics a quantity that has a definite value at some instant has no definite value at all at subsequent instants; this was discussed in more detail in § 1.

For this reason the concept of the time derivative has to be given a different definition in quantum mechanics. A natural course is to define the derivative $\dot{f}$ of a quantity f as that quantity whose mean value equals the time derivative of the mean value $\bar{f}$. Thus, by definition, we have

$$\bar{\dot{f}} = \dot{\bar{f}}. \tag{9.1}$$

Taking this definition as the starting point, it is not hard to derive an expression for the quantum-mechanical operator $\hat{\dot{f}}$ that corresponds to the quantity $\dot{f}$:

$$\begin{aligned}\bar{\dot{f}} = \dot{\bar{f}} = \frac{d}{dt}\int \Psi^* \hat{f} \Psi \, dq &= \\ = \int \Psi^* \frac{\partial \hat{f}}{\partial t} \Psi \, dq &+ \int \frac{\partial \Psi^*}{\partial t} \hat{f} \Psi \, dq + \int \Psi^* \hat{f} \frac{\partial \Psi}{\partial t} \, dq.\end{aligned}$$

In this expression $\partial \hat{f}/\partial t$ stands for the operator that results from differentiating the operator $\hat{f}$ with respect to time, on which the latter may depend parametrically. Substituting for the derivatives $\partial\Psi/\partial t$, $\partial\Psi^*/\partial t$ their expressions according to (8.1), we obtain

$$\bar{\dot{f}} = \int \Psi^* \frac{\partial \hat{f}}{\partial t} \Psi \, dq + \frac{i}{\hbar} \int (\hat{H}^* \Psi^*) \hat{f} \Psi \, dq - \frac{i}{\hbar} \int \Psi^* \hat{f} (\hat{H} \Psi) \, dq.$$

Since the operator $\hat{H}$ is Hermitian,

$$\int (\hat{H}^* \Psi^*)(\hat{f} \Psi) \, dq = \int \Psi^* \hat{H} \hat{f} \Psi \, dq;$$

thus, we have

$$\bar{\dot{f}} = \int \Psi^* \left(\frac{\partial \hat{f}}{\partial t} + \frac{i}{\hbar} \hat{H} \hat{f} - \frac{i}{\hbar} \hat{f} \hat{H} \right) \Psi \, dq.$$

Since, on the other hand, by the definition of mean values we must have $\bar{\dot{f}} = \int \Psi^* \hat{\dot{f}} \Psi \, dq$, it is seen that the expression in parentheses under the integral is the required operator $\hat{\dot{f}}$[1]:

$$\hat{\dot{f}} = \frac{\partial \hat{f}}{\partial t} + \frac{i}{\hbar} (\hat{H} \hat{f} - \hat{f} \hat{H}). \tag{9.2}$$

[1] In classical mechanics, for the total time derivative of a quantity f that is a function of the generalized

If the operator $\hat{f}$ does not depend explicitly on time, then, up to a factor, $\hat{\dot{f}}$ reduces to the commutator of the operator $\hat{f}$ with the Hamiltonian.

The physical quantities whose operators have no explicit dependence on time and which, moreover, commute with the Hamiltonian—so that $\hat{\dot{f}} = 0$—form a very important category. Such quantities are called *conserved*. For these, $\bar{\dot{f}} = \dot{\bar{f}} = 0$, i.e. $\bar{f} = \text{const}$. Put differently, the quantity's mean value stays constant in time. One can assert, moreover, that if the quantity f has a definite value in a given state (i.e. if the wave function is an eigenfunction of the operator $\hat{f}$), then at later instants as well it will have a definite value—and the very same one.

§ 10. Stationary states

The Hamiltonian of a closed system (and also of a system in a constant, but not a variable, external field) cannot contain time explicitly. This follows because all instants of time are equivalent for such a physical system. Since, on the other hand, every operator commutes with itself, we conclude that the Hamilton function is conserved for systems not situated in a variable external field. A conserved Hamilton function is called the energy. The meaning of energy conservation in quantum mechanics is that if the energy has a definite value in a given state, this value remains constant in time.

States possessing definite energy values are called the *stationary states* of the system. They are described by wave functions Ψ_n that are eigenfunctions of the Hamiltonian operator, satisfying $\hat{H}\Psi_n = E_n\Psi_n$, where E_n are the energy eigenvalues. Accordingly, the wave equation (8.1) for the function Ψ_n,

$$i\hbar\frac{\partial\Psi_n}{\partial t} = \hat{H}\Psi_n = E_n\Psi_n,$$

can be integrated directly with respect to time, giving

$$\Psi_n = \exp\left(-\frac{i}{\hbar}E_n t\right)\psi_n(q), \tag{10.1}$$

---

coordinates q_i and momenta p_i of the system, we have

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \sum_i\left(\frac{\partial f}{\partial q_i}\dot{q}_i + \frac{\partial f}{\partial p_i}\dot{p}_i\right).$$

Substituting, according to Hamilton's equations, $\dot{q}_i = \partial H/\partial p_i$, $\dot{p}_i = -\partial H/\partial q_i$, we obtain

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + [H, f], \qquad [H, f] \equiv \sum_i\left(\frac{\partial f}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial H}{\partial q_i}\right);$$

$[H, f]$ is the so-called *Poisson bracket* for the quantities f and H (see I, § 42). Comparing with expression (9.2), we see that in the transition to the classical limit the operator $i(\hat{H}\hat{f} - \hat{f}\hat{H})$ becomes zero to the first approximation, as it should, and in the next approximation (in $\hbar$) becomes the quantity $\hbar[H, f]$.

This result is also valid for any two quantities f and g: in the limit the operator $i(\hat{f}\hat{g} - \hat{g}\hat{f})$ becomes the quantity $\hbar[f, g]$, where $[f, g]$ is the Poisson bracket

$$[f, g] \equiv \sum_i\left(\frac{\partial g}{\partial q_i}\frac{\partial f}{\partial p_i} - \frac{\partial g}{\partial p_i}\frac{\partial f}{\partial q_i}\right).$$

This is a consequence of the fact that we can always, formally, conceive of a system whose Hamiltonian coincides with $\hat{g}$.

where ψ_n is a function of the coordinates alone. This determines the time dependence of the wave functions of stationary states.

We shall use the lower-case letter ψ for stationary-state wave functions without the time factor. These functions and the energy eigenvalues themselves are determined by

$$\hat{H}\psi = E\psi. \tag{10.2}$$

Among all stationary states, the one possessing the lowest energy is termed the *normal*, or *ground*, state of the system.

The expansion of any wave function Ψ over the stationary-state wave functions has the form

$$\Psi = \sum_n a_n \exp\left(-\frac{i}{\hbar}E_n(t)\right)\psi_n(q). \tag{10.3}$$

As usual, the squares $|a_n|^2$ give the probabilities of the various energy values of the system.

The coordinate probability distribution in a stationary state is determined by $|\Psi_n|^2 = |\psi_n|^2$ and is therefore independent of time. The same is true of the mean value $\bar{f} = \int \Psi_n^* \hat{f}\Psi_n, dq = \int \psi_n^* \hat{f}\psi_n, dq$ of any physical quantity f whose operator does not depend explicitly on time.

As stated above, the operator of every conserved quantity commutes with the Hamiltonian. Thus every conserved physical quantity can be measured simultaneously with the energy.

Among the stationary states there may be states corresponding to one and the same energy eigenvalue (or *energy level*) but differing in the values of other physical quantities. A level to which several distinct stationary states correspond is said to be *degenerate*. Physically, degeneracy is possible because the energy does not, in general, constitute by itself a complete set of physical quantities. The energy levels of a system are, in general, degenerate if there are two conserved quantities f and g whose operators do not commute. Let ψ be the wave function of a stationary state in which f, as well as the energy, has a definite value. Then $\hat{g}\psi$ cannot coincide with ψ up to a constant factor; otherwise g too would have a definite value, which is impossible because f and g cannot be measured simultaneously. On the other hand, $\hat{g}\psi$ is a Hamiltonian eigenfunction with the same energy E as ψ:

$$\hat{H}(\hat{g}\psi) = \hat{g}\hat{H}\psi = E(\hat{g}\psi).$$

Thus more than one eigenfunction corresponds to E, and the level is degenerate.

Any linear combination of wave functions belonging to the same degenerate energy level is also an eigenfunction with that energy. The choice of eigenfunctions for a degenerate eigenvalue is therefore not unique. Arbitrarily chosen eigenfunctions of a degenerate level are not in general mutually orthogonal, but suitable linear combinations can always be chosen to form a mutually orthogonal and normalized set[2].

These statements apply not only to energy eigenfunctions but to the eigenfunctions of any operator. Only functions belonging to different eigenvalues are automatically

[2]This can be done in infinitely many ways: a linear transformation of n functions involves n^2 independent coefficients, but the normalization and orthogonality conditions number only $n(n+1)/2$, which is fewer than n^2.

orthogonal; functions belonging to the same degenerate eigenvalue are not in general orthogonal.

If $\hat{H} = \hat{H}_1 + \hat{H}_2$, where one part contains only coordinates q_1 and the other only q_2, the eigenfunctions of $\hat{H}$ can be written as products of eigenfunctions of $\hat{H}_1$ and $\hat{H}_2$, and the energy eigenvalues are sums of the eigenvalues of these operators. The energy spectrum may be discrete or continuous. A stationary state in the discrete spectrum always corresponds to *finite motion*: neither the system nor any part of it recedes to infinity. Indeed, $\int |\Psi|^2, dq$ is finite over all space for a discrete-spectrum eigenfunction. Thus $|\Psi|^2$ decreases sufficiently rapidly and vanishes at infinity. Equivalently, the probability of infinite coordinate values is zero; the system performs finite motion, or is in a *bound* state.

For continuous-spectrum wave functions, $\int |\Psi|^2, dq$ diverges. $|\Psi|^2$ then determines coordinate probabilities only up to proportionality. The divergence is always associated with $|\Psi|^2$ failing to vanish at infinity, or vanishing too slowly. The integral over the region outside any arbitrarily large but finite closed surface therefore still diverges: the system, or some part of it, is at infinity. A superposition of continuous-spectrum stationary states may have a convergent integral and be localized in a finite region, but this region spreads without bound with time, and ultimately the system recedes to infinity.

Indeed, an arbitrary continuous-spectrum superposition is

$$\Psi = \int a_E \exp\left(-\frac{i}{\hbar}Et\right) \psi_E(q)\, dE,$$

and

$$|\Psi|^2 = \iint a_E a^*_{E'} \exp\left(\frac{i}{\hbar}(E' - E)t\right) \psi_E(q)\psi^*_{E'}(q)\, dE\, dE'.$$

If this expression is averaged over an interval T and then T is allowed to tend to infinity, the oscillating factors average to zero, and with them the entire integral. Thus the time-averaged probability of finding the system at any specified point of configuration space vanishes; this is possible only if the motion occupies all infinite space[3].

Thus stationary states of the continuous spectrum correspond to infinite motion of the system.

§ 11. Matrices

Assume for convenience that the system in question possesses a discrete energy spectrum (every relation derived below extends directly to the continuous-spectrum case as well). Let $\Psi = \sum_n a_n \Psi_n$ denote the expansion of an arbitrary wave function over the stationary-state wave functions Ψ_n. If this expansion is substituted into the definition (3.8) of the

[3]For a superposition of discrete-spectrum functions one would instead have

$$\overline{|\Psi|^2} = \overline{\sum_{n,m} a_n a^*_m \exp\left\{\frac{i}{\hbar}(E_m - E_n)t\right\} \psi_n \psi^*_m} = \sum_n |a_n \psi_n(q)|^2,$$

so the probability density remains finite under time averaging.

mean value of some quantity f, we obtain

$$\bar{f} = \sum_n \sum_m a_n^* a_m f_{nm}(t), \tag{11.1}$$

where $f_{nm}(t)$ denote the integrals

$$f_{nm}(t) = \int \Psi_n^* \hat{f} \Psi_m \, dq. \tag{11.2}$$

The set of quantities $f_{nm}(t)$ for all possible n, m is called the *matrix* of the quantity f, and each $f_{nm}(t)$ is spoken of as a *matrix element*, corresponding to the transition from state m to state n[4].

The time dependence of the matrix elements $f_{nm}(t)$ is determined (if the operator $\hat{f}$ does not contain t explicitly) by the time dependence of the functions Ψ_n. Substituting for them the expressions (10.1), we find that

$$f_{nm}(t) = f_{nm} e^{i\omega_{nm} t}, \tag{11.3}$$

where

$$\omega_{nm} = \frac{E_n - E_m}{\hbar} \tag{11.4}$$

is the transition frequency between states n and m, and the quantities

$$f_{nm} = \int \psi_n^* \hat{f} \psi_m \, dq \tag{11.5}$$

constitute the time-independent matrix of the quantity f, which is what one ordinarily uses[5].

One obtains the matrix elements of the derivative $\dot{f}$ by differentiating those of the quantity f with respect to time; this is an immediate consequence of the relation

$$\bar{\dot{f}} = \dot{\bar{f}} = \sum_n \sum_m a_n^* a_m \dot{f}_{nm}(t). \tag{11.6}$$

In view of (11.3) we thus have for the matrix elements of $\dot{f}$:

$$\dot{f}_{nm}(t) = i\omega_{nm} f_{nm}(t) \tag{11.7}$$

or (cancelling the time factor $\exp(i\omega_{nm}t)$ on both sides) for the time-independent matrix elements:

$$(\dot{f})_{nm} = i\omega_{nm} f_{nm} = \frac{i}{\hbar}(E_n - E_m) f_{nm}. \tag{11.8}$$

For simplicity of notation in the formulas, we derive below all relations for the time-independent matrix elements; exactly the same relations hold also for matrices that depend on time.

[4]The matrix representation of physical quantities was introduced by *Heisenberg* (*W. Heisenberg*) in 1925, even before Schrödinger's discovery of the wave equation. "Matrix mechanics" was subsequently developed by *Born, Heisenberg and Jordan* (*M. Born, P. Jordan*).

[5]In connection with the indeterminacy of the phase factor in the normalized wave functions (see § 2), the matrix elements f_{nm} (and $f_{nm}(t)$) are likewise defined only to within factors of the form $\exp[i(\alpha_m - \alpha_n)]$. Here too this indeterminacy does not affect the physical results.

For the matrix elements of the complex conjugate quantity f^* of f, taking into account the definition of the adjoint operator, we obtain

$$(f^*)_{nm} = \int \psi_n^* \hat{f}^+ \psi_m \, dq = \int \psi_n^* \tilde{f}^* \psi_m \, dq = \int \psi_m \hat{f}^* \psi_n^* \, dq,$$

i.e.

$$(f^*)_{nm} = (f_{mn})^*. \tag{11.9}$$

For real physical quantities, which are usually all that we consider, we therefore have

$$f_{nm} = f_{mn}^*. \tag{11.10}$$

Such matrices, like the operators corresponding to them, are called *Hermitian*.

Matrix elements with $n = m$ are called *diagonal*. These elements do not depend on time at all, and from (11.10) it is clear that they are real. The element f_{nn} is the mean value of f in the state Ψ_n.

The rule of *matrix multiplication* is readily derived. As a preliminary step, observe that the formula

$$\hat{f}\psi_n = \sum_m f_{mn}\psi_m. \tag{11.11}$$

holds. This is nothing other than the expansion of the function $\hat{f}\psi_n$ in the functions ψ_m with coefficients determined according to the general rule (3.5). Bearing this formula in mind, we write

$$\hat{f}\hat{g}\psi_n = \hat{f}(\hat{g}\psi_n) = \sum_k g_{kn}\hat{f}\psi_k = \sum_{k,m} g_{kn} f_{mk}\psi_m.$$

Since, on the other hand, we must have $\hat{f}\hat{g}\psi_n = \sum_m (fg)_{mn}\psi_m$, we obtain

$$(fg)_{mn} = \sum_k f_{mk} g_{kn}. \tag{11.12}$$

This rule matches the standard mathematical prescription for multiplying matrices: one multiplies the rows of the first factor by the columns of the second.

Specifying a matrix is equivalent to specifying the operator itself. In particular, it makes it possible in principle to determine the eigenvalues of a given physical quantity and the eigenfunctions corresponding to them.

Let us consider the values of all quantities at some definite instant of time and expand an arbitrary wave function Ψ in the eigenfunctions of the Hamiltonian, i.e. in the time-independent wave functions ψ_m of the stationary states:

$$\Psi = \sum_m c_m \psi_m, \tag{11.13}$$

where the expansion coefficients are denoted by c_m. Let us substitute this expansion into the equation $\hat{f}\Psi = f\Psi$, which determines the eigenvalues and eigenfunctions of the quantity f. We have

$$\sum_m c_m \hat{f}\psi_m = f \sum_m c_m \psi_m.$$

Multiply this equation on both sides by ψ_n^* and integrate over dq. Each of the integrals $\int \psi_n^* \hat{f} \psi_m \, dq$ on the left-hand side of the equality is the corresponding matrix element f_{nm}. On the right-hand side, all integrals $\int \psi_n^* \psi_m \, dq$ with $m \neq n$ vanish by virtue of the orthogonality of the functions ψ_m, and $\int \psi_n^* \psi_n \, dq = 1$ by virtue of their normalization[6]:

$$\sum_m f_{nm} c_m = f c_n, \tag{11.14}$$

or $\sum_m (f_{nm} - f \delta_{nm}) c_m = 0$, where $\delta_{nm} = 1$ for $n = m$ and $\delta_{nm} = 0$ for $n \neq m$.

Thus we have obtained a system of homogeneous linear algebraic equations (with unknowns c_m). As is well known, such a system has nonzero solutions only if the determinant formed from the coefficients in the equations vanishes, i.e. under the condition

$$|f_{nm} - f \delta_{nm}| = 0. \tag{11.15}$$

The roots of this equation (where f plays the role of the unknown) give the possible values of the quantity f. The set of quantities c_m satisfying equations (11.14) upon setting f equal to one of these values yields the corresponding eigenfunction. Suppose that in the definition (11.5) for the matrix elements of f we adopt, as the functions ψ_n, the eigenfunctions of this very quantity; owing to the equation $\hat{f}\psi_n = f_n \psi_n$ we obtain

$$f_{nm} = \int \psi_n^* \hat{f} \psi_m \, dq = f_m \int \psi_n^* \psi_m \, dq.$$

Since the functions ψ_m are orthogonal and normalized, it follows that $f_{nm} = 0$ for $n \neq m$ and $f_{mm} = f_m$. Thus only diagonal matrix elements turn out to be nonzero, each of them being equal to the corresponding eigenvalue of the quantity f; a matrix in which only these elements are nonzero is said to be reduced to *diagonal form*. In particular, in the ordinary representation—where the stationary-state wave functions serve as the ψ_n—the energy matrix is diagonal (along with the matrices of all other physical quantities that have definite values in the stationary states). More broadly, when the matrix of a quantity f is constructed using the eigenfunctions of an operator g, one speaks of it as the matrix of f in the representation in which g is diagonal. Everywhere where it is not otherwise specified, by the matrix of a physical quantity we shall henceforth understand the matrix in the ordinary representation, in which the energy is diagonal. All that has been said above concerning the time dependence of matrices refers, of course, only to this ordinary representation[7].

The matrix representation of operators makes it possible to prove the theorem cited in § 4: if two operators commute, they possess a common complete system of eigenfunctions.

[6]In accordance with the general rule (§ 5), the set of coefficients c_n of the expansion (11.13) can be regarded as the wave function in the "energy representation" (the variable here being the index n, which enumerates the energy eigenvalues). The matrix f_{nm} then plays the role of the operator $\hat{f}$ in this representation, whose action on the wave function is given by the expression on the left-hand side of equation (11.14). The formula $\bar{f} = \sum_n \sum_m c_n^* (f_{nm} c_m)$ then corresponds to the general formula for the mean value of a quantity expressed through its operator and the wave function of the state in question.

[7]Keeping in mind that the energy matrix is diagonal, one readily verifies that the equality (11.8) is the operator relation (9.2) written in matrix form.

Let $\hat{f}$ and $\hat{g}$ be two such operators. From $\hat{f}\hat{g} = \hat{g}\hat{f}$ and the rule of matrix multiplication (11.12) it follows that

$$\sum_k f_{mk} g_{kn} = \sum_k g_{mk} f_{kn}.$$

Taking as the system of functions ψ_n, in terms of which the matrix elements are computed, the eigenfunctions of the operator $\hat{f}$, we have $f_{mk} = 0$ for $m \neq k$, so that the written equality reduces to $f_m g_{mn} = g_{mn} f_n$, or $g_{mn}(f_m - f_n) = 0$. If all the eigenvalues f_n of the quantity f are distinct, then for all $m \neq n$ we have $f_m - f_n \neq 0$, so that we must have $g_{mn} = 0$. Thus the matrix g_{mn} likewise turns out to be diagonal, i.e. the functions ψ_n are eigenfunctions also of the physical quantity g. If, however, among the values f_n there are equal ones, then the matrix elements g_{mn} corresponding to each such group of functions ψ_n will, in general, be nonzero. However, linear combinations of the functions ψ_n corresponding to one eigenvalue of the quantity f are likewise its eigenfunctions; one can always choose these combinations in such a way as to make the corresponding nondiagonal matrix elements g_{mn} vanish, and thus in this case too we obtain a system of functions that are eigenfunctions simultaneously for the operators $\hat{f}$ and $\hat{g}$.

Let us note a useful formula

$$\left(\frac{\partial H}{\partial \lambda}\right)_{nn} = \frac{\partial E_n}{\partial \lambda}, \tag{11.16}$$

where λ — a parameter on which the Hamiltonian $\hat{H}$ depends (and with it the energy eigenvalues E_n). Indeed, differentiating the equation $(\hat{H} - E_n)\psi_n = 0$ with respect to λ and then multiplying it from the left by ψ_n^*, we obtain

$$\psi_n^*(\hat{H} - E_n)\frac{\partial \psi_n}{\partial \lambda} = \psi_n^*\left(\frac{\partial E_n}{\partial \lambda} - \frac{\partial \hat{H}}{\partial \lambda}\right)\psi_n.$$

On integrating over dq, the left-hand side of this equality vanishes, since

$$\int \psi_n^*(\hat{H} - E_n)\frac{\partial \psi_n}{\partial \lambda}\, dq = \int \frac{\partial \psi_n}{\partial \lambda}(\hat{H} - E_n)^*\psi_n^*\, dq$$

by virtue of the Hermiticity of the operator $\hat{H}$. The right-hand side gives the desired equality.

In the modern literature, a system of notation (introduced by *Dirac*) is widely used, in which matrix elements are written as[8]

$$\langle n|f|m\rangle. \tag{11.17}$$

This symbol is composed of the notation for the quantity f and the symbols $|m\rangle$ and $|n\rangle$, denoting respectively the initial and final states. If there is a complete set of state functions $|n_1\rangle, |n_2\rangle, \ldots$, then the expansion coefficients of the wave function of the state $|m\rangle$ are denoted by $\langle n_i|m\rangle$:

$$\langle n_i|m\rangle = \int \psi_{n_i}^* \psi_m\, dq. \tag{11.18}$$

[8]We will use both ways of denoting matrix elements in this book. The notation (11.17) is especially convenient when each index has to be written as a combination of several letters.

§ 12. Transformation of matrices

The matrix elements of one and the same physical quantity may be defined with respect to different sets of wave functions. These may be, for example, wave functions of stationary states described by different sets of physical quantities, or wave functions of stationary states of the same system placed in different external fields. The question therefore arises of transforming matrices from one representation to another.

Let $\psi_n(q)$ and $\psi'_n(q)$ $(n = 1, 2, \ldots)$ be two complete systems of orthonormal functions. They are related by a linear transformation

$$\psi'_n = \sum_m S_{mn}\psi_m, \tag{12.1}$$

which is simply the expansion of the functions ψ'_n in the complete system ψ_m. This transformation can be written in operator form as

$$\psi'_n = \hat{S}\psi_n. \tag{12.2}$$

The operator $\hat{S}$ must satisfy a certain condition if the functions ψ'_n are to be orthonormal whenever the functions ψ_n are. Indeed, substituting (12.2) into $\int \psi'^*_m \psi'_n \, dq = \delta_{mn}$ and using the definition (3.14) of the transposed operator, we obtain

$$\int (\hat{S}\psi_m)^* \hat{S}\psi_n \, dq = \int \psi^*_m \tilde{S}^* \hat{S}\psi_n \, dq = \delta_{mn}.$$

For this equality to hold for every m, n, we must have $\tilde{S}^*\hat{S} = 1$, or

$$\tilde{S}^* \equiv \hat{S}^+ = \hat{S}^{-1}, \tag{12.3}$$

i.e. the inverse operator coincides with the adjoint. Operators having this property are called *unitary*. By virtue of this property the transformation $\psi_n = \hat{S}^{-1}\psi'_n$, inverse to (12.1), is given by

$$\psi_n = \sum_m S^*_{nm}\psi'_m. \tag{12.4}$$

Writing $\hat{S}^+\hat{S} = 1$ or $\hat{S}\hat{S}^+ = 1$ in matrix form gives the unitarity conditions

$$\sum_l S^*_{lm} S_{ln} = \delta_{mn} \tag{12.5}$$

or

$$\sum_l S^*_{ml} S_{nl} = \delta_{mn}. \tag{12.6}$$

Now consider a physical quantity f and write its matrix elements in the "new" representation, i.e. with respect to the functions ψ'_n. They are given by

$$\int \psi'^*_m \hat{f} \psi'_n \, dq = \int (\tilde{S}^*\psi^*_m)(\hat{f}\hat{S}\psi_n) \, dq = \int \psi^*_m \hat{S}^{-1} \hat{f} \hat{S}\psi_n \, dq.$$

It follows that the matrix of $\hat{f}$ in the new representation coincides with the matrix of the operator

$$\hat{f}' = \hat{S}^{-1}\hat{f}\hat{S} \tag{12.7}$$

in the old representation[9]. The *trace* of a matrix is defined as the sum of its diagonal elements and is written Tr f[10]:

$$\operatorname{Tr} f = \sum_n f_{nn}. \tag{12.8}$$

First note that the trace of a product of two matrices does not depend on the order of the factors:

$$\operatorname{Tr}(fg) = \operatorname{Tr}(gf). \tag{12.9}$$

Indeed, by the rule for matrix multiplication,

$$\operatorname{Tr}(fg) = \sum_n \sum_k f_{nk} g_{kn} = \sum_k \sum_n g_{kn} f_{nk} = \operatorname{Tr}(gf).$$

Similarly, the trace of a product of several matrices is unchanged by a cyclic permutation of the factors; thus

$$\operatorname{Tr}(fgh) = \operatorname{Tr}(hfg) = \operatorname{Tr}(ghf). \tag{12.10}$$

The most important property of the trace is its independence of the system of functions relative to which the matrix elements are defined. Indeed,

$$\operatorname{Tr} f' = \operatorname{Tr}(S^{-1} f S) = \operatorname{Tr}(S S^{-1} f) = \operatorname{Tr} f. \tag{12.11}$$

Also, a unitary transformation leaves invariant the sum of the squared moduli of the transformed functions. From (12.6),

$$\sum_i |\psi_i'|^2 = \sum_{k,l,i} S_{ki} S_{li}^* \psi_k \psi_l^* = \sum_{k,l} \delta_{kl} \psi_k \psi_l^* = \sum_k |\psi_k|^2. \tag{12.12}$$

Every unitary operator can be represented as

$$\hat{S} = e^{i\hat{R}}, \tag{12.13}$$

where $\hat{R}$ is Hermitian; indeed, $\hat{R}^+ = \hat{R}$ gives $\hat{S}^+ = e^{-i\hat{R}^+} = e^{-i\hat{R}} = \hat{S}^{-1}$.

We also note the expansion

$$\hat{f}' = \hat{S}^{-1} \hat{f} \hat{S} = \hat{f} + \{\hat{f}, i\hat{R}\} + \frac{1}{2}\{\{\hat{f}, i\hat{R}\}, i\hat{R}\} + \ldots, \tag{12.14}$$

which is readily verified directly by expanding the factors $\exp(\pm i\hat{R})$ in powers of $\hat{R}$. This expansion can be useful when $\hat{R}$ is proportional to a small parameter, so that (12.14) becomes an expansion in powers of that parameter.

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[9]If $\{\hat{f}, \hat{g}\} = -i\hbar\hat{c}$ is the commutation rule for two operators, then after transformation (12.7) we obtain $\{\hat{f}', \hat{g}'\} = -i\hbar\hat{c}'$, so the rule is unchanged. It was noted in the note on p. 46 that c is the quantum analogue of the classical Poisson bracket $[f, g]$. Poisson brackets in classical mechanics, however, are invariant under canonical variable transformations (see Vol. I, § 45). One may therefore say that unitary transformations in quantum mechanics serve as the counterpart of canonical transformations in classical mechanics.

[10]The source's notation comes from the German *Spur*, trace. The notation Tr, from the English *trace*, is also used. Consideration of a matrix trace of course assumes convergence of the sum over n.

§ 13. Heisenberg representation of operators

In the mathematical apparatus of quantum mechanics as here presented, the operators corresponding to the various physical quantities act on functions of the coordinates and as such ordinarily contain no explicit dependence on time. The dependence of the mean values of physical quantities on time arises only through the time dependence of the wave function of the state, in accordance with the formula

$$\bar{f}(t) = \int \Psi^*(q,t)\hat{f}\Psi(q,t)\,dq. \tag{13.1}$$

The apparatus of quantum mechanics can, however, also be formulated in a somewhat different, equivalent form, in which the dependence on time is transferred from the wave functions to the operators. Although in this book we shall not make use of such a representation of operators—the so-called Heisenberg representation, as distinct from the Schrödinger representation—we shall formulate it here, with later applications in relativistic theory in view.

Let us introduce the unitary operator (cf. (12.13))

$$\hat{S} = \exp\left(-\frac{i}{\hbar}\hat{H}t\right), \tag{13.2}$$

where $\hat{H}$ is the Hamiltonian of the system. By definition its eigenfunctions coincide with the eigenfunctions of the operator $\hat{H}$, i.e. with the wave functions $\psi_n(q)$ of the stationary states, and moreover

$$\hat{S}\psi_n(q) = \exp\left(-\frac{i}{\hbar}E_n t\right)\psi_n(q) = \Psi_n(q,t). \tag{13.3}$$

It follows that expansion (10.3) of any wave function Ψ over the stationary-state wave functions can be recast in operator form as

$$\Psi(q,t) = \hat{S}\Psi(q,0), \tag{13.4}$$

i.e. the action of the operator $\hat{S}$ carries the wave function of the system at some initial instant of time over into the wave function at an arbitrary instant of time. On introducing, in accordance with (12.7), the time-dependent operator

$$\hat{f}(t) = \hat{S}^{-1}\hat{f}\hat{S}, \tag{13.5}$$

we obtain

$$\bar{f}(t) = \int \Psi^*(q,0)\hat{f}(t)\Psi(q,0)\,dq, \tag{13.6}$$

i.e. we bring the formula for the mean value of the quantity f (which serves as the definition of the operators) into a form in which the dependence on time is wholly transferred to the operator.

It is obvious that the matrix elements of the operator (13.5), with respect to the wave functions of stationary states, coincide with the time-dependent matrix elements $f_{nm}(t)$ defined by formula (11.3).

Finally, on differentiating expression (13.5) with respect to time (assuming here that the operators $\hat{f}$ and $\hat{H}$ themselves contain no t), we obtain the equation

$$\frac{\partial \hat{f}(t)}{\partial t} = \frac{i}{\hbar}\left[\hat{H}\hat{f}(t) - \hat{f}(t)\hat{H}\right], \tag{13.7}$$

which is analogous to formula (9.2) but has a somewhat different meaning: expression (9.2) constitutes the definition of the operator $\hat{\dot{f}}$ corresponding to the physical quantity $\dot{f}$, whereas the left-hand side of equation (13.7) contains the time derivative of the operator of the quantity f itself.

§ 14. The density matrix

Describing a system by its wave function represents the fullest description that quantum mechanics allows, in the sense indicated at the end of § 1.

States not admitting such a description arise when the system considered is part of a larger closed system. Suppose the closed system as a whole is in a state described by $\Psi(q, x)$, where x denotes the coordinates of the system considered and q the remaining coordinates of the closed system. In general this function does not factor into functions of x alone and q alone, so that the system has no wave function of its own[11].

Let f be a physical quantity pertaining to our system. Its operator acts on the x coordinates only, not on q. Its mean value is

$$\bar{f} = \iint \Psi^*(q, x)\hat{f}\Psi(q, x)\, dq\, dx. \tag{14.1}$$

Introduce the function

$$\rho(x, x') = \int \Psi(q, x)\Psi^*(q, x')\, dq, \tag{14.2}$$

where integration is only over q; it is called the system's *density matrix*. From (14.2),

$$\rho^*(x, x') = \rho(x', x). \tag{14.3}$$

The "diagonal elements"

$$\rho(x, x) = \int |\Psi(q, x)|^2\, dq$$

determine the coordinate probability distribution.

In terms of the density matrix,

$$\bar{f} = \int [\hat{f}\rho(x, x')]_{x'=x}\, dx. \tag{14.4}$$

Here $\hat{f}$ acts only on the variables x in $\rho(x, x')$; after its action has been evaluated, one sets $x' = x$. Thus the density matrix determines the mean of every quantity characterizing

[11] For $\Psi(q, x)$ to factor in this way at a given instant, the measurement that produced the state must describe completely and separately the system considered and the remainder of the closed system. For the factorization to persist, these two parts must also not interact (see § 2). Neither condition is assumed here.

the system and hence also the probabilities of its possible values. A state having no wave function can therefore be described by a density matrix. The density matrix contains none of the coordinates q that do not belong to the system, although it naturally depends in substance on the state of the closed system as a whole.

The density-matrix description is the most general form of quantum-mechanical description of systems. The wave-function description is the special case $\rho(x, x') = \Psi(x)\Psi^*(x')$. There is an important distinction. For a state possessing a wave function (a *pure* state), there is always a complete system of measurements whose results can be predicted with certainty (mathematically, Ψ is an eigenfunction of some operator). For states possessing only a density matrix (*mixed* states), no complete system of measurements has uniquely predictable results.

Suppose the system is closed, or became closed at some instant. We derive an equation governing the time dependence of its density matrix, analogous to the wave equation. The desired linear differential equation for $\rho(x, x', t)$ must also hold in the special case

$$\rho(x, x', t) = \Psi(x, t)\Psi^*(x', t).$$

Differentiating and using (8.1),

$$\begin{aligned} i\hbar\frac{\partial\rho}{\partial t} &= i\hbar\Psi^*(x', t)\frac{\partial\Psi(x, t)}{\partial t} + i\hbar\Psi(x, t)\frac{\partial\Psi^*(x', t)}{\partial t} \\ &= \Psi^*(x', t)\hat{H}\Psi(x, t) - \Psi(x, t)\hat{H}'^*\Psi^*(x', t). \end{aligned}$$

Here $\hat{H}$ acts on functions of x, while $\hat{H}'$ is the same operator acting on functions of x'. Bringing the other factors under the respective operator signs gives

$$i\hbar\frac{\partial\rho(x, x', t)}{\partial t} = (\hat{H} - \hat{H}'^*)\rho(x, x', t). \tag{14.5}$$

Let $\Psi_n(x, t)$ be the stationary-state wave functions. Expanding the density matrix in these functions gives the double series

$$\rho(x, x', t) = \sum_m \sum_n a_{mn}\psi_n^*(x')\psi_m(x)\exp\left[\frac{i}{\hbar}(E_n - E_m)t\right]. \tag{14.6}$$

This plays the same role for the density matrix as (10.3) does for the wave function. In place of the single set a_n there is the double set a_{mn}, which, like the density matrix, is Hermitian:

$$a_{mn} = a_{nm}^*. \tag{14.7}$$

Substitution in (14.4) gives

$$\bar{f} = \sum_m \sum_n a_{mn} \int \Psi_n^*(x, t)\hat{f}\Psi_m(x, t)\, dx,$$

or

$$\bar{f} = \sum_m \sum_n a_{mn}f_{nm}(t) = \sum_m \sum_n a_{mn}f_{nm}\exp\left[\frac{i}{\hbar}(E_n - E_m)t\right]. \tag{14.8}$$

This is analogous to (11.1)[12].

The a_{mn} must satisfy inequalities. Since the diagonal elements $\rho(x,x)$ are probabilities, they must be positive. Hence the quadratic form

$$\sum_n \sum_m a_{mn}\xi_m\xi_n^*$$

must be positive definite for arbitrary complex ξ_n. The standard conditions for quadratic forms therefore apply; in particular,

$$a_{nn} > 0, \tag{14.9}$$

and for every a_{mm}, a_{nn}, a_{mn},

$$a_{mm}a_{nn} \geq |a_{mn}|^2. \tag{14.10}$$

The pure case, in which the density matrix is a product of functions, corresponds to

$$a_{mn} = a_m a_n^*. \tag{14.11}$$

A simple criterion distinguishes a pure from a mixed state. In the pure case,

$$(a^2)_{mn} = \sum_k a_{mk}a_{kn} = \sum_k a_m a_k^* a_k a_n^* = a_m a_n^* \sum_k |a_k|^2 = a_m a_n^*,$$

and therefore

$$(a^2)_{mn} = a_{mn}. \tag{14.12}$$

Thus the square of the density matrix coincides with the matrix itself.

§ 15. Momentum

Consider a closed system of particles in no external field. Since all positions of the system as a whole are equivalent, its Hamiltonian is unchanged by an arbitrary parallel displacement. It is enough to require this for every infinitesimal displacement; it then holds for finite displacements as well.

Under an infinitesimal displacement $\delta\mathbf{r}$, every radius vector $\mathbf{r}_a$ acquires the same increment. Thus $\mathbf{r}_a \to \mathbf{r}_a + \delta\mathbf{r}$, and

$$\psi(\mathbf{r}_1 + \delta\mathbf{r}, \mathbf{r}_2 + \delta\mathbf{r}, \ldots) = \left(1 + \delta\mathbf{r} \cdot \sum_a \boldsymbol{\nabla}_a\right)\psi(\mathbf{r}_1, \mathbf{r}_2, \ldots).$$

Here a labels the particles and $\boldsymbol{\nabla}_a$ differentiates with respect to $\mathbf{r}_a$. The expression in parentheses is the operator of an infinitesimal displacement.

If a transformation does not change the Hamiltonian, transforming $\hat{H}\psi$ gives the same result as first transforming ψ and then applying $\hat{H}$. If $\hat{O}$ produces the transformation, then $\hat{O}(\hat{H}\psi) = \hat{H}(\hat{O}\psi)$, so $\hat{O}\hat{H} - \hat{H}\hat{O} = 0$: $\hat{H}$ commutes with $\hat{O}$.

[12] The quantities a_{mn} constitute the density matrix in the energy representation. Description of states by such a matrix was introduced independently by Landau and Bloch (*F. Bloch*) in 1927.

Here $\hat{O}$ is the infinitesimal-displacement operator. The unit operator commutes with every operator, and the constant $\delta\mathbf{r}$ may be taken outside $\hat{H}$; hence

$$\left(\sum_a \boldsymbol{\nabla}_a\right)\hat{H} - \hat{H}\left(\sum_a \boldsymbol{\nabla}_a\right) = 0. \tag{15.1}$$

An explicitly time-independent operator commuting with the Hamiltonian corresponds to a conserved quantity. The quantity whose conservation follows from the homogeneity of space is momentum (cf. Vol. I, § 7). Thus (15.1) expresses momentum conservation; $\sum_a \boldsymbol{\nabla}_a$ corresponds, up to a constant factor, to the total momentum, and each term to the momentum of one particle.

The factor relating the momentum operator to $\boldsymbol{\nabla}$ follows from the classical limit and equals $-i\hbar$:

$$\hat{\mathbf{p}} = -i\hbar\boldsymbol{\nabla}, \tag{15.2}$$

or

$$\hat{p}_x = -i\hbar\frac{\partial}{\partial x}, \qquad \hat{p}_y = -i\hbar\frac{\partial}{\partial y}, \qquad \hat{p}_z = -i\hbar\frac{\partial}{\partial z}.$$

Indeed, using (6.1),

$$\hat{\mathbf{p}}\Psi = -i\hbar\boldsymbol{\nabla}\Psi = \Psi\boldsymbol{\nabla}S.$$

Thus classically the operator acts by multiplication by $\boldsymbol{\nabla}S$, which is the classical momentum (see Vol. I, § 43).

The operator (15.2) is Hermitian, as required. For functions $\psi(x)$ and $\varphi(x)$ vanishing at infinity, integration by parts gives

$$\int \varphi\hat{p}_x\psi\,dx = -i\hbar\int \varphi\frac{\partial\psi}{\partial x}\,dx = i\hbar\int \psi\frac{\partial\varphi}{\partial x}\,dx = \int \psi\hat{p}_x^*\varphi\,dx,$$

which is the Hermiticity condition.

Since differentiations with respect to different variables commute,

$$\hat{p}_x\hat{p}_y - \hat{p}_y\hat{p}_x = 0, \quad \hat{p}_x\hat{p}_z - \hat{p}_z\hat{p}_x = 0, \quad \hat{p}_y\hat{p}_z - \hat{p}_z\hat{p}_y = 0. \tag{15.3}$$

Thus all three momentum components may have definite values simultaneously.

Their eigenfunctions and eigenvalues satisfy

$$-i\hbar\boldsymbol{\nabla}\psi = \mathbf{p}\psi, \tag{15.4}$$

with solutions

$$\psi = \text{const}\cdot e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}. \tag{15.5}$$

The three components completely determine the wave function and form one possible complete set. Their eigenvalues form a continuous spectrum from $-\infty$ to ∞.

By (5.4), $\int \psi_{\mathbf{p}'}^*\psi_{\mathbf{p}},dV$ over all space should equal $\delta(\mathbf{p}'-\mathbf{p})$[13]. For later applications it is more natural to normalize to the delta function of the momentum difference divided by $2\pi\hbar$:

$$\int \psi_{\mathbf{p}'}^*\psi_{\mathbf{p}},dV = \delta\left(\frac{\mathbf{p}'-\mathbf{p}}{2\pi\hbar}\right),$$

[13]For a vector argument, $\delta(\mathbf{a}) = \delta(a_x)\delta(a_y)\delta(a_z)$.

or equivalently

$$\int \psi^*_{\mathbf{p}'}\psi_{\mathbf{p}}, dV = (2\pi\hbar)^3\delta(\mathbf{p}' - \mathbf{p}). \tag{15.6}$$

Each one-dimensional factor obeys, for example, $\delta[(p'_x - p_x)/(2\pi\hbar)] = 2\pi\hbar\delta(p'_x - p_x)$. Integration uses[14]

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{i\alpha\xi}\, d\xi = \delta(\alpha). \tag{15.7}$$

It follows that const = 1 in (15.5)[15]:

$$\psi_{\mathbf{p}} = e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}. \tag{15.8}$$

Expansion of an arbitrary $\psi(\mathbf{r})$ in momentum eigenfunctions is its Fourier integral:

$$\psi(\mathbf{r}) = \int a(\mathbf{p})\psi_{\mathbf{p}}(\mathbf{r})\frac{d^3p}{(2\pi\hbar)^3} = \int a(\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}\frac{d^3p}{(2\pi\hbar)^3}. \tag{15.9}$$

Here $d^3p = dp_x dp_y dp_z$, and

$$a(\mathbf{p}) = \int \psi(\mathbf{r})\psi^*_{\mathbf{p}}(\mathbf{r})\, dV = \int \psi(\mathbf{r})e^{-i\mathbf{p}\cdot\mathbf{r}/\hbar}\, dV. \tag{15.10}$$

The function $a(\mathbf{p})$ is the momentum-representation wave function; $|a(\mathbf{p})|^2 d^3p/(2\pi\hbar)^3$ is the probability that the momentum lies in d^3p. Just as $\hat{\mathbf{p}}$ represents momentum in coordinate space, introduce $\hat{\mathbf{r}}$ for the coordinates in momentum space, defined by

$$\bar{\mathbf{r}} = \int a^*(\mathbf{p})\hat{\mathbf{r}}a(\mathbf{p})\frac{d^3p}{(2\pi\hbar)^3}. \tag{15.11}$$

But also $\bar{\mathbf{r}} = \int \psi^*\mathbf{r}\psi, dV$. Substituting (15.9) and integrating by parts,

$$\mathbf{r}\psi(\mathbf{r}) = \int i\hbar e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}\frac{\partial a(\mathbf{p})}{\partial\mathbf{p}}\frac{d^3p}{(2\pi\hbar)^3},$$

and hence

$$\bar{\mathbf{r}} = \int i\hbar a^*(\mathbf{p})\frac{\partial a(\mathbf{p})}{\partial\mathbf{p}}\frac{d^3p}{(2\pi\hbar)^3}.$$

Comparison with (15.11) gives

$$\hat{\mathbf{r}} = i\hbar\frac{\partial}{\partial\mathbf{p}}. \tag{15.12}$$

In this representation momentum is simply multiplication by $\mathbf{p}$.

[14]The formula means that its left-hand side has the defining property (5.8) of the delta function. Substitution in (5.8) gives the Fourier formula

$$f(a) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x)e^{i\xi(x-a)}\, dx\, \frac{d\xi}{2\pi}.$$

[15]With this normalization $|\psi|^2 = 1$: the function is normalized to "one particle per unit volume." This agreement is not accidental; cf. the note on p. 221.

Finally, express the operator of translation through an arbitrary finite distance $\mathbf{a}$ in terms of $\hat{\mathbf{p}}$. By definition, $\hat{T}_{\mathbf{a}}\psi(\mathbf{r}) = \psi(\mathbf{r}+\mathbf{a})$. Expanding $\psi(\mathbf{r}+\mathbf{a})$ in a Taylor series gives

$$\psi(\mathbf{r}+\mathbf{a}) = \psi(\mathbf{r}) + \mathbf{a}\cdot\frac{\partial\psi(\mathbf{r})}{\partial\mathbf{r}} + \ldots$$

or, on introducing $\hat{\mathbf{p}} = -i\hbar\boldsymbol{\nabla}$,

$$\psi(\mathbf{r}+\mathbf{a}) = \left[1 + \frac{i}{\hbar}\mathbf{a}\cdot\hat{\mathbf{p}} + \frac{1}{2}\left(\frac{i}{\hbar}\mathbf{a}\cdot\hat{\mathbf{p}}\right)^2 + \ldots\right]\psi(\mathbf{r}).$$

Thus

$$\hat{T}_{\mathbf{a}} = \exp\left(\frac{i}{\hbar}\mathbf{a}\cdot\hat{\mathbf{p}}\right). \tag{15.13}$$

This is the required *finite-displacement operator*.

§ 16. Uncertainty Relations

Let us derive the commutation rules for the momentum and coordinate operators. Successive differentiation with respect to one of the variables x, y, z and multiplication by another of them gives a result independent of the order of these operations. Hence

$$\hat{p}_x y - y\hat{p}_x = 0, \qquad \hat{p}_x z - z\hat{p}_x = 0, \tag{16.1}$$

and similarly for $\hat{p}_y$ and $\hat{p}_z$.

To derive the commutation rule for $\hat{p}_x$ and x, write

$$(\hat{p}_x x - x\hat{p}_x)\psi = -i\hbar\frac{\partial}{\partial x}(x\psi) + i\hbar x\frac{\partial\psi}{\partial x} = -i\hbar\psi.$$

We see that the action of $\hat{p}_x x - x\hat{p}_x$ reduces to multiplication of the function by $-i\hbar$; the same is of course true for the commutation of $\hat{p}_y$ with y and of $\hat{p}_z$ with z. Thus[16]

$$\hat{p}_x x - x\hat{p}_x = -i\hbar, \qquad \hat{p}_y y - y\hat{p}_y = -i\hbar, \qquad \hat{p}_z z - z\hat{p}_z = -i\hbar. \tag{16.2}$$

Relations (16.1) and (16.2) can be written together as

$$\hat{p}_i x_k - x_k\hat{p}_i = -i\hbar\delta_{ik}, \qquad i, k = x, y, z. \tag{16.3}$$

Before examining the physical meaning of these relations and their consequences, we give two formulas useful below. If $f(\mathbf{r})$ is a function of the coordinates, then

$$\hat{\mathbf{p}}f(\mathbf{r}) - f(\mathbf{r})\hat{\mathbf{p}} = -i\hbar\nabla f. \tag{16.4}$$

Indeed,

$$(\hat{\mathbf{p}}f - f\hat{\mathbf{p}})\psi = -i\hbar[\nabla(f\psi) - f\nabla\psi] = -i\hbar\psi\nabla f.$$

[16] These relations, discovered in matrix form by *Heisenberg* in 1925, served as the starting point in the creation of quantum mechanics.

An analogous relation holds for the commutator with a function of the momentum operator:

$$f(\mathbf{p})\hat{\mathbf{r}} - \hat{\mathbf{r}}f(\mathbf{p}) = -i\hbar\frac{\partial f}{\partial \mathbf{p}}. \tag{16.5}$$

It may be derived in the same way as (16.4) by working in the momentum representation and using (15.12) for the coordinate operators.

Relations (16.1) and (16.2) show that a particle coordinate along one axis can have a definite value simultaneously with the momentum components along the other two axes; a coordinate and the momentum component along that same axis do not exist simultaneously. In particular, a particle cannot occupy a definite point in space while also having a definite momentum $\mathbf{p}$.

Let the particle occupy a finite spatial region whose dimensions along the three axes are of order Δx, Δy, Δz. Let its mean momentum be $\mathbf{p}_0$. Mathematically, this means that the wave function has the form $\psi = u(\mathbf{r})e^{i\mathbf{p}_0\cdot\mathbf{r}/\hbar}$, where $u(\mathbf{r})$ is appreciably different from zero only within the specified region.

Expand ψ in eigenfunctions of the momentum operator, that is, as a Fourier integral. The coefficients $a(\mathbf{p})$ in this expansion are given by (15.10) as integrals of functions of the form $u(\mathbf{r})e^{i(\mathbf{p}_0-\mathbf{p})\cdot\mathbf{r}/\hbar}$. For such an integral to be appreciably different from zero, the periods of the oscillating factor $e^{i(\mathbf{p}_0-\mathbf{p})\cdot\mathbf{r}/\hbar}$ must not be small compared with the dimensions Δx, Δy, and Δz of the region where $u(\mathbf{r})$ differs from zero. Thus $a(\mathbf{p})$ is appreciably different from zero only for values of $\mathbf{p}$ satisfying $(p_{0x} - p_x)\Delta x/\hbar \lesssim 1, \ldots$ Since $|a(\mathbf{p})|^2$ determines the probabilities of the various momentum values, the intervals of p_x, p_y, p_z in which $a(\mathbf{p})$ is nonzero are precisely the intervals in which the particle's momentum components may lie in the state considered. Denoting these intervals by Δp_x, Δp_y, and Δp_z, we therefore have

$$\Delta p_x \Delta x \sim \hbar, \qquad \Delta p_y \Delta y \sim \hbar, \qquad \Delta p_z \Delta z \sim \hbar. \tag{16.6}$$

These *uncertainty relations* were established by *Heisenberg* in 1927.

We see that the more accurately a particle coordinate is known (the smaller Δx is), the greater is the uncertainty Δp_x in the momentum component along the same axis, and conversely. In particular, if a particle occupies a strictly definite point of space ($\Delta x = \Delta y = \Delta z = 0$), then $\Delta p_x = \Delta p_y = \Delta p_z = \infty$. All momentum values are then equally probable. Conversely, if the particle has a strictly definite momentum $\mathbf{p}$, all its positions in space are equally probable (this also follows directly from the wave function (15.8), whose squared modulus is entirely independent of the coordinates).

If the coordinate and momentum uncertainties are characterized by their root-mean-square fluctuations,

$$\delta x = \sqrt{\overline{(x-\bar{x})^2}}, \qquad \delta p_x = \sqrt{\overline{(p_x-\bar{p}_x)^2}},$$

an exact bound can be given for the least possible value of their product (*H. Weyl*).

Consider the one-dimensional case, a packet with wave function $\psi(x)$ depending on only one coordinate; for simplicity, suppose that the mean values of x and p_x in this state are zero. Begin with the evident inequality

$$\int_{-\infty}^{\infty} \left| \alpha x \psi + \frac{d\psi}{dx} \right|^2 dx \geq 0,$$

where α is an arbitrary real constant. In evaluating this integral, note that

$$\int x^2 |\psi|^2 \, dx = (\delta x)^2,$$

$$\int \left(x \frac{d\psi^*}{dx} \psi + x\psi^* \frac{d\psi}{dx} \right) dx = \int x \frac{d|\psi|^2}{dx} \, dx = - \int |\psi|^2 \, dx = -1,$$

$$\int \frac{d\psi^*}{dx} \frac{d\psi}{dx} \, dx = - \int \psi^* \frac{d^2\psi}{dx^2} \, dx = \frac{1}{\hbar^2} \int \psi^* \hat{p}_x^2 \psi \, dx = \frac{1}{\hbar^2} (\delta p_x)^2,$$

and obtain

$$\alpha^2 (\delta x)^2 - \alpha + \frac{(\delta p_x)^2}{\hbar^2} \geq 0.$$

For this quadratic trinomial in α to be nonnegative for every value of α, its discriminant must be negative. It follows that

$$\delta x \, \delta p_x \geq \hbar/2. \tag{16.7}$$

The least possible value of the product is $\hbar/2$.

This value is attained by wave packets described by functions of the form

$$\psi = \frac{1}{(2\pi)^{1/4} \sqrt{\delta x}} \exp \left(\frac{i}{\hbar} p_0 x - \frac{x^2}{4(\delta x)^2} \right), \tag{16.8}$$

where p_0 and δx are constants. The probabilities of the various coordinate values in such a state are

$$|\psi|^2 = \frac{1}{\sqrt{2\pi} \, \delta x} \exp \left(- \frac{x^2}{2(\delta x)^2} \right),$$

that is, they have a Gaussian distribution about the origin (mean value $\bar{x} = 0$), with root-mean-square fluctuation δx. The wave function in the momentum representation is

$$a(p_x) = \int_{-\infty}^{\infty} \psi(x) \exp \left(-i \frac{p_x x}{\hbar} \right) dx.$$

Evaluation of the integral gives an expression of the form

$$a(p_x) = \text{const} \cdot \exp \left(- \frac{(\delta x)^2 (p_x - p_0)^2}{\hbar^2} \right).$$

The probability distribution of the momentum values, $|a(p_x)|^2$, is also Gaussian, about the mean $\bar{p}_x = p_0$, and has root-mean-square fluctuation $\delta p_x = \hbar/(2\delta x)$. Thus the product $\delta p_x \delta x$ is exactly $\hbar/2$.

Finally, let us derive one more useful relation. Let f and g be two physical quantities whose operators obey the commutation rule

$$\hat{f}\hat{g} - \hat{g}\hat{f} = -i\hbar\hat{c}, \tag{16.9}$$

where $\hat{c}$ is the operator of a physical quantity c. The factor $\hbar$ has been introduced on the right because in the classical limit ($\hbar \to 0$) all operators of physical quantities reduce to multiplication by those quantities and commute with one another. Thus, in the "quasiclassical" case, the right-hand side of (16.9) may be taken as zero in the first approximation. In the next approximation, $\hat{c}$ may be replaced by the operator of simple multiplication by c. We then have

$$\hat{f}\hat{g} - \hat{g}\hat{f} = -i\hbar c.$$

This equation is exactly analogous to $\hat{p}_x x - x\hat{p}_x = -i\hbar$, except that the quantity $\hbar c$ occurs in place of the constant $\hbar$[17]. By analogy with $\Delta x \Delta p_x \sim \hbar$, we therefore conclude that in the quasiclassical case the quantities f and g obey the uncertainty relation

$$\Delta f \Delta g \sim \hbar c. \tag{16.10}$$

In particular, if one of the quantities is the energy ($f \equiv H$) and the operator of the other (g) has no explicit time dependence, then by (9.2) $c = \dot{g}$, and the quasiclassical uncertainty relation is

$$\Delta E \Delta g \sim \hbar \dot{g}. \tag{16.11}$$

[17]The classical quantity c is the Poisson bracket of f and g (see the note on p. 46).

CHAPTER III

The Schrödinger Equation

§ 17. The Schrödinger equation

The Hamiltonian determines the form of a physical system's wave equation and consequently acquires fundamental significance in the entire mathematical apparatus of quantum mechanics.

For a free particle, the Hamiltonian is already established by the general requirements associated with the homogeneity and isotropy of space and with the Galilean relativity principle. In classical mechanics these requirements imply a quadratic dependence of the particle energy on its momentum: $E = p^2/2m$, where the constant m is called the mass of the particle (see I, § 4). In quantum mechanics the same requirements lead to the same relation for the eigenvalues of energy and momentum, quantities which for a free particle are conserved and can be measured simultaneously.

For the relation $E = p^2/2m$ to hold for every eigenvalue of energy and momentum, however, it must also hold for their operators:

$$\hat{H} = \frac{1}{2m}\left(\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2\right). \tag{17.1}$$

Substitution of (15.2) here gives the Hamiltonian of a freely moving particle in the form

$$\hat{H} = -\frac{\hbar^2}{2m}\Delta, \tag{17.2}$$

where $\Delta = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$ is the Laplace operator. The Hamiltonian of a system of noninteracting particles is equal to the sum of the Hamiltonians of the individual particles:

$$\hat{H} = -\frac{\hbar^2}{2}\sum_a \frac{\Delta_a}{m_a}, \tag{17.3}$$

where the index a numbers the particles; Δ_a is the Laplace operator in which differentiation is performed with respect to the coordinates of the ath particle. In classical (nonrelativistic) mechanics, particle interaction is described by an additive term in the Hamilton function, the potential energy of interaction $U(\mathbf{r}_1, \mathbf{r}_2, \ldots)$, which is a function of the particle coordinates. Particle interaction in quantum mechanics is likewise described by adding the same function to the Hamiltonian of the system[1]:

$$\hat{H} = -\frac{\hbar^2}{2}\sum_a \frac{\Delta_a}{m_a} + U(\mathbf{r}_1, \mathbf{r}_2, \ldots); \tag{17.4}$$

[1]This assertion is not, of course, a logical consequence of the fundamental principles of quantum mechanics and must be regarded as a consequence of experimental data.

the first term may be regarded as the kinetic-energy operator and the second as the potential-energy operator. In particular, the Hamiltonian for one particle in an external field is

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + U(x, y, z) = -\frac{\hbar^2}{2m}\Delta + U(x, y, z), \tag{17.5}$$

where $U(x, y, z)$ is the potential energy of the particle in the external field.

Substitution of expressions (17.2)–(17.5) in the general equation (8.1) yields the wave equations for the corresponding systems. Here we write out the wave equation for a particle in an external field:

$$i\hbar\frac{\partial\Psi}{\partial t} = -\frac{\hbar^2}{2m}\Delta\Psi + U(x, y, z)\Psi. \tag{17.6}$$

Equation (10.2), determining the stationary states, assumes the form

$$\frac{\hbar^2}{2m}\Delta\psi + [E - U(x, y, z)]\psi = 0. \tag{17.7}$$

Equations (17.6) and (17.7) were established by *Schrödinger* in 1926 and are known as the *Schrödinger equations*.

For a free particle equation (17.7) becomes

$$\frac{\hbar^2}{2m}\Delta\psi + E\psi = 0. \tag{17.8}$$

This equation admits solutions that remain finite over all space for any positive value of the energy E. In states with definite directions of motion these solutions are the eigenfunctions of the momentum operator, with $E = p^2/2m$. The full (time-dependent) wave functions of these stationary states have the form

$$\Psi = \text{const} \cdot \exp\left[-\frac{i}{\hbar}(Et - \mathbf{p} \cdot \mathbf{r})\right]. \tag{17.9}$$

Each of these functions, a *plane wave*, describes a state in which the particle possesses definite energy E and momentum $\mathbf{p}$. The frequency of this wave is $E/\hbar$, and its wave vector is $\mathbf{k} = \mathbf{p}/\hbar$; the associated wavelength $\lambda = 2\pi\hbar/p$ is termed the *de Broglie wavelength of the particle*[2].

The energy spectrum of a freely moving particle is therefore continuous, extending from zero to $+\infty$. Each of these eigenvalues, except only $E = 0$, is degenerate, and the degeneracy is of infinite multiplicity. Indeed, every nonzero value of E corresponds to an infinite set of eigenfunctions (17.9), differing in the directions of the vector $\mathbf{p}$ while its absolute magnitude remains the same.

Let us trace how the limiting transition to classical mechanics occurs in the Schrödinger equation, considering for simplicity just one particle in an external field. Substitution in the Schrödinger equation (17.6) of the limiting expression (6.1), $\Psi = ae^{iS/\hbar}$, for the wave function gives, after the differentiations are performed,

$$a\frac{\partial S}{\partial t} - i\hbar\frac{\partial a}{\partial t} + \frac{a}{2m}(\nabla S)^2 - \frac{i\hbar}{2m}a\Delta S - \frac{i\hbar}{m}\nabla S \cdot \nabla a - \frac{\hbar^2}{2m}\Delta a + Ua = 0.$$

[2] The concept of a wave associated with a particle was first introduced by *de Broglie* (*L. de Broglie*) in 1924.

This equation contains purely real and purely imaginary terms (recall that S and a are real); equating each group separately to zero gives two equations:

$$\frac{\partial S}{\partial t} + \frac{1}{2m}(\nabla S)^2 + U - \frac{\hbar^2}{2ma}\Delta a = 0,$$

$$\frac{\partial a}{\partial t} + \frac{a}{2m}\Delta S + \frac{1}{m}\nabla S \cdot \nabla a = 0.$$

Neglecting the term containing $\hbar^2$ in the first of these equations, we obtain

$$\frac{\partial S}{\partial t} + \frac{1}{2m}(\nabla S)^2 + U = 0, \tag{17.10}$$

that is, as expected, the classical Hamilton–Jacobi equation for the action S of the particle. We note in passing that, as $\hbar \to 0$ classical mechanics is accurate up to and including quantities of first order in $\hbar$ (rather than zeroth order).

After multiplication by $2a$, the second equation obtained above can be rewritten as

$$\frac{\partial a^2}{\partial t} + \nabla\cdot\left(a^2\frac{\nabla S}{m}\right) = 0. \tag{17.11}$$

This equation has a transparent physical meaning: a^2 is the probability density for finding the particle at one or another place in space ($|\Psi|^2 = a^2$); $\nabla S/m = \mathbf{p}/m$ is the classical velocity $\mathbf{v}$ of the particle. Equation (17.11) is therefore nothing other than the continuity equation, showing that the probability density is "transported" according to the laws of classical mechanics, with the classical velocity $\mathbf{v}$ at every point.

Problem

Find the transformation law of the wave function under a Galilean transformation.

Solution. Perform the transformation on the wave function for the free motion of a particle (a plane wave). Since every function Ψ can be expanded in plane waves, this will also determine the transformation law for an arbitrary wave function.

The plane waves in the reference frames K and K' (where K' moves with velocity $\mathbf{V}$ relative to K) are

$$\Psi(\mathbf{r}, t) = \text{const}\cdot\exp[i(\mathbf{p}\cdot\mathbf{r} - Et)/\hbar],$$

$$\Psi'(\mathbf{r}', t) = \text{const}\cdot\exp[i(\mathbf{p}'\cdot\mathbf{r}' - E't)/\hbar],$$

where $\mathbf{r} = \mathbf{r}' + \mathbf{V}t$, while the particle's momenta and energies in the two frames are related by

$$\mathbf{p} = \mathbf{p}' + m\mathbf{V}, \qquad E = E' + \mathbf{V}\cdot\mathbf{p}' + \frac{mV^2}{2}$$

(see I, § 8). Substitution of these expressions in Ψ gives

$$\begin{aligned}\Psi(\mathbf{r}, t) &= \Psi'(\mathbf{r}', t)\exp\left[\frac{i}{\hbar}\left(m\mathbf{V}\cdot\mathbf{r}' + \frac{mV^2}{2}t\right)\right]\\ &= \Psi'(\mathbf{r} - \mathbf{V}t, t)\exp\left[\frac{i}{\hbar}\left(m\mathbf{V}\cdot\mathbf{r} - \frac{mV^2}{2}t\right)\right].\end{aligned} \tag{1}$$

Written this way, the formula no longer contains quantities characterizing the free motion of the particle; it therefore furnishes the sought-after general rule governing how the wave function of an arbitrary state of the particle transforms. For a system of particles, the exponent of the exponential in (1) must contain a summation over all the particles.

§ 18. Fundamental properties of the Schrödinger equation

The conditions to be satisfied by solutions of the Schrödinger equation are quite general in nature. The wave function, to begin with, must be single-valued and continuous throughout all of space. The continuity requirement remains in force even when the field $U(x, y, z)$ itself has surfaces of discontinuity. On any such surface, the wave function together with its derivatives must still be continuous. The latter are not continuous, however, if the potential energy U becomes infinite beyond some surface. A particle cannot enter at all a region of space where $U = \infty$, that is, $\psi = 0$ must hold everywhere in that region. Continuity of ψ requires it to vanish on the boundary of this region; in this case, however, the derivatives of ψ generally undergo a jump.

If the field $U(x, y, z)$ nowhere becomes infinite, the wave function must also be finite throughout space. The same condition must be observed when U becomes infinite at some point, provided it does not do so too rapidly, namely as $1/r^s$ with $s < 2$ (see also § 35).

Let $U_{\min}$ be the minimum value of the function $U(x, y, z)$. Since the particle's Hamiltonian is the sum of two terms, the kinetic- and potential-energy operators $\hat{T}$ and U, the mean energy in an arbitrary state is the sum $\bar{E} = \bar{T} + \bar{U}$. But all eigenvalues of the operator $\hat{T}$ (which coincides with the Hamiltonian of a free particle) are positive, and therefore the mean value $\bar{T} > 0$ as well. Taking account also of the evident inequality $\bar{U} > U_{\min}$, we find that $\bar{E} > U_{\min}$. Since this inequality holds for every state, it plainly holds for all energy eigenvalues as well:

$$E_n > U_{\min}. \tag{18.1}$$

Suppose a particle moves through a force field that goes to zero at large distances; following standard convention, we fix $U(x, y, z)$ so that it too tends to zero at infinity. One readily verifies that the eigenvalue spectrum for negative energies is then discrete: every state with $E < 0$ in a field that vanishes at infinity is a bound state. Indeed, in stationary states of the continuous spectrum, which correspond to infinite motion, the particle is at infinity (see § 10). At sufficiently large distances the presence of the field may be neglected and the particle's motion may be regarded as free; in free motion, however, the energy can only be positive.

Conversely, the positive eigenvalues form a continuous spectrum and correspond to infinite motion; for $E > 0$ the Schrödinger equation generally does not have (in the field under consideration) solutions for which the integral $\int |\psi|^2 \, dV$ converges[3].

We draw attention to the fact that, in quantum mechanics, a particle in finite motion may also be present in regions of space where $E < U$. Although the probability $|\psi|^2$

[3]From a purely mathematical standpoint, however, it must be qualified that, for certain forms of the function $U(x, y, z)$ (having no physical significance), a discrete set of values may drop out of the continuous spectrum.

of finding the particle tends rapidly to zero deeper inside such a region, it nevertheless differs from zero at every finite distance. This represents a fundamental departure from classical mechanics, where a particle is altogether unable to penetrate a region with $U > E$. In classical mechanics, penetration into this region is impossible because, for $E < U$, the kinetic energy would become negative, meaning that the velocity would be imaginary. In quantum mechanics, too, the kinetic-energy eigenvalues are positive; nevertheless, no contradiction arises here, because if the measurement process localizes the particle at some definite point of space, that same process disturbs the particle's state in such a way that it ceases altogether to have any definite kinetic energy.

If $U(x, y, z) > 0$ throughout space (with $U \to 0$ at infinity), inequality (18.1) gives $E_n > 0$. On the other hand, since the spectrum must be continuous for $E > 0$, it follows that in the case under consideration the discrete spectrum is absent altogether, that is, only infinite motion of the particle is possible.

Suppose that at some point, chosen as the coordinate origin, U tends to $-\infty$ according to the law

$$U \approx -\alpha/r^s, \qquad \alpha > 0. \tag{18.2}$$

Consider a wave function that is finite in a certain small region of radius r_0 around the origin and zero outside it. The uncertainty in the particle's coordinate values in such a wave packet is of order r_0; the uncertainty in the momentum value is therefore $\sim \hbar/r_0$. The mean kinetic energy in this state is of order $\hbar^2/mr_0^2$, and the mean potential energy is of order $-\alpha/r_0^s$. Suppose first that $s > 2$. Then, for sufficiently small r_0, the sum $\hbar^2/mr_0^2 - \alpha/r_0^s$ takes negative values of arbitrarily large absolute magnitude. But if the mean energy can take such values, this means in any event that there exist negative energy eigenvalues of arbitrarily large absolute magnitude. Energy levels with large $|E|$ correspond to motion of the particle in a very small region of space around the origin. The "normal" state will correspond to the particle being located at the origin itself, that is, the particle will "fall" into the point $r = 0$.

If $s < 2$, on the other hand, the energy cannot take negative values of arbitrarily large absolute magnitude. The diagonal spectrum starts from a certain finite negative value, and the particle does not fall to the center. We note that in classical mechanics a particle can in principle fall to the center in any attractive field (that is, for any positive s). The case $s = 2$ will be considered separately in § 35.

Next, let us investigate the character of the energy spectrum as a function of the behaviour of the field at large distances. Suppose that as $r \to \infty$ the potential energy, while negative, tends to zero according to the power law (18.2) (in this formula r is now large). Consider a wave packet "filling" a spherical shell of large radius r_0 and thickness $\Delta r \ll r_0$. Then again the order of magnitude of the kinetic energy is $\hbar^2/m(\Delta r)^2$, and that of the potential energy is $-\alpha/r_0^s$. Let r_0 increase while Δr is increased at the same time (so that Δr grows in proportion to r_0). If $s < 2$, then for sufficiently large r_0 the sum $\hbar^2/m(\Delta r)^2 - \alpha/r_0^s$ becomes negative. It follows that stationary states with negative energy exist in which the particle may be found with appreciable probability at large distances from the origin. But this means that negative energy levels of arbitrarily small absolute magnitude exist (it must be remembered that wave functions decay rapidly in the region of space where $U > E$). Thus, in the case under consideration, the discrete spectrum contains an infinite set of levels which crowd together towards the level $E = 0$.

If, however, the field falls off at infinity as $-1/r^s$ with $s > 2$, there are no negative levels of arbitrarily small absolute magnitude. The discrete spectrum ends at a level with nonzero absolute magnitude, so that the total number of levels is finite.

The Schrödinger equation for the stationary-state wave functions ψ, together with the conditions placed on its solutions, is real. Its solutions can therefore always be chosen real[4]. The eigenfunctions of nondegenerate energy values automatically prove to be real up to an immaterial phase factor. Indeed, ψ^* satisfies the same equation as ψ and is therefore also an eigenfunction for the same value of the energy; hence, if that value is nondegenerate, ψ and ψ^* must be essentially identical, that is, they may differ at most by a constant factor whose modulus equals unity. Wave functions that correspond to one and the same degenerate level need not be real, but a set of real functions can always be obtained by a suitable choice of their linear combinations.

The full (time-dependent) wave functions Ψ, however, are determined by an equation whose coefficients contain i. This equation nevertheless retains its form if t is replaced by $-t$ and complex conjugation is performed at the same time[5]. The functions Ψ can therefore always be chosen so that Ψ and Ψ^* differ only in the sign of time.

As is known, the equations of classical mechanics are unchanged under *time reversal*, i.e. upon reversing the sign of time. In quantum mechanics, as we see, symmetry with respect to the two directions of time manifests itself in the invariance of the wave equation under reversal of the sign of t accompanied by the replacement of Ψ with Ψ^*. It must be remembered, however, that here this symmetry applies only to the equations and not to the concept of measurement itself, which has a fundamental role in quantum mechanics (as discussed in detail in § 7).

§ 19. Flux Density

In classical mechanics a particle's velocity $\mathbf{v}$ is related to its momentum by $\mathbf{p} = m\mathbf{v}$. As expected, quantum mechanics has the same relation between the corresponding operators. This follows at once by calculating $\hat{\mathbf{v}} = \hat{\dot{\mathbf{r}}}$ from the general rule (9.2) for differentiating operators with respect to time:

$$\hat{\mathbf{v}} = \frac{i}{\hbar}\left(\hat{H}\mathbf{r} - \mathbf{r}\hat{H}\right).$$

Using (17.5) for $\hat{H}$ together with (16.5), we obtain

$$\hat{\mathbf{v}} = \hat{\mathbf{p}}/m. \tag{19.1}$$

The same relations plainly hold between the eigenvalues of velocity and momentum, and between their mean values in any state.

Like momentum, the velocity of a particle cannot have a definite value simultaneously with its coordinates. Velocity multiplied by an infinitesimal time element dt, however,

[4] This is not true for systems in a magnetic field.

[5] It is assumed that the potential energy U has no explicit dependence on time: the system is either closed or is in a constant (nonmagnetic) field.

determines the particle's displacement during dt. Thus the nonexistence of velocity simultaneously with coordinates means that if a particle occupies a definite point of space at one instant, it no longer has a definite position at the next, infinitesimally close instant.

We record a useful formula for the operator $\hat{\dot{f}}$, the time derivative of a quantity $f(\mathbf{r})$ which is a function of the particle's radius vector. Since f commutes with $U(\mathbf{r})$, we find

$$\hat{\dot{f}} = \frac{i}{\hbar}(\hat{H}f - f\hat{H}) = \frac{i}{2m\hbar}(\hat{\mathbf{p}}^2 f - f\hat{\mathbf{p}}^2),$$

while (16.4) gives

$$\hat{\mathbf{p}}^2 f = \hat{\mathbf{p}}\,(f\hat{\mathbf{p}} - i\hbar\nabla f)\,, \qquad f\hat{\mathbf{p}}^2 = (\hat{\mathbf{p}}f + i\hbar\nabla f)\,\hat{\mathbf{p}}.$$

Hence the desired expression is

$$\hat{\dot{f}} = \frac{1}{2m}\,(\hat{\mathbf{p}}\,\nabla f + \nabla f\cdot\hat{\mathbf{p}})\,. \tag{19.2}$$

Next, let us find the acceleration operator. We have

$$\hat{\dot{\mathbf{v}}} = \frac{i}{\hbar}(\hat{H}\hat{\mathbf{v}} - \hat{\mathbf{v}}\hat{H}) = \frac{i}{m\hbar}(\hat{H}\hat{\mathbf{p}} - \hat{\mathbf{p}}\hat{H}) = \frac{i}{m\hbar}(U\hat{\mathbf{p}} - \hat{\mathbf{p}}U).$$

Using (16.4), we find

$$m\hat{\dot{\mathbf{v}}} = -\nabla U. \tag{19.3}$$

In form, this operator equation coincides exactly with the equation of motion (Newton's equation) of classical mechanics.

The integral $\int_V |\Psi|^2\,dV$ over a finite volume V is the probability of finding the particle in that volume. Let us calculate its time derivative:

$$\frac{d}{dt}\int_V |\Psi|^2\,dV = \int_V \left(\Psi\frac{\partial\Psi^*}{\partial t} + \Psi^*\frac{\partial\Psi}{\partial t}\right)dV = \frac{i}{\hbar}\int_V \left(\Psi\hat{H}^*\Psi^* - \Psi^*\hat{H}\Psi\right)dV.$$

Substitution of

$$\hat{H} = \hat{H}^* = -\frac{\hbar^2}{2m}\Delta + U(x,y,z)$$

and use of the identity

$$\Psi\Delta\Psi^* - \Psi^*\Delta\Psi = \nabla\cdot(\Psi\nabla\Psi^* - \Psi^*\nabla\Psi)$$

give

$$\frac{d}{dt}\int_V |\Psi|^2\,dV = -\int_V \nabla\cdot\mathbf{j}\,dV,$$

where $\mathbf{j}$ denotes the vector[6]

$$\mathbf{j} = \frac{i\hbar}{2m}(\Psi\nabla\Psi^* - \Psi^*\nabla\Psi) = \frac{1}{2m}(\Psi\hat{\mathbf{p}}^*\Psi^* + \Psi^*\hat{\mathbf{p}}\Psi). \tag{19.4}$$

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[6] If ψ is written as $|\psi|e^{i\alpha}$, then

$$\mathbf{j} = \frac{\hbar}{m}|\psi|^2\nabla\alpha. \tag{19.4a}$$

By Gauss's theorem, the integral of $\nabla \cdot \mathbf{j}$ can be transformed into an integral over the closed surface bounding V:

$$\frac{d}{dt}\int_V |\Psi|^2\, dV = -\oint \mathbf{j}\, d\mathbf{f}. \tag{19.5}$$

It follows that $\mathbf{j}$ may be called the *probability-flux-density vector*, or simply the *flux density*. Its surface integral is the probability that the particle crosses that surface per unit time. The vector $\mathbf{j}$ and the probability density $|\Psi|^2$ obey

$$\frac{\partial |\Psi|^2}{\partial t} + \nabla \cdot \mathbf{j} = 0, \tag{19.6}$$

an equation analogous to the classical continuity equation.

The free-motion wave function—the plane wave (17.9)—can be normalized so as to describe a particle flux of unit density (a flux in which, on average, one particle per unit time passes through a unit area of its cross-section). The function is

$$\Psi = \frac{1}{\sqrt{v}} \exp\left[-\frac{i}{\hbar}(Et - \mathbf{p}\cdot\mathbf{r})\right], \tag{19.7}$$

where v is the particle's speed. Indeed, substituting this function into (19.4) gives $\mathbf{j} = \mathbf{p}/mv$, the unit vector in the direction of motion. It is useful to show how the mutual orthogonality of wave functions of states with different energies follows directly from the Schrödinger equation. Let ψ_m and ψ_n be two such functions; they satisfy

$$-\frac{\hbar^2}{2m}\Delta\psi_m + U\psi_m = E_m\psi_m,$$

$$-\frac{\hbar^2}{2m}\Delta\psi_n^* + U\psi_n^* = E_n\psi_n^*.$$

Multiply the first equation by ψ_n^* and the second by ψ_m, and subtract them term by term. This gives

$$(E_m - E_n)\psi_m\psi_n^* = \frac{\hbar^2}{2m}(\psi_m\Delta\psi_n^* - \psi_n^*\Delta\psi_m) = \frac{\hbar^2}{2m}\nabla\cdot(\psi_m\nabla\psi_n^* - \psi_n^*\nabla\psi_m).$$

Integrating both sides over all space, the right-hand side vanishes after application of Gauss's theorem, and we obtain

$$(E_m - E_n)\int \psi_m\psi_n^*\, dV = 0.$$

Since $E_m \neq E_n$ by assumption, the required orthogonality relation follows:

$$\int \psi_m\psi_n^*\, dV = 0.$$

§ 20. The Variational Principle

The Schrödinger equation in its general form, $\hat{H}\psi = E\psi$, can be obtained from the variational principle

$$\delta \int \psi^*(\hat{H} - E)\psi \, dq = 0. \tag{20.1}$$

Since ψ is complex, variations with respect to ψ and ψ^* may be made independently. Variation with respect to ψ^* gives

$$\int \delta\psi^*(\hat{H} - E)\psi \, dq = 0,$$

and the arbitrariness of $\delta\psi^*$ then yields the required equation $\hat{H}\psi = E\psi$. Variation with respect to ψ gives nothing new. Indeed, varying ψ and using the Hermitian character of $\hat{H}$, we have

$$\int \psi^*(\hat{H} - E)\delta\psi \, dq = \int \delta\psi(\hat{H}^* - E)\psi^* \, dq = 0,$$

which yields the complex-conjugate equation $\hat{H}^*\psi^* = E\psi^*$.

The variational principle (20.1) requires an unconstrained extremum of the integral. It may be put in another form by treating E as a Lagrange multiplier in the problem of finding a constrained extremum of

$$\delta \int \psi^* \hat{H}\psi \, dq = 0 \tag{20.2}$$

subject to the supplementary condition

$$\int \psi\psi^* \, dq = 1. \tag{20.3}$$

The minimum value of the integral (20.2), subject to (20.3), is the first energy eigenvalue, that is, the energy E_0 of the normal state. The function ψ which realizes this minimum is correspondingly the normal-state wave function ψ_0[7]. The wave functions ψ_n $(n > 0)$ of the succeeding stationary states correspond only to extrema, not to true minima of the integral.

To obtain from the minimum condition for (20.2) the wave function ψ_1 and energy E_1 of the state immediately above the normal state, the trial functions ψ must be restricted not only by the normalization condition (20.3), but also by orthogonality to the normal-state wave function ψ_0: $\int \psi\psi_0 \, dq = 0$. In general, suppose that the wave functions $\psi_0, \psi_1, \ldots, \psi_{n-1}$ of the first n states are known (the states being ordered by increasing energy). The wave function of the next state minimizes (20.2) under the supplementary conditions

$$\int \psi^2 \, dq = 1, \qquad \int \psi\psi_m \, dq = 0, \qquad m = 0, 1, 2, \ldots, n-1. \tag{20.4}$$

[7] For the remainder of this section we take the wave functions ψ to be real, as they can always be chosen when no magnetic field is present.

We give here several general theorems which may be proved from the variational principle[8].

The normal-state wave function ψ_0 does not vanish (or, as one says, has no nodes) at any finite values of the coordinates[9]. In other words, it has the same sign throughout space. It follows that the wave functions ψ_n ($n > 0$) of the other stationary states, being orthogonal to ψ_0, necessarily have nodal points (if ψ_n also had a constant sign, the integral $\int \psi_0\psi_n \, dq$ could not vanish).

The absence of nodes in ψ_0 also implies that the normal energy level cannot be degenerate. Suppose otherwise, and let ψ_0 and ψ_0' be two different eigenfunctions belonging to the level E_0. Every linear combination $c\psi_0 + c'\psi_0'$ would also be an eigenfunction; but by a suitable choice of c and c', this function can always be made to vanish at any prescribed point of space, producing an eigenfunction with a node.

If motion is confined to a bounded region of space, then $\psi = 0$ on its boundary (see § 18). To determine the energy levels one must use the variational principle to find the minimum of (20.2) with this boundary condition. In this case the theorem that the normal-state wave function has no nodes states that ψ_0 vanishes nowhere inside the region.

Notice that increasing the size of the region of motion lowers every energy level E_n. This follows directly because enlarging the region enlarges the class of trial functions available for minimizing the integral, so its minimum value can only decrease. For discrete-spectrum states of a system of particles, the expression

$$\int \psi \hat{H} \psi \, dq = \int \left[-\sum_a \frac{\hbar^2}{2m_a} \psi \Delta_a \psi + U\psi^2 \right] dq$$

can be transformed into a form more convenient for actual variation. In the first term of the integrand, write

$$\psi \Delta_a \psi = \nabla_a \cdot (\psi \nabla_a \psi) - (\nabla_a \psi)^2.$$

The integral of $\nabla_a \cdot (\psi \nabla_a \psi)$ over dV_a becomes an integral over a closed surface at infinity. Since discrete-spectrum wave functions vanish sufficiently rapidly at infinity, this integral disappears. Thus

$$\int \psi \hat{H} \psi \, dq = \int \left[\sum_a \frac{\hbar^2}{2m_a} (\nabla_a \psi)^2 + U\psi^2 \right] dq. \tag{20.5}$$

§ 21. General Properties of One-Dimensional Motion

When a particle's potential energy depends on just one coordinate x, the wave function may be sought as a product of a function of y, z and a function of x alone. The

[8]Proofs of the theorems about zeros of eigenfunctions (see also the next section) may be found in: *M. A. Lavrent'ev and L. A. Lyusternik*, Course of the Calculus of Variations, Moscow: Gostekhizdat, 1950, Ch. IX; *R. Courant and D. Hilbert*, Methods of Mathematical Physics, Moscow: Gostekhizdat, 1951, Vol. I, Ch. VI.

[9]This theorem (and the consequences drawn from it below) is not valid in general for wave functions of systems containing several identical particles; see the end of § 63.

first is determined by the free-motion Schrödinger equation, and the second by the one-dimensional Schrödinger equation

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}[E - U(x)]\psi = 0. \tag{21.1}$$

The problem of motion in a field with potential energy $U(x, y, z) = U_1(x) + U_2(y) + U_3(z)$, separated into a sum of functions each depending on only one coordinate, plainly reduces to the same kind of one-dimensional equations. In §§ 22–24 we shall consider several specific examples of such "one-dimensional" motion. First we establish some of its general properties.

We begin by showing that every discrete-spectrum energy level in a one-dimensional problem is nondegenerate. Suppose the contrary, and let ψ_1 and ψ_2 be two different eigenfunctions belonging to the same energy value. Since both satisfy the same equation (21.1),

$$\frac{\psi_1''}{\psi_1} = \frac{2m}{\hbar^2}(U - E) = \frac{\psi_2''}{\psi_2},$$

or $\psi_2\psi_1'' - \psi_1\psi_2'' = 0$ (a prime denotes differentiation with respect to x). Integration gives

$$\psi_1\psi_2' - \psi_2\psi_1' = \text{const}. \tag{21.2}$$

Since $\psi_1 = \psi_2 = 0$ at infinity, the constant must be zero, and hence

$$\psi_1\psi_2' - \psi_2\psi_1' = 0,$$

or $\psi_1'/\psi_1 = \psi_2'/\psi_2$. A second integration gives $\psi_1 = \text{const} \cdot \psi_2$; the two functions are therefore essentially identical.

For the discrete-spectrum wave functions $\psi_n(x)$, the following result, known as the *oscillation theorem*, can be stated: the function $\psi_n(x)$ corresponding to the $(n + 1)$th eigenvalue E_n in increasing order vanishes n times at finite x[10].

Suppose that $U(x)$ tends to finite limits as $x \to \pm\infty$ (it need not in any way be monotonic). Take $U(+\infty)$ as the zero of energy, so that $U(+\infty) = 0$, and denote $U(-\infty)$ by U_0, with $U_0 > 0$. The discrete spectrum lies in the range of energies for which the particle cannot escape to infinity. The energy must therefore be less than both limiting values and, in particular, negative:

$$E < 0. \tag{21.3}$$

Of course $E > U_{\min}$ must also hold in every case, so $U(x)$ must possess at least one minimum with $U_{\min} < 0$.

Now consider the positive energies below U_0:

$$0 < E < U_0. \tag{21.4}$$

Here the spectrum is continuous and the particle motion in the corresponding stationary states is infinite, with the particle escaping toward $x = +\infty$. Every energy eigenvalue in

[10]If the particle can occupy only a bounded interval of the x axis, the zeros of $\psi_n(x)$ inside that interval are to be counted.

this part of the spectrum is likewise nondegenerate. To see this, it suffices to observe that the proof given above for the discrete spectrum requires ψ_1 and ψ_2 to vanish at only one of the two infinities (here they vanish as $x \to -\infty$).

At sufficiently large positive x, $U(x)$ can be neglected in the Schrödinger equation (21.1):

$$\psi'' + \frac{2m}{\hbar^2} E\psi = 0.$$

This equation has real solutions in the form of a standing plane wave,

$$\psi = a \cos(kx + \delta), \tag{21.5}$$

where a and δ are constants, and the "wave vector" is $k = p/\hbar = \sqrt{2mE}/\hbar$. This formula gives the asymptotic form, as $x \to +\infty$, of the wave functions for the nondegenerate energy levels in the continuous-spectrum interval (21.4). At large negative x, the Schrödinger equation becomes

$$\psi'' - \frac{2m}{\hbar^2}(U_0 - E)\psi = 0.$$

The solution which does not become infinite as $x \to -\infty$ is

$$\psi = be^{\kappa x}, \qquad \kappa = \frac{1}{\hbar}\sqrt{2m(U_0 - E)}. \tag{21.6}$$

This is the asymptotic wave function as $x \to -\infty$. The wave function thus decays exponentially into the region in which $E < U_0$.

Finally, when

$$E > U_0, \tag{21.7}$$

the spectrum is continuous and the motion is infinite in both directions. Every level in this part of the spectrum is doubly degenerate. The reason is that the corresponding wave functions are determined by the second-order equation (21.1), and both independent solutions meet the required conditions at infinity (whereas, for example, in the preceding case one solution became infinite as $x \to -\infty$ and had to be rejected). As $x \to +\infty$, the asymptotic wave function is

$$\psi = a_1 e^{ikx} + a_2 e^{-ikx}, \tag{21.8}$$

and similarly as $x \to -\infty$. The term containing e^{ikx} corresponds to a particle moving to the right, and that containing e^{-ikx} to a particle moving to the left.

Suppose $U(x)$ is even, $U(-x) = U(x)$. The Schrödinger equation (21.1) is then unchanged on reversing the sign of the coordinate. Hence, if $\psi(x)$ is a solution, $\psi(-x)$ is also a solution and differs from $\psi(x)$ only by a constant factor: $\psi(-x) = c\psi(x)$. Reversing x once more gives $\psi(x) = c^2\psi(x)$, so $c = \pm 1$. Thus, for a potential energy symmetric about $x = 0$, stationary-state wave functions may be either even, $\psi(-x) = \psi(x)$, or odd, $\psi(-x) = -\psi(x)$[11]. In particular, the ground-state wave function is even: it cannot have a node, whereas an odd function necessarily vanishes at $x = 0$, since $\psi(0) = -\psi(0) = 0$.

[11] This argument assumes that the stationary state is nondegenerate, that is, that its motion is not infinite in both directions. Otherwise the two wave functions belonging to the given energy level may transform into one another when the sign of x is reversed. Even then, the stationary-state wave functions are not necessarily even or odd, but definite parity can always be achieved by forming suitable linear combinations of the two solutions.

There is a simple method for normalizing one-dimensional-motion wave functions in the continuous spectrum, which determines the normalization coefficient directly from the asymptotic wave function at large $|x|$.

Consider the wave function for motion infinite in one direction ($x \to +\infty$). The normalization integral diverges as $x \to \infty$ (at $x \to -\infty$ the function decays exponentially, so the integral converges rapidly). To determine the normalization constant, one may therefore replace ψ by its asymptotic value for large positive x and take any finite value of x, say zero, as the lower limit of integration; this is equivalent to discarding a finite contribution beside a divergent one. We now show that a wave function normalized by

$$\int \psi_p^* \psi_{p'}\, dx = \delta\left(\frac{p - p'}{2\pi\hbar}\right) = 2\pi\hbar\delta(p - p') \tag{21.9}$$

(where p is the particle momentum at infinity) must have the asymptotic form (21.5) with $a = 2$:

$$\psi_p \approx 2\cos(kx + \delta) = e^{i(kx+\delta)} + e^{-i(kx+\delta)}. \tag{21.10}$$

Since we are not seeking to verify mutual orthogonality of functions corresponding to different p, when substituting (21.10) into the normalization integral we take p and p' arbitrarily close and may put $\delta = \delta'$ (in general, δ is a function of p). We then retain in the integrand only the terms which diverge when $p = p'$; equivalently, we drop terms containing factors $e^{\pm i(k+k')x}$. We thus obtain

$$\int \psi_p^* \psi_{p'}\, dx = \int_0^\infty e^{i(k'-k)x}\, dx + \int_0^\infty e^{-i(k'-k)x}\, dx = \int_{-\infty}^\infty e^{i(k'-k)x}\, dx,$$

which, by (15.7), agrees with (21.9). According to (5.14), conversion to normalization by an energy delta function is effected by multiplying ψ_p by

$$\left(\frac{d(p/2\pi\hbar)}{dE}\right)^{1/2} = \frac{1}{\sqrt{2\pi\hbar v}},$$

where v is the particle's speed at infinity. Thus

$$\psi_E = \frac{1}{\sqrt{2\pi\hbar v}}\psi_p = \frac{1}{\sqrt{2\pi\hbar v}}\left(e^{i(kx+\delta)} + e^{-i(kx+\delta)}\right). \tag{21.11}$$

Notice that each of the two traveling waves into which the standing wave (21.11) separates has flux density $1/(2\pi\hbar)$.

We may therefore state the following rule for normalizing, to an energy delta function, the wave function of motion infinite in one direction: write its asymptotic expression as the sum of two plane waves traveling in opposite directions, and choose the normalization coefficient so that the flux density in the wave traveling toward the origin (or away from it) equals $1/(2\pi\hbar)$.

The analogous argument gives the corresponding rule for wave functions of motion infinite in both directions. Such a wave function is normalized to an energy delta function when the sum of the fluxes in the waves traveling toward the origin from the positive and negative sides of the x axis equals $1/(2\pi\hbar)$.

§ 22. The Potential Well

As a simple example of one-dimensional motion, consider motion in a rectangular *potential well*, that is, in a field with the function $U(x)$ shown in Fig. 1: $U(x) = 0$ for $0 < x < a$, and $U(x) = U_0$ for $x < 0$ and $x > a$. It is evident in advance that the spectrum is discrete when $E < U_0$, whereas for $E > U_0$ there is a continuous spectrum of doubly degenerate levels.

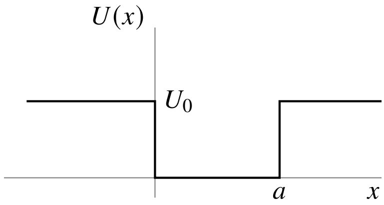

Fig. 1

In the region $0 < x < a$, the Schrödinger equation is

$$\psi'' + \frac{2m}{\hbar^2}E\psi = 0, \tag{22.1}$$

where a prime denotes differentiation with respect to x; outside the well it is

$$\psi'' + \frac{2m}{\hbar^2}(E - U_0)\psi = 0. \tag{22.2}$$

At $x = 0, a$, the solutions of these equations must pass continuously into one another, with continuous first derivative; at $x = \pm\infty$, the solution of (22.2) must remain finite (and for the discrete spectrum, $E < U_0$, it must vanish).

For $E < U_0$, the solution of (22.2) that vanishes at infinity is $\psi = \text{const} \cdot e^{\mp\kappa x}$, where

$$\kappa = \frac{1}{\hbar}\sqrt{2m(U_0 - E)} \tag{22.3}$$

(the minus and plus signs in the exponent refer respectively to the regions $x > a$ and $x < 0$). The probability $|\psi|^2$ of finding the particle decays exponentially into the region where $E < U(x)$. Instead of requiring ψ and ψ' to be continuous at the boundary of the potential well, it is convenient to require continuity of ψ and of the logarithmic derivative ψ'/ψ. With (22.3), the boundary condition becomes

$$|\psi'|/\psi = \mp\kappa. \tag{22.4}$$

Determining the energy levels for a well of arbitrary depth U_0 is left aside here (see Problem 2); we analyze in full only the limiting case of infinitely high walls ($U_0 \to \infty$).

When $U_0 = \infty$, motion occurs only on the interval bounded by $x = 0, a$ and, as stated in § 18, the boundary condition at these points is

$$\psi = 0. \tag{22.5}$$

(This condition also follows readily from the general condition (22.4). Indeed, when $U_0 \to \infty$, we also have $\kappa \to \infty$, and hence $\psi'/\psi \to \infty$; since ψ' cannot become infinite, it follows that $\psi = 0$.) Seek a solution of (22.1) inside the well in the form

$$\psi = c \sin(kx + \delta), \qquad k = \frac{\sqrt{2mE}}{\hbar}. \tag{22.6}$$

The condition $\psi = 0$ at $x = 0$ gives $\delta = 0$. The same condition at $x = a$ then gives $\sin ka = 0$, whence $ka = n\pi$ (n takes the positive integral values beginning with unity[12]), or

$$E_n = \frac{\pi^2\hbar^2}{2ma^2} n^2, \qquad n = 1, 2, 3, \ldots \tag{22.7}$$

The energy levels of a particle inside the potential well are thus determined. After normalization, the stationary-state wave functions are

$$\psi_n = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi n}{a} x\right). \tag{22.8}$$

From these results one can write down directly the energy levels of a particle in a rectangular "potential box," that is, for three-dimensional motion in a field whose potential energy is $U = 0$ for $0 < x < a$, $0 < y < b$, $0 < z < c$, and $U = \infty$ outside this region. Namely, the levels are the sums

$$E_{n_1 n_2 n_3} = \frac{\pi^2\hbar^2}{2m}\left(\frac{n_1^2}{a^2} + \frac{n_2^2}{b^2} + \frac{n_3^2}{c^2}\right), \qquad n_1, n_2, n_3 = 1, 2, 3, \ldots, \tag{22.9}$$

and the corresponding wave functions are the products

$$\psi_{n_1 n_2 n_3} = \sqrt{\frac{8}{abc}} \sin\left(\frac{\pi n_1}{a} x\right) \sin\left(\frac{\pi n_2}{b} y\right) \sin\left(\frac{\pi n_3}{c} z\right). \tag{22.10}$$

Notice that, according to (22.7) or (22.9), the energy of the ground state is of order $E_0 \sim \hbar^2/ml^2$, where l denotes the linear dimensions of the region in which the particle moves. This result agrees with the uncertainty relations: for a coordinate uncertainty of order l, the momentum uncertainty, and with it the order of the momentum itself, is $\sim \hbar/l$; the corresponding energy is $\sim (\hbar/l)^2/m$.

Problem

1. Determine the probability distribution of the possible momentum values for the normal state of a particle in an infinitely deep rectangular potential well.

Solution. The coefficients $a(p)$ in the expansion of the function ψ_1 (22.8) in momentum eigenfunctions are

$$a(p) = \int \psi_p^* \psi_1 \, dx = \sqrt{\frac{2}{a}} \int_0^a \sin\left(\frac{\pi}{a} x\right) \exp\left(-i\frac{px}{\hbar}\right) dx.$$

[12] For $n = 0$, one would obtain identically $\psi = 0$.

Evaluating the integral and taking the square of its modulus gives the required probability distribution

$$|a(p)|^2 \frac{dp}{2\pi\hbar} = \frac{4\pi\hbar^3 a}{\left(p^2a^2 - \pi^2\hbar^2\right)^2} \cos^2\left(\frac{pa}{2\hbar}\right) dp.$$

PROBLEM

2. Determine the energy levels for the potential well shown in Fig. 2.

SOLUTION. The discrete part of the spectrum is the range $E < U_1$, which is the range we consider. In the region $x < 0$, the wave function is

$$\psi = c_1 e^{\kappa_1 x}, \qquad \kappa_1 = \frac{1}{\hbar}\sqrt{2m(U_1 - E)},$$

and in the region $x > a$ it is

$$\psi = c_2 e^{-\kappa_2 x}, \qquad \kappa_2 = \frac{1}{\hbar}\sqrt{2m(U_2 - E)}.$$

Inside the well ($0 < x < a$), seek ψ in the form

$$\psi = c\sin(kx + \delta), \qquad k = \sqrt{2mE}/\hbar.$$

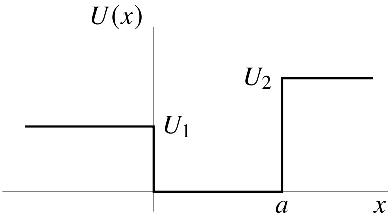

Fig. 2

Continuity of ψ'/ψ at the boundaries of the well gives

$$k\cot\delta = \kappa_1 = \sqrt{\frac{2m}{\hbar^2}U_1 - k^2},$$

$$k\cot(ak + \delta) = -\kappa_2 = -\sqrt{\frac{2m}{\hbar^2}U_2 - k^2},$$

or

$$\sin\delta = \frac{k\hbar}{\sqrt{2mU_1}}, \qquad \sin(ka + \delta) = -\frac{k\hbar}{\sqrt{2mU_2}}.$$

Eliminating δ, we obtain the transcendental equation

$$ka = n\pi - \arcsin\frac{k\hbar}{\sqrt{2mU_1}} - \arcsin\frac{k\hbar}{\sqrt{2mU_2}} \tag{1}$$

(where $n = 1, 2, 3, \ldots$, and the values of arcsin are taken between 0 and $\pi/2$), whose roots determine the energy levels $E = k^2\hbar^2/2m$. In general there is one root for each n; the values of n number the levels in increasing order.

Since the argument of arcsin cannot exceed unity, it is clear that the values of k can lie only in the interval from zero to $\sqrt{2mU_1}/\hbar$. The left-hand side of (1) is a monotonically increasing function of k, while the right-hand side is monotonically decreasing. Consequently, for a root of (1) to exist, its right-hand side at $k = \sqrt{2mU_1}/\hbar$ must be less than its left-hand side. In particular, the inequality

$$a\frac{\sqrt{2mU_1}}{\hbar} \geq \frac{\pi}{2} - \arcsin\sqrt{\frac{U_1}{U_2}}, \tag{2}$$

obtained for $n = 1$, is the condition for the well to contain at least one energy level. Thus, for given $U_1 \neq U_2$, there always exist sufficiently small values of the width a for which the well has no discrete energy level at all. When $U_1 = U_2$, condition (2) is plainly always satisfied.

When $U_1 = U_2 \equiv U_0$ (a symmetric well), equation (1) reduces to

$$\arcsin\frac{\hbar k}{\sqrt{2mU_0}} = \frac{n\pi - ka}{2}. \tag{3}$$

Introducing the variable $\xi = ka/2$, for odd n we obtain

$$\cos\xi = \pm\gamma\xi, \qquad \gamma = \frac{\hbar}{a}\sqrt{\frac{2}{mU_0}}, \tag{4}$$

where the roots for which $\tan\xi > 0$ must be selected. For even n we obtain

$$\sin\xi = \pm\gamma\xi, \tag{5}$$

where the roots for which $\tan\xi < 0$ must be selected. The energy levels are found from the roots of these two equations as $E = 2\xi^2\hbar^2/ma^2$; for $\gamma \neq 0$ only finitely many levels exist.

Consider as a special case a shallow well, in which $U_0 \ll \hbar^2/ma^2$, we have $\gamma \gg 1$, and equation (5) has no roots at all. Equation (4), however, has one root (with the upper sign on its right-hand side), equal to $\xi \approx (1/\gamma)(1 - 1/2\gamma^2)$. The well therefore contains only one energy level,

$$E_0 \approx U_0 - \frac{ma^2}{2\hbar^2}U_0^2,$$

lying close to its "top."

Problem

3. Determine the pressure exerted on the walls of a rectangular "potential box" by a particle inside it.

Solution. On a wall normal to the x axis, the force is given by the mean of the derivative $-\partial H/\partial a$ of the particle's Hamilton function with respect to the length of the box along the x axis; division of this force by the wall area bc gives the pressure. By (11.16), the required mean value is found by differentiating the energy eigenvalue (22.9). The resulting pressure is

$$p^{(x)} = \frac{\pi^2\hbar^2}{ma^3bc}n_1^2.$$

§ 23. The Linear Oscillator

Consider a particle executing small one-dimensional oscillations (the so-called *linear oscillator*). Its potential energy is $m\omega^2 x^2/2$, where in classical mechanics ω is the natural frequency of the oscillations. The oscillator Hamiltonian is consequently

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2 x^2}{2}. \tag{23.1}$$

Since the potential energy tends to infinity at $x = \pm\infty$, the particle can execute only finite motion. Accordingly, the entire energy spectrum of the oscillator is discrete.

Let us determine the oscillator's energy levels by the matrix method[13]. We begin with the equations of motion in the form (19.3), which in this case give

$$\hat{\ddot{x}} + \omega^2 x = 0. \tag{23.2}$$

In matrix form this equation is

$$(\ddot{x})_{mn} + \omega^2 x_{mn} = 0.$$

For the matrix elements of the acceleration, (11.8) gives $(\ddot{x})_{mn} = i\omega_{mn}(\dot{x})_{mn} = -\omega_{mn}^2 x_{mn}$. Hence

$$(\omega_{mn}^2 - \omega^2) x_{mn} = 0.$$

Thus all the matrix elements x_{mn} vanish except those for which $\omega_{mn} = \pm\omega$. Number all stationary states so that the frequencies $\pm\omega$ correspond to the transitions $n \to n \mp 1$, that is, $\omega_{n,n\mp 1} = \pm\omega$. The only nonzero matrix elements are then $x_{n,n\mp 1}$.

We shall suppose that the wave functions ψ_n have been chosen real. Since x is a real quantity, every matrix element x_{mn} is then real as well. The Hermiticity condition (11.10) now makes the matrix x_{mn} symmetric:

$$x_{mn} = x_{nm}.$$

To calculate the nonzero coordinate matrix elements, use the commutation rule

$$\hat{\dot{x}}\,\hat{x} - \hat{x}\,\hat{\dot{x}} = -i\frac{\hbar}{m},$$

written in matrix form as

$$(\dot{x}x)_{mn} - (x\dot{x})_{mn} = -\frac{i\hbar}{m}\delta_{mn}.$$

Using the matrix-multiplication rule (11.12), for $m = n$ this gives

$$i\sum_l (\omega_{nl} x_{nl} x_{ln} - x_{nl}\omega_{ln} x_{ln}) = 2i\sum_l \omega_{nl} x_{nl}^2 = -\frac{i\hbar}{m}.$$

Only the terms with $l = n \pm 1$ are nonzero in this sum, and therefore

$$(x_{n+1,n})^2 - (x_{n,n-1})^2 = \frac{\hbar}{2m\omega}. \tag{23.3}$$

[13] This was done by Heisenberg (1925), before Schrödinger's discovery of the wave equation.

This equality shows that the quantities $(x_{n+1,n})^2$ form an arithmetic progression, unbounded above but necessarily bounded below, since it can contain only positive terms. Thus far we have established only the relative ordering of the state labels n, not their absolute values; we may therefore choose arbitrarily the value of n assigned to the oscillator's first—the normal—state. Set it equal to zero. Correspondingly, $x_{0,-1}$ must be regarded as identically zero, and successive application of (23.3) for $n = 0, 1, \ldots$ gives

$$(x_{n,n-1})^2 = \frac{n\hbar}{2m\omega}.$$

The final expression for the nonzero coordinate matrix elements is therefore[14]

$$x_{n,n-1} = x_{n-1,n} = \sqrt{\frac{n\hbar}{2m\omega}}. \tag{23.4}$$

The matrix of $\hat{H}$ is diagonal, and its matrix elements H_{nn} are the required oscillator energy eigenvalues E_n. To calculate them, write

$$\begin{aligned} H_{nn} = E_n &= \frac{m}{2}\left[(\dot{x}^2)_{nn} + \omega^2 (x^2)_{nn}\right] \\ &= \frac{m}{2}\left[\sum_l i\omega_{nl} x_{nl} i\omega_{ln} x_{ln} + \omega^2 \sum_l x_{nl} x_{ln}\right] = \frac{m}{2}\sum_l (\omega^2 + \omega_{nl}^2) x_{ln}^2. \end{aligned}$$

Only the terms $l = n \pm 1$ are nonzero in the sum. Substitution of (23.4) gives

$$E_n = (n + 1/2)\hbar\omega, \qquad n = 0, 1, 2, \ldots \tag{23.5}$$

Thus the oscillator energy levels are separated by equal intervals $\hbar\omega$. The energy of the normal state ($n = 0$) is $\hbar\omega/2$; we emphasize that it is nonzero.

The result (23.5) may also be obtained by solving the Schrödinger equation. For the oscillator this equation is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}\left(E - \frac{m\omega^2 x^2}{2}\right)\psi = 0. \tag{23.6}$$

It is convenient here to replace x by a dimensionless variable ξ, defined by

$$\xi = \sqrt{\frac{m\omega}{\hbar}}\,x. \tag{23.7}$$

We then obtain

$$\psi'' + \left(\frac{2E}{\hbar\omega} - \xi^2\right)\psi = 0. \tag{23.8}$$

(A prime here denotes differentiation with respect to ξ.)

For large ξ, the term $2E/\hbar\omega$ may be omitted in comparison with ξ^2; the equation $\psi'' = \xi^2\psi$ has the asymptotic integrals $\psi = e^{\pm\xi^2/2}$. (Indeed, differentiation of this

[14] We choose the undetermined phases α_n (see the note on p. 52) so that every matrix element in (23.4) has a plus sign before the root. A phase convention of this kind can always be arranged when a matrix has nonzero entries solely for transitions connecting states whose labels are adjacent.

function gives $\psi'' = \xi^2\psi$ when lower-order terms in ξ are neglected.) Since the wave function ψ must remain finite at $\xi = \pm\infty$, the minus sign must be chosen in the exponent. It is therefore natural to make the substitution

$$\psi = e^{-\xi^2/2}\chi(\xi) \tag{23.9}$$

in (23.8). For $\chi(\xi)$ we obtain the equation (introducing the notation $2E/\hbar\omega - 1 = 2n$; since we know in advance that $E > 0$, we have $n > -1/2$)

$$\chi'' - 2\xi\chi' + 2n\chi = 0, \tag{23.10}$$

where χ must be finite for every finite ξ and, at $\xi = \pm\infty$, may become infinite no faster than a finite power of ξ (so that ψ tends to zero).

Such solutions of (23.10) exist only for integer positive values of n, including zero (see § a of the mathematical supplements); this gives the energy eigenvalues (23.5), already known to us. For the different integral values of n, the corresponding solutions of (23.10) are

$$\chi = \text{const} \cdot H_n(\xi),$$

where $H_n(\xi)$ are the so-called Hermite polynomials, polynomials of degree n in ξ defined by

$$H_n(\xi) = (-1)^n e^{\xi^2} \frac{d^n e^{-\xi^2}}{d\xi^n}. \tag{23.11}$$

Choosing the constant so that ψ_n obeys the normalization condition

$$\int_{-\infty}^{+\infty} \psi_n^2(x)\, dx = 1,$$

we obtain (see (a.7))

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} \exp\left(-\frac{m\omega}{2\hbar}x^2\right) H_n\left(x\sqrt{\frac{m\omega}{\hbar}}\right). \tag{23.12}$$

Thus, the normal-state wave function is

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left(-\frac{m\omega}{2\hbar}x^2\right). \tag{23.13}$$

As expected, it has no zeros at finite x.

Evaluation of the integrals $\int_{-\infty}^{+\infty} \psi_n\psi_m\xi\, d\xi$ determines the coordinate matrix elements; the calculation naturally gives the same values (23.4).

We conclude by demonstrating how the wave functions ψ_n can be computed via the matrix method. In the matrices of the operators $\hat{\dot{x}} \pm i\omega\hat{x}$, the only nonzero elements are

$$(\dot{x} - i\omega x)_{n-1,n} = -(\dot{x} + i\omega x)_{n,n-1} = -i\sqrt{\frac{2\omega\hbar n}{m}}. \tag{23.14}$$

Using the general formula (11.11) and recalling that $\psi_{-1} \equiv 0$, we obtain

$$(\hat{\dot{x}} - i\omega x)\psi_0 = 0.$$

Substitution of $\hat{x} = -i(\hbar/m)d/dx$ then gives

$$\frac{d\psi_0}{dx} = -\frac{m\omega}{\hbar}x\psi_0,$$

whose normalized solution is (23.13). Further, since

$$(\hat{\dot{x}} + i\omega\hat{x})\psi_{n-1} = (\dot{x} + i\omega x)_{n,n-1}\psi_n = i\sqrt{\frac{2\omega\hbar n}{m}}\psi_n,$$

we obtain the recurrence formula

$$\begin{aligned}\psi_n &= \sqrt{\frac{m}{2\omega\hbar n}}\left(-\frac{\hbar}{m}\frac{d}{dx} + \omega x\right)\psi_{n-1} = \frac{1}{\sqrt{2n}}\left(-\frac{d}{d\xi} + \xi\right)\psi_{n-1}\\ &= -\frac{1}{\sqrt{2n}}\exp\frac{\xi^2}{2}\frac{d}{d\xi}\left(\exp\left(-\frac{\xi^2}{2}\right)\psi_{n-1}\right),\end{aligned}$$

whose n-fold application to (23.13) gives (23.12) for the normalized functions ψ_n.

Problem

1. Determine the probability distribution of the possible momentum values for the oscillator.

Solution. Rather than expanding a stationary-state wave function in momentum eigenfunctions, for the oscillator it is simpler to begin directly with the Schrödinger equation in the momentum representation. Substituting the coordinate operator (15.12), $\hat{x} = i\hbar d/dp$, into (23.1), we obtain the Hamiltonian in the momentum representation

$$\hat{H} = \frac{p^2}{2m} - \frac{m\omega^2\hbar^2}{2}\frac{d^2}{dp^2}.$$

The corresponding Schrödinger equation $\hat{H}a(p) = Ea(p)$ for the momentum-representation wave function $a(p)$ is

$$\frac{d^2a(p)}{dp^2} + \frac{2}{m\omega^2\hbar^2}\left(E - \frac{p^2}{2m}\right)a(p) = 0.$$

This equation has exactly the same form as (23.6), so its solutions can be written down directly by analogy with (23.12). The required probability distribution is therefore

$$|a_n(p)|^2\frac{dp}{2\pi\hbar} = \frac{1}{2^n n!\sqrt{\pi m\omega\hbar}}\exp\left(-\frac{p^2}{m\omega\hbar}\right)H_n^2\left(\frac{p}{\sqrt{m\omega\hbar}}\right)dp.$$

Problem

2. Use the uncertainty relation (16.7) to determine a lower bound on the possible energy values of the oscillator.

Solution. Noting that $\overline{x^2} = \bar{x}^2 + (\delta x)^2$ and $\overline{p^2} = \bar{p}^2 + (\delta p)^2$, and using (16.7), for the oscillator's mean energy we have

$$\bar{E} = \frac{m\omega^2}{2}\overline{x^2} + \frac{\overline{p^2}}{2m} \geq \frac{m\omega^2}{2}(\delta x)^2 + \frac{1}{2m}(\delta p)^2 \geq \frac{m\omega^2\hbar^2}{8(\delta p)^2} + \frac{(\delta p)^2}{2m}.$$

Finding the minimum of this expression as a function of δp, we obtain the lower bound for the mean values, and hence for all possible energy values: $E \geq \hbar\omega/2$.

Problem

3. Find the wave functions of linear-oscillator states that minimize the uncertainty relation, that is, states in which the root-mean-square fluctuations of coordinate and momentum in the wave packet are related by $\delta p\, \delta x = \hbar/2$ (E. Schrödinger, 1926)[15].

Solution. The required wave functions must have the form

$$\Psi(x,t) = \frac{1}{(2\pi)^{1/4}(\delta x)^{1/2}} \exp\left\{\frac{i\bar{p}x}{\hbar} - \frac{(x-\bar{x})^2}{4(\delta x)^2} - i\varphi(t)\right\}. \tag{1}$$

At every instant their coordinate dependence corresponds to (16.8), where $\bar{x} = \bar{x}(t)$ and $\bar{p} = \bar{p}(t) = m\dot{\bar{x}}(t)$ are the mean coordinate and momentum. By (19.3), for the linear oscillator ($U = m\omega^2x^2/2$) we have $\dot{p} = -m\omega^2 x$, and hence for the mean values $\dot{\bar{p}} = -m\omega^2\bar{x}$, or

$$\ddot{\bar{x}} + \omega^2\bar{x} = 0, \tag{2}$$

that is, $\bar{x}(t)$ satisfies the classical equation of motion. The constant factor in (1) is fixed by the normalization condition $\int_{-\infty}^{\infty} |\Psi|^2 dx = 1$; besides this factor, Ψ may contain a phase factor with a time-dependent phase $\varphi(t)$. To find the unknown δx and the function $\varphi(t)$, we substitute (1) into the wave equation

$$-\frac{\hbar^2}{2m}\frac{\partial^2\Psi}{\partial x^2} + \frac{m\omega^2x^2}{2}\Psi = i\hbar\frac{\partial\Psi}{\partial t}.$$

Taking account of (2), the substitution gives

$$\left(\frac{x^2}{2} - x\bar{x}\right)\left(\frac{m^2\omega^2}{\hbar^2} - \frac{1}{4(\delta x)^4}\right) + \left[\frac{m^2\dot{\bar{x}}^2}{2\hbar^2} - \frac{\bar{x}^2}{8(\delta x)^4} + \frac{1}{4(\delta x)^2} - \frac{m}{\hbar}\dot{\varphi}(t)\right] = 0.$$

It follows that $(\delta x)^2 = \hbar/(2m\omega)$ and then

$$\dot{\varphi} = \frac{m}{2\hbar}(\dot{\bar{x}}^2 - \omega^2\bar{x}^2) + \frac{\omega}{2}, \qquad \varphi = \frac{1}{2\hbar}\bar{p}\bar{x} + \frac{\omega}{2}t.$$

[15]These states are called *coherent*.

Thus, finally,

$$\Psi(x,t) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left\{\frac{i\bar{p}x}{\hbar} - \frac{m\omega(x-\bar{x})^2}{2\hbar}\right\} \exp\left\{-\frac{i\omega t}{2} - \frac{i\bar{p}\bar{x}}{2\hbar}\right\}. \tag{3}$$

For $\bar{x} = 0$, $\bar{p} = 0$, this function becomes $\psi_0(x)e^{-i\omega t/2}$, the wave function of the oscillator ground state.

The oscillator's mean energy in a coherent state is

$$\bar{E} = \frac{\overline{p^2}}{2m} + \frac{m\omega^2\overline{x^2}}{2} = \frac{\bar{p}^2}{2m} + \frac{m\omega^2\bar{x}^2}{2} + \frac{\hbar\omega}{2} = \hbar\omega\left(\bar{n} + \frac{1}{2}\right); \tag{4}$$

the quantity $\bar{n}$ introduced here is the mean "number of quanta" $\hbar\omega$ in the state. We see that a coherent state is specified completely by giving a dependence $\bar{x}(t)$ that satisfies the classical equation (2). Its general form can be written

$$\frac{m\omega\bar{x} + i\bar{p}}{\sqrt{2m\hbar\omega}} = ae^{-i\omega t}, \qquad |a|^2 = \bar{n}. \tag{5}$$

The function (3) may be expanded in the wave functions of the oscillator's stationary states:

$$\Psi = \sum_{n=0}^{\infty} a_n\Psi_n, \qquad \Psi_n(x,t) = \psi_n(x)\exp\left\{-i\left(n+\frac{1}{2}\right)\omega t\right\}.$$

The coefficients of this expansion are[16]

$$a_n = \int_{-\infty}^{\infty} \Psi_n^*\Psi\, dx. \tag{6}$$

It follows that the probability for the oscillator to be in its nth state is

$$w_n = |a_n|^2 = e^{-\bar{n}}\frac{\bar{n}^n}{n!}, \tag{7}$$

that is, it is given by the familiar Poisson distribution.

Problem

4. Determine the energy levels of a particle moving in a field with the potential energy shown in Fig. 3 (Ph. Morse, 1929).

Solution. The spectrum of positive energy eigenvalues is continuous (and the levels are nondegenerate), while the spectrum of negative values is discrete.

The Schrödinger equation is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - Ae^{-2\alpha x} + 2Ae^{-\alpha x})\psi = 0.$$

Introduce the new variable

$$\xi = \frac{2\sqrt{2mA}}{\alpha\hbar}e^{-\alpha x}$$

[16] Compare the calculation in Problem 1 of § 41.

(which ranges from zero to $+\infty$), and the notation (we consider the discrete spectrum, so $E < 0$)

$$s = \frac{\sqrt{-2mE}}{\alpha\hbar}, \qquad n = \frac{\sqrt{2mA}}{\alpha\hbar} - \left(s + \frac{1}{2}\right). \tag{1}$$

The Schrödinger equation then becomes

$$\psi'' + \frac{1}{\xi}\psi' + \left(-\frac{1}{4} + \frac{n + s + 1/2}{\xi} - \frac{s^2}{\xi^2}\right)\psi = 0.$$

As $\xi \longrightarrow \infty$, ψ behaves asymptotically as $e^{\pm\xi/2}$, while as $\xi \longrightarrow 0$ it is proportional to $\xi^{\pm s}$. Finiteness requires the solution behaving as $e^{-\xi/2}$ at $\xi \longrightarrow \infty$ and as ξ^s at $\xi \longrightarrow 0$.

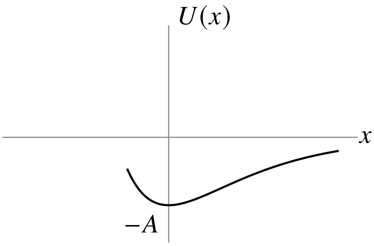

Fig. 3

Make the substitution

$$\psi = e^{-\xi/2}\xi^s w(\xi),$$

which gives for w the equation

$$\xi w'' + (2s + 1 - \xi)w' + nw = 0, \tag{2}$$

to be solved subject to the conditions that w is finite at $\xi = 0$, and that as $\xi \longrightarrow \infty$ it becomes infinite no faster than a finite power of ξ. Equation (2) is the confluent hypergeometric equation (see § d of the mathematical supplements),

$$w = F(-n, 2s + 1, \xi).$$

The solution satisfying the required condition is obtained for a nonnegative integer n (and F then reduces to a polynomial). From the definitions (1), the energy levels are consequently

$$-E_n = A\left[1 - \frac{\alpha\hbar}{\sqrt{2mA}}\left(n + \frac{1}{2}\right)\right]^2,$$

where n takes integer positive values beginning at zero and ending at the greatest value for which

$$\frac{\sqrt{2mA}}{\alpha\hbar} > n + \frac{1}{2}$$

(so that the parameter s is positive, in agreement with its definition). Thus the discrete spectrum contains a finite sequence of levels. If

$$\frac{\sqrt{2mA}}{\alpha\hbar} < \frac{1}{2},$$

there is no discrete spectrum at all.

PROBLEM

5. The same problem for $U = -U_0/\cosh^2 \alpha x$ (Fig. 4).

SOLUTION. The positive-energy spectrum is continuous and the negative-energy spectrum is discrete; we consider the latter. The Schrödinger equation is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}\left(E + \frac{U_0}{\cosh^2 \alpha x}\right)\psi = 0.$$

Make the change of variable $\xi = \tanh \alpha x$ and introduce the notation

$$\varepsilon = \frac{\sqrt{-2mE}}{\hbar\alpha}, \qquad \frac{2mU_0}{\alpha^2\hbar^2} = s(s+1), \qquad s = \frac{1}{2}\left(-1 + \sqrt{1 + \frac{8mU_0}{\alpha^2\hbar^2}}\right).$$

We then obtain

$$\frac{d}{d\xi}\left[(1-\xi^2)\frac{d\psi}{d\xi}\right] + \left[s(s+1) - \frac{\varepsilon^2}{1-\xi^2}\right]\psi = 0.$$

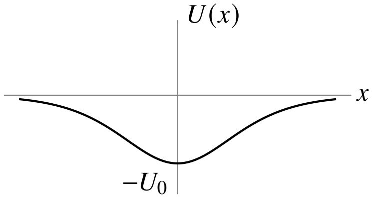

Fig. 4

This is the equation of the generalized Legendre functions. Reduce it to hypergeometric form by the substitution

$$\psi = (1-\xi^2)^{\varepsilon/2} w(\xi)$$

and the temporary change of variable $\frac{1}{2}(1-\xi) = u$:

$$u(1-u)w'' + (\varepsilon+1)(1-2u)w' - (\varepsilon - s)(\varepsilon + s + 1)w = 0.$$

The solution finite at $\xi = 1$ (that is, at $x = \infty$) is

$$\psi = (1-\xi^2)^{\varepsilon/2} F[\varepsilon - s, \varepsilon + s + 1, \varepsilon + 1, (1-\xi)/2].$$

For ψ to remain finite also at $\xi = -1$ (that is, at $x = -\infty$), we must have $\varepsilon - s = -n$, where $n = 0, 1, 2, \dots$ (then F is a polynomial of degree n, finite at $\xi = -1$).

Thus the energy levels are fixed by $s - \varepsilon = n$, which gives

$$E_n = -\frac{\hbar^2\alpha^2}{8m}\left[-(1+2n) + \sqrt{1 + \frac{8mU_0}{\alpha^2\hbar^2}}\right]^2.$$

There is a finite number of levels, determined by the condition $\varepsilon > 0$, that is, $n < s$.

§ 24. Motion in a Uniform Field

Consider the motion of a particle in a uniform external field. Choose the direction of the field as the x axis, and let F be the force exerted by the field on the particle; in an electric field of strength E, this force is $F = eE$, where e is the particle's charge.

The potential energy of the particle in a uniform field has the form $U = -Fx + \text{const}$. Choosing the constant so that $U = 0$ at $x = 0$, we have $U = -Fx$. The Schrödinger equation for the problem is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E + Fx)\psi = 0. \tag{24.1}$$

Since U tends to $+\infty$ as $x \to -\infty$ and to $-\infty$ as $x \to \infty$, it is evident in advance that the energy levels form a continuous spectrum filling the entire interval from $-\infty$ to $+\infty$. All these eigenvalues are nondegenerate and correspond to motion that is finite on the side $x = -\infty$ and infinite in the direction $x \to +\infty$.

In place of the coordinate x, introduce the dimensionless variable

$$\xi = \left(x + \frac{E}{F}\right)\left(\frac{2mF}{\hbar^2}\right)^{1/3}. \tag{24.2}$$

Equation (24.1) then becomes

$$\psi'' + \xi\psi = 0. \tag{24.3}$$

This equation contains no energy parameter at all. Consequently, once a solution satisfying the required finiteness conditions has been found, it provides an eigenfunction for every value of the energy.

The solution of (24.3) that is finite for all x is (see § b of the mathematical supplements)

$$\psi(\xi) = A\Phi(-\xi), \tag{24.4}$$

where

$$\Phi(\xi) = \frac{1}{\sqrt{\pi}}\int_0^\infty \cos\left(\frac{u^3}{3} + u\xi\right) du$$

is the so-called Airy function, and A is a normalization factor to be determined below.

As $\xi \to -\infty$, the function $\psi(\xi)$ tends exponentially to zero. The asymptotic expression determining $\psi(\xi)$ for negative ξ of large absolute magnitude is (see (b.4))

$$\psi(\xi) \approx \frac{A}{2|\xi|^{1/4}} \exp\left(-\frac{2}{3}|\xi|^{3/2}\right). \tag{24.5}$$

For large positive ξ, the asymptotic expression for $\psi(\xi)$ is instead (see (b.5)[17]):

$$\psi(\xi) = \frac{A}{\xi^{1/4}} \sin\left(\frac{2}{3}\xi^{3/2} + \frac{\pi}{4}\right). \tag{24.6}$$

[17]We note in anticipation that the asymptotic expressions (24.5) and (24.6) correspond precisely to the quasiclassical forms of the wave function in classically forbidden and allowed regions, respectively (§ 47).

Following the general rule (5.4) for normalizing eigenfunctions of a continuous spectrum, put the functions (24.4) into a form normalized to a delta function of the energy:

$$\int_{-\infty}^{+\infty} \psi(\xi)\psi(\xi')\,dx = \delta(E'-E). \tag{24.7}$$

In § 21 a simple method was given for determining the normalization coefficient from the asymptotic form of the wave functions. Following that method, write (24.6) as a sum of two traveling waves:

$$\psi(\xi) \approx \frac{A}{2\xi^{1/4}}\left\{\exp\left[i\left(\frac{2}{3}\xi^{3/2}-\frac{\pi}{4}\right)\right]+\exp\left[-i\left(\frac{2}{3}\xi^{3/2}-\frac{\pi}{4}\right)\right]\right\}.$$

The flux density calculated for each of these two terms is

$$v\left(\frac{A}{2\xi^{1/4}}\right)^2 = \sqrt{\frac{2}{m}(E+Fx)}\left(\frac{A}{2\xi^{1/4}}\right)^2 = A^2\frac{(2\hbar F)^{1/3}}{4m^{2/3}}.$$

Equating it to $1/(2\pi\hbar)$, we find

$$A = \frac{(2m)^{1/3}}{\pi^{1/2}F^{1/6}\hbar^{2/3}}. \tag{24.8}$$

Problem

Determine the momentum-representation wave functions for a particle in a uniform field.

Solution. The Hamiltonian in the momentum representation is

$$\widehat{H} = \frac{p^2}{2m} - i\hbar F\frac{d}{dp},$$

so the Schrödinger equation for the wave function $a(p)$ is

$$-i\hbar F\frac{da}{dp} + \left(\frac{p^2}{2m}-E\right)a = 0.$$

Solving this equation gives the required functions

$$a_E(p) = \frac{1}{\sqrt{2\pi\hbar F}}\exp\left\{\frac{i}{\hbar F}\left(Ep-\frac{p^3}{6m}\right)\right\}.$$

These functions are normalized by the condition

$$\int_{-\infty}^{+\infty} a_E^*(p)a_{E'}(p)\,dp = \delta(E'-E).$$

§ 25. Transmission Coefficient

We consider particle motion in a field of the type depicted in Fig. 5: $U(x)$ increases monotonically from one constant limiting value ($U = 0$ as $x \to -\infty$) to another ($U = U_0$ as $x \to +\infty$). According to classical mechanics, a particle with energy $E < U_0$ moving in such a field from left to right arrives at the *potential wall*, reflects off it, and proceeds in the reverse direction; for $E > U_0$, however, the particle keeps traveling in its original direction with a reduced velocity. Quantum mechanics introduces a new phenomenon: even when $E > U_0$, the particle may be reflected from the potential wall. In principle, the reflection probability is to be calculated as follows.

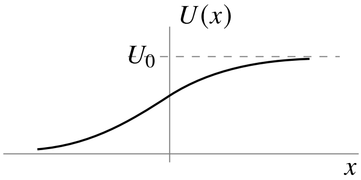

Figure 5

Let the particle move from left to right. At large positive values of x, the wave function must describe a particle that has passed "over the wall" and is moving in the positive direction of the x axis; that is, it must have the asymptotic form

$$\text{as } x \to \infty: \qquad \psi \approx A e^{ik_2 x}, \qquad k_2 = \frac{1}{\hbar}\sqrt{2m(E - U_0)} \tag{25.1}$$

(A is a constant). Having found a solution of the Schrödinger equation that satisfies this limiting condition, we calculate its asymptotic expression as $x \to -\infty$. This is a linear combination of two solutions of the free-motion equation and therefore has the form

$$\text{as } x \to -\infty: \qquad \psi \approx e^{ik_1 x} + B e^{-ik_1 x}, \qquad k_1 = \frac{1}{\hbar}\sqrt{2mE}. \tag{25.2}$$

The first term describes the particle impinging on the wall (ψ is taken normalized so that this term has unit coefficient); the second term depicts the particle reflected from the wall. The flux density is proportional to k_1 in the incident wave, to $k_1|B|^2$ in the reflected wave, and to $k_2|A|^2$ in the transmitted wave. Define the particle's *transmission coefficient* D as the ratio of the flux density in the transmitted wave to that in the incident wave:

$$D = \frac{k_2}{k_1}|A|^2. \tag{25.3}$$

Similarly, the *reflection coefficient* R may be defined as the ratio of the reflected flux density to the incident flux density; evidently $R = 1 - D$:

$$R = |B|^2 = 1 - \frac{k_2}{k_1}|A|^2 \tag{25.4}$$

(this relation between A and B follows automatically from the constancy of the flux along the x axis).

If a particle moves from left to right with energy $E < U_0$, then k_2 is purely imaginary and the wave function decreases exponentially as $x \to +\infty$. The reflected flux equals the incident flux; that is, the particle is totally reflected from the potential wall. We stress, however, that the probability for the particle to be located in the region $E < U$ remains nonzero even in this case, though it decreases rapidly with growing x.

For the general case of an arbitrary stationary state (energy $E > U_0$), the asymptotic form of the wave function both as $x \to -\infty$ and as $x \to +\infty$ is a sum of two waves propagating in the two directions along the x axis:

$$\begin{aligned} \psi &= A_1 e^{ik_1 x} + B_1 e^{-ik_1 x} \quad \text{as } x \to -\infty, \\ \psi &= A_2 e^{ik_2 x} + B_2 e^{-ik_2 x} \quad \text{as } x \to +\infty. \end{aligned} \tag{25.5}$$

Since both expressions are asymptotic forms of one and the same solution of a linear differential equation, the coefficients A_1, B_1 and A_2, B_2 are linearly related. Let $A_2 = \alpha A_1 + \beta B_1$, where α and β are constants (in general complex) that depend on the particular form of the field $U(x)$. The analogous relation for B_2 may then be written by using the fact that the Schrödinger equation is real. By this fact, if ψ is a solution of the given Schrödinger equation, the complex-conjugate function ψ^* is also a solution of the same equation.

The asymptotic form

$$\begin{aligned} \psi^* &= A_1^* e^{-ik_1 x} + B_1^* e^{ik_1 x} \quad \text{as } x \to -\infty, \\ \psi^* &= A_2^* e^{-ik_2 x} + B_2^* e^{ik_2 x} \quad \text{as } x \to +\infty \end{aligned}$$

differs from (25.5) only in the notation of the constant coefficients; hence $B_2^* = \alpha B_1^* + \beta A_1^*$, or $B_2 = \alpha^* B_1 + \beta^* A_1$. Thus the coefficients in (25.5) are related by equations of the form

$$A_2 = \alpha A_1 + \beta B_1, \qquad B_2 = \beta^* A_1 + \alpha^* B_1. \tag{25.6}$$

The condition that the flux be constant along the x axis gives the following relation for the coefficients in (25.5):

$$k_1\big(|A_1|^2 - |B_1|^2\big) = k_2\big(|A_2|^2 - |B_2|^2\big).$$

On expressing A_2 and B_2 here in terms of A_1 and B_1 according to (25.6), we obtain

$$|\alpha|^2 - |\beta|^2 = \frac{k_1}{k_2}. \tag{25.7}$$

The relations (25.6) can be used to show that the reflection coefficients are identical (at a given energy $E > U_0$) for particles propagating in the positive or negative direction along the x axis. Indeed, the first case is obtained by setting $B_2 = 0$ in the functions (25.5); then $B_1/A_1 = -\beta^*/\alpha^*$. In the second case we put $A_1 = 0$, whereupon $A_2/B_2 = \beta/\alpha^*$. The corresponding reflection coefficients are

$$R_1 = \left|\frac{B_1}{A_1}\right|^2 = \left|\frac{\beta^*}{\alpha^*}\right|^2, \qquad R_2 = \left|\frac{A_2}{B_2}\right|^2 = \left|\frac{\beta}{\alpha^*}\right|^2,$$

and it is clear that $R_1 = R_2$.

The quantities $B_1/A_1 = -\beta^*/\alpha^*$ and $A_2/B_2 = \beta/\alpha^*$ are naturally called the *reflection amplitudes* for motion in the positive and negative directions, respectively. These amplitudes have equal moduli but may differ by a phase factor.

Problem

1. For a particle with energy $E > U_0$, calculate the coefficient of reflection from a rectangular potential wall (Fig. 6).

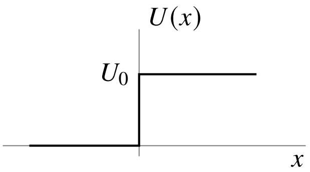

Figure 6

Solution. Throughout the region $x > 0$ the wave function has the form (25.1), and in the region $x < 0$ it has the form (25.2). The constants A and B are determined by the conditions that ψ and $d\psi/dx$ be continuous at $x = 0$:

$$1 + B = A, \qquad k_1(1 - B) = k_2 A,$$

whence

$$A = \frac{2k_1}{k_1 + k_2}, \qquad B = \frac{k_1 - k_2}{k_1 + k_2}.$$

The reflection coefficient is (25.4) [18]

$$R = \left(\frac{k_1 - k_2}{k_1 + k_2}\right)^2 = \left(\frac{p_1 - p_2}{p_1 + p_2}\right)^2.$$

At $E = U_0$ ($k_2 = 0$), R becomes unity; as $E \to \infty$, it tends to zero as $R = (U_0/4E)^2$.

Problem

2. Determine the transmission coefficient of a particle through a rectangular potential barrier (Fig. 7).

Solution. Let $E > U_0$, and let the incident particle move from left to right. The wave function then has the following forms in the different regions:

$$\begin{aligned} &\psi = e^{ik_1x} + Ae^{-ik_1x} && \text{for } x < 0, \\ &\psi = Be^{ik_2x} + B'e^{-ik_2x} && \text{for } 0 < x < a, \\ &\psi = Ce^{ik_1x} && \text{for } x > a. \end{aligned}$$

(On the side $x > a$ there must be only the transmitted wave, propagating in the positive direction of the x axis.) The constants A, B, B', and C are determined by the conditions

[18] In the limiting case of classical mechanics the reflection coefficient must go to zero. However, the resulting expression does not contain a quantum constant at all. This seeming contradiction is resolved as follows. The classical limit corresponds to the situation where the particle's de Broglie wavelength $\lambda \sim \hbar/p$ is small in comparison with the characteristic dimensions of the problem, that is, with the distances over which the field $U(x)$ changes appreciably. In the schematic example under consideration this distance is zero (at $x = 0$), so that the limiting transition cannot be made.

that ψ and $d\psi/dx$ be continuous at the points $x = 0, a$. The transmission coefficient is $D = k_1|C|^2/k_1 = |C|^2$. Calculation gives

$$D = \frac{4k_1^2k_2^2}{(k_1^2 - k_2^2)^2 \sin^2 ak_2 + 4k_1^2k_2^2}.$$

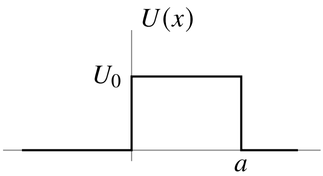

Figure 7

When $E < U_0$, k_2 is purely imaginary; the corresponding expression for D is obtained by replacing k_2 by $i\kappa_2$, where $\hbar\kappa_2 = \sqrt{2m(U_0 - E)}$:

$$D = \frac{4k_1^2\kappa_2^2}{(k_1^2 + \kappa_2^2)^2 \sinh^2(a\kappa_2) + 4k_1^2\kappa_2^2}.$$

Problem

3. Determine the reflection coefficient of a particle from the potential wall defined by

$$U(x) = U_0/(1 + e^{-\alpha x})$$

(see Fig. 5); the particle energy is $E > U_0$.

Solution. The Schrödinger equation is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}\left(E - \frac{U_0}{1 + e^{-\alpha x}}\right)\psi = 0.$$

We must find the solution that, as $x \to +\infty$, has the form

$$\psi = \text{const} \cdot e^{ik_2 x}.$$

Introduce the new variable

$$\xi = -e^{-\alpha x}$$

(which ranges from $-\infty$ to 0), and seek a solution in the form

$$\psi = \xi^{-ik_2/\alpha} w(\xi),$$

where $w(\xi)$ tends to a constant as $\xi \to 0$ (that is, as $x \to \infty$). For $w(\xi)$ we obtain an equation of hypergeometric type,

$$\xi(1 - \xi)w'' + \left(1 - \frac{2i}{\alpha}k_2\right)(1 - \xi)w' + \frac{1}{\alpha^2}(k_2^2 - k_1^2)w = 0,$$

whose solution is the hypergeometric function

$$w = F\left[-\frac{i}{\alpha}(k_1 - k_2), -\frac{i}{\alpha}(k_1 + k_2), -\frac{2i}{\alpha}k_2 + 1, \xi\right]$$

(a constant factor is omitted). As $\xi \to 0$, this function tends to 1 and thus satisfies the stated condition.

The asymptotic form of ψ as $\xi \to -\infty$ (that is, as $x \to -\infty$) is [19]

$$\begin{aligned}\psi &\approx \xi^{-ik_2/\alpha}\left[C_1(-\xi)^{i(k_2-k_1)/\alpha} + C_2(-\xi)^{i(k_1+k_2)/\alpha}\right]\\ &= (-1)^{ik_2/\alpha}\left[C_1 e^{ik_1x} + C_2 e^{-ik_1x}\right],\end{aligned}$$

where

$$\begin{aligned}C_1 &= \frac{\Gamma(-(2i/\alpha)k_1)\Gamma(-(2i/\alpha)k_2 + 1)}{\Gamma(-(i/\alpha)(k_1 + k_2))\Gamma(-(i/\alpha)(k_1 + k_2) + 1)},\\ C_2 &= \frac{\Gamma((2i/\alpha)k_1)\Gamma(-(2i/\alpha)k_2 + 1)}{\Gamma((i/\alpha)(k_1 - k_2))\Gamma((i/\alpha)(k_1 - k_2) + 1)}.\end{aligned}$$

The required reflection coefficient is $R = |C_2/C_1|^2$; calculation using the familiar formula

$$\Gamma(x)\Gamma(1-x) = \frac{\pi}{\sin \pi x}$$

gives

$$R = \left\{\frac{\sinh[(\pi/\alpha)(k_1 - k_2)]}{\sinh[(\pi/\alpha)(k_1 + k_2)]}\right\}^2.$$

At $E = U_0$ ($k_2 = 0$), R becomes unity, while as $E \to \infty$ it tends to zero according to

$$R = \left(\frac{\pi U_0}{\alpha\hbar}\right)^2 \frac{2m}{E} \exp\left(-\frac{4\pi}{\alpha\hbar}\sqrt{2mE}\right).$$

In the limiting transition to classical mechanics, R vanishes, as it should.

Problem

4. Determine the transmission coefficient of a particle through the potential barrier defined by

$$U(x) = \frac{U_0}{\cosh^2(\alpha x)}$$

(Fig. 8); the particle energy is $E < U_0$.

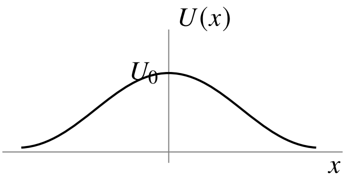

Figure 8

[19] See formula (e.6), in each of whose two terms only the first term of the expansion is to be retained; that is, the hypergeometric functions of $1/z$ are to be replaced by unity.

SOLUTION. The Schrödinger equation for this problem is obtained from the example considered in Problem 5 of § 23 by changing the sign of U_0, with the energy E now taken to be positive.

The same method gives the solution

$$\psi = (1-\xi^2)^{-ik/2\alpha} F\left(-\frac{ik}{\alpha} - s, -\frac{ik}{\alpha} + s + 1, -\frac{ik}{\alpha} + 1, \frac{1-\xi}{2}\right), \tag{1}$$

where

$$\xi = \tanh(\alpha x), \qquad k = \frac{1}{\hbar}\sqrt{2mE}, \qquad s = \frac{1}{2}\left(-1 + \sqrt{1 - \frac{8mU_0}{\alpha^2\hbar^2}}\right).$$

This solution already satisfies the condition that as $x \to \infty$ (that is, as $\xi \to 1$, $(1-\xi) \approx 2e^{-2\alpha x}$) the wave function contain only the transmitted wave ($\sim e^{ikx}$). The asymptotic expression for the wave function as $x \to -\infty$ ($\xi \to -1$) is obtained by transforming the hypergeometric function via formula (e.7):

$$\psi \sim e^{-ikx}\frac{\Gamma(ik/\alpha)\Gamma(1-ik/\alpha)}{\Gamma(-s)\Gamma(1+s)} + e^{ikx}\frac{\Gamma(-ik/\alpha)\Gamma(1-ik/\alpha)}{\Gamma(-ik/\alpha - s)\Gamma(-ik/\alpha + s + 1)}. \tag{2}$$

Taking the squared modulus of the ratio of the coefficients in this function, we obtain the following expression for the transmission coefficient $D = 1 - R$:

$$D = \frac{\sinh^2(\pi k/\alpha)}{\sinh^2(\pi k/\alpha) + \cos^2\left(\frac{\pi}{2}\sqrt{1 - \frac{8mU_0}{\hbar^2\alpha^2}}\right)} \quad \text{when} \quad \frac{8mU_0}{\hbar^2\alpha^2} < 1,$$

$$D = \frac{\sinh^2(\pi k/\alpha)}{\sinh^2(\pi k/\alpha) + \cosh^2\left(\frac{\pi}{2}\sqrt{\frac{8mU_0}{\hbar^2\alpha^2} - 1}\right)} \quad \text{when} \quad \frac{8mU_0}{\hbar^2\alpha^2} > 1.$$

The first of these formulas also applies when $U_0 < 0$, that is, when the particle passes not over a potential barrier but over a potential well. It is noteworthy that then $D = 1$ if $1 + (8m|U_0|/\hbar^2\alpha^2) = (2n+1)^2$; thus, for certain values of the well depth $|U_0|$, particles passing over it are not reflected. This is already clear from expression (2), in which the term containing e^{-ikx} is entirely absent when s is a positive integer.

PROBLEM

5. Determine how the transmission coefficient tends to zero as $E \to 0$, assuming that the potential energy $U(x)$ decreases rapidly at distances $|x| \gg a$, where a is the characteristic size of the interaction region.

SOLUTION. In the range of distances $k|x| \ll 1$, the energy E may be neglected in the Schrödinger equation. If also $|x| \gg a$, the potential energy may be neglected as well, and the equation reduces to

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = 0,$$

whose solution we write as

$$\psi = a_1 + b_1 x, \quad x < 0, \qquad \psi = a_2 + b_2 x, \quad x > 0. \tag{1}$$

Solving the equation at distances $|x| \sim a$ determines the relation among a_1, b_1 and a_2, b_2. This relation is linear and has the form

$$a_1 = \rho a_2 + \mu b_2, \qquad b_1 = \nu a_2 + \tau b_2. \tag{2}$$

The coefficients ρ, μ, ν, and τ are real and independent of the energy, since the energy no longer enters the equation [20]. Solution (1) must agree with the first two terms in the expansions of (25.1) and (25.2) in powers of x, whence

$$a_1 = 1 + B, \qquad b_1 = ik(1 - B), \qquad a_2 = A, \qquad b_2 = ikA.$$

Substituting these expressions in (2) and solving the equations for A, we find, at small k, $A \approx 2ik/\nu$, whence

$$D \approx \frac{4k^2}{\nu^2} \sim E.$$

Thus the transmission coefficient tends to zero in proportion to the particle energy. This general relation is, of course, satisfied in the examples considered in Problems 2 and 4.

[20]By constancy of the flux, they satisfy $\rho\tau - \mu\nu = 1$.

CHAPTER IV

Angular Momentum

§ 26. Angular momentum

In deriving momentum conservation in § 15, we used the homogeneity of space for a closed system of particles. Besides being homogeneous, space is also isotropic: every direction in it is equivalent. The Hamiltonian of a closed system must therefore remain unchanged when the system as a whole is rotated through an arbitrary angle about an arbitrary axis. It is enough to impose this requirement for an arbitrary infinitesimal rotation.

Let $\delta\boldsymbol{\varphi}$ be the vector of an infinitesimal rotation. Its magnitude is the angle of rotation $\delta\varphi$, and its direction is the axis about which the rotation is made. Under such a rotation, the changes $\delta\mathbf{r}_a$ of the particles' radius vectors $\mathbf{r}_a$ are

$$\delta\mathbf{r}_a = \delta\boldsymbol{\varphi}\times\mathbf{r}_a.$$

An arbitrary function $\psi(\mathbf{r}_1, \mathbf{r}_2, \ldots)$ is transformed into

$$\begin{aligned}\psi(\mathbf{r}_1+\delta\mathbf{r}_1, \mathbf{r}_2+\delta\mathbf{r}_2, \ldots) &= \psi(\mathbf{r}_1, \mathbf{r}_2, \ldots) + \sum_a \delta\mathbf{r}_a \boldsymbol{\nabla}_a\psi = \\ &= \psi(\mathbf{r}_1, \mathbf{r}_2, \ldots) + \sum_a (\delta\boldsymbol{\varphi}\times\mathbf{r}_a)\cdot\boldsymbol{\nabla}_a\psi = \\ &= \left(1+\delta\boldsymbol{\varphi}\cdot\sum_a(\mathbf{r}_a\times\boldsymbol{\nabla}_a)\right)\psi(\mathbf{r}_1, \mathbf{r}_2, \ldots).\end{aligned}$$

The expression

$$1+\delta\boldsymbol{\varphi}\cdot\sum_a(\mathbf{r}_a\times\boldsymbol{\nabla}_a)$$

is the operator of an infinitesimal rotation. The fact that such a rotation does not change the system's Hamiltonian is expressed (cf. § 15) by the commutation of the rotation operator with $\hat{H}$. Since $\delta\boldsymbol{\varphi}$ is a constant vector, this condition reduces to

$$\left(\sum_a(\mathbf{r}_a\times\boldsymbol{\nabla}_a)\right)\hat{H} - \hat{H}\left(\sum_a(\mathbf{r}_a\times\boldsymbol{\nabla}_a)\right) = 0, \tag{26.1}$$

which expresses a conservation law.

The quantity conserved in a closed system as a consequence of the isotropy of space is the system's *angular momentum* (cf. Vol. I, § 9). Thus, apart from a constant factor, the operator $\sum_a(\mathbf{r}_a\times\boldsymbol{\nabla}_a)$ must correspond to the total angular momentum of the system's motion, while each term $(\mathbf{r}_a\times\boldsymbol{\nabla}_a)$ must correspond to the angular momentum of one particle.

The proportionality factor must be chosen as $-i\hbar$. The resulting particle angular-momentum operator $-i\hbar(\mathbf{r}\times\boldsymbol{\nabla}) = \mathbf{r}\times\hat{\mathbf{p}}$ then agrees exactly with the classical expression $\mathbf{r}\times\mathbf{p}$. Henceforth we shall always measure angular momentum in units of $\hbar$. We denote the operator of the angular momentum of one particle, defined in this way, by $\hat{\mathbf{l}}$, and that of the entire system by $\hat{\mathbf{L}}$. Thus the particle angular-momentum operator is

$$\hbar\hat{\mathbf{l}} = \mathbf{r}\times\hat{\mathbf{p}} = -i\hbar(\mathbf{r}\times\boldsymbol{\nabla}) \tag{26.2}$$

or, in components,

$$\hbar\hat{l}_x = y\hat{p}_z - z\hat{p}_y, \qquad \hbar\hat{l}_y = z\hat{p}_x - x\hat{p}_z, \qquad \hbar\hat{l}_z = x\hat{p}_y - y\hat{p}_x.$$

For a system in an external field, angular momentum is not generally conserved. It may nevertheless be conserved when the field has a suitable symmetry. In a centrally symmetric field, for example, all spatial directions from the center are equivalent, so angular momentum about that center is conserved. When the field has axial symmetry, the same reasoning shows that what is conserved is the angular-momentum component directed along the symmetry axis. All these conservation laws of classical mechanics remain valid in quantum mechanics.

If a system's angular momentum is not conserved, it has no definite value in a stationary state. In such cases the mean angular momentum in a given stationary state can sometimes be of interest. Its mean value is readily seen to vanish in every nondegenerate stationary state. Indeed, the energy is unchanged when the sign of time is reversed; and since only one stationary state corresponds to the given energy level, replacing t by $-t$ must leave the state of the system unchanged . The mean values of all quantities, including angular momentum, must therefore also remain unchanged. But angular momentum reverses sign under time reversal, which would give $\bar{\mathbf{L}} = -\bar{\mathbf{L}}$; hence $\bar{\mathbf{L}} = 0$. A direct calculation confirms this: the integral $\int \psi^*\hat{\mathbf{L}}\psi\, dq$, which by definition gives the mean $\bar{\mathbf{L}}$, can be evaluated using the fact that nondegenerate wave functions are real (see the end of § 18). One then finds that

$$\bar{\mathbf{L}} = -i\hbar\int \psi^*\left(\sum_a(\mathbf{r}_a\times\boldsymbol{\nabla}_a)\right)\psi\, dq$$

is purely imaginary. Since $\bar{\mathbf{L}}$ must of course be real, it follows that $\bar{\mathbf{L}} = 0$.

We now determine the commutation rules between the angular-momentum operators and the coordinate and momentum operators. From (16.2) we readily obtain

$$\begin{aligned} \{\hat{l}_x, x\} &= 0, \quad \{\hat{l}_x, y\} = iz, \quad \{\hat{l}_x, z\} = -iy,\\ \{\hat{l}_y, y\} &= 0, \quad \{\hat{l}_y, z\} = ix, \quad \{\hat{l}_y, x\} = -iz,\\ \{\hat{l}_z, z\} &= 0, \quad \{\hat{l}_z, x\} = iy, \quad \{\hat{l}_z, y\} = -ix. \end{aligned} \tag{26.3}$$

For example,

$$\hat{l}_x y - y\hat{l}_x = \frac{1}{\hbar}(y\hat{p}_z - z\hat{p}_y)y - y(y\hat{p}_z - z\hat{p}_y)\frac{1}{\hbar} = -\frac{z}{\hbar}\{\hat{p}_y, y\} = iz.$$

All the relations (26.3) may be written in tensor form as

$$\{\hat{l}_i, x_k\} = i e_{ikl} x_l, \tag{26.4}$$

where e_{ikl} is the antisymmetric unit tensor of third rank[1] , with summation understood over each pair of repeated “dummy” indices. The angular-momentum and momentum operators satisfy analogous commutation relations:

$$\{\hat{l}_i, \hat{p}_k\} = i e_{ikl} \hat{p}_l. \tag{26.5}$$

These formulas make it easy to obtain the commutation rules among the operators for the components of angular momentum. We have

$$\begin{aligned}\hbar(\hat{l}_x\hat{l}_y - \hat{l}_y\hat{l}_x) &= \hat{l}_x(z\hat{p}_x - x\hat{p}_z) - (z\hat{p}_x - x\hat{p}_z)\hat{l}_x = \\ &= (\hat{l}_x z - z\hat{l}_x)\hat{p}_x - x(\hat{l}_x\hat{p}_z - \hat{p}_z\hat{l}_x) = -iy\hat{p}_x + ix\hat{p}_y = i\hbar\hat{l}_z.\end{aligned}$$

Thus

$$\{\hat{l}_y, \hat{l}_z\} = i\hat{l}_x, \qquad \{\hat{l}_z, \hat{l}_x\} = i\hat{l}_y, \qquad \{\hat{l}_x, \hat{l}_y\} = i\hat{l}_z \tag{26.6}$$

or

$$\{\hat{l}_i, \hat{l}_k\} = i e_{ikl} \hat{l}_l. \tag{26.7}$$

Exactly the same relations hold for the operators $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$ of the system’s total angular momentum. Indeed, since the angular-momentum operators of different particles commute with one another, we have, for example,

$$\sum_a \hat{l}_{ay} \sum_a \hat{l}_{az} - \sum_a \hat{l}_{az} \sum_a \hat{l}_{ay} = \sum_a (\hat{l}_{ay}\hat{l}_{az} - \hat{l}_{az}\hat{l}_{ay}) = i \sum_a \hat{l}_{ax}.$$

Therefore

$$\{\hat{L}_y, \hat{L}_z\} = i\hat{L}_x, \qquad \{\hat{L}_z, \hat{L}_x\} = i\hat{L}_y, \qquad \{\hat{L}_x, \hat{L}_y\} = i\hat{L}_z. \tag{26.8}$$

Relations (26.8) show that the three components of angular momentum cannot have definite values simultaneously (except when all three components vanish at once; see below). In this respect angular momentum differs essentially from momentum, whose three components can be measured simultaneously.

From $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$ we form the operator for the square of the magnitude of the angular-momentum vector:

$$\hat{\mathbf{L}}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2. \tag{26.9}$$

[1]The antisymmetric unit tensor of third rank e_{ikl} (also called the unit axial tensor) is defined to be antisymmetric in all three indices, with $e_{123} = 1$. Evidently, of its 27 components, only the 6 for which i, k, l form a permutation of the numbers 1, 2, 3 are nonzero. A component is $+1$ if the permutation i, k, l is obtained from 1, 2, 3 by an even number of pairwise interchanges (transpositions), and is -1 for an odd number. Clearly,

$$e_{ikl}e_{ikm} = 2\delta_{lm}, \qquad e_{ikl}e_{ikl} = 6.$$

The components of the vector $\mathbf{C} = \mathbf{A} \times \mathbf{B}$, the vector product of $\mathbf{A}$ and $\mathbf{B}$, can be expressed through e_{ikl} as

$$C_i = e_{ikl} A_k B_l.$$

This operator commutes with each of $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$:

$$\{\hat{\mathbf{L}}^2, \hat{L}_x\} = 0, \qquad \{\hat{\mathbf{L}}^2, \hat{L}_y\} = 0, \qquad \{\hat{\mathbf{L}}^2, \hat{L}_z\} = 0. \tag{26.10}$$

Indeed, using (26.8), we find, for example,

$$\begin{aligned}
&\{\hat{L}_x^2, \hat{L}_z\} = \hat{L}_x\{\hat{L}_x, \hat{L}_z\} + \{\hat{L}_x, \hat{L}_z\}\hat{L}_x = -i(\hat{L}_x\hat{L}_y + \hat{L}_y\hat{L}_x),\\
&\{\hat{L}_y^2, \hat{L}_z\} = i(\hat{L}_x\hat{L}_y + \hat{L}_y\hat{L}_x),\\
&\{\hat{L}_z^2, \hat{L}_z\} = 0.
\end{aligned}$$

Adding these equalities gives the last relation in (26.10).

Physically, (26.10) means that the square of the angular momentum (that is, its magnitude) can have a definite value simultaneously with one of its components.

Instead of $\hat{L}_x$ and $\hat{L}_y$, it is often convenient to use their complex combinations

$$\hat{L}_+ = \hat{L}_x + i\hat{L}_y, \qquad \hat{L}_- = \hat{L}_x - i\hat{L}_y. \tag{26.11}$$

Direct calculation from (26.8) readily gives the following commutation rules for these combinations:

$$\{\hat{L}_+, \hat{L}_-\} = 2\hat{L}_z, \qquad \{\hat{L}_z, \hat{L}_+\} = \hat{L}_+, \qquad \{\hat{L}_z, \hat{L}_-\} = -\hat{L}_-. \tag{26.12}$$

It is also easy to verify that

$$\hat{\mathbf{L}}^2 = \hat{L}_+\hat{L}_- + \hat{L}_z^2 - \hat{L}_z = \hat{L}_-\hat{L}_+ + \hat{L}_z^2 + \hat{L}_z. \tag{26.13}$$

Finally, we give the frequently used expressions for the angular-momentum operator of one particle in spherical coordinates. Introducing these coordinates through the usual relations

$$x = r\sin\theta\cos\varphi, \qquad y = r\sin\theta\sin\varphi, \qquad z = r\cos\theta,$$

a straightforward calculation gives

$$\hat{l}_z = -i\frac{\partial}{\partial\varphi}, \tag{26.14}$$

$$\hat{l}_\pm = e^{\pm i\varphi}\left(\pm\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\varphi}\right). \tag{26.15}$$

Substitution in (26.13) gives the square of the particle's angular-momentum operator in the form

$$\hat{\mathbf{l}}^2 = -\left[\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2} + \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)\right]. \tag{26.16}$$

Notice that, apart from a factor, this is the angular part of the Laplace operator.

§ 27. Eigenvalues of angular momentum

In finding the eigenvalues of a particle's angular-momentum projection onto a chosen direction, one may orient the polar axis along that direction and employ the spherical-coordinate form of the operator. Using (26.14), the equation $\hat{l}_z\psi = l_z\psi$ takes the form

$$-i\frac{\partial\psi}{\partial\varphi} = l_z\psi. \tag{27.1}$$

Its solution is

$$\psi = f(r,\theta)e^{il_z\varphi},$$

where $f(r,\theta)$ is an arbitrary function of r and θ. For ψ to be single-valued, it must be periodic in φ, with period 2π. Hence we find[2]

$$l_z = m, \qquad m = 0,\ \pm 1,\ \pm 2,\ldots \tag{27.2}$$

Thus the eigenvalues l_z are the positive and negative integers, including zero. We denote the factor depending on φ that is characteristic of the eigenfunctions of $\hat{l}_z$ by

$$\Phi_m(\varphi) = \frac{1}{\sqrt{2\pi}}e^{im\varphi}. \tag{27.3}$$

These functions are normalized so that

$$\int_0^{2\pi}\Phi_m^*(\varphi)\Phi_{m'}(\varphi)\,d\varphi = \delta_{mm'}. \tag{27.4}$$

The eigenvalues of the z component of the total angular momentum of a system are evidently likewise positive and negative integers:

$$L_z = M, \qquad M = 0,\ \pm 1,\ \pm 2,\ldots \tag{27.5}$$

(this follows because $\hat{L}_z$ is the sum of the mutually commuting operators $\hat{l}_z$ of the individual particles).

Since the direction of the z axis has not been distinguished in advance, the same result clearly holds for L_x, L_y, and, in general, for the component of angular momentum along any direction: each can take only integer values. At first sight this result may seem paradoxical, particularly if it is applied to two infinitely close directions. One must bear in mind, however, that the only common eigenfunction of the operators $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$ corresponds to the simultaneous values

$$L_x = L_y = L_z = 0;$$

in this case the angular-momentum vector, and therefore its projection on every direction, is zero. If at least one of the eigenvalues L_x, L_y, L_z is nonzero, the corresponding operators have no common eigenfunctions. Put differently, no state exists in which definite nonzero values are possessed simultaneously by two or three angular-momentum

[2] The customary notation of the eigenvalues of the angular-momentum projection by the letter m—the same letter that also denotes the particle mass—cannot in practice lead to confusion.

components referred to distinct directions; one can therefore only discuss the integer nature of a single component at a time.

Stationary states of a system that differ only in the value of M have the same energy. This follows already from general considerations, since the direction of the z axis has not been distinguished beforehand. Thus the energy levels of a system with conserved nonzero angular momentum are in any event degenerate[3].

We now turn to finding the eigenvalues of the square of angular momentum and show how they can be obtained solely from the commutation rules (26.8). Let ψ_M denote the wave functions of stationary states having the same value of $\mathbf{L}^2$, belonging to one degenerate energy level, and differing in the value of M[4].

First, since the two directions of the z axis are physically equivalent, every possible positive value $M = |M|$ has a corresponding negative value $M = -|M|$. Let L (a nonnegative integer) denote the largest possible value of $|M|$ for a given $\mathbf{L}^2$. The very existence of such an upper bound follows from the fact that the difference $\hat{\mathbf{L}}^2 - \hat{L}_z^2 = \hat{L}_x^2 + \hat{L}_y^2$ is the operator of the necessarily nonnegative physical quantity $L_x^2 + L_y^2$, and hence its eigenvalues cannot be negative.

Applying the operator $\hat{L}_z \hat{L}_\pm$ to an eigenfunction ψ_M of $\hat{L}_z$, and using the commutation rules (26.12), we obtain

$$\hat{L}_z \hat{L}_\pm \psi_M = (M \pm 1)\hat{L}_\pm \psi_M. \tag{27.6}$$

It follows that $\hat{L}_\pm \psi_M$ is, apart from a normalization constant, an eigenfunction corresponding to the value $M \pm 1$ of L_z:

$$\psi_{M+1} = \text{const} \cdot \hat{L}_+ \psi_M, \qquad \psi_{M-1} = \text{const} \cdot \hat{L}_- \psi_M. \tag{27.7}$$

If $M = L$ is put in the first of these equalities, then we must have identically

$$\hat{L}_+ \psi_L = 0, \tag{27.8}$$

since, by definition, there are no states with $M > L$. Applying $\hat{L}_-$ to this equality and using (26.13), we obtain

$$\hat{L}_- \hat{L}_+ \psi_L = \left(\hat{\mathbf{L}}^2 - \hat{L}_z^2 - \hat{L}_z\right)\psi_L = 0.$$

But since the ψ_M are common eigenfunctions of $\hat{\mathbf{L}}^2$ and $\hat{L}_z$,

$$\hat{\mathbf{L}}^2 \psi_L = \mathbf{L}^2 \psi_L, \qquad \hat{L}_z^2 \psi_L = L^2 \psi_L, \qquad \hat{L}_z \psi_L = L \psi_L,$$

[3] This circumstance is a special case of the general theorem stated in § 10 on the degeneracy of levels in the presence of at least two conserved quantities whose operators do not commute. Here those quantities are the components of angular momentum.

[4] Here it is understood that there is no additional degeneracy by which the same energy corresponds to different values of the square of angular momentum. This is true for the discrete spectrum (apart from the so-called "accidental degeneracy" in the Coulomb field; see § 36), and is, generally speaking, not true for energy levels in the continuous spectrum. Even when there is additional degeneracy, however, the eigenfunctions can always be chosen to represent states with definite values of $\mathbf{L}^2$; from them one can then select states with the same values of E and $\mathbf{L}^2$. The mathematical basis for this is that simultaneous diagonalization of the matrices of mutually commuting operators is always achievable. In analogous cases below, for brevity, we shall speak as if there were no additional degeneracy, with the understanding that, in view of what has just been said, the results obtained do not in fact depend on this assumption.

so the resulting equation gives

$$\mathbf{L}^2 = L(L+1). \tag{27.9}$$

Formula (27.9) determines the required eigenvalues of the square of angular momentum; L takes all nonnegative integer values. For a given value of L, the angular-momentum component $L_z = M$ can take the values

$$M = L, L-1, \dots, -L, \tag{27.10}$$

that is, $2L+1$ distinct values in all. An energy level corresponding to angular momentum L is therefore $(2L+1)$-fold degenerate; this is usually called directional degeneracy of angular momentum. A state with zero angular momentum, $L = 0$ (so that all three of its components vanish), is nondegenerate. We note that the wave function of such a state is spherically symmetric. This is already clear from the fact that its change under any infinitesimal rotation, given by $\hat{\mathbf{L}}\psi$, vanishes in this case. For brevity we shall often use the customary phrase "angular momentum L" of the system, meaning angular momentum whose square equals $L(L+1)$; the z component of angular momentum is usually called simply its "projection."

We shall denote the angular momentum of one particle by the lowercase letter l; thus for it we write (27.9) as

$$\mathbf{l}^2 = l(l+1). \tag{27.11}$$

Let us calculate the matrix elements of L_x and L_y in the representation in which, together with the energy, $\mathbf{L}^2$ and L_z are diagonal (*M. Born, W. Heisenberg, P. Jordan*, 1926). We note that, since $\hat{L}_x$ and $\hat{L}_y$ commute with the Hamiltonian, their matrices are diagonal with respect to energy; that is, every matrix element for a transition between states of different energies (and different angular momenta L) is zero. It is therefore sufficient to consider matrix elements for transitions within the group of states with different values of M corresponding to a single degenerate energy level.

From (27.7) one sees that the matrix of $\hat{L}_+$ has nonzero elements only for transitions $M-1 \to M$, and that of $\hat{L}_-$ only for $M \to M-1$. With this in mind, equating the diagonal matrix elements of the two sides of (26.13) gives[5]

$$L(L+1) = \langle M|L_+|M-1\rangle\langle M-1|L_-|M\rangle + M^2 - M.$$

Since the operators $\hat{L}_x$ and $\hat{L}_y$ are Hermitian,

$$\langle M-1|L_-|M\rangle = \langle M|L_+|M-1\rangle^*,$$

we rewrite this equality as

$$\left|\langle M|L_+|M-1\rangle\right|^2 = L(L+1) - M(M-1) = (L-M+1)(L+M),$$

whence[6]

$$\langle M|L_+|M-1\rangle = \langle M-1|L_-|M\rangle = \sqrt{(L+M)(L-M+1)}. \tag{27.12}$$

[5] In the notation for matrix elements, for brevity, we omit all indices with respect to which they are diagonal (including the index L).

[6] The choice of sign in this formula is consistent with the choice of phase factors in the angular-momentum eigenfunctions.

For the nonzero matrix elements of L_x and L_y themselves, this gives

$$\begin{aligned}\langle M|L_x|M-1\rangle &= \langle M-1|L_x|M\rangle = \frac{1}{2}\sqrt{(L+M)(L-M+1)},\\ \langle M|L_y|M-1\rangle &= -\langle M-1|L_y|M\rangle = -\frac{i}{2}\sqrt{(L+M)(L-M+1)}.\end{aligned} \tag{27.13}$$

We draw attention to the absence of diagonal elements in the matrices of L_x and L_y. Since a diagonal matrix element gives the mean value of a quantity in the corresponding state, this means that in states with definite values of L_z the mean values are $\bar{L}_x = \bar{L}_y = 0$. Thus, if the projection of angular momentum on some direction in space has a definite value, the whole vector $\bar{\mathbf{L}}$ lies along that same direction.

§ 28. Angular-Momentum Eigenfunctions

Specifying the values of l and m does not determine a particle's wave function completely. Indeed, the expressions for the operators of these quantities in spherical coordinates contain only the angles θ and φ, so their eigenfunctions may include an arbitrary factor depending on r. Here we shall consider only the angular part of the wave function characteristic of angular-momentum eigenfunctions. Denote it by $Y_{lm}(\theta, \varphi)$ and impose the normalization condition

$$\int |Y_{lm}|^2 \, do = 1$$

($do = \sin\theta \, d\theta \, d\varphi$ is the element of solid angle).

The calculation below shows that the problem of finding the common eigenfunctions of $\hat{\mathbf{l}}^2$ and $\hat{l}_z$ permits separation of the variables θ and φ. These functions may therefore be sought in the form

$$Y_{lm} = \Phi_m(\varphi)\Theta_{lm}(\theta), \tag{28.1}$$

where $\Phi_m(\varphi)$ are the eigenfunctions of $\hat{l}_z$ defined by (27.3). Since the functions Φ_m have already been normalized by (27.4), the functions Θ_{lm} must satisfy

$$\int_0^\pi |\Theta_{lm}|^2 \sin\theta \, d\theta = 1. \tag{28.2}$$

Functions Y_{lm} with different values of l or m are automatically mutually orthogonal:

$$\int_0^{2\pi}\int_0^\pi Y^*_{l'm'} Y_{lm} \sin\theta \, d\theta \, d\varphi = \delta_{ll'}\delta_{mm'}, \tag{28.3}$$

because they are eigenfunctions of angular-momentum operators belonging to different eigenvalues. The functions $\Phi_m(\varphi)$ are also separately orthogonal (see (27.4)), since they are eigenfunctions of $\hat{l}_z$ belonging to its different eigenvalues m. The functions $\Theta_{lm}(\theta)$ by themselves, however, are not eigenfunctions of any angular-momentum operator; they are mutually orthogonal for different l, but not for different m.

The most direct way to calculate the required functions is to solve the eigenfunction problem for $\hat{\mathbf{l}}^2$ written in spherical coordinates (formula (26.16)). The equation $\hat{\mathbf{l}}^2\psi = l^2\psi$ reads

$$\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2\psi}{\partial\varphi^2} + l(l+1)\psi = 0.$$

Substitution of ψ in the form (28.1) gives the following equation for Θ_{lm}:

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta_{lm}}{d\theta}\right) - \frac{m^2}{\sin^2\theta}\Theta_{lm} + l(l+1)\Theta_{lm} = 0. \tag{28.4}$$

This equation is well known from the theory of spherical harmonics. It has solutions satisfying the finiteness and single-valuedness conditions for nonnegative integral values $l \geq |m|$, in agreement with the angular-momentum eigenvalues obtained above by the matrix method. The corresponding solutions are the associated Legendre polynomials $P_l^m(\cos\theta)$ (see § c of the mathematical supplements). Normalizing the solution by (28.2), we obtain[7]

$$\Theta_{lm} = (-1)^m i^l \sqrt{\frac{(2l+1)(l-m)!}{2(l+m)!}}\, P_l^m(\cos\theta). \tag{28.5}$$

Here $m \geq 0$ is assumed. For negative m, define Θ_{lm} by

$$\Theta_{l,-|m|} = (-1)^m \Theta_{l|m|}, \tag{28.6}$$

that is, for $m < 0$, Θ_{lm} is given by (28.5) with $|m|$ in place of m and with the factor $(-1)^m$ omitted. Thus, from the mathematical point of view, the angular-momentum eigenfunctions are spherical harmonics with a particular normalization. For later reference, we write their complete expression, incorporating all the definitions just given:

$$Y_{lm}(\theta,\varphi) = (-1)^{(m+|m|)/2} i^l \left[\frac{2l+1}{4\pi}\frac{(l-|m|)!}{(l+|m|)!}\right]^{1/2} P_l^{|m|}(\cos\theta)e^{im\varphi}. \tag{28.7}$$

In particular,

$$Y_{l0} = i^l\sqrt{\frac{2l+1}{4\pi}}P_l(\cos\theta). \tag{28.8}$$

Functions differing only in the sign of m are evidently related by

$$(-1)^{l-m}Y_{l,-m} = Y^*_{lm}. \tag{28.9}$$

For $l = 0$ (and consequently $m = 0$), the spherical harmonic reduces to a constant. In other words, the wave functions of particle states with zero angular momentum depend only on r and hence possess complete spherical symmetry, in accordance with the general statement made in § 27.

For a fixed m, the values of l beginning with $|m|$ enumerate, in ascending order, the successive eigenvalues of $\mathbf{l}^2$. The general theorem on zeros of eigenfunctions (§ 21) therefore implies that Θ_{lm} vanishes at $l - |m|$ different values of θ; in other words, its nodal lines are $l - |m|$ "circles of latitude" on the sphere. For the complete angular

[7] The normalization condition does not, of course, determine the choice of phase factor. The convention used in this book is the one most natural from the standpoint of the general theory of angular-momentum addition; it differs from the convention usually adopted by a factor i^l. The advantages of this choice will be apparent from the notes on pp. 278, 527, and 533.

functions, if real factors $\cos m\varphi$ or $\sin m\varphi$ are chosen instead of $e^{\pm i|m|\varphi}$[8], they also have $|m|$ "meridian circles" as nodal lines. The total number of nodal lines is therefore l.

Finally, let us show how the functions Θ_{lm} can be calculated by the matrix method. The procedure is analogous to the calculation of the oscillator wave functions in § 23. We start from (27.8), $\hat{l}_+ Y_{ll} = 0$. Using (26.15) for the operator $\hat{l}_+$ and substituting

$$Y_{ll} = \frac{1}{\sqrt{2\pi}} e^{il\varphi} \Theta_{ll}(\theta),$$

we obtain for Θ_{ll} the equation

$$\frac{d\Theta_{ll}}{d\theta} - l \cot\theta \cdot \Theta_{ll} = 0,$$

whence $\Theta_{ll} = \text{const} \cdot \sin^l \theta$. Determining the constant from the normalization condition gives

$$\Theta_{ll} = (-i)^l \sqrt{\frac{(2l+1)!}{2}} \frac{1}{2^l l!} \sin^l \theta. \tag{28.10}$$

Next, using (27.12), write

$$\hat{l}_- Y_{l,m+1} = (l_-)_{m,m+1} Y_{lm} = \sqrt{(l-m)(l+m+1)} Y_{lm}.$$

Repeated application of this formula gives

$$\sqrt{\frac{(l-m)!}{(l+m)!}} Y_{lm} = \frac{1}{\sqrt{(2l)!}} \hat{l}_-^{l-m} Y_{ll}.$$

The right-hand side is readily evaluated with the aid of (26.15) for $\hat{l}_-$, according to which

$$\hat{l}_- \left[f(\theta) e^{im\varphi} \right] = e^{i(m-1)\varphi} \sin^{1-m}\theta \frac{d}{d\cos\theta} \left(f \sin^m \theta \right).$$

Repeated application of this formula gives

$$\hat{l}_-^{l-m} e^{il\varphi} \Theta_{ll} = e^{im\varphi} \sin^{-m}\theta \frac{d^{l-m}}{(d\cos\theta)^{l-m}} \left(\sin^l \theta \cdot \Theta_{ll} \right).$$

Finally, using these relations together with (28.10) for Θ_{ll}, we obtain

$$\Theta_{lm}(\theta) = (-i)^l \sqrt{\frac{2l+1}{2} \frac{(l+m)!}{(l-m)!}} \frac{1}{2^l l! \sin^m \theta} \frac{d^{l-m}}{(d\cos\theta)^{l-m}} \sin^{2l}\theta, \tag{28.11}$$

which agrees with (28.5).

[8]Each such function represents a state in which l_z has no definite value, but may take the values $\pm m$ with equal probability.

§ 29. Matrix elements of vectors

Return once more to a closed system of particles[9]. Let f stand for some scalar physical quantity of this system and $\hat{f}$ for its operator. Any scalar is invariant under a rotation of the coordinate system. Hence the scalar operator $\hat{f}$ is unchanged by a rotation; that is, it commutes with the rotation operator. We know, however, that the operator of an infinitesimal rotation coincides, apart from a constant factor, with the angular-momentum operator. Therefore

$$\{\hat{f}, \hat{\mathbf{L}}\} = 0. \tag{29.1}$$

It follows from the commutation of $\hat{f}$ with the angular-momentum operator that, for transitions between states with definite L and M, the matrix of f is diagonal in these indices. Moreover, specifying M determines only the orientation of the system relative to the coordinate axes, while the value of a scalar quantity does not depend on this orientation at all. Consequently, the matrix elements $\langle n'LM|f|nLM\rangle$ do not depend on M (the letter n provisionally denotes the collection of all quantum numbers, other than L and M, that specify the state of the system). A formal proof follows from the commutation of $\hat{f}$ and $\hat{L}_+$:

$$\hat{f}\hat{L}_+ - \hat{L}_+\hat{f} = 0. \tag{29.2}$$

Take the matrix element of this equality for the transition $n, L, M \to n', L, M+1$. Since the matrix of L_+ has only elements for transitions $n, L, M \to n, L, M+1$, we find

$$\begin{aligned} \langle n', L, M+1|f|n, L, M+1\rangle\langle n, L, M+1|L_+|n, L, M\rangle = \\ = \langle n', L, M+1|L_+|n', L, M\rangle\langle n', L, M|f|n, L, M\rangle, \end{aligned}$$

and, since the matrix elements of L_+ do not depend on the index n,

$$\langle n', L, M+1|f|n, L, M+1\rangle = \langle n', L, M|f|n, L, M\rangle, \tag{29.3}$$

which shows that all the elements $\langle n', L, M|f|n, L, M\rangle$ with different M (and identical remaining indices) are equal.

Applied to the Hamiltonian itself, this result gives the already known independence of the stationary-state energy from M, that is, the $(2L+1)$-fold degeneracy of the energy levels.

Next, let $\mathbf{A}$ be some vector physical quantity characterizing the closed system. Under a rotation of the coordinate system (in particular, an infinitesimal rotation, that is, under the action of the angular-momentum operator), the components of the vector transform into one another. The commutation of the operators $\hat{L}_i$ with the operators $\hat{A}_i$ must therefore again yield components of the same vector $\hat{A}_i$. Which components occur can be found by noting that, in the special case where $\mathbf{A}$ is the radius vector of a particle, the result must reduce to (26.4). We thus obtain the commutation rules

$$\{\hat{L}_i, \hat{A}_k\} = ie_{ikl}\hat{A}_l. \tag{29.4}$$

[9] Everything derived in this section applies equally to a particle in a centrally symmetric field (more broadly, to any situation in which the system's total angular momentum is conserved).

From these relations one can derive a number of results about the structure of the component matrices of the vector **A** (*M. Born, W. Heisenberg, P. Jordan*, 1926). First, they make it possible to find the *selection rules*, which determine the transitions for which matrix elements may be nonzero. We shall not, however, give the corresponding rather cumbersome calculations here, since it will be shown below (§ 107) that these rules are in fact direct consequences of the general transformation properties of vector quantities and can be obtained from them essentially without calculation. Here we merely state the rules without derivation.

Matrix elements of every component of a vector can be nonzero only for transitions in which the angular momentum L changes by no more than one:

$$L \to L, L \pm 1. \tag{29.5}$$

There is also an additional selection rule forbidding transitions between any two states with $L = 0$. This rule is an evident consequence of the complete spherical symmetry of states with zero angular momentum.

The selection rules for the angular-momentum projection M differ among the components of the vector. Specifically, matrix elements may be nonzero for transitions with the following changes of M:

$$\begin{aligned} &M \to M+1 \quad &&\text{for } A_+ = A_x + iA_y, \\ &M \to M-1 \quad &&\text{for } A_- = A_x - iA_y, \\ &M \to M \quad &&\text{for } A_z. \end{aligned} \tag{29.6}$$

It is also possible to determine in general form the dependence of the matrix elements of a vector on M. We again give these important and frequently used formulas without derivation, since they too are special cases of more general relations, applying to arbitrary tensor quantities, which will be derived in § 107. The nonzero matrix elements of A_z are given by

$$\begin{aligned} \langle n'LM|A_z|nLM\rangle &= \frac{M}{\sqrt{L(L+1)(2L+1)}}\langle n'L\|A\|nL\rangle, \\ \langle n'LM|A_z|n, L-1, M\rangle &= \sqrt{\frac{L^2-M^2}{L(2L-1)(2L+1)}}\langle n'L\|A\|n, L-1\rangle, \\ \langle n', L-1, M|A_z|nLM\rangle &= \sqrt{\frac{L^2-M^2}{L(2L-1)(2L+1)}}\langle n', L-1\|A\|nL\rangle. \end{aligned} \tag{29.7}$$

Here the symbol

$$\langle n'L'\|A\|nL\rangle$$

denotes a so-called *reduced matrix element*, a quantity independent of the quantum number M[10]. They are related by

$$\langle n'L'\|A\|nL\rangle = \langle nL\|A\|n'L'\rangle^*, \tag{29.8}$$

[10]The appearance in (29.7) and (29.9) of denominators depending on L accords with the general notation introduced in § 107. The usefulness of these denominators is seen, in particular, in the simple form taken by (29.12) for the matrix elements of the scalar product of two vectors.

One should regard the notation for the reduced matrix element as an indivisible entity (in contrast to what was said about the matrix element notation (11.17)).

which follows directly from the Hermiticity of $\hat{A}_z$.

The matrix elements of A_- and A_+ can be expressed through the same reduced elements. The nonzero matrix elements of A_- are

$$\begin{aligned}
\langle n', L, M-1|A_-|nLM\rangle &= \sqrt{\frac{(L-M+1)(L+M)}{L(L+1)(2L+1)}}\langle n'L\|A\|nL\rangle,\\
\langle n', L, M-1|A_-|n, L-1, M\rangle &= \sqrt{\frac{(L-M+1)(L-M)}{L(2L-1)(2L+1)}}\langle n'L\|A\|n, L-1\rangle,\\
\langle n', L-1, M-1|A_-|nLM\rangle &= -\sqrt{\frac{(L+M-1)(L+M)}{L(2L-1)(2L+1)}}\langle n', L-1\|A\|nL\rangle.
\end{aligned} \tag{29.9}$$

The matrix elements of A_+ need no separate formulas, since the reality of A_x and A_y gives

$$\langle n'L'M'|A_+|nLM\rangle = \langle nLM|A_-|n'L'M'\rangle^*. \tag{29.10}$$

Let us write out the formula that relates the matrix elements of the scalar **AB** to the reduced matrix elements of the two vectors **A** and **B**. The calculation is conveniently made by writing the operator $\hat{\mathbf{A}}\hat{\mathbf{B}}$ as

$$\hat{\mathbf{A}}\hat{\mathbf{B}} = \frac{1}{2}(\hat{A}_+\hat{B}_- + \hat{A}_-\hat{B}_+) + \hat{A}_z\hat{B}_z. \tag{29.11}$$

The matrix of **AB**, like that of any scalar, is diagonal in L and M. Calculation using (29.7)–(29.9) gives

$$\langle n'LM|\mathbf{AB}|nLM\rangle = \frac{1}{2L+1}\sum_{n'', L''}\langle n'L\|A\|n''L''\rangle\langle n''L''\|B\|nL\rangle, \tag{29.12}$$

where L'' takes the values $L, L\pm 1$.

For reference, we give the reduced matrix elements of the vector **L** itself. Comparing (29.9) and (27.12), we find

$$\begin{aligned}
\langle L\|L\|L\rangle &= \sqrt{L(L+1)(2L+1)},\\
\langle L-1\|L\|L\rangle &= \langle L\|L\|L-1\rangle = 0.
\end{aligned} \tag{29.13}$$

A quantity frequently encountered in applications is the unit vector **n** along the radius vector of a particle. Let us find its reduced matrix elements. It is sufficient to calculate, for example, the matrix elements of $n_z = \cos\theta$ at zero angular-momentum projection, $m = 0$. We have

$$\langle l-1, 0|n_z|l0\rangle = \int_0^\pi \Theta^*_{l-1,0}\cos\theta\cdot\Theta_{l0}\sin\theta\, d\theta$$

with the functions Θ_{l0} from (28.11). Evaluation of the integral gives[11]

$$\langle l-1, 0|n_z|l0\rangle = \frac{il}{\sqrt{(2l-1)(2l+1)}}.$$

[11]The calculation is performed by integrating by parts $(l-1)$ times with respect to $d\cos\theta$. For the general formula for integrals of this kind, see (107.14).

The matrix elements for transitions $l \to l$ are zero (as they are for any polar vector belonging to a single particle; see below, (30.8)). Comparison with (29.7) now gives

$$\langle l-1\|n\|l\rangle = -\langle l\|n\|l-1\rangle = i\sqrt{l}, \qquad \langle l\|n\|l\rangle = 0. \tag{29.14}$$

Problem

Average the tensor $n_i n_k - (1/3)\delta_{ik}$ (where $\mathbf{n}$ is the unit vector along the particle's radius vector) over a state with a definite value of $|\mathbf{l}|$ but no preferred orientation (that is, l_z is unspecified).

Solution. The required mean value is an operator that can be expressed only in terms of $\hat{\mathbf{l}}$. We seek it in the form

$$\overline{n_i n_k} - \frac{1}{3}\delta_{ik} = a\left[\hat{l}_i\hat{l}_k + \hat{l}_k\hat{l}_i - \frac{2}{3}\delta_{ik}l(l+1)\right];$$

this is the most general form of a symmetric second-rank tensor with zero trace constructed from the components of $\hat{\mathbf{l}}$. To determine the constant a, multiply this equality on the left by $\hat{l}_i$ and on the right by $\hat{l}_k$, with summation over i and k. Since the vector $\mathbf{n}$ is perpendicular to the vector $\hbar\hat{\mathbf{l}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$, we have $n_i\hat{l}_i = 0$. We replace the product $\hat{l}_i\hat{l}_i\hat{l}_k\hat{l}_k = (\hat{\mathbf{l}}^2)^2$ by its eigenvalue $l^2(l+1)^2$, and transform the product $\hat{l}_i\hat{l}_k\hat{l}_i\hat{l}_k$ using the commutation relations (26.7) as follows:

$$\begin{aligned}
\hat{l}_i\hat{l}_k\hat{l}_i\hat{l}_k &= \hat{l}_i\hat{l}_i\hat{l}_k\hat{l}_k - ie_{ikl}\hat{l}_i\hat{l}_l\hat{l}_k \\
&= (\hat{\mathbf{l}}^2)^2 - \frac{i}{2}e_{ikl}\hat{l}_i(\hat{l}_l\hat{l}_k - \hat{l}_k\hat{l}_l) \\
&= (\hat{\mathbf{l}}^2)^2 + \frac{1}{2}e_{ikl}e_{lkm}\hat{l}_i\hat{l}_m = (\hat{\mathbf{l}}^2)^2 - \hat{\mathbf{l}}^2 \\
&= l^2(l+1)^2 - l(l+1)
\end{aligned}$$

(we have used $e_{ikl}e_{mkl} = 2\delta_{im}$). Straightforward simplification then gives

$$a = -\frac{1}{(2l-1)(2l+3)}.$$

§ 30. Parity of a state

Besides translations and rotations of the coordinate system (invariance under which expresses, respectively, the homogeneity and isotropy of space), there is one more operation that preserves the Hamiltonian of a closed system: the *inversion transformation*, which reverses the sign of every coordinate at once (equivalently, it flips the direction of all axes, turning a right-handed frame into a left-handed one and conversely). Invariance of the Hamiltonian under inversion is the mathematical expression of the mirror symmetry of space[12]. In classical mechanics, the invariance of the Hamilton function under

[12] Inversion likewise leaves invariant the Hamiltonian of particles moving in a centrally symmetric field (provided the coordinate origin coincides with the center of that field).

inversion gives no new conservation laws. In quantum mechanics, however, the situation is essentially different.

Introduce the inversion operator $\hat{P}$[13], whose action on a wave function $\psi(\mathbf{r})$ consists in reversing the signs of the coordinates:

$$\hat{P}\psi(\mathbf{r}) = \psi(-\mathbf{r}). \tag{30.1}$$

The eigenvalues P of this operator, defined by the equation

$$\hat{P}\psi(\mathbf{r}) = P\psi(\mathbf{r}), \tag{30.2}$$

are easily found. Two successive applications of the inversion operator give the identity: the arguments of the function are left entirely unchanged. In other words, $\hat{P}^2\psi = P^2\psi = \psi$, so that $P^2 = 1$, whence

$$P = \pm 1. \tag{30.3}$$

An eigenfunction of the inversion operator is therefore either left entirely unaltered or acquires an overall sign change. Wave functions (and the corresponding states) of the first type are called *even*, and those of the second type *odd*.

The invariance of the Hamiltonian under inversion (that is, the commutation of $\hat{H}$ and $\hat{P}$) therefore expresses the *law of conservation of parity*: if a state of a closed system has definite parity (if it is even or odd), that parity is conserved in time[14].

The angular-momentum operator is likewise invariant under inversion: inversion reverses the signs both of the coordinates and of the operators for differentiation with respect to them, and hence the operator (26.2) remains unchanged. Put differently, inversion and angular momentum commute, so that a definite parity can coexist with definite values of L and of the projection M. Moreover, all states that differ only in M have the same parity. This is already evident from the independence of the properties of a closed system from its orientation in space; formally, it can be proved from the commutation relation $\hat{L}_+\hat{P} - \hat{P}\hat{L}_+ = 0$ by the same method used to obtain (29.3) from (29.2). There are definite *parity selection rules* for the matrix elements of various physical quantities.

Consider scalar quantities first. Here one must distinguish *true scalars*, which remain entirely unchanged under inversion, from *pseudoscalars*, which change sign under inversion (the scalar product of an axial vector and a polar vector is a pseudoscalar). The operator $\hat{f}$ of a true scalar commutes with $\hat{P}$. It follows that if the matrix of P is diagonal, then f is likewise diagonal with respect to parity: matrix elements vanish except for the $g \longrightarrow g$ and $u \longrightarrow u$ transitions (here g and u label even and odd states). For the operator of a pseudoscalar quantity, on the other hand, $\hat{P}\hat{f} = -\hat{f}\hat{P}$; the operators $\hat{P}$ and $\hat{f}$ "anticommute." The matrix element of this equality for a transition $g \longrightarrow g$ is

$$P_{gg}f_{gg} = -f_{gg}P_{gg},$$

and since $P_{gg} = 1$, we have $f_{gg} = 0$; in the same way one finds $f_{uu} = 0$. Thus the matrix of a pseudoscalar quantity has nonzero elements only for transitions in which the parity

[13]From the English word *parity*.

[14]To avoid misunderstanding, we recall that the discussion concerns nonrelativistic theory. Interactions exist in nature, and are treated in relativistic theory, that violate parity conservation.

changes. The selection rules for matrix elements of scalar quantities are therefore

$$\begin{aligned} &\text{true scalars:} && g \to g, \quad u \to u, \\ &\text{pseudoscalars:} && g \to u, \quad u \to g. \end{aligned} \tag{30.4}$$

These rules can also be obtained in another way, directly from the definition of matrix elements. Consider, for example, the integral $f_{ug} = \int \psi_u^* \hat{f} \psi_g \, dq$, where ψ_g is even and ψ_u is odd. On reversing the signs of all coordinates, the integrand changes sign if f is a true scalar. On the other hand, an integral over all space cannot change when the variables of integration are merely renamed. Hence $f_{ug} = -f_{ug}$, so that $f_{ug} = 0$.

The selection rules for vector quantities are obtained in the same way. One must remember that ordinary, polar vectors reverse sign under inversion, whereas axial vectors remain unchanged (an example is the angular-momentum vector, the cross product $\mathbf{r} \times \mathbf{p}$ of two polar vectors). Taking this into account, we find

$$\begin{aligned} &\text{polar vectors:} && g \to u, \quad u \to g, \\ &\text{axial vectors:} && g \to g, \quad u \to u. \end{aligned} \tag{30.5}$$

Let us determine the parity of a one-particle state with angular momentum l. Inversion ($x \to -x$, $y \to -y$, $z \to -z$) consists, in spherical coordinates, of the transformation

$$r \to r, \qquad \theta \to \pi - \theta, \qquad \varphi \to \varphi + \pi. \tag{30.6}$$

The angular dependence of the particle's wave function is given by the spherical function Y_{lm}, which, apart from a constant immaterial here, has the form $P_l^m(\cos\theta)e^{im\varphi}$. When φ is replaced by $\varphi + \pi$, the factor $e^{im\varphi}$ is multiplied by $(-1)^m$; when θ is replaced by $\pi - \theta$, $P_l^m(\cos\theta)$ becomes $P_l^m(-\cos\theta) = (-1)^{l-m} P_l^m(\cos\theta)$. Thus the entire function is multiplied by $(-1)^l$ (which is independent of m, in agreement with the statement above); hence the parity of a state with a given l is

$$P = (-1)^l. \tag{30.7}$$

Accordingly, even values of l yield even states and odd values yield odd ones.

For a vector physical quantity associated with a single particle, nonzero matrix elements can arise only in transitions with $l \to l, l \pm 1$ (§ 29). Combining this fact with (30.7) and with the preceding statements concerning parity changes in vector matrix elements, we conclude that matrix elements of vector quantities belonging to an individual particle are nonzero only for the transitions

$$\begin{aligned} &\text{polar vectors:} && l \to l \pm 1, \\ &\text{axial vectors:} && l \to l. \end{aligned} \tag{30.8}$$

§ 31. Addition of angular momenta

Consider a system consisting of two weakly interacting parts. When the interaction is entirely neglected, the conservation of angular momentum holds for each part independently, so that the angular momenta $\mathbf{L}_1$ and $\mathbf{L}_2$ of the two parts can be summed to give the total angular momentum $\mathbf{L}$ of the system. In the next approximation, once the weak interaction is included, the conservation laws for $\mathbf{L}_1$ and $\mathbf{L}_2$ cease to hold rigorously; nonetheless, the quantum numbers L_1 and L_2, which fix the squares of those angular momenta, remain "good" and can still serve for an approximate characterization of the system's state. In a pictorial, classical treatment of the angular momenta, one may say that in this approximation $\mathbf{L}_1$ and $\mathbf{L}_2$ precess about the direction of $\mathbf{L}$ while their magnitudes remain unchanged.

The consideration of such systems raises the question of the *rule for adding angular momenta*. What values of L are possible for given L_1 and L_2? The addition rule for the angular-momentum projections is evident: from $\hat{L}_z = \hat{L}_{1z} + \hat{L}_{2z}$ it follows that

$$M = M_1 + M_2. \tag{31.1}$$

There is no equally simple relation for the operators of the squares of the angular momenta, and we derive their "addition rule" as follows.

If the quantities $\mathbf{L}_1^2$, $\mathbf{L}_2^2$, L_{1z}, and L_{2z} are chosen as a complete set of physical quantities[15], each state is specified by the numbers L_1, L_2, M_1, and M_2. For given L_1 and L_2, the numbers M_1 and M_2 take, respectively, $2L_1 + 1$ and $2L_2 + 1$ values, so there are altogether $(2L_1 + 1)(2L_2 + 1)$ different states with the same L_1 and L_2. We denote the wave functions of the states in this description by $\varphi_{L_1 L_2 M_1 M_2}$.

Instead of the four quantities just stated, one can choose the four quantities $\mathbf{L}_1^2$, $\mathbf{L}_2^2$, $\mathbf{L}^2$, and L_z as a complete set. Each state is then characterized by the numbers L_1, L_2, L, and M (we denote the corresponding wave functions by $\psi_{L_1 L_2 L M}$). For given L_1 and L_2 there must, of course, still be $(2L_1 + 1)(2L_2 + 1)$ different states; that is, for fixed L_1 and L_2, the pair L, M can take $(2L_1 + 1)(2L_2 + 1)$ pairs of values. These values can be found by the following argument.

Adding the different allowed values of M_1 and M_2 gives the corresponding values of M:

$$\begin{array}{ccl}
M_1 & M_2 & \qquad M \\
\hline
L_1 & L_2 & \quad L_1 + L_2 \\
\begin{array}{c} L_1 \\ L_1 - 1 \end{array} & \begin{array}{c} L_2 - 1 \\ L_2 \end{array} & \left.\vphantom{\begin{array}{c} L_1 \\ L_1 \end{array}}\right\} L_1 + L_2 - 1 \\
\begin{array}{c} L_1 - 1 \\ L_1 \\ L_1 - 2 \end{array} & \begin{array}{c} L_2 - 1 \\ L_2 - 2 \\ L_2 \end{array} & \left.\vphantom{\begin{array}{c} L_1 \\ L_1 \\ L_1 \end{array}}\right\} L_1 + L_2 - 2 \\
\multicolumn{3}{c}{\cdots\cdots\cdots\cdots\cdots\cdots}
\end{array}$$

The largest possible value of M is evidently $M = L_1 + L_2$, and it corresponds to one state φ (one pair of values M_1, M_2). Consequently, the largest possible M in the states ψ, and hence the largest L, is also $L_1 + L_2$. Next, there are two states φ with $M = L_1 + L_2 - 1$. There must therefore also be two states ψ with this value of M; one is a state with

[15] Together with a number of other quantities which, along with the four stated here, make up a complete set. These remaining quantities play no part in the following argument; for conciseness we do not mention them at all and provisionally call the system of the four stated quantities complete.

$L = L_1 + L_2$ (and $M = L - 1$), and the other has $L = L_1 + L_2 - 1$ (with $M = L$). For $M = L_1 + L_2 - 2$ there are three different states φ. This means that, in addition to $L = L_1 + L_2$ and $L = L_1 + L_2 - 1$, the value $L = L_1 + L_2 - 2$ is also possible.

This argument can be continued in the same way as long as decreasing M by one increases by one the number of states with that value of M. It is easy to see that this continues until M reaches $|L_1 - L_2|$. As M is reduced further, the number of states ceases to increase and remains equal to $2L_2 + 1$ (if $L_2 \le L_1$). Thus $|L_1 - L_2|$ is the smallest possible value of L.

We therefore obtain the result that, for given L_1 and L_2, L can take the values

$$L = L_1 + L_2, L_1 + L_2 - 1, \ldots, |L_1 - L_2|, \tag{31.2}$$

altogether $2L_2 + 1$ different values when $L_2 \le L_1$. One readily checks that these indeed give $(2L_1 + 1)(2L_2 + 1)$ different pairs of values M, L. It is important to note that, apart from the $2L + 1$ different values of M for a given L, each possible value of L in (31.2) corresponds to just one state.

This result can be represented pictorially by the so-called *vector model*. Introduce two vectors $\mathbf{L}_1$ and $\mathbf{L}_2$ of lengths L_1 and L_2. The values of L are represented by the integer-valued lengths of vectors $\mathbf{L}$ obtained by vector addition of $\mathbf{L}_1$ and $\mathbf{L}_2$; the largest value, $L_1 + L_2$, is obtained when $\mathbf{L}_1$ and $\mathbf{L}_2$ are parallel, and the smallest, $|L_1 - L_2|$, when they are antiparallel.

In states with definite angular momenta L_1, L_2, and definite total angular momentum L, the scalar products $\mathbf{L}_1\mathbf{L}_2$, $\mathbf{L}\mathbf{L}_1$, and $\mathbf{L}\mathbf{L}_2$ also have definite values. These values are readily found. To calculate $\mathbf{L}_1\mathbf{L}_2$, write $\hat{\mathbf{L}} = \hat{\mathbf{L}}_1 + \hat{\mathbf{L}}_2$, or, on squaring and transposing terms,

$$2\hat{\mathbf{L}}_1\hat{\mathbf{L}}_2 = \hat{\mathbf{L}}^2 - \hat{\mathbf{L}}_1^2 - \hat{\mathbf{L}}_2^2.$$

Replacing the operators on the right by their eigenvalues gives the eigenvalue of the operator on the left:

$$\mathbf{L}_1\mathbf{L}_2 = \frac{1}{2}\{L(L+1) - L_1(L_1+1) - L_2(L_2+1)\}. \tag{31.3}$$

Similarly, we find

$$\mathbf{L}\mathbf{L}_1 = \frac{1}{2}\{L(L+1) + L_1(L_1+1) - L_2(L_2+1)\}. \tag{31.4}$$

Let us now establish the *rule for combining parities*. For a system that consists of two independent parts, the wave function Ψ equals the product of the individual wave functions Ψ_1 and Ψ_2. Hence, if the latter two have the same parity (both change sign, or both remain unchanged, when all coordinates change sign), the wave function of the whole system is even. If, on the other hand, Ψ_1 and Ψ_2 have opposite parities, Ψ is odd. These statements can be written as

$$P = P_1 P_2, \tag{31.5}$$

where P is the parity of the whole system, and P_1 and P_2 are the parities of its parts. This rule plainly generalizes directly to a system consisting of any number of noninteracting parts.

In particular, for a system of particles in a centrally symmetric field (with the interaction between the particles treated as weak), the parity of the state of the whole system is

$$P = (-1)^{l_1+l_2+\dots} \tag{31.6}$$

(see (30.7)). Note that the exponent involves the algebraic sum of the particle angular momenta, a quantity that, generally speaking, does not coincide with their "vector sum," i.e., with the angular momentum L of the system.

If a closed system decays into parts under the action of forces within the system itself, its total angular momentum and parity must be conserved. This may make the decay impossible even when it is energetically possible.

For example, consider an atom in an even state with angular momentum $L = 0$, whose energy would permit it to break up into a free electron and an ion occupying an odd state, also with $L = 0$. It is easy to show that this process is in fact impossible (one calls it *forbidden*). Indeed, by conservation of angular momentum the free electron would also have to possess zero angular momentum and would therefore be in an even state ($P = (-1)^0 = +1$). But the state ion+free electron would then be odd, whereas the initial state of the atom was even.

CHAPTER V

Motion in a Centrally Symmetric Field

§ 32. Motion in a central field

As in classical mechanics, the quantum-mechanical problem of two interacting particles can be reduced to a one-particle problem. For masses m_1, m_2 interacting through $U(r)$, where r is their separation,

$$\hat{H} = -\frac{\hbar^2}{2m_1}\Delta_1 - \frac{\hbar^2}{2m_2}\Delta_2 + U(r). \tag{32.1}$$

Here Δ_1, Δ_2 are the Laplace operators in the particle coordinates. Replace $\mathbf{r}_1, \mathbf{r}_2$ by

$$\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1, \qquad \mathbf{R} = \frac{m_1\mathbf{r}_1 + m_2\mathbf{r}_2}{m_1 + m_2}. \tag{32.2}$$

Thus $\mathbf{r}$ is the relative radius vector and $\mathbf{R}$ the center-of-mass radius vector. Direct calculation gives

$$\hat{H} = -\frac{\hbar^2}{2(m_1 + m_2)}\Delta_R - \frac{\hbar^2}{2m}\Delta + U(r), \tag{32.3}$$

where $m = m_1 m_2/(m_1 + m_2)$ is the reduced mass. The Hamiltonian separates into two independent parts. Accordingly, $\psi(\mathbf{r}_1, \mathbf{r}_2) = \varphi(\mathbf{R})\psi(\mathbf{r})$: the first factor describes free motion of the center of mass, of mass $m_1 + m_2$, and the second describes relative motion as that of mass m in $U(r)$.

The Schrödinger equation in a central field is

$$\Delta\psi + \frac{2m}{\hbar^2}[E - U(r)]\psi = 0. \tag{32.4}$$

In spherical coordinates this becomes

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2}\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial\psi}{\partial\theta}) + \frac{1}{\sin^2\theta}\frac{\partial^2\psi}{\partial\varphi^2}\right] + \frac{2m}{\hbar^2}[E - U(r)]\psi = 0. \tag{32.5}$$

Introducing the squared-angular-momentum operator (26.16),[1]

$$\frac{\hbar^2}{2m}\left[-\frac{1}{r^2}\frac{\partial}{\partial r}(r^2\frac{\partial\psi}{\partial r}) + \frac{\hat{\mathbf{l}}^2}{r^2}\psi\right] + U(r)\psi = E\psi. \tag{32.6}$$

[1] If the radial momentum operator is defined by

$$\hat{p}_r\psi = -i\hbar\frac{1}{r}\frac{\partial}{\partial r}(r\psi) = -i\hbar(\frac{\partial}{\partial r} + \frac{1}{r})\psi,$$

then

$$\hat{H} = \frac{1}{2m}(\hat{p}_r^2 + \frac{\hbar^2\hat{\mathbf{l}}^2}{r^2}) + U(r),$$

formally the same as the classical Hamilton function in spherical coordinates.

Angular momentum is conserved in a central field. For stationary states with definite l, m, these values fix the angular dependence. Seek

$$\psi = R(r)Y_{lm}(\theta, \varphi), \tag{32.7}$$

where Y_{lm} are spherical functions. Since $\hat{\mathbf{l}}^2 Y_{lm} = l(l+1)Y_{lm}$, the radial function obeys

$$\frac{1}{r^2}\frac{d}{dr}(r^2\frac{dR}{dr}) - \frac{l(l+1)}{r^2}R + \frac{2m}{\hbar^2}[E - U(r)]R = 0. \tag{32.8}$$

This contains no $l_z = m$, consistently with the $(2l+1)$-fold directional degeneracy. Put

$$R(r) = \frac{\chi(r)}{r}. \tag{32.9}$$

Then

$$\frac{d^2\chi}{dr^2} + [\frac{2m}{\hbar^2}(E - U) - \frac{l(l+1)}{r^2}]\chi = 0. \tag{32.10}$$

If $U(r)$ is finite everywhere, ψ and hence R must remain finite at the origin, so

$$\chi(0) = 0. \tag{32.11}$$

This condition in fact also holds for fields diverging at the origin (§ 35).

Equation (32.10) is formally the one-dimensional Schrödinger equation with

$$U_l(r) = U(r) + \frac{\hbar^2}{2m}\frac{l(l+1)}{r^2}, \tag{32.12}$$

the sum of $U(r)$ and the centrifugal energy

$$\frac{\hbar^2 l(l+1)}{2mr^2} = \frac{\hbar^2\hat{\mathbf{l}}^2}{2mr^2}.$$

Thus the central-field problem reduces to one-dimensional motion on a region bounded on one side by the condition at $r = 0$. The normalization is likewise one-dimensional:

$$\int_0^\infty |R|^2 r^2 dr = \int_0^\infty |\chi|^2 dr.$$

Energy levels for one-dimensional motion in such a region are nondegenerate (§ 21), so E completely determines the radial solution of (32.10). The angular part is fixed by l, m. Hence E, l, m completely determine the wave function: energy, squared angular momentum, and its projection form a complete set of physical quantities. The one-dimensional reduction permits use of the oscillation theorem (§ 21). At fixed l, order the discrete energy eigenvalues increasingly and number them $n_r = 0, 1, \ldots$. Then n_r is the number of finite-r nodes of the radial function, excluding $r = 0$, and is called the *radial quantum number*. The numbers l and m are also called the *azimuthal* and *magnetic quantum numbers*.

States with different l are conventionally denoted by

$$\begin{array}{c|cccccccc|c} l & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & \ldots \\ & s & p & d & f & g & h & i & k & \ldots \end{array} \tag{32.13}$$

The normal state in a central field is always an s state: for $l \neq 0$ the angular wave function necessarily has nodes, whereas the normal-state wave function has none. The lowest possible eigenvalue at fixed l also rises with l, since angular momentum adds the positive term $\hbar^2 l(l+1)/(2mr^2)$ to the Hamiltonian.

To find the radial function near the origin, assume

$$\lim_{r\to 0} U(r)r^2 = 0. \tag{32.14}$$

Take the leading term as $R = \text{const}\cdot r^s$. From (32.8), after multiplication by r^2 and passage to $r \to 0$,

$$\frac{d}{dr}(r^2\frac{dR}{dr}) - l(l+1)R = 0,$$

so $s(s+1) = l(l+1)$ and

$$s = l \quad \text{or} \quad s = -(l+1).$$

The latter solution is inadmissible, since it diverges at $r = 0$. Therefore

$$R_l \approx \text{const}\cdot r^l. \tag{32.15}$$

The probability of finding the particle between r and $r + dr$ is determined by $r^2|R|^2$ and is proportional to $r^{2(l+1)}$; it vanishes at the origin more rapidly for larger l.

§ 33. Spherical waves

The plane wave

$$\psi_{\mathbf{p}} = \text{const}\cdot\exp(i\mathbf{pr}/\hbar)$$

describes a free particle with definite momentum $\mathbf{p}$ and energy $E = p^2/2m$. We now consider stationary free-particle states having definite angular momentum and projection as well as energy. It is convenient to use the wave vector

$$k = \frac{p}{\hbar} = \frac{\sqrt{2mE}}{\hbar}. \tag{33.1}$$

The state with angular momentum l and projection m has

$$\psi_{klm} = R_{kl}(r)Y_{lm}(\theta,\varphi), \tag{33.2}$$

where

$$R''_{kl} + \frac{2}{r}R'_{kl} + [k^2 - \frac{l(l+1)}{r^2}]R_{kl} = 0. \tag{33.3}$$

The continuum wave functions satisfy

$$\int \psi^*_{k'l'm'}\psi_{klm}dV = \delta_{ll'}\delta_{mm'}\delta((k'-k)/2\pi).$$

Angular functions ensure orthogonality in l, m; the radial functions must obey

$$\int_0^\infty r^2 R_{k'l}R_{kl}dr = \delta((k'-k)/2\pi) = 2\pi\delta(k'-k). \tag{33.4}$$

For normalization on the energy scale instead, $\int r^2 R_{E'l} R_{El} dr = \delta(E' - E)$, (5.14) gives

$$R_{El} = R_{kl} (\frac{1}{2\pi}\frac{dk}{dE})^{1/2} = \frac{1}{\hbar}\sqrt{\frac{m}{2\pi k}} R_{kl}. \tag{33.5}$$

For $l = 0$, (33.3) reads $d^2(rR_{k0})/dr^2 + k^2 r R_{k0} = 0$. The finite solution normalized by (33.4) is

$$R_{k0} = 2\frac{\sin kr}{r}. \tag{33.6}$$

For $l \neq 0$, put

$$R_{kl} = r^l \chi_{kl}. \tag{33.7}$$

Then

$$\chi''_{kl} + \frac{2(l+1)}{r}\chi'_{kl} + k^2 \chi_{kl} = 0.$$

Differentiation gives

$$\chi'''_{kl} + \frac{2(l+1)}{r}\chi''_{kl} + [k^2 - \frac{2(l+1)}{r^2}]\chi'_{kl} = 0.$$

Putting $\chi'_{kl} = r\chi_{k,l+1}$ yields the required equation for $\chi_{k,l+1}$, so

$$\chi_{k,l+1} = \frac{1}{r}\chi'_{kl}. \tag{33.8}$$

Hence

$$\chi_{kl} = (\frac{1}{r}\frac{d}{dr})^l \chi_{k0},$$

where $\chi_{k0} = R_{k0}$ from (33.6), up to an arbitrary constant. Thus

$$R_{kl} = (-1)^l \, 2\frac{r^l}{k^l}(\frac{1}{r}\frac{d}{dr})^l \frac{\sin kr}{r}. \tag{33.9}$$

The factor k^{-l} normalizes the function, and $(-1)^l$ is convenient. Equivalently,

$$R_{kl} = \sqrt{\frac{2\pi k}{r}} J_{l+1/2}(kr) = 2k j_l(kr), \tag{33.10}$$

where the spherical Bessel functions are

$$j_l(x) = \sqrt{\frac{\pi}{2x}} J_{l+1/2}(x). \tag{33.11}$$

2

[2] The first few are

$$j_0 = \frac{\sin x}{x}, \quad j_1 = \frac{\sin x}{x^2} - \frac{\cos x}{x}, \quad j_2 = (\frac{3}{x^3} - \frac{1}{x}) \sin x - \frac{3\cos x}{x^2}.$$

Another convention in the literature differs from (33.11) by a factor x.

At large r, the slowest-decaying term in (33.9) comes from differentiating the sine l times. Every operation $-d/dr$ shifts its argument by $-\pi/2$, giving

$$R_{kl} \approx \frac{2}{r}\sin(kr - \frac{\pi l}{2}). \tag{33.12}$$

Normalization may be performed from the asymptotic form (§ 21); comparison with (33.6) confirms the coefficient in (33.9). Near the origin, expansion of $\sin kr$ and retention of the term giving the lowest power after differentiation yields[3]

$$(\frac{1}{r}\frac{d}{dr})^l \frac{\sin kr}{r} \approx (\frac{1}{r}\frac{d}{dr})^l \frac{(-1)^l (kr)^{2l+1}}{r(2l+1)!} = \frac{(-1)^l k^{2l+1}}{(2l+1)!!}.$$

Therefore

$$R_{kl} \approx \frac{2k^{l+1}}{(2l+1)!!} r^l, \tag{33.13}$$

in agreement with (32.15).

Scattering theory also uses wave functions which are not finite in the usual sense, but describe particles emitted from or incident on the center. For $l = 0$, replace the standing wave (33.6) by an outgoing or incoming wave:

$$R^{\pm}_{k0} = \frac{A}{r} e^{\pm ikr}. \tag{33.14}$$

For arbitrary l,

$$R^{\pm}_{kl} = (-1)^l A \frac{r^l}{k^l} (\frac{1}{r}\frac{d}{dr})^l \frac{e^{\pm ikr}}{r}. \tag{33.15}$$

In terms of Hankel functions,

$$R^{\pm}_{kl} = \pm iA \sqrt{\frac{k\pi}{2r}} H^{(1,2)}_{l+1/2}(kr), \tag{33.16}$$

of the first and second kind for + and − respectively. Their asymptotic forms are

$$R^{\pm}_{kl} \approx \frac{A}{r} \exp[\pm i(kr - \frac{\pi l}{2})], \tag{33.17}$$

and near the origin

$$R^{\pm}_{kl} \approx A \frac{(2l-1)!!}{k^l} r^{-l-1}. \tag{33.18}$$

Normalize these functions to emission or absorption of one particle per unit time. Far away, each small part of a spherical wave is locally plane, with current density $j = v\psi\psi^*$ and $v = k\hbar/m$. The condition $\oint j df = 1$, or $\int jr^2 do = 1$, with do the solid-angle element, and the unchanged angular normalization require

$$A = \frac{1}{\sqrt{v}} = \sqrt{\frac{m}{k\hbar}}. \tag{33.19}$$

[3] !! denotes the product of all integers of the same parity up to and including the stated one.

An asymptotic expression like (33.12) also holds for positive-energy motion in any field decreasing sufficiently rapidly with distance.[4] At large r, neglecting both the field and centrifugal energy leaves

$$\frac{1}{r}\frac{d^2(rR_{kl})}{dr^2}+k^2R_{kl}=0.$$

Its general solution is

$$R_{kl}\approx 2\frac{\sin(kr-l\pi/2+\delta_l)}{r}, \tag{33.20}$$

where δ_l is the *phase shift*; the coefficient corresponds to $k/2\pi$ normalization.[5] The exact boundary condition of finiteness at the origin fixes δ_l. These phases depend on both l and k and are essential characteristics of continuum eigenfunctions.

Problem

1. Find the energy levels for $l=0$ motion in the spherical square well $U(r)=-U_0$ for $r<a$, $U(r)=0$ for $r>a$.

Solution. For $l=0$ the wave functions depend only on r. Inside,

$$\frac{1}{r}\frac{d^2}{dr^2}(r\psi)+k^2\psi=0, \qquad k=\frac{1}{\hbar}\sqrt{2m(U_0-|E|)},$$

and the finite solution is $\psi=A\sin(kr)/r$. Outside,

$$\frac{1}{r}\frac{d^2}{dr^2}(r\psi)-\kappa^2\psi=0, \qquad \kappa=\frac{1}{\hbar}\sqrt{2m|E|},$$

and the decaying solution is $\psi=A'e^{-\kappa r}/r$. Continuity of the logarithmic derivative of $r\psi$ at a gives

$$k\cot ka=-\kappa=-\sqrt{\frac{2mU_0}{\hbar^2}-k^2}, \tag{1}$$

or

$$\sin ka=\pm\sqrt{\frac{\hbar^2}{2ma^2U_0}}\,ka. \tag{2}$$

These equations implicitly determine the levels, using roots for which $\cot ka<0$. The first $l=0$ level is also the deepest level of all, the particle's normal state.

If the well is too shallow, there are no negative-energy levels. Graphically, the roots of $\pm\sin x=\alpha x$ are intersections of $y=\alpha x$ with $y=\pm\sin x$, retaining only portions with $\cot x<0$, shown solid in Fig. 9.

[4]It will be shown in § 124 that the field must decrease faster than $1/r$.

[5]The term $-l\pi/2$ is included so that $\delta_l=0$ in the absence of a field. Since the overall sign of a wave function is immaterial, δ_l is defined modulo π, not 2π, and can always be put between 0 and π.

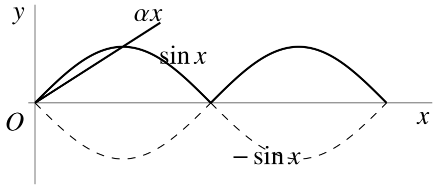

Fig. 9

For excessive α (small U_0) there are no intersections. The first appears at $\alpha = 2/\pi$, $x = \pi/2$. With $\alpha = \hbar/\sqrt{2ma^2U_0}$ and $x = ka$, the minimum depth is

$$U_{0\,\min} = \frac{\pi^2\hbar^2}{8ma^2}. \tag{3}$$

It increases as a decreases. At appearance, $ka = \pi/2$ and $E_1 = 0$. With further deepening, E_1 falls; for small $\Delta = U_0/U_{0\,\min} - 1$,

$$-E_1 = (\pi^2/16)U_{0\,\min}\Delta^2. \tag{4}$$

2. Find the order of levels with different l in a very deep spherical well, $U_0 \gg \hbar^2/ma^2$ (*W. Elsasser*, 1933).

Solution. As $U_0 \to \infty$, the boundary condition is $\psi = 0$ at the wall (§ 22). Using (33.10) inside gives

$$J_{l+1/2}(ka) = 0.$$

Its roots determine the levels above the well bottom, $U_0 - |E| = \hbar^2k^2/2m$. Starting from the ground state, their order is

$$1s,\ 1p,\ 1d,\ 2s,\ 1f,\ 2p,\ 1g,\ 2d,\ 1h,\ 3s,\ 2f, \ldots$$

The preceding numeral orders levels having the same l.[6]

3. Find the order in which levels with different l appear as the well depth U_0 increases.

Solution. At first appearance a new level has $E = 0$. Outside, the solution vanishing at infinity is $R_l = \text{const}\cdot r^{-(l+1)}$. Continuity of R_l, R_l' implies

$$(r^{l+1}R_l)' = 0 \qquad \text{at } r = a.$$

This is equivalent[7] to $R_{l-1} = 0$, hence

$$J_{l-1/2}\left(\frac{a}{\hbar}\sqrt{2mU_0}\right) = 0;$$

for $l = 0$, replace $J_{l-1/2}$ by cos. The order of appearance is

$$1s,\ 1p,\ 1d,\ 2s,\ 1f,\ 2p,\ 1g,\ 2d,\ 3s,\ 1h,\ 2f, \ldots$$

Differences from the deep-well ordering begin only for rather high levels.

[6] This notation is used for particle levels in nuclei (§ 118).

[7] From (33.7), (33.8), $(r^{-l}R_l)' \sim r^{-l}R_{l+1}$. Since (33.3) is invariant under $l \mapsto -l-1$, also $(r^{l+1}R_{-l-1})' \sim r^{l+1}R_{-l}$. Since R_{-l} and R_{l-1} obey the same equation, finally $(r^{l+1}R_l)' \sim r^{l+1}R_{l-1}$, as used here.

4. Find the energy levels of the spatial oscillator $U = \frac{1}{2}m\omega^2 r^2$, their degeneracies, and the allowed orbital angular momenta.

SOLUTION. Separation in Cartesian coordinates gives three linear oscillators, hence

$$E = \hbar\omega(n_1 + n_2 + n_3 + 3/2) \equiv \hbar\omega(n + 3/2).$$

The degeneracy is the number of ways of writing n as a sum of three nonnegative integers,[8] namely

$$\frac{(n+1)(n+2)}{2}.$$

The stationary wave functions are

$$\psi_{n_1 n_2 n_3} = \text{const} \cdot e^{-\alpha^2 r^2/2} H_{n_1}(\alpha x) H_{n_2}(\alpha y) H_{n_3}(\alpha z), \tag{5}$$

where $\alpha = \sqrt{m\omega/\hbar}$. Since H_n gains $(-1)^n$ under a coordinate sign change, (5) has parity $(-1)^n$. Linear combinations at fixed $n_1 + n_2 + n_3 = n$ yield

$$\psi_{nlm} = \text{const} \cdot r^l e^{-\alpha^2 r^2/2} Y_{lm} F(-(n-l)/2, l + 3/2, \alpha^2 r^2), \tag{6}$$

where $|m| = 0, 1, \ldots, l$. The values of l are $0, 2, \ldots, n$ for even n and $1, 3, \ldots, n$ for odd n, as also follows by matching parities $(-1)^n$ and $(-1)^l$. Thus the level sequence is

$$(1s),\ (1p),\ (1d, 2s),\ (1f, 2p),\ (1g, 2d, 3s), \ldots,$$

with mutually degenerate states in parentheses.[9]

§ 34. Expansion of a plane wave

Consider a free particle moving with definite momentum $p = k\hbar$ along the positive z axis. Its wave function is

$$\psi = \text{const} \cdot e^{ikz}.$$

Expand it in the free-motion functions ψ_{klm} with definite angular momentum. Since $E = k^2\hbar^2/2m$ is definite, only functions with the same k enter. Since e^{ikz} is axially symmetric about z, only functions independent of φ, with $m = 0$, can occur. Thus

$$e^{ikz} = \sum_{l=0}^{\infty} a_l \psi_{kl0} = \sum_{l=0}^{\infty} a_l R_{kl} Y_{l0}.$$

Substitution of (28.8) and (33.9) gives

$$e^{ikz} = \sum_{l=0}^{\infty} C_l P_l(\cos\theta) \left(\frac{r}{k}\right)^l \left(\frac{1}{r}\frac{d}{dr}\right)^l \frac{\sin kr}{kr} \qquad (z = r\cos\theta).$$

Determine C_l by comparing coefficients of $(r\cos\theta)^n$ in the power-series expansions. On the right this term occurs only in the nth summand: for $l > n$ the radial expansion

[8] Equivalently, the number of ways to distribute n identical balls among three boxes.

[9] Note the degeneracy of levels with different l; see the note on p. 164.

begins with higher powers of r, while for $n > l$, $P_l(\cos\theta)$ has lower powers of $\cos\theta$. The coefficient of $\cos^l\theta$ in $P_l(\cos\theta)$ is $(2l)!/2^l(l!)^2$ (see (c.1)). By (33.13), the relevant term is

$$(-1)^l C_l \frac{(2l)!(kr\cos\theta)^l}{2^l(l!)^2\, 1\cdot 3\ldots(2l+1)}.$$

The corresponding term in $\exp(ikr\cos\theta)$ is

$$\frac{(ikr\cos\theta)^l}{l!}.$$

Hence $C_l = (-i)^l(2l+1)$, and

$$e^{ikz} = \sum_{l=0}^{\infty}(-i)^l(2l+1)P_l(\cos\theta)\left(\frac{r}{k}\right)^l\left(\frac{1}{r}\frac{d}{dr}\right)^l\frac{\sin kr}{kr}. \tag{34.1}$$

At large distances,

$$e^{ikz} \approx \frac{1}{kr}\sum_{l=0}^{\infty} i^l(2l+1)P_l(\cos\theta)\sin\left(kr - \frac{l\pi}{2}\right). \tag{34.2}$$

In (34.1), z is along the wave vector $\mathbf{k}$. The expansion also has a coordinate-independent form. The addition theorem for spherical functions (c.11) expresses $P_l(\cos\theta)$ through spherical functions of the directions of $\mathbf{k}$ and $\mathbf{r}$:

$$e^{i\mathbf{kr}} = 4\pi\sum_{l=0}^{\infty}\sum_{m=-l}^{l} i^l j_l(kr)Y_{lm}^{*}\left(\frac{\mathbf{k}}{k}\right)Y_{lm}\left(\frac{\mathbf{r}}{r}\right). \tag{34.3}$$

Since $j_l(kr)$ depends only on kr, the symmetry in $\mathbf{k}$ and $\mathbf{r}$ is explicit; which spherical function is conjugated is immaterial.

Normalize e^{ikz} to unit probability-current density: one particle crosses unit area per unit time. The function is

$$\psi = \frac{1}{\sqrt{v}}e^{ikz} = \sqrt{\frac{m}{k\hbar}}e^{ikz} \tag{34.4}$$

(v is the speed; see (19.7)). Multiplying (34.1) by $\sqrt{m/k\hbar}$ and using $\psi^{\pm}_{klm} = R^{\pm}_{kl}Y_{lm}$ gives

$$\psi = \sum_{l=0}^{\infty}\sqrt{\pi(2l+1)}\,\frac{1}{ik\hbar}(\psi^{+}_{kl0} - \psi^{-}_{kl0}).$$

The squared coefficient of ψ^{-}_{kl0} (or ψ^{+}_{kl0}) gives the probability that a particle incident on the center (or leaving it) has angular momentum l. Since the incident flux density is unity, this probability has dimensions of area and can be interpreted as the target area in the xy plane for particles with angular momentum l. Denoting it by σ_l,

$$\sigma_l = \frac{\pi}{k^2}(2l+1). \tag{34.5}$$

For large l, summing over a range Δl, where $1 \ll \Delta l \ll l$, gives

$$\sum_{\Delta l} \sigma_l \approx \frac{\pi}{k^2} \cdot 2l\Delta l = 2\pi \frac{l\hbar^2}{p^2} \Delta l.$$

With the classical relation $\hbar l = \rho p$, where ρ is the impact parameter, this becomes

$$2\pi\rho\Delta\rho,$$

the classical result. This is not accidental: for large l the motion is semiclassical (§ 49).

Problem

PROBLEM

Expand a plane wave in wave functions with definite projection m of angular momentum on the y axis and definite momentum projection p_y on that axis.

SOLUTION. Introduce cylindrical coordinates y, ρ, φ with axis y. The functions have the form $Q_m(\rho)e^{im\varphi}e^{ip_y y/\hbar}$. Measuring φ from z,

$$e^{ikz} = e^{ik\rho\cos\varphi} = \sum_{m=-\infty}^{\infty} Q_m(\rho)e^{im\varphi}$$

(here $p_y = 0$), whence

$$Q_m(\rho) = \frac{1}{2\pi}\int_0^{2\pi} \exp[i(k\rho\cos\varphi - m\varphi)]\, d\varphi = i^m J_m(k\rho),$$

where J_m is a Bessel function. For $k\rho \gg 1$,

$$Q_m(\rho) \approx i^m \sqrt{\frac{2}{\pi k\rho}} \sin\left[k\rho - \frac{\pi}{2}\left(m - \frac{1}{2}\right)\right].$$

§ 35. Fall of a particle to the center

To clarify certain features of quantum-mechanical motion, it is useful to study a case which, although it has no immediate physical realization, involves a particle moving in a field whose potential energy becomes infinite at a point, the origin, according to $U(r) \approx -\beta/r^2$ with $\beta > 0$. The form of the field far from the origin will not matter. We saw in § 18 that this case lies precisely between those admitting ordinary stationary states and those in which the particle falls to the origin.

Near the origin the Schrödinger equation is

$$R'' + \frac{2}{r}R' + \frac{\gamma}{r^2}R = 0, \tag{35.1}$$

where $R(r)$ is the radial wave function and

$$\gamma = \frac{2m\beta}{\hbar^2} - l(l+1). \tag{35.2}$$

All terms of lower order in $1/r$ have been omitted. The energy E is assumed finite, so its term has likewise been dropped.

Put $R \sim r^s$. Then s satisfies

$$s(s+1) + \gamma = 0,$$

with roots

$$s_1 = -\frac{1}{2} + \sqrt{\frac{1}{4} - \gamma}, \qquad s_2 = -\frac{1}{2} - \sqrt{\frac{1}{4} - \gamma}. \tag{35.3}$$

For the subsequent analysis, isolate a small region of radius r_0 about the origin and replace $-\gamma/r^2$ inside it by the constant $-\gamma/r_0^2$. After determining the wave functions in this cut-off field, we shall examine the limit $r_0 \to 0$.

First suppose $\gamma < 1/4$. Then s_1 and s_2 are real negative numbers, with $s_1 > s_2$. For $r > r_0$, the general small-r solution is

$$R = Ar^{s_1} + Br^{s_2}, \tag{35.4}$$

where A and B are constants. For $r < r_0$, the solution finite at the origin of

$$R'' + \frac{2}{r}R' + \frac{\gamma}{r_0^2}R = 0$$

is

$$R = C\frac{\sin kr}{r}, \qquad k = \frac{\sqrt{\gamma}}{r_0}. \tag{35.5}$$

At $r = r_0$, both R and R' must be continuous. One condition may conveniently be written as continuity of the logarithmic derivative of rR, which gives

$$\frac{A(s_1+1)r_0^{s_1} + B(s_2+1)r_0^{s_2}}{Ar_0^{s_1+1} + Br_0^{s_2+1}} = k\cot kr_0,$$

or

$$\frac{A(s_1+1)r_0^{s_1} + B(s_2+1)r_0^{s_2}}{Ar_0^{s_1} + Br_0^{s_2}} = \sqrt{\gamma}\cot\sqrt{\gamma}.$$

Solving for B/A yields an expression of the form

$$\frac{B}{A} = \text{const}\cdot r_0^{s_1 - s_2}. \tag{35.6}$$

As $r_0 \to 0$, $B/A \to 0$ because $s_1 > s_2$. Of the two solutions of (35.1) that diverge at the origin, one must therefore choose the less rapidly divergent:

$$R = A\frac{1}{r^{|s_1|}}. \tag{35.7}$$

Now let $\gamma > 1/4$. The roots are complex:

$$s_1 = -\frac{1}{2} + i\sqrt{\gamma - \frac{1}{4}}, \qquad s_2 = s_1^*.$$

Repeating the argument again gives (35.6), which now becomes

$$\frac{B}{A} = \text{const} \cdot r_0^{i\sqrt{4\gamma-1}}. \tag{35.8}$$

This expression has no definite limit as $r_0 \to 0$, so a direct limiting procedure is impossible. Taking (35.8) into account, a real solution has the general form

$$R = \text{const} \cdot \frac{1}{\sqrt{r}} \cos\left(\sqrt{\gamma - \frac{1}{4}} \ln \frac{r}{r_0} + \text{const}\right). \tag{35.9}$$

The number of zeros of this function grows without bound as r_0 decreases. On the one hand, (35.9) applies at sufficiently small r for every finite particle energy E; on the other, the ground-state wave function must have no zeros. We therefore conclude that the ground state in this field corresponds to $E = -\infty$. In every discrete-spectrum state the particle is found mainly where $E > U$. Thus as $E \to -\infty$ it is confined to an infinitesimal region about the origin: the particle falls to the center.

The critical field U_{cr} at which fall to the center becomes possible corresponds to $\gamma = 1/4$. The smallest coefficient of $-1/r^2$ occurs for $l = 0$, and hence

$$U_{\text{cr}} = -\frac{\hbar^2}{8mr^2}. \tag{35.10}$$

Formula (35.3), with s_1, shows that the admissible Schrödinger solution near a point where $U \sim 1/r^2$ diverges no faster than $1/\sqrt{r}$ as $r \to 0$. If the field becomes infinite more slowly than $1/r^2$, then near the origin $U(r)$ may be neglected relative to the remaining terms. The free-motion solutions result, namely $\psi \sim r^l$ (§ 33). If instead the field diverges faster than $1/r^2$, as does $-1/r^s$ with $s > 2$, the wave function near the origin is proportional to $r^{s/4-1}$; see the problem in § 49. In every such case $r\psi$ vanishes at $r = 0$.

Next consider a field that at large distances decreases as $U \approx -\beta/r^2$ but is arbitrary at small distances. Suppose first that $\gamma < 1/4$. Only finitely many negative-energy levels can then exist[10]. Indeed, at $E = 0$ the large-distance Schrödinger equation is (35.1), with general solution (35.4). This function has no zeros for $r \neq 0$. All zeros of the required radial wave function therefore lie at finite distances from the origin, and their number is finite. In other words, the ordinal number of the level $E = 0$ that closes the discrete spectrum is finite.

For $\gamma > 1/4$, by contrast, the discrete spectrum contains infinitely many negative-energy levels. At $E = 0$ the large-distance wave function has the form (35.9), with infinitely many zeros, so its ordinal number is infinite. Finally, let $U = -\beta/r^2$ throughout space. If $\gamma > 1/4$, the particle falls to the center. If $\gamma < 1/4$, there are no negative-energy levels at all. The $E = 0$ wave function is then of the form (35.7) everywhere and has no zeros at finite distances; it therefore corresponds to the lowest level for the specified l.

[10] At small r the field is assumed such that the particle does not fall to the center.

§ 36. Motion in a Coulomb field (spherical coordinates)

An especially important case of motion in a central field is motion in a *Coulomb field*,

$$U = \pm\frac{\alpha}{r},$$

where $\alpha > 0$. We first consider attraction and write $U = -\alpha/r$. General considerations already show that the negative-energy spectrum is discrete, with infinitely many levels, whereas the positive-energy spectrum is continuous.

Equation (32.8) for the radial functions becomes

$$\frac{d^2R}{dr^2} + \frac{2}{r}\frac{dR}{dr} - \frac{l(l+1)}{r^2}R + \frac{2m}{\hbar^2}\left(E + \frac{\alpha}{r}\right)R = 0. \tag{36.1}$$

For the relative motion of two attracting particles, m denotes their reduced mass.

Calculations in a Coulomb field are conveniently made in special *Coulomb units*. We choose as the units of mass, length, and time

$$m, \qquad \frac{\hbar^2}{m\alpha}, \qquad \frac{\hbar^3}{m\alpha^2}.$$

All other units follow; the energy unit is $m\alpha^2/\hbar^2$. These units are used throughout this and the next section unless stated otherwise.[11] In these units (36.1) is

$$\frac{d^2R}{dr^2} + \frac{2}{r}\frac{dR}{dr} - \frac{l(l+1)}{r^2}R + 2\left(E + \frac{1}{r}\right)R = 0. \tag{36.2}$$

Discrete spectrum. Introduce

$$n = \frac{1}{\sqrt{-2E}}, \qquad \rho = \frac{2r}{n}. \tag{36.3}$$

For negative energy n is real and positive. Equation (36.2) becomes

$$R'' + \frac{2}{\rho}R' + \left[-\frac{1}{4} + \frac{n}{\rho} - \frac{l(l+1)}{\rho^2}\right]R = 0, \tag{36.4}$$

where primes mean differentiation with respect to ρ.

At small ρ the admissible solution is proportional to ρ^l (see (32.15)). At large ρ, omission of the $1/\rho$ and $1/\rho^2$ terms gives $R'' = R/4$, hence $R = e^{\pm\rho/2}$. The solution vanishing at infinity therefore behaves as $e^{-\rho/2}$. Put

$$R = \rho^l e^{-\rho/2}\omega(\rho). \tag{36.5}$$

Then

$$\rho\omega'' + (2l + 2 - \rho)\omega' + (n - l - 1)\omega = 0. \tag{36.6}$$

[11] For an electron of mass $m = 9.11 \cdot 10^{-28}$ g and $\alpha = e^2$, the Coulomb units coincide with the *atomic units*. The atomic length unit is $\hbar^2/me^2 = 0.529 \cdot 10^{-8}$ cm, the *Bohr radius*. The atomic energy unit is $me^4/\hbar^2 = 4.36 \cdot 10^{-11}$ erg $= 27.21$ eV; half of it is the *rydberg*, Ry. The atomic charge unit is $e = 4.80 \cdot 10^{-10}$ electrostatic units. Formally, formulas are converted to atomic units by putting $e = m = \hbar = 1$. For $\alpha = Ze^2$, Coulomb and atomic units differ.

The solution must grow no faster than a finite power at infinity and remain finite at the origin. The solution satisfying the latter condition is the confluent hypergeometric function

$$\omega = F(-n + l + 1, 2l + 2, \rho) \tag{36.7}$$

(see mathematical appendix § d).[12] The condition at infinity holds only when $-n + l + 1$ is a nonpositive integer, so that (36.7) is a polynomial of degree $n - l - 1$; otherwise it grows as e^{ρ} (see (d.14)). Thus n is a positive integer and

$$n \geq l + 1. \tag{36.8}$$

From (36.3),

$$E = -\frac{1}{2n^2}, \qquad n = 1, 2, \ldots. \tag{36.9}$$

This determines the discrete energy levels. Infinitely many lie between the normal level $E_1 = -1/2$ and zero. Their spacings decrease with n, and they accumulate at $E = 0$, where the discrete and continuous spectra meet. In ordinary units,[13]

$$E = -\frac{m\alpha^2}{2\hbar^2 n^2}. \tag{36.10}$$

The integer n is the *principal quantum number*; the radial quantum number of § 32 is $n_r = n - l - 1$. At fixed n,

$$l = 0, 1, \ldots, n - 1. \tag{36.11}$$

Since the energy depends only on n, states with different l but the same n have equal energies. Each eigenvalue is therefore degenerate not only in the magnetic quantum number m, as in every central field, but also in l. This latter, *accidental* or *Coulomb*, degeneracy is peculiar to the Coulomb field. Since each l has $2l + 1$ values of m, the degeneracy of level n is

$$\sum_{l=0}^{n-1} (2l + 1) = n^2. \tag{36.12}$$

The stationary-state wave functions follow from (36.5) and (36.7). For integer parameters the confluent hypergeometric function is, apart from a factor, a generalized Laguerre polynomial (appendix § d), so

$$R_{nl} = \text{const} \cdot \rho^l e^{-\rho/2} L_{n+l}^{2l+1}(\rho).$$

[12] The second solution of (36.6) diverges as ρ^{-2l-1} when $\rho \to 0$.

[13] Formula (36.10) was first obtained by N. Bohr in 1913, before quantum mechanics. W. Pauli derived it in quantum mechanics by the matrix method in 1926, and Schrödinger derived it from the wave equation a few months later.

With $\int_0^\infty R_{nl}^2 r^2\,dr = 1$, the final form is[14]

$$\begin{aligned} R_{nl} &= -\frac{2}{n^2}\sqrt{\frac{(n-l-1)!}{[(n+l)!]^3}}\,e^{-r/n}(2r/n)^l L_{n+l}^{2l+1}(2r/n) \\ &= \frac{2}{n^{l+2}(2l+1)!}\sqrt{\frac{(n+l)!}{(n-l-1)!}}(2r)^l e^{-r/n}F(-n+l+1, 2l+2, 2r/n). \end{aligned} \tag{36.13}$$

(For the normalization integral see appendix § f, integral (f.6).[15]) Near the origin,

$$R_{nl} \approx r^l \frac{2^{l+1}}{n^{2+l}(2l+1)!}\sqrt{\frac{(n+l)!}{(n-l-1)!}}. \tag{36.14}$$

At large distances,

$$R_{nl} \approx (-1)^{n-l-1}\frac{2^n r^{n-1}e^{-r/n}}{n^{n+1}\sqrt{(n+l)!(n-l-1)!}}. \tag{36.15}$$

The normal-state function R_{10} decays over distances $r \sim 1$, or $r \sim \hbar^2/m\alpha$ in ordinary units.

Mean powers of r are $\overline{r^k} = \int_0^\infty r^{k+2}R_{nl}^2 dr$. The general result follows from (f.7); some first values are

$$\begin{aligned} \bar{r} &= \tfrac{1}{2}[3n^2 - l(l+1)], \qquad & \overline{r^2} &= \tfrac{n^2}{2}[5n^2+1-3l(l+1)], \\ \overline{r^{-1}} &= \frac{1}{n^2}, & \overline{r^{-2}} &= \frac{1}{n^3(l+1/2)}. \end{aligned} \tag{36.16}$$

Continuous spectrum. Positive energies form a continuous spectrum from zero to infinity. Every eigenvalue is infinitely degenerate, with all integer $l \geq 0$ and all allowed m. Now n and ρ are purely imaginary:

$$n = -\frac{i}{\sqrt{2E}} = -\frac{i}{k}, \qquad \rho = 2ikr, \tag{36.17}$$

where $k = \sqrt{2E}$.[16] The continuum radial eigenfunctions are

$$R_{kl} = \frac{C_{kl}}{(2l+1)!}(2kr)^l e^{-ikr}F(i/k+l+1, 2l+2, 2ikr), \tag{36.18}$$

[14] The first few functions are

$$R_{10} = 2e^{-r}, \quad R_{20} = \frac{1}{\sqrt{2}}e^{-r/2}(1-r/2), \quad R_{21} = \frac{1}{2\sqrt{6}}e^{-r/2}r,$$
$$R_{30} = \frac{2}{3\sqrt{3}}e^{-r/3}(1-2r/3+2r^2/27),$$
$$R_{31} = \frac{8}{27\sqrt{6}}e^{-r/3}r(1-r/6), \quad R_{32} = \frac{4}{81\sqrt{30}}e^{-r/3}r^2.$$

[15] It may also be evaluated by substituting (d.13) for the Laguerre polynomials and integrating by parts, as for the Legendre-polynomial integral (c.8).

[16] One could instead choose the complex conjugates $n = i/k$, $\rho = -2ikr$; the real functions R_{kl} are independent of this choice.

or, as a complex integral (appendix § d),

$$R_{kl} = C_{kl}(2kr)^l e^{-ikr} \frac{1}{2\pi i} \oint e^{\xi} \left(1 - \frac{2ikr}{\xi}\right)^{-i/k-l-1} \xi^{-2l-2} d\xi, \tag{36.19}$$

over the contour in Fig. 10.[17] With $\xi = 2ikr(t + 1/2)$ this becomes

$$R_{kl} = C_{kl} \frac{(-2kr)^{-l-1}}{2\pi} \oint e^{2ikrt} (t + \tfrac{1}{2})^{i/k-l-1} (t - \tfrac{1}{2})^{-i/k-l-1} dt. \tag{36.20}$$

The path encircles $t = \pm 1/2$ positively; this form directly shows that R_{kl} is real.

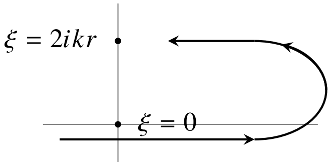

Fig. 10

Using (d.14), the asymptotic expansion is

$$R_{kl} = C_{kl} \frac{e^{-\pi/2k}}{kr} \operatorname{Re}\left\{ \frac{\exp[-i(kr - (\pi/2)(l+1) + (1/k)\ln 2kr)]}{\Gamma(l + 1 - i/k)} \times G(l + 1 + i/k, i/k - l, -2ikr) \right\}. \tag{36.21}$$

For normalization on the $k/2\pi$ scale, condition (33.4),

$$C_{kl} = 2ke^{\pi/2k} |\Gamma(l + 1 - i/k)|. \tag{36.22}$$

Indeed, at large r this gives

$$R_{kl} \approx \frac{2}{r} \sin(kr + k^{-1} \ln 2kr - \pi l/2 + \delta_l), \qquad \delta_l = \arg \Gamma(l + 1 - i/k). \tag{36.23}$$

This agrees with the general form (33.20), apart from the logarithmic phase; since $\ln r$ grows slowly compared with r, it does not affect the divergent normalization integral.

The gamma-function modulus in (36.22) is elementary. From $\Gamma(z+1) = z\Gamma(z)$ and $\Gamma(z)\Gamma(1-z) = \pi/\sin \pi z$, one finds

$$\Gamma(l + 1 + i/k) = (l + i/k) \cdots (1 + i/k)(i/k)\Gamma(i/k),$$
$$\Gamma(l + 1 - i/k) = (l - i/k) \cdots (1 - i/k)\Gamma(1 - i/k),$$

and

$$|\Gamma(l + 1 - i/k)| = \sqrt{\frac{\pi}{k}} \prod_{s=1}^{l} \sqrt{s^2 + k^{-2}} (\sinh(\pi/k))^{-1/2}.$$

[17] It may be replaced by any closed loop encircling $\xi = 0$ and $\xi = 2ikr$ positively. For integral l, the function $V(\xi) = \xi^{-n-l}(\xi - 2ikr)^{n-l}$ (appendix § d) returns to its original value around such a contour.

Thus

$$C_{kl} = \left[\frac{8\pi k}{1 - e^{-2\pi/k}}\right]^{1/2} \prod_{s=1}^{l} \sqrt{s^2 + \frac{1}{k^2}}, \tag{36.24}$$

with the product replaced by 1 at $l = 0$. The $E = 0$ radial function follows by taking $k \to 0$:

$$\begin{aligned} F(i/k + l + 1, 2l + 2, 2ikr) \to F(i/k, 2l + 2, 2ikr) & \\ = 1 - \frac{2r}{(2l+1)!} + \frac{(2r)^2}{(2l+2)(2l+3)2!} - \cdots & \\ = (2l+1)!(2r)^{-l-1/2} J_{2l+1}(\sqrt{8r}), & \end{aligned}$$

where J_{2l+1} is a Bessel function, while $C_{kl} \approx \sqrt{8\pi}\, k^{-l+1/2}$. Hence

$$\left.\frac{R_{kl}}{\sqrt{k}}\right|_{k\to 0} = \sqrt{\frac{4\pi}{r}} J_{2l+1}(\sqrt{8r}). \tag{36.25}$$

At large r,[18]

$$\left.\frac{R_{kl}}{\sqrt{k}}\right|_{k\to 0} = \left(\frac{8}{r^3}\right)^{1/4} \sin(\sqrt{8r} - l\pi - \pi/4). \tag{36.26}$$

The factor $\sqrt{k}$ disappears on changing to energy-scale normalization, from R_{kl} to R_{El} by (33.5); R_{El} remains finite as $E \to 0$.

For the repulsive field $U = \alpha/r$, only a continuous positive-energy spectrum exists. Formally changing r to $-r$ in the attractive-field equation gives

$$\begin{aligned} R_{kl} &= \frac{C_{kl}}{(2l+1)!}(2kr)^l e^{ikr} F(i/k + l + 1, 2l + 2, -2ikr), \\ C_{kl} &= 2ke^{-\pi/2k}|\Gamma(l + 1 + i/k)| = \left(\frac{8\pi k}{e^{2\pi/k} - 1}\right)^{1/2} \prod_{s=1}^{l} \sqrt{s^2 + \frac{1}{k^2}}. \end{aligned} \tag{36.27}$$

At large r,

$$R_{kl} \approx \frac{2}{r} \sin(kr - k^{-1} \ln 2kr - l\pi/2 + \delta_l), \qquad \delta_l = \arg\Gamma(l + 1 + i/k). \tag{36.28}$$

Origin of the Coulomb degeneracy. Classical motion in an attractive Coulomb field has the special conserved quantity

$$\mathbf{A} = \frac{\mathbf{r}}{r} - \mathbf{p} \times \mathbf{l} = \text{const} \tag{36.29}$$

(see I, § 15). Its quantum operator is

$$\hat{\mathbf{A}} = \frac{\mathbf{r}}{r} - \frac{1}{2}(\hat{\mathbf{p}} \times \hat{\mathbf{l}} - \hat{\mathbf{l}} \times \hat{\mathbf{p}}), \tag{36.30}$$

which commutes with $\hat{H} = \hat{\mathbf{p}}^2/2 - 1/r$.

[18] This function corresponds to the semiclassical approximation (§ 49) in $(l + 1/2)^2 \ll r \ll k^{-2}$.

Direct calculation gives

$$[\hat{l}_i, \hat{A}_k] = ie_{ikl}\hat{A}_l, \qquad [\hat{A}_i, \hat{A}_k] = -2i\hat{H}e_{ikl}\hat{l}_l. \tag{36.31}$$

The mutual noncommutativity of the $\hat{A}_i$ means that A_x, A_y, A_z cannot all be definite. Each component, say $\hat{A}_z$, commutes with $\hat{l}_z$ but not with $\hat{\mathbf{l}}^2$. A new conserved quantity which cannot be measured together with the other conserved quantities causes the additional degeneracy (§ 10): this is the accidental degeneracy peculiar to the Coulomb field.

The same origin may be stated in terms of the Coulomb problem's enlarged quantum-mechanical symmetry (V. A. Fock, 1935). At fixed negative energy, replace $\hat{H}$ on the right of (36.31) by E and define $\hat{u}_i = \hat{A}_i/\sqrt{-2E}$. Then

$$[\hat{l}_i, \hat{u}_k] = ie_{ikl}\hat{u}_l, \qquad [\hat{u}_i, \hat{u}_k] = ie_{ikl}\hat{l}_l. \tag{36.32}$$

Together with $[\hat{l}_i, \hat{l}_k] = ie_{ikl}\hat{l}_l$, these are precisely the formal commutation rules for infinitesimal rotations in four-dimensional Euclidean space.[19] This is the symmetry of the Coulomb problem.[20]

The energy levels can again be derived from (36.32).[21] Introduce

$$\hat{\mathbf{j}}_1 = \tfrac{1}{2}(\hat{\mathbf{l}} + \hat{\mathbf{u}}), \qquad \hat{\mathbf{j}}_2 = \tfrac{1}{2}(\hat{\mathbf{l}} - \hat{\mathbf{u}}). \tag{36.33}$$

Then

$$[\hat{j}_{1i}, \hat{j}_{1k}] = ie_{ikl}\hat{j}_{1l}, \quad [\hat{j}_{2i}, \hat{j}_{2k}] = ie_{ikl}\hat{j}_{2l}, \quad [\hat{j}_{1i}, \hat{j}_{2k}] = 0. \tag{36.34}$$

These are the rules for two independent three-dimensional angular momenta. Thus the eigenvalues of $\hat{\mathbf{j}}_1^2$ and $\hat{\mathbf{j}}_2^2$ are $j_1(j_1+1)$ and $j_2(j_2+1)$, with $j_1, j_2 = 0, 1/2, 1, 3/2, \ldots$.[22] Directly from $\hat{\mathbf{u}}$ and $\hat{\mathbf{l}} = \mathbf{r} \times \hat{\mathbf{p}}$,

$$\hat{\mathbf{l}}\hat{\mathbf{u}} = \hat{\mathbf{u}}\hat{\mathbf{l}} = 0, \qquad \hat{\mathbf{l}}^2 + \hat{\mathbf{u}}^2 = -1 - \frac{1}{2E} \qquad \hat{\mathbf{l}}^2 + \hat{\mathbf{u}}^2 = -1 - \frac{1}{2E}.$$

Hence

$$\hat{\mathbf{j}}_1^2 = \hat{\mathbf{j}}_2^2 = -\tfrac{1}{4}(1 + 1/2E) = j(j+1),$$

so $E = -1/[2(2j+1)^2]$. Writing

$$2j + 1 = n, \qquad n = 1, 2, 3, \ldots, \tag{36.35}$$

gives $E = -1/(2n^2)$. The degeneracy is $(2j_1+1)(2j_2+1) = n^2$. Since $\hat{\mathbf{l}} = \hat{\mathbf{j}}_1 + \hat{\mathbf{j}}_2$, for $j_1 = j_2 = (n-1)/2$ the orbital momentum ranges from $l = 0$ to $2j = n - 1$.[23]

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[19] Here $\hat{l}_x, \hat{l}_y, \hat{l}_z$ generate rotations in the yz, xz, xy planes of Cartesian coordinates x, y, z, u, and $\hat{u}_x, \hat{u}_y, \hat{u}_z$ generate rotations in the xu, yu, zu planes.

[20] It appears explicitly in momentum-space wave functions; see V. A. Fock, Izv. AN SSSR, physical series, 1935, no. 2, p. 169; Zs. f. Physik 98 (1935), 145.

[21] This derivation is essentially Pauli's (1926).

[22] We anticipate § 54 in using the possible integral and half-integral values of j.

[23] Accidental degeneracy between different l also occurs in the central field $U = m\omega^2 r^2/2$ (the spatial

Problem

1. Find the probability distribution of momentum values in the ground state of the hydrogen atom.

Solution. [24] The ground-state wave function is

$$\psi = R_{10}Y_{00} = \frac{1}{\sqrt{\pi}}e^{-r}.$$

In the **p**-representation,

$$a(\mathbf{p}) = \int \psi(\mathbf{r})e^{-i\mathbf{pr}}\,dV.$$

Using spherical coordinates with polar axis along **p** gives

$$a(\mathbf{p}) = \frac{8\sqrt{\pi}}{(1+p^2)^2},$$

and the probability density in momentum space is $|a(\mathbf{p})|^2/(2\pi)^3$.

2. Find the mean potential of the field produced by the nucleus and the electron in the hydrogen ground state.

Solution. The mean potential φ_e of the "electron cloud" is most simply found as the spherical solution of Poisson's equation with charge density $\rho = -|\psi|^2$:

$$\frac{1}{r}\frac{d^2}{dr^2}(r\varphi_e) = 4e^{-2r}.$$

Choosing the integration constants so that $\varphi_e(0)$ is finite and $\varphi_e(\infty) = 0$, and adding the nuclear potential, gives

$$\varphi = 1/r + \varphi_e(r) = (1/r+1)e^{-2r}.$$

Thus $\varphi \approx 1/r$ for $r \ll 1$, while $\varphi \approx e^{-2r}$ for $r \gg 1$.

oscillator; Problem 4 of § 33), again because of an enlarged Hamiltonian symmetry. In $\hat{H} = \hat{\mathbf{p}}^2/2m + m\omega^2 r^2/2$, both $\hat{p}_i$ and x_i enter as sums of squares. With

$$\hat{a}_i = \frac{m\omega x_i + i\hat{p}_i}{\sqrt{2m\hbar\omega}}, \qquad \hat{a}_i^+ = \frac{m\omega x_i - i\hat{p}_i}{\sqrt{2m\hbar\omega}},$$

one obtains $\hat{H} = \hbar\omega[\hat{\mathbf{a}}^+\hat{\mathbf{a}} + 3/2]$. This is invariant under arbitrary unitary transformations of $\hat{a}_i^+$ and $\hat{a}_j$, a group larger than the three-dimensional rotation group. The quantum peculiarity of the Coulomb and oscillator fields has its classical counterpart: particle trajectories are closed in these, and only these, fields.

[24] Atomic units are used in Problems 1 and 2.

3. Find the energy levels of a particle moving in the central field $U = A/r^2 - B/r$ (Fig. 11).

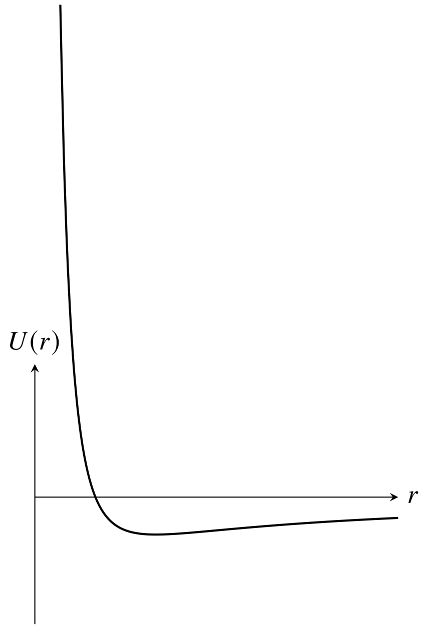


Fig. 11

SOLUTION. Positive energies form a continuous spectrum and negative energies a discrete one; consider the latter. The radial equation is

$$\frac{d^2R}{dr^2} + \frac{2}{r}\frac{dR}{dr} + \frac{2m}{\hbar^2}\left(E - \frac{\hbar^2 l(l+1)}{2mr^2} - \frac{A}{r^2} + \frac{B}{r}\right)R = 0. \tag{1}$$

Put $\rho = 2\sqrt{-mE}\,r/\hbar$ and define

$$\frac{2mA}{\hbar^2} + l(l+1) = s(s+1), \tag{2}$$

$$\frac{B}{\hbar}\sqrt{\frac{m}{-2E}} = n. \tag{3}$$

Then

$$R'' + \frac{2}{\rho}R' + (-\tfrac{1}{4} + n/\rho - s(s+1)/\rho^2)R = 0,$$

formally identical to (36.4). Hence

$$R = \rho^s e^{-\rho/2} F(-n + s + 1, 2s + 2, \rho),$$

where $n - s - 1 = p$ is a nonnegative integer and s is the positive root of (2). Formula (3) gives

$$-E_p = \frac{2B^2 m}{\hbar^2}[2p + 1 + \sqrt{(2l+1)^2 + 8mA/\hbar^2}]^{-2}.$$

4. Repeat the calculation for $U = A/r^2 + Br^2$ (Fig. 12).

Solution. There is only a discrete spectrum. The radial equation is

$$\frac{d^2R}{dr^2} + \frac{2}{r}\frac{dR}{dr} + \frac{2m}{\hbar^2}(E - \hbar^2 l(l+1)/(2mr^2) - A/r^2 - Br^2)R = 0.$$

Set

$$\xi = \frac{\sqrt{2mB}}{\hbar}r^2,$$

and define

$$l(l+1) + \frac{2mA}{\hbar^2} = 2s(2s+1), \qquad \sqrt{\frac{2m}{B}}\frac{E}{\hbar} = 4(n+s) + 3.$$

Then

$$\xi R'' + \frac{3}{2}R' + [n + s + \tfrac{3}{4} - \xi/4 - s(s+1/2)/\xi]R = 0.$$

The desired solution behaves as $e^{-\xi/2}$ at infinity and as ξ^s at the origin, where

$$s = \frac{1}{4}[-1 + \sqrt{(2l+1)^2 + 8mA/\hbar^2}].$$

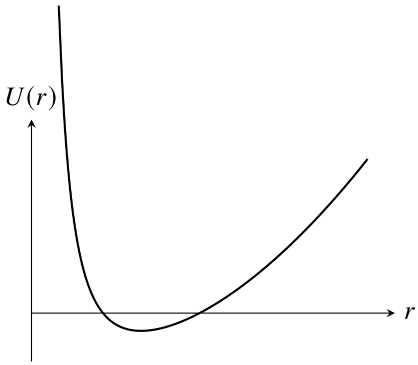


Fig. 12

Writing $R = e^{-\xi/2}\xi^s\omega$ gives

$$\xi\omega'' + (2s + 3/2 - \xi)\omega' + n\omega = 0,$$

so

$$\omega = F(-n, 2s + 3/2, \xi),$$

where n is a nonnegative integer. The equally spaced levels are

$$E_n = \hbar\sqrt{\frac{B}{2m}}[4n + 2 + \sqrt{(2l+1)^2 + 8mA/\hbar^2}], \qquad n = 0, 1, 2, \ldots.$$

§ 37. Motion in a Coulomb field (parabolic coordinates)

For motion in any centrally symmetric field, the Schrödinger equation can always be separated in spherical coordinates. In a Coulomb field, separation is also possible in the so-called parabolic coordinates. The solution of the Coulomb-field problem in these coordinates is useful in studying a number of problems in which a particular direction in space is singled out, for example by an external electric field in addition to the Coulomb field (§ 77). The parabolic coordinates ξ, η, and φ are defined by

$$x = \sqrt{\xi\eta}\cos\varphi, \qquad y = \sqrt{\xi\eta}\sin\varphi, \qquad z = \frac{1}{2}(\xi - \eta),$$
$$r = \sqrt{x^2 + y^2 + z^2} = \frac{1}{2}(\xi + \eta) \tag{37.1}$$

or, conversely,

$$\xi = r + z, \qquad \eta = r - z, \qquad \varphi = \arctan\frac{y}{x}; \tag{37.2}$$

ξ and η range from 0 to ∞, while φ ranges from 0 to 2π. The surfaces ξ = const and η = const are paraboloids of revolution whose axis is the z axis and whose focus is at the origin. This coordinate system is orthogonal. The line element is

$$dl^2 = \frac{\xi + \eta}{4\xi}\,d\xi^2 + \frac{\xi + \eta}{4\eta}\,d\eta^2 + \xi\eta\,d\varphi^2, \tag{37.3}$$

and the volume element is

$$dV = \frac{1}{4}(\xi + \eta)\,d\xi\,d\eta\,d\varphi. \tag{37.4}$$

Equation (37.3) gives the following expression for the Laplace operator:

$$\Delta = \frac{4}{\xi + \eta}\left[\frac{\partial}{\partial\xi}\left(\xi\frac{\partial}{\partial\xi}\right) + \frac{\partial}{\partial\eta}\left(\eta\frac{\partial}{\partial\eta}\right)\right] + \frac{1}{\xi\eta}\frac{\partial^2}{\partial\varphi^2}. \tag{37.5}$$

For a particle in the attractive Coulomb field

$$U = -\frac{1}{r} = -\frac{2}{\xi + \eta},$$

the Schrödinger equation becomes

$$\frac{4}{\xi + \eta}\left[\frac{\partial}{\partial\xi}\left(\xi\frac{\partial\psi}{\partial\xi}\right) + \frac{\partial}{\partial\eta}\left(\eta\frac{\partial\psi}{\partial\eta}\right)\right] + \frac{1}{\xi\eta}\frac{\partial^2\psi}{\partial\varphi^2} + 2\left(E + \frac{2}{\xi + \eta}\right)\psi = 0. \tag{37.6}$$

We seek the eigenfunctions ψ in the form

$$\psi = f_1(\xi) f_2(\eta) e^{im\varphi}, \tag{37.7}$$

where m is the magnetic quantum number. Substitute this expression into (37.6), after multiplying that equation by $(\xi + \eta)/4$, and separate the variables ξ and η. The resulting equations for f_1 and f_2 are

$$\frac{d}{d\xi}\left(\xi\frac{df_1}{d\xi}\right) + \left[\frac{E}{2}\xi - \frac{m^2}{4\xi} + \beta_1\right] f_1 = 0,$$
$$\frac{d}{d\eta}\left(\eta\frac{df_2}{d\eta}\right) + \left[\frac{E}{2}\eta - \frac{m^2}{4\eta} + \beta_2\right] f_2 = 0, \tag{37.8}$$

where the “separation parameters” β_1 and β_2 are related by

$$\beta_1 + \beta_2 = 1. \tag{37.9}$$

Consider the discrete energy spectrum ($E < 0$). In place of E, ξ, and η, introduce the quantities

$$n = \frac{1}{\sqrt{-2E}}, \qquad \rho_1 = \xi\sqrt{-2E} = \frac{\xi}{n}, \qquad \rho_2 = \frac{\eta}{n}, \tag{37.10}$$

after which the equation for f_1 takes the form

$$\frac{d^2 f_1}{d\rho_1^2} + \frac{1}{\rho_1}\frac{df_1}{d\rho_1} + \left[-\frac{1}{4} + \frac{1}{\rho_1}\left(\frac{|m|+1}{2} + n_1\right) - \frac{m^2}{4\rho_1^2}\right] f_1 = 0, \tag{37.11}$$

with an identical equation for f_2; here we have also introduced the notation

$$n_1 = -\frac{|m|+1}{2} + n\beta_1, \qquad n_2 = -\frac{|m|+1}{2} + n\beta_2. \tag{37.12}$$

Proceeding as for equation (36.4), we find that at large ρ_1, f_1 behaves as $e^{-\rho_1/2}$, and at small ρ_1 as $\rho_1^{|m|/2}$. We therefore seek a solution of (37.11) in the form

$$f_1(\rho_1) = e^{-\rho_1/2}\rho_1^{|m|/2} w_1(\rho_1)$$

(and analogously for f_2), obtaining for w_1 the equation

$$\rho_1 w_1'' + (|m| + 1 - \rho_1)w_1' + n_1 w_1 = 0.$$

This is again the confluent hypergeometric equation. The solution satisfying the finiteness conditions is

$$w_1 = F(-n_1, |m| + 1, \rho_1),$$

and n_1 must be a nonnegative integer.

Thus, in parabolic coordinates every stationary state of the discrete spectrum is specified by three integers: the “parabolic quantum numbers” n_1 and n_2, and the magnetic quantum number m. From (37.9) and (37.12), the number n (the “principal quantum number”) is

$$n = n_1 + n_2 + |m| + 1. \tag{37.13}$$

The energy levels are, of course, the same as in (36.9).

For a given n, the number $|m|$ can take the n values from 0 through $n-1$. With n and $|m|$ fixed, n_1 takes the $n - |m|$ values from 0 through $n - |m| - 1$. For each prescribed $|m|$, functions with $m = \pm|m|$ may also be chosen. Hence the total number of distinct states for a given n is

$$2\sum_{m=1}^{n-1}(n-m) + (n-0) = n^2,$$

in agreement with the result obtained in § 36.

The discrete-spectrum wave functions $\psi_{n_1 n_2 m}$ must obey the normalization condition

$$\int |\psi_{n_1 n_2 m}|^2 \, dV = \frac{1}{4} \int_0^\infty \int_0^\infty \int_0^{2\pi} |\psi_{n_1 n_2 m}|^2 (\xi + \eta) \, d\varphi \, d\xi \, d\eta = 1. \tag{37.14}$$

The normalized functions are

$$\psi_{n_1 n_2 m} = \frac{\sqrt{2}}{n^2} f_{n_1 m}\left(\frac{\xi}{n}\right) f_{n_2 m}\left(\frac{\eta}{n}\right) \frac{e^{im\varphi}}{\sqrt{2\pi}}, \tag{37.15}$$

$$f_{pm}(\rho) = \frac{1}{|m|!}\sqrt{\frac{(p+|m|)!}{p!}} F(-p, |m|+1, \rho) e^{-\rho/2} \rho^{|m|/2}. \tag{37.16}$$

Unlike the wave functions in spherical coordinates, those in parabolic coordinates are not symmetric about the plane $z = 0$. When $n_1 > n_2$, the particle is more likely to be found on the $z > 0$ side than on the $z < 0$ side; for $n_1 < n_2$, the reverse holds.

The continuous spectrum ($E > 0$) corresponds to a continuous spectrum of real values of the parameters β_1, β_2 in equations (37.8) (still, of course, related by (37.9)); we shall not write the corresponding wave functions here. Regarded as equations for the "eigenvalues" β_1 and β_2, equations (37.8) also possess, when $E > 0$, a spectrum of complex values. The associated wave functions will be given in § 135, where they will be used to solve the problem of scattering in a Coulomb field.

The existence of the stationary states $|n_1 n_2 m\rangle$ is tied to the additional conservation law (36.29). In these states, besides the energy, the quantities $l_z = m$ and A_z have definite values. Evaluation of the diagonal matrix elements of the operator $\hat{A}_z$ gives

$$A_z = \frac{n_1 - n_2}{n}. \tag{37.17}$$

Here $u_z = n_1 - n_2$, while the projections of the "angular momenta" $\mathbf{j}_1$ and $\mathbf{j}_2$ are

$$j_{1z} = \frac{m + n_1 - n_2}{2} \equiv \mu_1, \qquad j_{2z} = \frac{m - n_1 + n_2}{2} \equiv \mu_2. \tag{37.18}$$

These properties of the states $|n_1 n_2 m\rangle$ (or, equivalently, $|n\mu_1\mu_2\rangle$) make it easy to relate their wave functions to those of the states $|nlm\rangle$. Since $\mathbf{l} = \mathbf{j}_1 + \mathbf{j}_2$, a change between the two descriptions reduces to constructing wave functions for the addition of two angular momenta (considered below in § 106). In terms of the "angular momenta" $\mathbf{j}_1$ and $\mathbf{j}_2$, the states $|nlm\rangle$ and $|n_1 n_2 m\rangle$ are written as $|j_1 j_2 l m\rangle$ and $|j_1 j_2 \mu_1 \mu_2\rangle$, where, by (36.35) and (37.13),

$$j_1 = j_2 = \frac{n-1}{2} = \frac{n_1 + n_2 + |m|}{2}. \tag{37.19}$$

The general formulas (106.9)–(106.11) give

$$\begin{aligned} \psi_{nlm} &= \sum_{\mu_1 + \mu_2 = m} \langle lm|\mu_1\mu_2\rangle \psi_{n\mu_1\mu_2}, \\ \psi_{n\mu_1\mu_2} &= \sum_{l=0}^{n-1} \langle l, \mu_1 + \mu_2|\mu_1\mu_2\rangle \psi_{nlm} \end{aligned} \tag{37.20}$$

(*D. Park*, 1960).

CHAPTER VI

Perturbation Theory

§ 38. Time-independent perturbations

An exact solution of the Schrödinger equation can be found in only a comparatively small number of the simplest cases. Most problems in quantum mechanics lead to equations too complicated for exact solution. Frequently, however, a problem contains quantities of different orders of magnitude, including small quantities whose omission simplifies the problem enough to make an exact solution possible. The first step is then to solve this simplified problem exactly; the second is to calculate approximately the corrections due to the small terms that were omitted. The general method for calculating these corrections is called *perturbation theory*.

Suppose that the Hamiltonian of the physical system is

$$\hat{H} = \hat{H}_0 + \hat{V},$$

where $\hat{V}$ is a small correction, the *perturbation*, to the "unperturbed" operator $\hat{H}_0$. In § 38 and § 39 we consider perturbations $\hat{V}$ with no explicit time dependence; the same is assumed for $\hat{H}_0$. The conditions under which $\hat{V}$ may be regarded as small in comparison with $\hat{H}_0$ will be established below.

For the discrete spectrum, the problem may be stated as follows. The eigenfunctions $\psi^{(0)}$ and eigenvalues of $\hat{H}_0$ are assumed known; that is, the exact solutions of

$$\hat{H}_0\psi^{(0)} = E^{(0)}\psi^{(0)} \tag{38.1}$$

are known. Approximate solutions are required for

$$\hat{H}\psi = (\hat{H}_0 + \hat{V})\psi = E\psi, \tag{38.2}$$

namely approximate expressions for the eigenfunctions ψ_n and eigenvalues E_n of the perturbed operator $\hat{H}$.

Throughout this section all eigenvalues of $\hat{H}_0$ will be assumed nondegenerate. To simplify the derivation, we first suppose that the energy levels form a discrete spectrum only.

It is convenient to use matrix form from the outset. Expand the required function ψ in the functions $\psi_m^{(0)}$:

$$\psi = \sum_m c_m \psi_m^{(0)}. \tag{38.3}$$

Substitution in (38.2) gives

$$\sum_m c_m (E_m^{(0)} + \hat{V})\psi_m^{(0)} = \sum_m c_m E \psi_m^{(0)}.$$

Multiplying by $\psi_k^{(0)*}$ and integrating, we find

$$(E - E_k^{(0)})c_k = \sum_m V_{km} c_m. \tag{38.4}$$

Here V_{km} is the matrix of $\hat{V}$ in the unperturbed functions:

$$V_{km} = \int \psi_k^{(0)*} \hat{V} \psi_m^{(0)}\, dq. \tag{38.5}$$

Seek the coefficients and energy as series

$$E = E^{(0)} + E^{(1)} + E^{(2)} + \dots, \qquad c_m = c_m^{(0)} + c_m^{(1)} + c_m^{(2)} + \dots,$$

where $E^{(1)}, c_m^{(1)}$ have the same order of smallness as $\hat{V}$, $E^{(2)}, c_m^{(2)}$ are of second order, and so forth.

To calculate the corrections to the nth eigenvalue and eigenfunction, set $c_n^{(0)} = 1$ and $c_m^{(0)} = 0$ for $m \neq n$. For the first approximation, put $E = E_n^{(0)} + E_n^{(1)}$ and $c_k = c_k^{(0)} + c_k^{(1)}$ in (38.4), retaining first-order terms only. The equation with $k = n$ gives

$$E_n^{(1)} = V_{nn} = \int \psi_n^{(0)*} \hat{V} \psi_n^{(0)}\, dq. \tag{38.6}$$

Thus the first-order correction to $E_n^{(0)}$ is the mean value of the perturbation in the state $\psi_n^{(0)}$.

For $k \neq n$, equation (38.4) gives

$$c_k^{(1)} = \frac{V_{kn}}{E_n^{(0)} - E_k^{(0)}}, \qquad k \neq n. \tag{38.7}$$

The coefficient $c_n^{(1)}$ remains arbitrary and must be chosen so that $\psi_n = \psi_n^{(0)} + \psi_n^{(1)}$ is normalized through first order. This requires $c_n^{(1)} = 0$. Indeed,

$$\psi_n^{(1)} = {\sum_m}' \frac{V_{mn}}{E_n^{(0)} - E_m^{(0)}} \psi_m^{(0)} \tag{38.8}$$

(the prime means that $m = n$ is omitted) is orthogonal to $\psi_n^{(0)}$; hence the integral of $|\psi_n^{(0)} + \psi_n^{(1)}|^2$ differs from unity only by a second-order quantity.

Formula (38.8) gives the first-order correction to the wave functions. It also shows the condition for applicability of the method:

$$|V_{mn}| \ll |E_n^{(0)} - E_m^{(0)}|. \tag{38.9}$$

Thus perturbation matrix elements must be small compared with the corresponding separations between unperturbed energy levels.

To find the second-order correction to $E_n^{(0)}$, substitute $E = E_n^{(0)} + E_n^{(1)} + E_n^{(2)}$ and $c_k = c_k^{(0)} + c_k^{(1)} + c_k^{(2)}$ in (38.4), and retain second-order terms. The equation with $k = n$ gives

$$E_n^{(2)} c_n^0 = {\sum_m}' V_{nm} c_m^{(1)},$$

and consequently

$$E_n^{(2)} = {\sum_m}' \frac{|V_{mn}|^2}{E_n^{(0)} - E_m^{(0)}}. \tag{38.10}$$

Here (38.7) was used, together with Hermiticity, $V_{mn} = V_{nm}^*$. The second-order correction to the ground-state energy is always negative: if $E_n^{(0)}$ is the lowest eigenvalue, every term in (38.10) is negative. Higher approximations may be calculated in the same manner.

These results extend directly to an operator $\hat{H}_0$ that also has a continuous spectrum, while the perturbed state itself still belongs to the discrete spectrum. The appropriate continuous-spectrum integrals are simply added to the discrete sums. Denote continuous-spectrum states by an index ν ranging continuously; ν stands for the complete set of quantities needed to specify a state. If these states are degenerate, as they nearly always are, energy alone does not specify one.[1] For example, (38.8) becomes

$$\psi_n^{(1)} = {\sum_m}' \frac{V_{mn}}{E_n^{(0)} - E_m^{(0)}} \psi_m^{(0)} + \int \frac{V_{\nu n}}{E_n^{(0)} - E_\nu^{(0)}} \psi_\nu^{(0)} \, d\nu. \tag{38.11}$$

It is also useful to give the perturbed matrix elements of a physical quantity f, calculated through first order with $\psi_n = \psi_n^{(0)} + \psi_n^{(1)}$ and (38.8):

$$f_{nm} = f_{nm}^{(0)} + {\sum_k}' \frac{V_{nk} f_{km}^{(0)}}{E_n^{(0)} - E_k^{(0)}} + {\sum_k}' \frac{V_{km} f_{nk}^{(0)}}{E_m^{(0)} - E_k^{(0)}}. \tag{38.12}$$

In the first sum $k \neq n$, and in the second $k \neq m$.

Problem

1. Find the second-order correction to the eigenfunctions.

Solution. The coefficients $c_k^{(2)}$ for $k \neq n$ are obtained from (38.4) with $k \neq n$, written through second order; $c_n^{(2)}$ is chosen so that $\psi_n = \psi_n^{(0)} + \psi_n^{(1)} + \psi_n^{(2)}$ is normalized through second order. The result is

$$\psi_n^{(2)} = {\sum_m}' {\sum_k}' \frac{V_{mk} V_{kn}}{\hbar^2 \omega_{nk} \omega_{nm}} \psi_m^{(0)} - {\sum_m}' \frac{V_{nn} V_{mn}}{\hbar^2 \omega_{nm}^2} \psi_m^{(0)}$$
$$- \frac{\psi_n^{(0)}}{2} {\sum_m}' \frac{|V_{mn}|^2}{\hbar^2 \omega_{nm}^2},$$

where

$$\omega_{nm} = \frac{1}{\hbar}(E_n^{(0)} - E_m^{(0)}).$$

2. Find the third-order correction to the energy eigenvalues.

[1]The wave functions $\psi_\nu^{(0)}$ must then be normalized to delta functions of the quantities ν.

Solution. Retaining the third-order terms in (38.4) with $k = n$ gives

$$E_n^{(3)} = \sum_k{}' \sum_m{}' \frac{V_{nm}V_{mk}V_{kn}}{\hbar^2\omega_{mn}\omega_{kn}} - V_{nn} \sum_m{}' \frac{|V_{nm}|^2}{\hbar^2\omega_{mn}^2}.$$

3. Find the energy levels of the anharmonic linear oscillator with Hamiltonian

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2 x^2}{2} + \alpha x^3 + \beta x^4.$$

Solution. Matrix elements of x^3 and x^4 follow directly from the rule for matrix multiplication and the elements of x in (23.4). The nonzero elements of x^3 are

$$(x^3)_{n-3,n} = (x^3)_{n,n-3} = \left(\frac{\hbar}{m\omega}\right)^{3/2} \sqrt{\frac{n(n-1)(n-2)}{8}},$$

$$(x^3)_{n-1,n} = (x^3)_{n,n-1} = \left(\frac{\hbar}{m\omega}\right)^{3/2} \sqrt{\frac{9n^3}{8}}.$$

There are no diagonal elements in this matrix, so the term αx^3 gives no first-order correction when regarded as a perturbation of the harmonic oscillator. Its second-order correction has the same order as the first-order correction from βx^4. The diagonal elements of x^4 are

$$(x^4)_{n,n} = \left(\frac{\hbar}{m\omega}\right)^2 \frac{3}{4}(2n^2 + 2n + 1).$$

Equations (38.6) and (38.10) then give

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right) - \frac{15}{4}\frac{\alpha^2}{\hbar\omega}\left(\frac{\hbar}{m\omega}\right)^3\left(n^2 + n + \frac{11}{30}\right) + \frac{3}{2}\beta\left(\frac{\hbar}{m\omega}\right)^2\left(n^2 + n + \frac{1}{2}\right).$$

4. An infinite spherical potential well undergoes a small volume-preserving deformation into a slightly prolate or oblate ellipsoid of revolution with semiaxes $a = b$ and c. Find the splitting of its particle energy levels (*A. B. Migdal*, 1959).

Solution. The boundary equation

$$\frac{x^2 + y^2}{a^2} + \frac{z^2}{c^2} = 1$$

becomes the sphere $x^2 + y^2 + z^2 = R^2$ under $x \to ax/R$, $y \to ay/R$, $z \to cz/R$. The particle Hamiltonian $\hat{H} = \hat{\mathbf{p}}^2/2M = -\hbar^2\Delta/2M$ (M is the mass; energy is measured from the bottom of the well) becomes $\hat{H} = \hat{H}_0 + \hat{V}$, with

$$\hat{H}_0 = -\frac{\hbar^2}{2M}\Delta, \qquad \hat{V} = -\frac{\hbar^2}{2M}\left[\left(\frac{R^2}{a^2} - 1\right)\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) + \left(\frac{R^2}{c^2} - 1\right)\frac{\partial^2}{\partial z^2}\right].$$

The ellipsoidal-well problem is thereby reduced to the spherical-well problem. If the ellipsoid differs only slightly from a sphere of radius $R = (a^2c)^{1/3}$, $\hat{V}$ is a small perturbation. Introducing the "ellipticity" β, $|\beta| \ll 1$, by

$$a \approx R\left(1 - \frac{\beta}{3}\right), \qquad c \approx R\left(1 + \frac{2\beta}{3}\right),$$

we obtain

$$\hat{V} = \frac{\beta}{3M}(\hat{\mathbf{p}}^2 - 3\hat{p}_z^2).$$

In first order, the change from the spherical-well levels is

$$\Delta E_{nlm} = E_{nlm} - E_{nl}^{(0)} = \langle nlm|V|nlm\rangle.$$

Here l, m are the angular momentum and its projection on the ellipsoid axis, and n numbers the spherical-well levels for fixed l, which do not depend on m. The expression $\mathbf{p}^2 - 3p_z^2$ is the zz component of the traceless irreducible tensor $\delta_{ik}\mathbf{p}^2 - 3p_ip_k$. By (107.2) and (107.6), $\langle nlm|V|nlm\rangle$ is proportional to $(-1)^m \begin{pmatrix} l & 2 & l \\ -m & 0 & m \end{pmatrix}$, and hence

$$\langle nlm|V|nlm\rangle = \left(1 - \frac{3m^2}{l(l+1)}\right)\langle nl0|V|nl0\rangle.$$

(The table of $3j$ symbols is on p. 530.) We next write

$$\begin{aligned}\langle nl0|V|nl0\rangle &= \frac{2}{3}\beta E_{nl}^{(0)} + \beta\frac{\hbar^2}{M}\left\langle nl0\left|\frac{\partial^2}{\partial z^2}\right|nl0\right\rangle \\ &= \frac{2}{3}\beta E_{nl}^{(0)} - \frac{\beta\hbar^2}{M}\int\left|\frac{\partial\psi_{nl0}}{\partial z}\right|^2 r^2\,dr\,do.\end{aligned}$$

The Schrödinger equation $\hat{H}_0\psi_{nlm} = E_{nl}^{(0)}\psi_{nlm}$ was used in the first term, and the second was integrated by parts. Using (28.11) for Y_{l0}, the derivative of $\psi_{nl0} = R_{nl}(r)Y_{l0}(\theta,\varphi)$ is

$$\begin{aligned}\frac{\partial}{\partial z}\psi_{nl0} &= \left(\cos\theta\frac{\partial}{\partial r} - \frac{\sin\theta}{r}\frac{\partial}{\partial\theta}\right)\psi_{nl0} \\ &= -\frac{i(l+1)}{[4(l+1)^2-1]^{1/2}}\left(R'_{nl} - \frac{l}{r}R_{nl}\right)Y_{l+1,0} \\ &\quad + \frac{il}{[4l^2-1]^{1/2}}\left(R'_{nl} + \frac{l+1}{r}R_{nl}\right)Y_{l-1,0}.\end{aligned}$$

The radial integrals are evaluated from

$$\int_0^\infty R_{nl}R'_{nl}r\,dr = -\frac{1}{2}\int_0^\infty R_{nl}^2\,dr,$$

$$\int_0^\infty R'^{;2}_{nl}r^2\,dr = \frac{2M}{\hbar^2}E_{nl}^{(0)} - l(l+1)\int_0^\infty R_{nl}^2\,dr,$$

by integration by parts and the radial Schrödinger equation (33.3),

$$R''_{nl} + \frac{2}{r}R'_{nl} - \frac{l(l+1)}{r^2}R_{nl} = -\frac{2M}{\hbar^2}E^{(0)}_{nl}.$$

Terms containing integrals of R^2_{nl} cancel, leaving

$$\Delta E_{nlm} = 4\beta \frac{l(l+1)}{(2l-1)(2l+3)}\left[\frac{m^2}{l(l+1)} - \frac{1}{3}\right]E^{(0)}_{nl}.$$

Finally,

$$\frac{1}{2l+1}\sum_{m=-l}^{l} E_{nlm} = E^{(0)}_{nl},$$

so the centre of gravity of the multiplet is unchanged.

§ 39. The secular equation

We now turn to the case in which the unperturbed operator $\hat{H}_0$ has degenerate eigenvalues. Let $\psi^{(0)}_n, \psi^{(0)}_{n'}, \ldots$ denote eigenfunctions belonging to the same energy eigenvalue $E^{(0)}_n$. As we know, the choice of these functions is not unique: they may be replaced by any s independent linear combinations of the same functions, where s is the degeneracy of the level $E^{(0)}_n$. This freedom disappears, however, when the wave functions are required to change only slightly under the applied small perturbation. For the moment, let $\psi^{(0)}_n$, $\psi^{(0)}_{n'}, \ldots$ denote arbitrarily chosen unperturbed eigenfunctions. The correct zeroth-order functions are linear combinations of the form

$$c^{(0)}_n \psi^{(0)}_n + c^{(0)}_{n'}\psi^{(0)}_{n'} + \ldots$$

The coefficients in these combinations and the first-order corrections to the eigenvalues are determined as follows.

Write equations (38.4) for $k = n, n', \ldots$, substituting in first order $E = E^{(0)}_n + E^{(1)}$. For the c_k it is sufficient to retain their zeroth-order values $c_n = c^{(0)}_n$, $c_{n'} = c^{(0)}_{n'}, \ldots$, while $c_m = 0$ for $m \neq n, n', \ldots$. We then obtain

$$E^{(1)}c^{(0)}_n = \sum_{n'} V_{nn'}c^{(0)}_{n'},$$

or

$$\sum_{n'}(V_{nn'} - E^{(1)}\delta_{nn'})c^{(0)}_{n'} = 0, \tag{39.1}$$

where n and n' run over all values labeling states that belong to the given unperturbed eigenvalue $E^{(0)}_n$. This homogeneous system of linear equations for the $c^{(0)}_n$ has nonzero solutions if the determinant of its coefficients vanishes. Thus

$$|V_{nn'} - E^{(1)}\delta_{nn'}| = 0. \tag{39.2}$$

This equation has degree s in $E^{(1)}$ and, in general, has s distinct real roots. These roots are the required first-order corrections to the eigenvalues. Equation (39.2) is called the *secular equation*[2]. The sum of its roots is the sum of the diagonal matrix elements V_{nn}, $V_{n'n'}, \ldots$; it is the coefficient of $E^{(1)s-1}$ in the equation.

Substituting the roots of (39.2) one by one into (39.1) and solving the resulting systems gives the coefficients $c_n^{(0)}$ and hence the correct zeroth-order eigenfunctions. As a result of the perturbation, the initially degenerate energy level is in general no longer degenerate, since the roots of (39.2) are in general different. The perturbation is said to lift the degeneracy. The lifting may be complete or partial; in the latter case the perturbed level remains degenerate, but with a lower multiplicity than before.

It may happen that, for some reason, all matrix elements for transitions within one group of mutually degenerate states $n, n', \ldots$ are especially small, or even vanish. It may then be useful, while retaining $V_{nn'}$ in first order, also to include in higher orders the matrix elements V_{nm}, with $m \neq n, n', \ldots$, for transitions to states of other energies. We shall do this by including V_{mn} through second order.

In (38.4) with $k = n$, set $E = E_n^{(0)} + E^{(1)}$ on the left-hand side; we retain the notation $E^{(1)}$ for the energy correction in the approximation now considered. Replace c_n by $c_n^{(0)}$. Since $c_m^{(0)} = 0$ for every $m \neq n, n', \ldots$, we have

$$E^{(1)} c_n^{(0)} = \sum_m V_{nm} c_m^{(1)} + \sum_{n'} V_{nn'} c_{n'}^{(0)}. \tag{39.3}$$

Equations (38.4) with $k = m \neq n, n', \ldots$, through first order, give

$$(E_n^{(0)} - E_m^{(0)}) c_m^{(1)} = \sum_{n'} V_{mn'} c_{n'}^{(0)},$$

and hence

$$c_m^{(1)} = \sum_{n'} \frac{V_{mn'}}{E_n^{(0)} - E_m^{(0)}} c_{n'}^{(0)}.$$

Substitution in (39.3) gives

$$E^{(1)} c_n^{(0)} = \sum_{n'} c_{n'}^{(0)} \left(V_{nn'} + \sum_m \frac{V_{nm} V_{mn'}}{E_n^{(0)} - E_m^{(0)}} \right).$$

This system now replaces (39.1). Its compatibility condition again gives a secular equation, differing from (39.2) by the replacement

$$V_{nn'} \longrightarrow V_{nn'} + \sum_m \frac{V_{nm} V_{mn'}}{E_n^{(0)} - E_m^{(0)}}. \tag{39.4}$$

[2]Also called the age equation (the French word *siècle* means an age); the name was borrowed from celestial mechanics.

Problem

1. Find the first-order corrections to the eigenvalue and the correct zeroth-order functions for a doubly degenerate level.

Solution. Equation (39.2) now has the form

$$\begin{vmatrix} V_{11} - E^{(1)} & V_{21} \\ V_{12} & V_{22} - E^{(1)} \end{vmatrix} = 0$$

(the indices 1 and 2 refer to two arbitrarily chosen unperturbed eigenfunctions $\psi_1^{(0)}$ and $\psi_2^{(0)}$ of the degenerate level). Solving it, we find

$$E^{(1)} = \frac{1}{2}[V_{11} + V_{22} \pm \hbar\omega^{(1)}], \tag{1}$$

where

$$\hbar\omega^{(1)} = \sqrt{(V_{11} - V_{22})^2 + 4|V_{12}|^2}$$

denotes the difference between the two values of the correction $E^{(1)}$. Solving (39.1) with these values of $E^{(1)}$, we obtain the coefficients in the normalized correct zeroth-order functions $\psi^{(0)} = c_1^{(0)}\psi_1^{(0)} + c_2^{(0)}\psi_2^{(0)}$:

$$\begin{aligned} c_1^{(0)} &= \left\{\frac{V_{12}}{2|V_{12}|}\left[1 \pm \frac{V_{11} - V_{22}}{\hbar\omega^{(1)}}\right]\right\}^{1/2}, \\ c_2^{(0)} &= \pm\left\{\frac{V_{21}}{2|V_{12}|}\left[1 \mp \frac{V_{11} - V_{22}}{\hbar\omega^{(1)}}\right]\right\}^{1/2}. \end{aligned} \tag{2}$$

Problem

2. Derive formulas for the first-order corrections to the eigenfunctions and the second-order corrections to the eigenvalues.

Solution. Let the functions $\psi_n^{(0)}$ be chosen as the correct zeroth-order functions. The matrix $V_{nn'}$ defined with these functions is evidently diagonal in the indices n, n' belonging to the same group of functions of a degenerate level. Its diagonal elements $V_{nn}, V_{n'n'}, \ldots$ equal the corresponding first-order corrections $E_n^{(1)}, E_{n'}^{(1)}, \ldots$.

Consider the perturbation of the eigenfunction $\psi_n^{(0)}$. In zeroth order, $E = E_n^{(0)}$, $c_n^{(0)} = 1$, and $c_m^{(0)} = 0$ for $m \neq n$. In first order, $E = E_n^{(0)} + V_{nn}$, $c_n = 1 + c_n^{(1)}$, and $c_m = c_m^{(1)}$. From the general system (38.4), take the equation with $k \neq n, n', \ldots$ and retain its first-order terms:

$$(E_n^{(0)} - E_k^{(0)})c_k^{(1)} = V_{kn}c_n^{(0)} = V_{kn},$$

whence

$$c_k^{(1)} = \frac{V_{kn}}{E_n^{(0)} - E_k^{(0)}} \quad \text{for} \quad k \neq n, n', \ldots \tag{1}$$

Next write the equation with $k = n'$, retaining its second-order terms:

$$E_n^{(1)} c_{n'}^{(1)} = V_{n'n'} c_{n'}^{(1)} + \sum_m{}' V_{n'm} c_m^{(1)}.$$

In the sum over m, the terms with $m = n, n', \ldots$ are omitted. Substituting $E_n^{(1)} = V_{nn}$ and (1) for $c_m^{(1)}$, we obtain, for $n' \neq n$,

$$c_{n'}^{(1)} = \frac{1}{V_{nn} - V_{n'n'}} \sum_m{}' \frac{V_{n'm} V_{mn}}{E_n^{(0)} - E_m^{(0)}}. \tag{2}$$

The coefficient $c_n^{(1)}$ is zero in this approximation. Formulas (1) and (2) determine the first-order correction $\psi_n^{(1)} = \sum_m c_m^{(1)} \psi_m^{(0)}$ to the eigenfunctions[3].

Finally, retaining the second-order terms in (38.4) with $k = n$, we obtain the second-order energy correction

$$E_n^{(2)} = \sum_m{}' \frac{V_{nm} V_{mn}}{E_n^{(0)} - E_m^{(0)}}, \tag{3}$$

which has the same formal appearance as (38.10).

Problem

3. At the initial time $t = 0$, a system is in a state $\psi_1^{(0)}$ belonging to a doubly degenerate level. Find the probability that at a later time t the system is in the other state $\psi_2^{(0)}$ of the same energy, when the transition is caused by a constant perturbation.

Solution. Construct the correct zeroth-order functions:

$$\psi = c_1 \psi_1 + c_2 \psi_2, \qquad \psi' = c_1' \psi_1 + c_2' \psi_2,$$

where c_1, c_2 and c_1', c_2' are the two pairs of coefficients given by (2) of Problem 1. Superscripts (0) have been omitted from all quantities for brevity. Conversely,

$$\psi_1 = \frac{c_2' \psi - c_2 \psi'}{c_1 c_2' - c_1' c_2}.$$

The functions ψ and ψ' belong to states with perturbed energies $E + E^{(1)}$ and $E + E^{(1)'}$, where $E^{(1)}$ and $E^{(1)'}$ are the two values of correction (1) in Problem 1. Introducing the time factors gives the time-dependent wave function

$$\Psi_1 = \frac{\exp[-(i/\hbar)Et]}{c_1 c_2' - c_1' c_2} \left[c_2' \psi \exp\left(-\frac{i}{\hbar} E^{(1)} t\right) - c_2 \psi' \exp\left(-\frac{i}{\hbar} E^{(1)'} t\right) \right].$$

[3] Notice that the requirement that the quantities (1) and (2) be small, and hence the condition for applicability of this perturbation method, still requires (38.9) only for transitions between states belonging to different energy levels. Transitions between states of the same degenerate level are accounted for by the secular equation in a certain sense exactly.

At $t = 0$, $\Psi_1 = \psi_1$. Expressing ψ and ψ' once more in terms of ψ_1 and ψ_2, we write Ψ_1 as a linear combination of ψ_1 and ψ_2 with time-dependent coefficients. The squared modulus of the coefficient of ψ_2 is the required transition probability w_{21}. Using (1) and (2) of Problem 1 gives

$$w_{21} = 2\frac{|V_{12}|^2}{(\hbar\omega^{(1)})^2}[1 - \cos(\omega^{(1)}t)].$$

Thus the probability oscillates periodically with frequency $\omega^{(1)}$. For times t small compared with the corresponding period, the expression in square brackets, and hence w_{21} itself, is proportional to t^2:

$$w_{21} = \frac{1}{\hbar^2}|V_{12}|^2 t^2.$$

This formula follows very simply by the method developed in the next section, using (40.4).

§ 40. Time-dependent perturbations

We now turn to perturbations with an explicit dependence on time. In this case there can be no question at all of corrections to the energy eigenvalues: when the Hamiltonian depends on time (as does the perturbed operator $\hat{H} = \hat{H}_0 + \hat{V}(t)$), energy is not conserved, and stationary states therefore do not exist. The problem is instead to calculate approximate wave functions from the wave functions of the unperturbed system's stationary states.

For this purpose we use the analogue of the familiar variation-of-constants method for solving linear differential equations (*P. A. M. Dirac*, 1926). Let $\Psi_k^{(0)}$ be the wave functions (including the time factor) of the unperturbed system's stationary states. An arbitrary solution of the unperturbed wave equation can then be written as $\Psi = \sum_k a_k \Psi_k^{(0)}$. We seek a solution of the perturbed equation

$$i\hbar\frac{\partial\Psi}{\partial t} = (\hat{H}_0 + \hat{V})\Psi \tag{40.1}$$

in the form

$$\Psi = \sum_k a_k(t)\Psi_k^{(0)}, \tag{40.2}$$

where the expansion coefficients are functions of time. Substitution of (40.2) into (40.1), with

$$i\hbar\frac{\partial\Psi_k^{(0)}}{\partial t} = \hat{H}_0\Psi_k^{(0)},$$

gives

$$i\hbar\sum_k \Psi_k^{(0)}\frac{da_k}{dt} = \sum_k a_k\hat{V}\Psi_k^{(0)}.$$

Multiplying both sides on the left by $\Psi_m^{(0)*}$ and integrating, we find

$$i\hbar\frac{da_m}{dt} = \sum_k V_{mk}(t)a_k, \tag{40.3}$$

where

$$V_{mk}(t) = \int \Psi_m^{(0)*} \hat{V} \Psi_k^{(0)}\, dq = V_{mk} e^{i\omega_{mk} t}, \qquad \omega_{mk} = \frac{E_m^{(0)} - E_k^{(0)}}{\hbar}$$

are the perturbation matrix elements including their time factor. If V itself depends explicitly on time, the quantities V_{mk} must of course also be regarded as functions of time.

Choose the wave function of the nth stationary state as the unperturbed wave function. The corresponding coefficients in (40.2) are $a_n^{(0)} = 1$ and $a_k^{(0)} = 0$ for $k \neq n$. To obtain the first approximation, write $a_k = a_k^{(0)} + a_k^{(1)}$ and put $a_k = a_k^{(0)}$ on the right-hand side of (40.3), which already contains the small quantities V_{mk}. Thus

$$i\hbar \frac{da_k^{(1)}}{dt} = V_{kn}(t). \tag{40.4}$$

To indicate the unperturbed function for which the correction is being calculated, introduce a second index on a_k and write

$$\Psi_n = \sum_k a_{kn}(t) \Psi_k^{(0)}.$$

The integrated form of (40.4) is accordingly

$$a_{kn}^{(1)} = -\frac{i}{\hbar} \int V_{kn}(t)\, dt = -\frac{i}{\hbar} \int V_{kn} e^{i\omega_{kn} t}\, dt. \tag{40.5}$$

This determines the wave functions in the first approximation.

Consider more closely the important case of a time-periodic perturbation

$$\hat{V} = \hat{F} e^{-i\omega t} + \hat{G} e^{i\omega t}, \tag{40.6}$$

where $\hat{F}$ and $\hat{G}$ do not depend on time. Hermiticity of V requires

$$\hat{F} e^{-i\omega t} + \hat{G} e^{i\omega t} = \hat{F}^+ e^{i\omega t} + \hat{G}^+ e^{-i\omega t},$$

whence $\hat{G} = \hat{F}^+$, or

$$G_{nm} = F_{mn}^*. \tag{40.7}$$

Using this relation, we have

$$V_{kn}(t) = V_{kn} e^{i\omega_{kn} t} = F_{kn} e^{i(\omega_{kn} - \omega)t} + F_{nk}^* e^{i(\omega_{kn} + \omega)t}. \tag{40.8}$$

Substitution in (40.5) and integration yield

$$a_{kn}^{(1)} = -\frac{F_{kn} e^{i(\omega_{kn} - \omega)t}}{\hbar(\omega_{kn} - \omega)} - \frac{F_{nk}^* e^{i(\omega_{kn} + \omega)t}}{\hbar(\omega_{kn} + \omega)}. \tag{40.9}$$

These expressions apply provided neither denominator vanishes[4]; that is, for every k (at fixed n),

$$E_k^{(0)} - E_n^{(0)} \neq \pm\hbar\omega. \tag{40.10}$$

[4]More precisely, neither may be so small that $a_{kn}^{(1)}$ ceases to be small in comparison with unity.

For a number of applications it is useful to know the matrix elements of an arbitrary quantity f, defined using the perturbed wave functions. To first order,

$$f_{nm}(t) = f_{nm}^{(0)}(t) + f_{nm}^{(1)}(t),$$

where

$$f_{nm}^{(0)}(t) = \int \Psi_n^{(0)*}\hat{f}\Psi_m^{(0)}\,dq = f_{nm}^{(0)}e^{i\omega_{nm}t},$$

$$f_{nm}^{(1)}(t) = \int \left(\Psi_n^{(0)*}\hat{f}\Psi_m^{(1)} + \Psi_n^{(1)*}\hat{f}\Psi_m^{(0)}\right) dq.$$

Inserting

$$\Psi_n^{(1)} = \sum_k a_{kn}^{(1)}\Psi_k^{(0)}$$

with the coefficients from (40.9), we obtain

$$\begin{aligned} f_{nm}^{(1)}(t) = -e^{i\omega_{nm}t}\sum_k \Bigg\{ &\left[\frac{f_{nk}^{(0)}F_{km}}{\hbar(\omega_{km}-\omega)} + \frac{f_{km}^{(0)}F_{nk}}{\hbar(\omega_{kn}+\omega)}\right]e^{-i\omega t} \\ &+ \left[\frac{f_{nk}^{(0)}F_{mk}^*}{\hbar(\omega_{km}+\omega)} + \frac{f_{km}^{(0)}F_{kn}^*}{\hbar(\omega_{kn}-\omega)}\right]e^{i\omega t}\Bigg\}. \end{aligned} \tag{40.11}$$

This formula applies if none of its terms becomes large, hence if all the frequencies ω_{kn} and ω_{km} remain sufficiently far from ω. At $\omega = 0$ it reduces to (38.12). The formulas above assume that the unperturbed energy levels form only a discrete spectrum. They extend immediately to the presence of a continuous spectrum as well (the states being perturbed still belonging to the discrete spectrum): one simply adds the corresponding integrals over the continuous spectrum to the sums over discrete levels. The denominators $\omega_{kn} \pm \omega$ in (40.9) and (40.11) must then remain nonzero as $E_k^{(0)}$ ranges through both the discrete and continuous spectra. If, as is usual, the continuous spectrum lies above all discrete levels, condition (40.10), for example, must be supplemented by

$$E_{\min}^{(0)} - E_n^{(0)} > \hbar\omega, \tag{40.12}$$

where $E_{\min}^{(0)}$ is the energy at the lower edge of the continuous spectrum.

Problem

1. Determine how the nth and mth solutions of the Schrödinger equation change under a periodic perturbation of the form (40.6), whose frequency ω satisfies $E_m^{(0)} - E_n^{(0)} = \hbar(\omega + \varepsilon)$, where ε is small.

Solution. The method developed above does not apply, since the coefficient $a_{mn}^{(1)}$ in (40.9) becomes large. Return to the exact equations (40.3), with $V_{mk}(t)$ given by (40.8). Evidently, the principal effect comes from terms on their right-hand sides whose time dependence is governed by the small frequency $\omega_{mn} - \omega$. Dropping all other terms gives

$$i\hbar\frac{da_m}{dt} = F_{mn}e^{i\varepsilon t}a_n, \qquad i\hbar\frac{da_n}{dt} = F_{mn}^*e^{-i\varepsilon t}a_m.$$

Put $a_n e^{i\varepsilon t} = b_n$. Then

$$i\hbar\dot{a}_m = F_{mn}b_n, \qquad i\hbar(\dot{b}_n - i\varepsilon b_n) = F^*_{mn}a_m,$$

and elimination of a_m gives

$$\ddot{b}_n - i\varepsilon\dot{b}_n + (1/\hbar^2)|F_{mn}|^2 b_n = 0.$$

Two independent solutions may be chosen as

$$a_n = Ae^{i\alpha_1 t}, \qquad a_m = -A\frac{\hbar\alpha_1}{F_{mn}}e^{i\alpha_2 t} \tag{1}$$

and

$$a_n = Be^{-i\alpha_2 t}, \qquad a_m = B\frac{\hbar\alpha_2}{F^*_{mn}}e^{-i\alpha_1 t}, \tag{2}$$

where A and B are constants (to be fixed by normalization), and

$$\alpha_1 = -\varepsilon/2 + \Omega, \quad \alpha_2 = \varepsilon/2 + \Omega, \quad \Omega = \sqrt{\varepsilon^2/4 + |\eta|^2}, \quad \eta = F_{mn}/\hbar.$$

Thus the perturbation transforms $\Psi_n^{(0)}$ and $\Psi_m^{(0)}$ into $a_n\Psi_n^{(0)} + a_m\Psi_m^{(0)}$, with a_n, a_m taken from (1) or (2).

Suppose that at $t = 0$ the system was in the state $\Psi_m^{(0)}$. At later times its state is the linear combination of the two solutions just found which reduces to $\Psi_m^{(0)}$ at $t = 0$:

$$\Psi = e^{i\varepsilon t/2}\left(\cos\Omega t - \frac{i\varepsilon}{2\Omega}\sin\Omega t\right)\Psi_m^{(0)} - \frac{i\eta^*}{\Omega}e^{-i\varepsilon t/2}\sin\Omega t \cdot \Psi_n^{(0)}. \tag{3}$$

The squared modulus of the coefficient of $\Psi_n^{(0)}$ is

$$\frac{|\eta|^2}{2\Omega^2}\left[1 - \cos(2\Omega t)\right]. \tag{4}$$

It is the probability that the system is in $\Psi_n^{(0)}$ at time t. This probability is periodic, with frequency 2Ω, and varies from zero to $|\eta|^2/\Omega^2$.

At exact resonance, $\varepsilon = 0$, (4) becomes

$$(1/2)\left[1 - \cos(2|\eta|t)\right].$$

It varies periodically between zero and unity: the system passes periodically from $\Psi_m^{(0)}$ to $\Psi_n^{(0)}$.

§ 41. Transitions produced by a perturbation acting for a finite time

Suppose that the perturbation $V(t)$ acts only during some finite time interval (or, alternatively, that $V(t)$ decreases sufficiently rapidly as $t \to \pm\infty$). Before the perturbation begins to act (or in the limit $t \to -\infty$), let the system be in the nth stationary state of the discrete spectrum. At any later instant its state is described by

$$\Psi = \sum_k a_{kn}\Psi_k^{(0)},$$

where, to first order,

$$
\begin{aligned}
a_{kn} = a_{kn}^{(1)} &= -\frac{i}{\hbar}\int_{-\infty}^{t} V_{kn}e^{i\omega_{kn}t}\,dt, \qquad k \neq n,\\
a_{nn} = 1 + a_{nn}^{(1)} &= 1 - \frac{i}{\hbar}\int_{-\infty}^{t} V_{nn}\,dt.
\end{aligned}
\tag{41.1}
$$

The integration limits in (40.5) have been selected so that every $a_{kn}^{(1)}$ vanishes as $t \to -\infty$. Once the action of the perturbation has ended (or as $t \to \infty$), the coefficients a_{kn} acquire constant values $a_{kn}(\infty)$, and the system is in the state whose wave function is

$$
\Psi = \sum_k a_{kn}(\infty)\Psi_k^{(0)}.
$$

This function again satisfies the unperturbed wave equation, though it differs from the original function $\Psi_n^{(0)}$. By the general rules, the squared modulus of $a_{kn}(\infty)$ gives the probability that the system has energy $E_k^{(0)}$, that is, that it is in the kth stationary state.

Thus a perturbation can carry the system from its original stationary state into any other one. The probability of a transition from the initial (ith) stationary state to the final (fth) stationary state is[5]

$$
w_{fi} = \frac{1}{\hbar^2}\left|\int_{-\infty}^{+\infty} V_{fi}e^{i\omega_{fi}t}\,dt\right|^2. \tag{41.2}
$$

Now consider a perturbation which, once it has arisen, continues to act indefinitely (while remaining small throughout). In other words, $V(t)$ tends to zero as $t \to -\infty$ and to a finite nonzero limit as $t \to \infty$. Formula (41.2) cannot be used directly here, since its integral diverges. From the physical standpoint, however, this divergence is immaterial and is easily removed. Integration by parts gives

$$
\begin{aligned}
a_{fi} &= -\frac{i}{\hbar}\int_{-\infty}^{t} V_{fi}e^{i\omega_{fi}t}\,dt\\
&= -\left.\frac{V_{fi}e^{i\omega_{fi}t}}{\hbar\omega_{fi}}\right|_{-\infty}^{t} + \int_{-\infty}^{t} \frac{\partial V_{fi}}{\partial t}\frac{e^{i\omega_{fi}t}}{\hbar\omega_{fi}}\,dt.
\end{aligned}
$$

The first term vanishes at the lower limit; at the upper limit it formally coincides with the expansion coefficients in (38.8). (The extra periodic factor occurs simply because a_{fi} are expansion coefficients of the full wave function Ψ, whereas the c_{fi} of § 38 are expansion coefficients of the time-independent function ψ.) It is consequently clear that the limit of this term as $t \to \infty$ describes only the change of the original wave function $\Psi_i^{(0)}$ caused by the "constant part" $V(+\infty)$ of the perturbation, and therefore has nothing to do with transitions into other states. The transition probability is instead given by the square of the second term:

$$
w_{fi} = \frac{1}{\hbar^2\omega_{fi}^2}\left|\int_{-\infty}^{+\infty} \frac{\partial V_{fi}}{\partial t}e^{i\omega_{fi}t}\,dt\right|^2. \tag{41.3}
$$

[5] For uniformity, whenever transition probabilities are discussed below, we shall denote the initial and final states by the indices i and f, respectively. We shall also write the indices of transition probabilities specifically in the order fi, consistently with the convention for matrix-element indices.

The formulas obtained also apply when a transition proceeds from a discrete state to a continuous-spectrum state. The only difference is that one then speaks of the probability of transition from the specified (ith) state into states for which the quantities ν_f (see the end of § 38) lie between ν_f and $\nu_f + d\nu_f$. Thus, for example, (41.2) must be written as

$$dw_{if} = \frac{1}{\hbar^2}\left|\int_{-\infty}^{\infty} V_{fi}e^{i\omega_{fi}t}\,dt\right|^2 d\nu_f. \tag{41.4}$$

If $V(t)$ varies little over time intervals of order $1/\omega_{fi}$, the integral in (41.2) or (41.3) is very small. In the limit of arbitrarily slow variation of the applied perturbation, the probability of every transition that changes the energy (that is, with nonzero ω_{fi}) tends to zero. Hence, when an applied perturbation changes sufficiently slowly (*adiabatically*), a system initially in a particular nondegenerate stationary state continues to remain in that same state (see also § 53).

In the opposite limit, when a perturbation is switched on very rapidly—that is, *suddenly*—the derivatives $\partial V_{fi}/\partial t$ become infinite at the "switching instant." In the integral of $(\partial V_{fi}/\partial t)e^{i\omega_{fi}t}$, the comparatively slow factor $e^{i\omega_{fi}t}$ may then be taken outside the integral at its value at that instant. The remaining integration is immediate and gives

$$w_{fi} = \frac{|V_{fi}|^2}{\hbar^2\omega_{fi}^2}. \tag{41.5}$$

Transition probabilities for sudden perturbations can also be found when the perturbation is not small. Let the system initially be in a state described by an eigenfunction $\psi_i^{(0)}$ of the original Hamiltonian $\hat{H}_0$. If the Hamiltonian changes suddenly (in a time short compared with the periods $1/\omega_{fi}$ of transitions from the given state i to other states), the wave function has no time to change and remains what it was before the perturbation. It is no longer, however, an eigenfunction of the system's new Hamiltonian $\hat{H}$; the state $\psi_i^{(0)}$ is therefore not stationary. By the general rules of quantum mechanics, the probabilities w_{fi} for transition into any of the new stationary states are determined by expanding $\psi_i^{(0)}$ in the eigenfunctions ψ_f of $\hat{H}$:

$$w_{fi} = \left|\int \psi_i^{(0)}\psi_f^*\,dq\right|^2. \tag{41.6}$$

We now show how this general formula reduces to (41.5) when the change $\hat{V} = \hat{H} - \hat{H}_0$ of the Hamiltonian is small. Multiply the equations

$$\hat{H}_0\psi_i^{(0)} = E_i^{(0)}\psi_i^{(0)}, \qquad \hat{H}^*\psi_f^* = E_f\psi_f^*$$

by ψ_f^* and $\psi_i^{(0)}$, respectively, integrate over dq, and subtract the two results term by term. Using also the self-adjointness of $\hat{H}$, we obtain

$$(E_f - E_i^{(0)})\int \psi_f^*\psi_i^{(0)}\,dq = \int \psi_f^*\hat{V}\psi_i^{(0)}\,dq.$$

If $\hat{V}$ is small, then to first order E_f may be replaced by the nearby unperturbed level $E_f^{(0)}$, and ψ_f on the right-hand side by the corresponding function $\psi_f^{(0)}$. It follows that

$$\int \psi_f^* \psi_i^{(0)}\, dq = \frac{1}{\hbar\omega_{fi}} \int \psi_f^{(0)*} \hat{V} \psi_i^{(0)}\, dq,$$

so that (41.6) becomes (41.5).

Problems

Problem

1. A uniform electric field is suddenly applied to a charged oscillator in its ground state. Find the probabilities for the perturbation to transfer the oscillator into its excited states.

Solution. Potential energy of an oscillator in a uniform field exerting force F upon it is

$$U(x) = \frac{m\omega^2}{2}x^2 - Fx = \frac{m\omega^2}{2}(x - x_0)^2 + \text{const},$$

where $x_0 = F/m\omega^2$. Thus it again has a purely oscillator form, but with its equilibrium position displaced. Consequently, the stationary-state wave functions of the perturbed oscillator are $\psi_k(x - x_0)$, where $\psi_k(x)$ are the oscillator functions (23.12), while the initial wave function is $\psi_0(x)$ from (23.13). Using these functions and expression (23.11) for the Hermite polynomials, we find

$$\int_{-\infty}^{+\infty} \psi_0^{(0)} \psi_k\, dx = \frac{(-1)^k}{\sqrt{2^k \pi k!}} e^{-\xi_0^2/2} \int_{-\infty}^{\infty} e^{-\xi\xi_0} \frac{d^k}{d\xi^k} e^{-\xi^2 + 2\xi\xi_0}\, d\xi,$$

where $\xi_0 = x_0\sqrt{m\omega/\hbar}$. Integrating by parts k times reduces the integral here to

$$\xi_0^k \int_{-\infty}^{\infty} \exp(-\xi^2 + \xi\xi_0)\, d\xi = \xi_0^k \sqrt{\pi} \exp\frac{\xi_0^2}{4}.$$

Formula (41.6) therefore gives the required transition probability as

$$w_{k0} = \frac{\bar{k}^k}{k!} e^{-\bar{k}}, \qquad \bar{k} = \frac{\xi_0^2}{2} = \frac{F^2}{2m\hbar\omega^3}.$$

As a function of the integer k, this is a Poisson distribution with mean $\bar{k}$.

The perturbative regime corresponds to small F such that $\bar{k} \ll 1$. The excitation probabilities are then small and fall rapidly as k increases; the largest is $w_{10} \approx \bar{k}$.

Conversely, when F is large ($\bar{k} \gg 1$), excitation occurs with overwhelming probability: the chance that the oscillator remains in its normal state is $w_{00} = e^{-\bar{k}}$.

Problem

2. The nucleus of an atom in its normal state receives a sudden impulse and thereby acquires velocity $\mathbf{v}$. Assume that the duration τ of the impulse is short both in comparison with the electronic periods and with a/v, where a is an atomic dimension. Determine the probability that this "shaking" excites the atom (*A. B. Migdal*, 1939).

Solution. Pass to the reference frame K' that moves with the nucleus after the impact. Because $\tau \ll a/v$, the nucleus may be treated as having hardly moved during the impact, and the electron coordinates in K' and in the original frame K therefore coincide immediately after the perturbation. The initial wave function in K' is

$$\psi_0' = \psi_0 \exp\left(-i\mathbf{q}\sum_a \mathbf{r}_a\right), \qquad \mathbf{q} = \frac{m\mathbf{v}}{\hbar},$$

where ψ_0 is the normal-state wave function for a stationary nucleus and the sum in the exponential extends over all Z electrons of the atom. According to (41.6), the probability sought for transition into the kth excited state is now

$$w_{k0} = \left|\left\langle k \left| \exp\left(-i\mathbf{q}\sum_a \mathbf{r}_a\right)\right| 0\right\rangle\right|^2.$$

In particular, for $qa \ll 1$, expand the exponential under the integral. The integral of $\psi_k^*\psi_0$ vanishes by orthogonality, and hence

$$w_{k0} = \left|\left\langle k \left| \mathbf{q}\sum_a \mathbf{r}_a \right| 0\right\rangle\right|^2.$$

Problem

3. Find the total probability of excitation and ionization of a hydrogen atom subjected to a sudden "shaking" (see the preceding problem).

Solution. The required probability can be evaluated as the difference

$$1 - w_{00} = 1 - \left|\int \psi_0^2 e^{-i\mathbf{q}\mathbf{r}}\, dV\right|^2,$$

where w_{00} is the probability that the atom remains in the ground state; $\psi_0 = (\pi a^3)^{-1/2} e^{-r/a}$ is the hydrogen ground-state wave function, and a is the Bohr radius. Evaluation of the integral gives

$$1 - w_{00} = 1 - \frac{1}{(1 + \frac{1}{4}q^2a^2)^4}.$$

In the limiting case $qa \ll 1$, this probability tends to zero as $1 - w_{00} \approx q^2a^2$; for $qa \gg 1$, it approaches unity as $1 - w_{00} \approx 1 - (2/qa)^8$.

Problem

4. Determine the probability that an electron is ejected from the K shell of an atom with large atomic number Z during nuclear β decay. Assume the β-particle velocity to be large relative to the velocity of a K electron (*A. B. Migdal*, 1941; *E. L. Feinberg*, 1939).

Solution. [6] Under the stated conditions, the time taken by the β particle to cross the K shell is short compared with the electron's orbital period; the change in nuclear charge may therefore be regarded as instantaneous. The perturbation is the change $V = 1/r$ in the nuclear field when its charge changes by an amount small compared with Z (namely, by 1). According to (41.5), the probability for either of the two K-shell electrons, whose energy is $E_0 = -Z^2/2$,[7] to enter a continuous-spectrum state of energy $E = k^2/2$ in the interval $dE = k\,dk$ is

$$dw = 2\frac{4|V_{0k}|^2}{(k^2+Z^2)^2}\,dk.$$

The integral defining V_{0k} receives its essential contribution from distances of order $1/Z$ near the nucleus. In this region a hydrogen-like expression may also be used for the continuous-spectrum wave function. The electron's final state must have $l = 0$, the same angular momentum as the initial state. From the function R_{10} and the function R_{k0} normalized on the $k/2\pi$ scale, obtained in § 36, together with formula (f.3) of the Mathematical Appendices, we find[8]

$$\left(\frac{1}{r}\right)_{0k} = \frac{4\sqrt{2\pi k}}{1-e^{-2\pi Z/k}}\,\frac{(1+ik/Z)^{iZ/k}(1-ik/Z)^{-iZ/k}}{1+k^2/Z^2},$$

and, since

$$|(1+i\alpha)^{i/\alpha}|^2 = \exp\left(-2\frac{\arctan\alpha}{\alpha}\right),$$

we finally obtain

$$dw = \frac{2^7}{Z^4(1+k^2/Z^2)^4}\,f\left(\frac{k}{Z}\right)k\,dk,$$

where

$$f(\alpha) = \frac{1}{1-e^{-2\pi/\alpha}}\exp\left(-4\frac{\arctan\alpha}{\alpha}\right).$$

The limiting values of $f(\alpha)$ are

$$f = e^{-4} \quad \text{for } \alpha \ll 1, \qquad f = \alpha/2\pi \quad \text{for } \alpha \gg 1.$$

The total probability of ionizing the K shell follows by integrating dw over all energies of the emitted electron. Numerical evaluation yields $w = 0.65Z^{-2}$.

[6] Atomic units are used in Problems 4 and 5.

[7] Here and below the K-electron states are treated as hydrogen-like (see § 74).

[8] It is convenient during the calculation to use Coulomb units, returning to atomic units in the final result.

Problem

5. Determine the probability that an electron is ejected from the K shell of a large-Z atom during nuclear α decay. The α particle moves slowly compared with a K electron, although its time of emergence from the nucleus is short compared with the electron's orbital period (*A. B. Migdal*, 1941; *J. Levinger*, 1953).

Solution. After the α particle has emerged, the perturbation acting upon the electron is adiabatic. The effect sought is therefore determined chiefly by times close to the nonadiabatic "switching instant," when the α particle, already outside the nucleus and moving freely, is still at distances small compared with the radius of the K orbit. The ionizing perturbation V is then the departure of the combined nuclear and α-particle field from the pure Coulomb field Z/r. For particles of atomic weights 4 and $A-4$, charges 2 and $Z-2$, and separation vt (where v is the relative speed of the nucleus and α particle), the dipole moment is

$$\frac{2(A-4)-(Z-2)4}{A}vt = \frac{2(A-2Z)}{A}vt.$$

Thus the dipole term in the field of the nucleus and the α particle is [9]

$$V = \frac{2(A-2Z)}{A}vt\frac{z}{r^3},$$

where the z axis points along $\mathbf{v}$. The matrix element of this perturbation can be reduced to that of z. Taking the matrix element of the electron's equation of motion $\ddot{z} = -Zz/r^3$, we obtain

$$\left(\frac{z}{r^3}\right)_{0k} = \frac{(E-E_0)^2}{Z}z_{0k}.$$

By (41.2), the transition probability sought for either of the two K-shell electrons is

$$dw = 2\left|\int_0^\infty V_{0k}e^{i(E_0-E)t}\,dt\right|^2 dk = \frac{8(A-2Z)^2v^2}{A^2Z^2}|z_{0k}|^2\frac{dk}{2\pi}.$$

To evaluate the integral, an additional damping factor $e^{-\lambda t}$, $\lambda > 0$, is inserted in the integrand, and $\lambda \to 0$ is then taken in the result. For the matrix element of $z = r\cos\theta$, note that the initial orbital angular momentum is $l = 0$, so $\cos\theta$ has a nonzero matrix element only for transition to a state with $l = 1$. In that case

$$|(\cos\theta)_{01}|^2 = \frac{1}{3}, \qquad |z_{0k}|^2 = \frac{1}{3}|r_{0k}|^2.$$

Evaluation of r_{0k} using the radial functions R_{00} and R_{k1} then gives

$$dw = \frac{2^{11}(A-2Z)^2v^2}{3A^2Z^6(1+k^2/Z^2)^5}f\left(\frac{k}{Z}\right)k\,dk,$$

where f is the function defined in Problem 4.

[9]When the difference $A-2Z$ is small, it may also be necessary to include the next, quadrupole term.

§ 42. Transitions under a periodic perturbation

A different type of result is obtained for the probability of transition to continuous-spectrum states under a periodic perturbation. Suppose that at an initial time $t = 0$ the system is in the ith stationary state of the discrete spectrum. Let the frequency ω of the periodic perturbation satisfy

$$\hbar\omega > E_{\min} - E_i^{(0)}, \tag{42.1}$$

where $E_{\min}$ is the energy at which the continuous spectrum begins.

It follows at once from the results of § 40 that the principal role is played by continuous-spectrum states whose energies E_f lie close to the "resonant" energy $E_i^{(0)} + \hbar\omega$, that is, states for which $\omega_{fi} - \omega$ is small. For the same reason, only the first term in the perturbation matrix element (40.8), whose frequency $\omega_{fi} - \omega$ is close to zero, need be retained. Substituting this term in (40.5) and integrating gives

$$a_{fi} = -\frac{i}{\hbar}\int_0^t V_{fi}(t)\,dt = -F_{fi}\frac{\exp[i(\omega_{fi}-\omega)t] - 1}{\hbar(\omega_{fi}-\omega)}. \tag{42.2}$$

The lower integration limit has been chosen so that $a_{fi} = 0$ at $t = 0$, in accordance with the initial condition.

The squared modulus of a_{fi} is

$$|a_{fi}|^2 = |F_{fi}|^2 \frac{4\sin^2 \dfrac{\omega_{fi}-\omega}{2}t}{\hbar^2(\omega_{fi}-\omega)^2}. \tag{42.3}$$

It is readily seen that, at large t, the function appearing here may be represented as proportional to t.

To see this, use the formula

$$\lim_{t\to\infty} \frac{\sin^2 \alpha t}{\pi t \alpha^2} = \delta(\alpha). \tag{42.4}$$

For $\alpha \neq 0$ the displayed limit is zero, while for $\alpha = 0$,

$$\frac{\sin^2 \alpha t}{t\alpha^2} = t,$$

so the limit is infinite. On integrating over $d\alpha$ from $-\infty$ to $+\infty$ and making the substitution $\alpha t = \xi$, we obtain

$$\frac{1}{\pi}\int_{-\infty}^{+\infty} \frac{\sin^2 \alpha t}{t\alpha^2}\,d\alpha = \frac{1}{\pi}\int_{-\infty}^{+\infty}\frac{\sin^2 \xi}{\xi^2}\,d\xi = 1.$$

Thus the function on the left-hand side of (42.4) satisfies all the requirements defining the delta function.

Accordingly, for large t we may write

$$|a_{fi}|^2 = \frac{1}{\hbar^2}|F_{fi}|^2 \pi t\,\delta\left(\frac{\omega_{fi}-\omega}{2}\right),$$

or, substituting $\hbar\omega_{fi} = E_f - E_i^{(0)}$ and using $\delta(ax) = \delta(x)/a$,

$$|a_{fi}|^2 = \frac{2\pi}{\hbar}|F_{fi}|^2\delta(E_f - E_i^{(0)} - \hbar\omega)t.$$

The quantity $|a_{fi}|^2 d\nu_f$ is the probability of transition from the initial state to states in a specified interval $d\nu_f$. At large t it is proportional to the interval of time elapsed since $t = 0$. The transition probability dw_{fi} per unit time is therefore[10]

$$dw_{fi} = \frac{2\pi}{\hbar}|F_{fi}|^2\delta(E_f - E_i^{(0)} - \hbar\omega)\, d\nu_f. \tag{42.5}$$

As expected, it is nonzero only for transitions to states with energy $E_f = E_i^{(0)} + \hbar\omega$. If the continuous-spectrum energy levels are nondegenerate, so that ν_f may be taken to mean the energy alone, the whole "interval" $d\nu_f$ reduces to a single state of energy $E = E_i^{(0)} + \hbar\omega$, and the probability of transition to this state is

$$w_{Ei} = \frac{2\pi}{\hbar}|F_{Ei}|^2. \tag{42.6}$$

It is also instructive to derive (42.5) in another way. Instead of switching the periodic perturbation on at the definite instant $t = 0$, let it grow slowly from $t = -\infty$ according to the exponential law $e^{\lambda t}$, where λ is positive and is ultimately allowed to tend to zero (*adiabatic switching*). The initial condition $a_{fi} = 0$ is then imposed at $t = -\infty$. The perturbation matrix element is now

$$V_{fi} = F_{fi}e^{i(\omega_{fi}-\omega)t+\lambda t},$$

and in place of (42.2) we write

$$a_{fi} = -\frac{i}{\hbar}\int_{-\infty}^{t} V_{fi}(t)\, dt = -F_{fi}\frac{\exp[i(\omega_{fi}-\omega)t + \lambda t]}{\hbar(\omega_{fi} - \omega - i\lambda)}. \tag{42.7}$$

Hence

$$|a_{fi}|^2 = \frac{1}{\hbar^2}|F_{fi}|^2\frac{e^{2\lambda t}}{(\omega_{fi}-\omega)^2 + \lambda^2}.$$

The transition probability per unit time is determined by

$$\frac{d}{dt}|a_{fi}|^2 = 2\lambda|a_{fi}|^2.$$

Now use the formula

$$\lim_{\lambda\to 0}\frac{\lambda}{\pi(\alpha^2 + \lambda^2)} = \delta(\alpha), \tag{42.8}$$

which is valid in the same sense as (42.4). Taking the limit $\lambda \to 0$ gives

$$\frac{d}{dt}|a_{fi}|^2 \longrightarrow \frac{2\pi}{\hbar^2}|F_{fi}|^2\delta(\omega_{fi} - \omega),$$

and we return once more to (42.5).

[10] It is readily verified that retaining the omitted second term in (40.8) would give additional expressions which, after division by t, tend to zero as $t \to +\infty$.

§ 43. Transitions within the continuous spectrum

One of the most important uses of perturbation theory is the calculation of transition probabilities within the continuous spectrum under a constant (time-independent) perturbation. As already noted, continuous-spectrum states are almost always degenerate. After choosing a definite set of unperturbed wave functions for a given energy level, the question can be posed as follows. The system is known to have occupied one of these states initially; we seek the probability that it passes into another state of the same energy. For transitions from an initial state i into states between ν_f and $\nu_f + d\nu_f$, (42.5), with $\omega = 0$ and a change of notation, gives directly

$$dw_{fi} = \frac{2\pi}{\hbar}|V_{fi}|^2\delta(E_f - E_i)\,d\nu_f. \tag{43.1}$$

As expected, this is nonzero only when $E_f = E_i$: a constant perturbation causes transitions solely between states of equal energy. For transitions from continuous-spectrum states, however, dw_{fi} cannot itself be treated as a transition probability. It does not even have the corresponding dimension, but has dimension 1/s. Expression (43.1) gives the number of transitions per unit time, and its dimension depends on the chosen normalization of the continuous-spectrum wave functions.[11]

We shall calculate the perturbed wave function which, before the perturbation begins to act, coincides with the original unperturbed function $\psi_i^{(0)}$. Following the procedure stated at the end of the preceding section, take the perturbation to be switched on adiabatically as $e^{\lambda t}$, with $\lambda \to 0$. Formula (42.7), after setting $\omega = 0$ and changing notation, gives

$$a_{fi}^{(1)} = V_{fi}\frac{\exp\left\{\dfrac{i}{\hbar}(E_f - E_i)t + \lambda t\right\}}{E_i - E_f + i\lambda}. \tag{43.2}$$

The perturbed wave function is

$$\Psi_i = \Psi_i^{(0)} + \int a_{fi}^{(1)}\Psi_f^{(0)}\,d\nu_f,$$

where the integration extends over the entire continuous spectrum.[12] Substitution of (43.2) yields

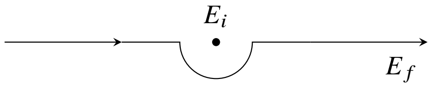
$$\Psi_i = \left[\psi_i^{(0)} + \int V_{fi}\psi_f^{(0)}\frac{d\nu_f}{E_i - E_f + i0}\right]\exp\left(-\frac{i}{\hbar}E_i t\right). \tag{43.3}$$

In the limit $\lambda \to 0$, the factor $e^{\lambda t}$ has been replaced by unity. The term $+i0$, meaning the limit of $i\lambda$ as the positive quantity λ tends to zero, specifies the integration prescription for E_f. Its differential occurs as a factor in $d\nu_f$, together with those of the other quantities characterizing continuous-spectrum states. Without $i\lambda$, the integrand in (43.3) would

[11] Phenomena covered by this theory include, for example, various collisions. In their initial and final states the system is an assemblage of free particles, while their mutual interaction serves as the perturbation. With a suitable wave- function normalization, (43.1) can then become a collision cross-section (see § 126).

[12] If a discrete spectrum is also present, the corresponding sum over its states must be added to the integral here and in the formulas below.

have a pole at $E_f = E_i$, and the integral would diverge near it. The term $i\lambda$ displaces this pole into the upper half-plane of the complex variable E_f. When $\lambda \to 0$, the pole returns to the real axis, but the integration path is now known to pass below it:



(43.4)

The time factor in (43.3) shows, as it should, that this function belongs to the same energy E_i as the original unperturbed function. Equivalently,

$$\psi_i = \psi_i^{(0)} + \int \frac{V_{fi}}{E_i - E_f + i0}\psi_f^{(0)}\, d\nu_f \tag{43.5}$$

satisfies the Schrödinger equation

$$(\hat{H}_0 + \hat{V})\psi_i = E_i\psi_i.$$

It is consequently natural that this expression agrees exactly with (38.8).[13]

The preceding calculation is the first approximation of perturbation theory. The second approximation is also readily obtained. One derives the next approximation for Ψ_i by the method of § 38, now knowing how the "divergent" integrals are to be taken. A direct calculation gives

$$\Psi_i = \left\{\psi_i^{(0)} + \int \left[V_{fi} + \int \frac{V_{f\nu}V_{\nu i}}{E_i - E_\nu + i0}\, d\nu\right] \frac{\psi_f^{(0)}\, d\nu_f}{E_i - E_f + i0}\right\} \exp\left(-\frac{i}{\hbar}E_i t\right).$$

Comparison with (43.3) lets us write the corresponding probability formula (more precisely, the number of transitions) by direct analogy with (43.1):

$$dw_{fi} = \frac{2\pi}{\hbar}\left|V_{fi} + \int \frac{V_{f\nu}V_{\nu i}}{E_i - E_\nu + i0}\, d\nu\right|^2 \delta(E_i - E_f)\, d\nu_f. \tag{43.6}$$

The matrix element V_{fi} may vanish for the transition in question. The first-order effect is then absent altogether, and (43.6) becomes

$$dw_{fi} = \frac{2\pi}{\hbar}\left|\int \frac{V_{f\nu}V_{\nu i}}{E_i - E_\nu}\, d\nu\right|^2 \delta(E_f - E_i)\, d\nu_f. \tag{43.7}$$

In applications of this formula, $E_\nu = E_i$ is usually not a pole of the integrand. The prescription for integration over dE_ν is then immaterial, and the path may be taken directly along the real axis.

States ν for which $V_{f\nu}$ and $V_{\nu i}$ are nonzero are often called *intermediate* states for the transition $i \to f$. In a pictorial description, the transition appears to take place in two stages, $i \to \nu$ and $\nu \to f$; this description must not, of course, be understood literally. A

[13]The integration prescription in (43.5) can also be fixed by requiring the large-distance asymptotic form of ψ_i to contain only an outgoing wave, and no incoming wave (see § 136).

transition $i \to f$ may require not one but several successive intermediate states. Formula (43.7) extends directly to such cases. If two intermediate states are needed, then

$$dw_{fi} = \frac{2\pi}{\hbar}\left|\int \frac{V_{f\nu'}V_{\nu'\nu}V_{\nu i}}{(E_i - E_{\nu'})(E_i - E_\nu)}\,d\nu\,d\nu'\right|^2 \delta(E_f - E_i)\,d\nu_f. \tag{43.8}$$

Finally, the mathematical meaning of integrals taken along a path such as (43.4) is expressed by

$$\int \frac{f(x)\,dx}{x - a - i0} = \mathrm{P}\int \frac{f(x)\,dx}{x - a} + i\pi f(a), \tag{43.9}$$

where the interval of integration on the real axis contains $x = a$. Indeed, detouring around the pole $x = a$ on a semicircle of radius ρ, the full integral is the sum of the real-axis integrals from the lower limit to $a - \rho$ and from $a + \rho$ to the upper limit, plus $i\pi$ times the residue at the pole. As $\rho \to 0$, the real-axis pieces combine into the integral over the full interval understood as a principal value, as indicated by the marked integral sign, and (43.9) follows. The same formula is also written symbolically as

$$\frac{1}{x - a - i0} = P\frac{1}{x - a} + i\pi\delta(x - a); \tag{43.10}$$

the symbol P means that the principal value is to be taken when $f(x)/(x - a)$ is integrated.

§ 44. The uncertainty relation for energy

Consider a system made up of two weakly interacting parts. Suppose that at a certain instant these parts are known to have definite energies, denoted by E and ε. After an interval Δt, the energies are measured again, giving values E' and ε' which generally differ from E and ε. We can readily determine the order of magnitude of the most probable value of $E' + \varepsilon' - E - \varepsilon$ found by the measurement.

According to (42.3), with $\omega = 0$, the probability that a time-independent perturbation transfers the system during time t from energy E to energy E' is proportional to

$$\frac{\sin^2\left(\dfrac{E' - E}{2\hbar}t\right)}{(E' - E)^2}.$$

It follows that the most probable difference $E' - E$ is of order $\hbar/t$.

Applying this result to the present case, with the interaction between the two parts as the perturbation, gives

$$|E + \varepsilon - E' - \varepsilon'|\Delta t \sim \hbar. \tag{44.1}$$

Thus the shorter the interval Δt, the larger the observed energy change. Its order of magnitude, $\hbar/\Delta t$, is importantly independent of the strength of the perturbation.

The energy change specified by (44.1) will be observed however weak the interaction between the two parts may be. This is a purely quantum result with a profound physical meaning. It shows that in quantum mechanics two measurements can test energy conservation only to an accuracy of order $\hbar/\Delta t$, where Δt is the time between them.

Relation (44.1) is often called the uncertainty relation for energy. It must, however, be emphasized that its meaning is essentially different from that of the coordinate–momentum relation $\Delta p \Delta x \sim \hbar$. In the latter, Δp and Δx are the uncertainties in momentum and coordinate at one and the same instant; the relation says that these two quantities cannot have strictly definite values simultaneously.

The energies E and ε, by contrast, can be measured at each given instant with arbitrary precision. The quantity $(E+\varepsilon)-(E'+\varepsilon')$ in (44.1) is the difference between two exactly measured energy values at two different times, not an uncertainty in the energy at a particular instant.

If E is regarded as the energy of a system and ε as the energy of a "measuring apparatus," their interaction energy can be taken into account only to an accuracy $\hbar/\Delta t$. Denote the errors in the measurements of the corresponding quantities by $\Delta E, \Delta\varepsilon, \ldots$. In the favorable case where ε and ε' are known exactly ($\Delta\varepsilon = \Delta\varepsilon' = 0$),

$$\Delta(E-E') \sim \frac{\hbar}{\Delta t}. \tag{44.2}$$

This relation has important consequences for momentum measurement. Measuring the momentum of a particle (take an electron for definiteness) involves a collision with another, "measuring," particle whose momenta before and after the collision may be regarded as known exactly.[14] Applying momentum conservation to this collision gives three equations, the components of one vector equation, for six unknowns: the components of the electron momentum before and after the collision. More equations might be sought by arranging a sequence of collisions with measuring particles and applying momentum conservation to each. But the number of unknowns also grows, since it includes the electron momenta between collisions; for any number of collisions there are plainly three more unknowns than equations. Momentum measurement must therefore use energy conservation at each collision in addition to momentum conservation. As we have seen, the energy law can be applied only to an accuracy of order $\hbar/\Delta t$, with Δt the interval between the beginning and end of the process under consideration.

For simplicity, consider an idealized thought experiment in which the "measuring particle" is a perfectly reflecting plane mirror. Only the momentum component normal to the mirror then matters. Momentum and energy conservation give, for determining the particle momentum P,

$$p' + P' - p - P = 0, \tag{44.3}$$

$$|\varepsilon' + E' - \varepsilon - E| \sim \frac{\hbar}{\Delta t} \tag{44.4}$$

(P, E are the particle's momentum and energy; p, ε are those of the mirror; unprimed and primed quantities refer respectively to times before and after the collision). The quantities $p, p', \varepsilon, \varepsilon'$ of the "measuring particle" may be treated as known exactly, so their errors vanish. The equations above then give

$$\Delta P = \Delta P', \qquad |\Delta E' - \Delta E| \sim \frac{\hbar}{\Delta t}.$$

[14] For the present argument, how the energy of the "measuring" particle becomes known is immaterial.

But

$$\Delta E = \frac{\partial E}{\partial P}\Delta P = v\Delta P,$$

and similarly

$$\Delta E' = v'\Delta P' = v'\Delta P.$$

Hence

$$|(v'_x - v_x)\Delta P_x| \sim \frac{\hbar}{\Delta t}. \tag{44.5}$$

The suffix x has been attached to the velocities and momentum to stress that the relation applies separately to each component.

This is the required relation. It shows that measuring an electron's momentum to a specified accuracy ΔP is unavoidably accompanied by a change in its velocity, and therefore in the momentum itself.

The shorter the measurement process, the greater this change. The velocity change can be made arbitrarily small only as $\Delta t \to \infty$, but a momentum measurement extending over a long time can in general have meaning only for a free particle. Here the nonrepeatability of momentum measurements at short intervals, and the two-sided nature of measurement in quantum mechanics, are especially clear: one must distinguish the measured value of a quantity from the value produced by the measuring process.[15]

The argument at the beginning of this section, based on perturbation theory, can also be approached through the decay of a system under some perturbation. Let E_0 be an energy level calculated while entirely neglecting the possibility of decay. Denote by τ the *lifetime* of this state, the reciprocal of its decay probability per unit time. The same method gives

$$|E_0 - E - \varepsilon| \sim \hbar/\tau, \tag{44.6}$$

where E and ε are the energies of the two parts into which the system has decayed. Their sum $E + \varepsilon$ provides a measure of the system's energy before decay. The relation therefore shows that the energy of a decaying system in a *quasistationary* state can be defined only to an accuracy of order $\hbar/\tau$. This quantity is customarily called the level *width* Γ. Thus

$$\Gamma \sim \hbar/\tau. \tag{44.7}$$

§ 45. Potential energy treated as a perturbation

The case in which the particle's entire potential energy in an external field can be treated as the perturbation calls for separate consideration. The unperturbed Schrödinger equation is then the equation for free motion,

$$\Delta\psi^{(0)} + k^2\psi^{(0)} = 0, \qquad k = \frac{\sqrt{2mE}}{\hbar} = \frac{p}{\hbar} \tag{45.1}$$

[15] Relation (44.5), as well as the clarification of the physical meaning of the energy uncertainty relation, is due to N. Bohr (1928).

whose solutions are plane waves. Since the energy spectrum of free motion is continuous, this is a special instance of perturbation theory in a continuous spectrum. Here it is more convenient to obtain the solution directly, without using the general formulas.

The equation for the first-order correction $\psi^{(1)}$ to the wave function is

$$\Delta\psi^{(1)} + k^2\psi^{(1)} = \frac{2mU}{\hbar^2}\psi^{(0)} \tag{45.2}$$

(U is the potential energy). As is known from electrodynamics, its solution may be written in the form of "retarded potentials," namely[16]

$$\psi^{(1)}(x, y, z) = -\frac{m}{2\pi\hbar^2}\int \psi^{(0)}U(x', y', z')e^{ikr}\frac{dV'}{r}, \tag{45.3}$$

$$dV' = dx'\,dy'\,dz', \qquad r^2 = (x-x')^2 + (y-y')^2 + (z-z')^2.$$

Let us determine the conditions under which the field U may be regarded as a perturbation. Perturbation theory is applicable when $\psi^{(1)} \ll \psi^{(0)}$. Let a give the order of magnitude of the size of the region in which the field differs appreciably from zero. Suppose first that the particle's energy is so low that ak is at most of order unity. For an order-of-magnitude estimate the factor e^{ikr} in the integrand of (45.3) is then immaterial, and the whole integral is of order $\psi^{(0)}|U|a^2$. Thus $\psi^{(1)} \sim (ma^2|U|/\hbar^2)\psi^{(0)}$, and the condition becomes

$$|U| \ll \frac{\hbar^2}{ma^2} \quad \text{when} \quad ka \lesssim 1. \tag{45.4}$$

Notice that $\hbar^2/ma^2$ has a simple physical meaning: it gives the order of magnitude of the kinetic energy that a particle confined to a volume of linear dimensions a would have (for its momentum would be $\sim \hbar/a$ by the uncertainty relation).

As a particular example, consider a potential well so shallow that (45.4) is satisfied. Such a well has no negative energy levels (*R. Peierls*, 1929), as was already seen in the problem to § 33 for the special case of a spherically symmetric well. Indeed, at $E = 0$ the unperturbed wave function reduces to a constant, which may conventionally be set equal to unity: $\psi^{(0)} = 1$. Since $\psi^{(1)} \ll \psi^{(0)}$, the wave function for motion in the well, $\psi = 1 + \psi^{(1)}$, plainly vanishes nowhere. An eigenfunction without nodes belongs to the normal state, so $E = 0$ remains the lowest possible energy of the particle.

Thus, if a well is not sufficiently deep, only infinite motion of the particle is possible: the well cannot "capture" it. This result is specifically quantum-mechanical. In classical mechanics a particle can undergo finite motion in any potential well.

It must be stressed that everything just said applies only to a three-dimensional well. A one- or two-dimensional well (that is, one whose field depends on only one or two coordinates) always has negative energy levels (see the problems to this section). The reason is that in one and two dimensions the perturbation theory under consideration is wholly inapplicable at zero (or very small) energy E. [17]

[16]This is a particular integral of equation (45.2). Any solution of the equation with no right-hand side (that is, of the unperturbed equation (45.1)) may still be added to it.

[17]In two dimensions, $\psi^{(1)}$ is expressed (as follows from the theory of the two-dimensional wave equation) by

At high energies, where $ka \gg 1$, the factor e^{ikr} in the integrand is essential and greatly reduces the magnitude of the integral. In this case solution (45.3) can be put into another form. For its derivation, however, it is simpler to return directly to equation (45.2). Choose the direction of unperturbed motion as the x axis. The unperturbed wave function is then $\psi^{(0)} = e^{ikx}$ (the constant factor is conventionally set to unity). Seek a solution of

$$\Delta\psi^{(1)} + k^2\psi^{(1)} = \frac{2m}{\hbar^2}Ue^{ikx}$$

in the form $\psi^{(1)} = e^{ikx}f$. Since k is assumed large, in $\Delta\psi^{(1)}$ it suffices to retain only terms in which the factor e^{ikx} is differentiated at least once. The equation for f is then

$$2ik\frac{\partial f}{\partial x} = \frac{2mU}{\hbar^2},$$

and hence

$$\psi^{(1)} = e^{ikx}f = -\frac{im}{\hbar^2 k}e^{ikx}\int U\,dx. \tag{45.5}$$

The integral is of order $|\psi^{(1)}| \sim m|U|a/\hbar^2 k$, so in this case perturbation theory is applicable under the condition

$$|U| \ll \frac{\hbar^2}{ma^2}ka = \frac{\hbar v}{a}, \qquad ka \gg 1 \tag{45.6}$$

($v = k\hbar/m$ is the particle's velocity). This condition is weaker than (45.4). Consequently, if the field may be treated as a perturbation at low particle energies, it certainly may be so treated at high energies; the converse, generally speaking, does not hold. [18]

The applicability of the perturbation theory developed here to the Coulomb field requires separate consideration. In the field $U = \alpha/r$ there is no finite region of space outside which U is substantially smaller than inside it. The required condition follows by replacing the parameter a in (45.6) with the variable distance r; this gives

$$\frac{\alpha}{\hbar v} \ll 1. \tag{45.7}$$

Thus the Coulomb field may be treated as a perturbation at high particle energies. [19]

Finally, let us derive a formula giving an approximate wave function for a particle whose energy E is everywhere much greater than the potential energy U (no other conditions are required). To the first approximation, the coordinate dependence of the

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an integral analogous to (45.3), with $i\pi H_0^{(1)}(kr)dx'\,dy'$ in place of $e^{ikr}dx'\,dy'\,dz'/r$ ($H_0^{(1)}$ is a Hankel function), and $r = \sqrt{(x'-x)^2 + (y'-y)^2}$. As $k \to 0$, the Hankel function, and with it the entire integral, tends logarithmically to infinity.

Similarly, in one dimension the integrand defining $\psi^{(1)}$ contains $2\pi i e^{ikr}dx'/k$ (where $r = |x'-x|$), and as $k \to 0$, $\psi^{(1)}$ tends to infinity as $1/k$.

[18] In one dimension the condition for applicability of perturbation theory is the inequality (45.6) for all ka. The derivation of (45.4) given above for three dimensions cannot be carried out in one dimension because of the divergence, noted in the footnote on p. 204, of the function $\psi^{(1)}$ constructed in that manner.

[19] It should be borne in mind that the integral (45.5) with $U = \alpha/r$ diverges logarithmically at large $x/\sqrt{y^2+z^2}$. The Coulomb-field wave function obtained by perturbation theory is therefore inapplicable inside a narrow cone about the x axis.

wave function is the same as in free motion, whose direction we choose as the x axis. Accordingly, write $\psi = e^{ikx}F$, where F is a function of the coordinates that varies slowly in comparison with the factor e^{ikx} (although in general it cannot be said to be close to unity). Substitution in the Schrödinger equation gives

$$2ik\frac{\partial F}{\partial x} = \frac{2m}{\hbar^2}UF, \tag{45.8}$$

whence

$$\psi = e^{ikx}F = \text{const}\cdot e^{ikx}\exp\left(-\frac{i}{\hbar v}\int U\,dx\right). \tag{45.9}$$

This is the required expression. It must be remembered, however, that it is not valid at excessively large distances. In (45.8) the term ΔF, which contains second derivatives of F, was omitted. At large distances $\partial^2 F/\partial x^2$ tends to zero together with $\partial F/\partial x$. The derivatives with respect to the transverse coordinates y, z, on the other hand, do not tend to zero, and they may be neglected only if $x \ll ka^2$.

Problems

Problem

1. Determine the energy level in a shallow one-dimensional potential well; condition (45.4) is assumed to hold.

Solution. Make the assumption, confirmed by the result, that $|E| \ll |U|$. Then E may be neglected on the right-hand side of the Schrödinger equation within the well,

$$\frac{d^2\psi}{dx^2} = \frac{2m}{\hbar^2}(U(x) - E)\psi,$$

and ψ may also be treated as a constant, which can be set equal to unity without loss of generality:

$$\frac{d^2\psi}{dx^2} = \frac{2m}{\hbar^2}U.$$

Integrate this equality with respect to x between two points $\pm x_1$ such that $a \ll x_1 \ll 1/\kappa$, where a is the width of the well and $\kappa = \sqrt{2m|E|}/\hbar$. Since the integral of $U(x)$ converges, the integration on the right may be extended over the whole interval from $-\infty$ to $+\infty$:

$$\left.\frac{d\psi}{dx}\right|_{-x_1}^{x_1} = \frac{2m}{\hbar^2}\int_{-\infty}^{+\infty} U\,dx. \tag{1}$$

Far from the well the wave function has the form $\psi = e^{\mp\kappa x}$. Substitution in (1) gives

$$-2\kappa = \frac{2m}{\hbar^2}\int_{-\infty}^{+\infty} U\,dx \qquad \text{or} \qquad |E| = \frac{m}{2\hbar^2}\left(\int_{-\infty}^{+\infty} U\,dx\right)^2.$$

In accord with the assumption made, the magnitude of the level is thus a small quantity of higher (second) order than the depth of the well.

PROBLEM

2. Determine the energy level in a shallow two-dimensional potential well $U(r)$ (r is the polar coordinate in the plane); the integral $\int_0^\infty rU\,dr$ is assumed to converge.

SOLUTION. Proceeding as in the preceding problem, within the well we obtain

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{d\psi}{dr}\right) = \frac{2m}{\hbar^2}U.$$

Integrating with respect to r from 0 to r_1 (where $a \ll r_1 \ll 1/\kappa$), we find

$$\left.\frac{d\psi}{dr}\right|_{r=r_1} = \frac{2m}{\hbar^2 r_1}\int_0^\infty rU(r)\,dr. \qquad (1)$$

Far from the well, the two-dimensional free-motion equation

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{d\psi}{dr}\right) + \frac{2m}{\hbar^2}E\psi = 0$$

has the solution (vanishing at infinity) $\psi = \text{const}\cdot H_0^{(1)}(i\kappa r)$. At small values of the argument, the leading term of this function is proportional to $\ln \kappa r$. Bearing this in mind, at $r \sim a$ equate the logarithmic derivatives of ψ evaluated inside the well (the right-hand side of (1)) and outside it. This gives

$$-\frac{1}{a\ln\kappa a} \approx \frac{2m}{\hbar^2 a}\int_0^\infty U(r)r\,dr,$$

and hence

$$|E| \sim \frac{\hbar^2}{ma^2}\exp\left[-\frac{\hbar^2}{m}\left|\int_0^\infty Ur\,dr\right|^{-1}\right].$$

Thus the energy level is exponentially small in comparison with the depth of the well.

CHAPTER VII

The Quasiclassical Case

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The semiclassical case

§ 46. The wave function in the semiclassical case

When the de Broglie wavelengths of the particles are small in comparison with the characteristic dimensions L that determine the conditions of a given problem, the system has nearly classical properties. (This is analogous to the passage from wave optics to geometrical optics as the wavelength tends to zero.)

We shall now examine the properties of semiclassical systems in greater detail. In the Schrödinger equation

$$\sum_a \frac{\hbar^2}{2m_a}\Delta_a\psi + (E-U)\psi = 0$$

make the formal substitution

$$\psi = \exp\left(\frac{i}{\hbar}\sigma\right). \tag{46.1}$$

The resulting equation for σ is

$$\sum_a \frac{1}{2m_a}(\nabla_a\sigma)^2 - \sum_a \frac{i\hbar}{2m_a}\Delta_a\sigma = E-U. \tag{46.2}$$

In keeping with the assumption that the properties of the system are nearly classical, seek σ as the series

$$\sigma = \sigma_0 + \frac{\hbar}{i}\sigma_1 + \left(\frac{\hbar}{i}\right)^2\sigma_2 + \cdots, \tag{46.3}$$

arranged in powers of $\hbar$.

Begin with the simplest case, one-dimensional motion of a single particle. Equation (46.2) then reduces to

$$\frac{1}{2m}\sigma'^2 - \frac{i\hbar}{2m}\sigma'' = E - U(x) \tag{46.4}$$

(a prime denotes differentiation with respect to the coordinate x).

In the first approximation put $\sigma = \sigma_0$ and omit the term containing $\hbar$ from the equation:

$$\frac{1}{2m}\sigma_0'^2 = E - U(x).$$

It follows that

$$\sigma_0 = \pm\int\sqrt{2m[E-U(x)]}\,dx.$$

The integrand is precisely the classical momentum $p(x)$ of the particle, expressed as a function of the coordinate. Defining $p(x)$ with the positive sign before the square root, we have

$$\sigma_0 = \pm\int p\,dx, \qquad p = \sqrt{2m(E-U)}, \tag{46.5}$$

as expected from the limiting expression (6.1) for the wave function.[1]

The neglect made in (46.4) is justified only when the second term on the left-hand side is small compared with the first. Thus $\hbar|\sigma''/\sigma'^2| \ll 1$, or

$$\left|\frac{d}{dx}\left(\frac{\hbar}{\sigma'}\right)\right| \ll 1.$$

To first order, (46.5) gives $\sigma' = p$, so this condition can be written

$$\left|\frac{d\lambda\!\!\!^{-}}{dx}\right| \ll 1, \tag{46.6}$$

where $\lambda\!\!\!^{-} = \lambda/2\pi$, while $\lambda(x) = 2\pi\hbar/p(x)$ is the particle's de Broglie wavelength expressed as a function of x through the classical function $p(x)$. We have thereby obtained a quantitative semiclassical condition: over a distance comparable with the wavelength itself, that wavelength must vary only slightly. The approximation fails in regions of space where this condition is not met.

Condition (46.6) may also be put in another form. Since

$$\frac{dp}{dx} = \frac{d}{dx}\sqrt{2m(E-U)} = -\frac{m}{p}\frac{dU}{dx} = \frac{mF}{p},$$

where $F = -dU/dx$ is the classical force acting on the particle in the external field, introduction of this force gives

$$\frac{m\hbar|F|}{p^3} \ll 1. \tag{46.7}$$

This shows that the semiclassical approximation fails when the particle's momentum is too small. In particular, it certainly fails near turning points, namely points at which classical mechanics would have the particle stop and then begin moving in the opposite direction. Such points are determined by $p(x) = 0$, or $E = U(x)$. As $p \to 0$, the de Broglie wavelength tends to infinity and plainly cannot be regarded as small.

It should be stressed, however, that condition (46.6) or (46.7) may by itself be insufficient to justify the semiclassical approximation. It was obtained by estimating different terms in the differential equation (46.4), and the discarded term contains the highest derivative. What must actually be small are successive terms in the expansion of the solution of this equation; the smallness of the discarded term in the equation need not ensure this. For example, suppose the solution for $\sigma(x)$ contains a term whose growth with x is nearly linear. The smallness of its second derivative does not prevent that term from accumulating a large value over a sufficiently great distance. Such a situation generally arises when the field extends over distances large compared with the characteristic length L on which it changes appreciably (see the remark below concerning (46.11)). The semiclassical approximation then cannot follow the wave function over large distances.

[1] As is well known, $\int p\,dx$ is the time-independent part of the action. The complete mechanical action of the particle is $S = -Et \pm \int p\,dx$. The term Et is absent from σ because we are considering the time-independent wave function ψ.

We proceed to calculate the next term of (46.3). Terms of first order in $\hbar$ in (46.4) give

$$\sigma_0'\sigma_1' + \frac{1}{2}\sigma_0'' = 0, \qquad \sigma_1' = -\frac{\sigma_0''}{2\sigma_0'} = -\frac{p'}{2p}.$$

Integration yields

$$\sigma_1 = -\frac{1}{2}\ln p \tag{46.8}$$

(the integration constant is omitted).

Substitution of this expression in (46.1) and (46.3) gives the wave function

$$\psi = \frac{C}{\sqrt{p}}\exp\left(\frac{i}{\hbar}\int p\,dx\right) + \frac{C'}{\sqrt{p}}\exp\left(-\frac{i}{\hbar}\int p\,dx\right). \tag{46.9}$$

The factor $1/\sqrt{p}$ has a simple interpretation. The probability of finding the particle between x and $x + dx$ is determined by $|\psi|^2$, and hence is principally proportional to $1/p$. This is just what one expects for a "semiclassical particle," since in classical motion the time spent by a particle in an interval dx is inversely proportional to its velocity (or momentum).

In classically inaccessible regions, where $E < U(x)$, the function $p(x)$ is purely imaginary and the exponential arguments are real. The general form of the wave-equation solution in such regions is

$$\psi = \frac{C}{\sqrt{|p|}}\exp\left(-\frac{1}{\hbar}\int |p|\,dx\right) + \frac{C'}{\sqrt{|p|}}\exp\left(\frac{1}{\hbar}\int |p|\,dx\right). \tag{46.10}$$

It must nevertheless be remembered that the accuracy of the semiclassical approximation does not permit exponentially small terms to be retained in a wave function against a background of exponentially large terms. In this sense, retaining both terms in (46.10) simultaneously is generally inadmissible.

Although higher-order terms in the wave function are usually unnecessary, we shall also obtain the next term in (46.3), in order to point out certain features concerning the accuracy of the semiclassical approximation.

The terms of order $\hbar^2$ in (46.4) give

$$\sigma_0'\sigma_2' + \frac{1}{2}\sigma_1'^2 + \frac{1}{2}\sigma_1'' = 0,$$

whence, on substituting (46.5) and (46.8) for σ_0 and σ_1,

$$\sigma_2' = \frac{p''}{4p^2} - \frac{3p'^2}{8p^3}.$$

On integrating (the first term being integrated by parts) and introducing the force $F = pp'/m$, we find

$$\sigma_2 = \frac{mF}{4p^3} + \frac{m^2}{8}\int \frac{F^2}{p^5}\,dx.$$

In the approximation under consideration the wave function is

$$\psi = \exp[(i/\hbar)\sigma] = \exp[(i/\hbar)\sigma_0 + \sigma_1](1 - i\hbar\sigma_2),$$

or

$$\psi = \frac{\text{const}}{\sqrt{p}}\left[1 - \frac{im\hbar F}{4p^3} - \frac{im^2\hbar}{8}\int \frac{F^2}{p^5}\,dx\right]\exp\left(\frac{i}{\hbar}\int p\,dx\right). \tag{46.11}$$

The imaginary correction terms in the prefactor are equivalent to a real correction in the phase of the wave function (that is, an addition to the integral $\hbar^{-1}\int p\,dx$ in its exponent). This correction is proportional to $\hbar$, and is therefore of order λ/L.

The second and third terms in the square brackets in (46.11) must be small compared with unity. For the former this requirement is identical to (46.7), whereas for the latter an estimate of the integral leads to (46.7) only if F^2 tends to zero sufficiently rapidly over distances of order L.

§ 47. Boundary conditions in the semiclassical case

Let $x = a$ be a turning point, so that $U(a) = E$, and suppose $U > E$ for every $x > a$; the region to the right of the turning point is then classically inaccessible. The wave function must decay into the interior of this region. At a sufficient distance from the turning point it has the form

$$\psi = \frac{C}{2\sqrt{|p|}}\exp\left(-\frac{1}{\hbar}\int_a^x |p|\,dx\right), \qquad x > a, \tag{47.1}$$

corresponding to the first term of (46.10). To the left of the turning point, the wave function must instead be represented by a real combination of the two semiclassical solutions (46.9) of the Schrödinger equation:

$$\psi = \frac{C_1}{\sqrt{p}}\exp\left(\frac{i}{\hbar}\int_a^x p\,dx\right) + \frac{C_2}{\sqrt{p}}\exp\left(-\frac{i}{\hbar}\int_a^x p\,dx\right), \qquad x < a. \tag{47.2}$$

To determine the coefficients in this combination, one must trace the change in the wave function from positive to negative values of $x - a$, beginning in the region where (47.1) applies. This path, however, passes through the neighborhood of the turning point, where the semiclassical approximation fails and the exact solution of the Schrödinger equation must be considered. For small $|x - a|$,

$$E - U(x) \approx F_0(x - a), \qquad F_0 = -\left.\frac{dU}{dx}\right|_{x=a} < 0; \tag{47.3}$$

in other words, this region presents the problem of motion in a constant field. The exact solution of the Schrödinger equation for that problem was found in § 24. The relation between the coefficients in (47.1) and (47.2) can be found by comparing with the asymptotic expressions (24.5) and (24.6) of that exact solution on the two sides of the turning point. It should be noted that (47.3) gives $p(x) = \sqrt{2mF_0(x-a)}$, and therefore

$$\frac{1}{\hbar}\int_a^x p\,dx = \frac{2}{3\hbar}\sqrt{2mF_0}(x-a)^{3/2};$$

this coincides with the expression in the argument of the exponential or sine in (24.5) or (24.6). The essential point in this reasoning is that the ranges of validity of expansion (47.3) and of the semiclassical approximation partly overlap. If the motion is semiclassical in almost the entire field region, as has been assumed, there are values of $|x-a|$ small enough for (47.3) to be valid and at the same time large enough to satisfy the semiclassical condition and permit the asymptotic forms (24.5), (24.6).[2]

There is, however, another and methodically more instructive procedure that avoids any need to invoke the exact solution. Formally regard $\psi(x)$ as a function of the complex variable x, and pass from positive to negative $x-a$ along a path lying wholly far from $x=a$, so that the semiclassical condition is formally satisfied everywhere on the path (A. Zwaan, 1929). We again consider values of $|x-a|$ for which (47.3) is valid as well. The wave function (47.1) then becomes

$$\psi(x) = \frac{C}{2(2m|F_0|)^{1/4}(x-a)^{1/4}} \exp\left\{-\frac{1}{\hbar}\sqrt{2m|F_0|}\int_a^x \sqrt{x-a}\,dx\right\}. \tag{47.4}$$

First trace how this function changes when the point $x=a$ is passed from right to left along a semicircle of radius ρ in the upper half of the complex x plane. On this semicircle

$$x-a=\rho e^{i\varphi}, \qquad \int_a^x \sqrt{x-a}\,dx = \frac{2}{3}\rho^{3/2}\left(\cos\frac{3\varphi}{2}+i\sin\frac{3\varphi}{2}\right),$$

where the phase φ changes from 0 to π. The modulus of the exponential factor in (47.4) first increases (for $0<\varphi<2\pi/3$), and then decreases to unity. At the end of the passage, the exponent becomes purely imaginary and is equal to

$$-\frac{i}{\hbar}\int_a^x \sqrt{2m|F_0|(a-x)}\,dx = -\frac{i}{\hbar}\int_a^x p(x)\,dx.$$

Meanwhile, the prefactor in (47.4) changes under the continuation according to

$$(x-a)^{-1/4} \longrightarrow (a-x)^{-1/4}e^{-i\pi/4}.$$

Thus the whole function (47.4) becomes the second term of (47.2), with $C_2 = (1/2)Ce^{-i\pi/4}$.

There is a simple explanation for why continuation through the upper half plane determines only C_2 in (47.2). If the change of (47.2) is followed along the same semicircle in the reverse direction, from left to right, its first term rapidly becomes exponentially small compared with the second at the beginning of the path. The semiclassical approximation cannot detect exponentially small terms in ψ against a large principal term; this is why the first term in (47.2) is “lost” during this continuation.

To determine C_1, the point must instead be passed from right to left along a semicircle in the lower half of the complex x plane. The same reasoning shows that (47.4) then becomes the first term in (47.2), with $C_1 = (1/2)Ce^{i\pi/4}$.

Consequently, the wave function (47.1) for $x>a$ corresponds for $x<a$ to

$$\psi = \frac{C}{\sqrt{p}}\cos\left(\frac{1}{\hbar}\int_a^x p\,dx+\frac{\pi}{4}\right).$$

[2]Indeed, expansion (47.3) applies for $|x-a| \ll L$, where L is the characteristic distance over which the field $U(x)$ changes. The semiclassical condition (46.7), on the other hand, requires $|x-a|^{3/2} \gg \hbar/\sqrt{m|F_0|}$. These requirements are compatible because semiclassical motion far from the turning point (that is, for $|x-a|\sim L$) means that $L^{3/2} \gg \hbar/\sqrt{m|F_0|}$.

The resulting connection rule may be written in a form independent of which side of the turning point contains the classically inaccessible region:

$$\frac{2C}{\sqrt{|p|}} \exp\left\{-\frac{1}{\hbar}\int_a^x |p|\,dx\right\} \longrightarrow \frac{C}{\sqrt{p}}\cos\left\{\frac{1}{\hbar}\int_a^x p\,dx - \frac{\pi}{4}\right\} \qquad \underset{U(x)>E}{} \qquad \underset{U(x)<E}{} \tag{47.5}$$

(H. A. Kramers, 1926). We emphasize once more a point already evident from the derivation: this rule is tied to a definite boundary condition imposed on one side of the turning point, and in this sense it is to be used only in one direction. Specifically, (47.5) was derived with the boundary condition $\psi \to 0$ into the interior of the classically inaccessible region; it must be applied in passing from that region to the classically allowed region, as indicated by the arrow in (47.5).[3]

If the classically accessible region is bounded at $x = a$ by an infinitely high potential wall, the boundary condition on the wave function at $x = a$ is $\psi = 0$ (see § 18). The semiclassical approximation then remains valid right up to the wall, and the wave function is

$$\begin{aligned} \psi &= \frac{C}{\sqrt{p}}\sin\left(\frac{1}{\hbar}\int_a^x p\,dx\right), && x < a, \\ \psi &= 0, && x > a. \end{aligned} \tag{47.6}$$

§ 48. The Bohr–Sommerfeld quantization rule

States in the discrete energy spectrum are semiclassical at large values of the quantum number n, the ordinal number of the state. Indeed, this number determines the number of nodes of the eigenfunction (see § 21). The distance between adjacent nodes, however, is of the same order as the de Broglie wavelength. At large n this distance is small, so the wavelength is small compared with the dimensions of the region of motion.

We derive the condition determining the quantum energy levels in the semiclassical case. Consider finite one-dimensional motion of a particle in a potential well; the classically accessible region $b \le x \le a$ is bounded by two turning points[4]. By rule (47.5), the boundary condition at $x = b$ gives, in the region to its right,

$$\psi = \frac{C}{\sqrt{p}}\cos\left(\frac{1}{\hbar}\int_b^x p\,dx - \frac{\pi}{4}\right). \tag{48.1}$$

Applying the same rule to the region to the left of $x = a$, we obtain the same wave function as

$$\psi = \frac{C'}{\sqrt{p}}\cos\left(\frac{1}{\hbar}\int_x^a p\,dx - \frac{\pi}{4}\right).$$

[3] A passage in the reverse direction loses its meaning in the sense that even a small change of the wave function on the right-hand side of (47.5) may produce an exponentially growing term in the function on the left-hand side.

[4] In classical mechanics a particle in such a field would execute periodic motion with period (the time for motion from b to a and back)

$$T = 2\int_b^a \frac{dx}{v}$$

(v is the particle's speed).

For these expressions to agree throughout the region, the sum of their phases (a constant) must be an integral multiple of π:

$$\frac{1}{\hbar}\int_b^a p\,dx - \frac{\pi}{2} = n\pi$$

(with $C' = (-1)^n C$). Hence

$$\frac{1}{2\pi\hbar}\oint p\,dx = n + \frac{1}{2}, \tag{48.2}$$

where $\oint p\,dx = 2\int_b^a p\,dx$ is taken over a complete period of the particle's classical motion. This condition determines the particle's stationary states in the semiclassical case. It corresponds to the Bohr–Sommerfeld quantization rule of the old quantum theory.

The quantity $I = (1/2\pi)\oint p\,dx$ is called the adiabatic invariant (see I, § 49), so (48.2) may be written

$$I(E) = \hbar(n + 1/2).$$

It was mentioned in § 41 that under a sufficiently slow, "adiabatic" change of the parameters, a system remains in the same quantum state, here a state with a certain n. In the semiclassical limit this statement is thus the classical theorem that the adiabatic invariant remains constant under a slow change of the parameters. It is readily seen that the integer n equals the number of zeros of the wave function and is therefore the ordinal number of the stationary state. Indeed, the phase in (48.1) grows from $-\pi/4$ at $x = b$ to $(n + 1/4)\pi$ at $x = a$, so the cosine vanishes n times in this interval (outside the interval $b \ll x \ll a$, the wave function decays monotonically and has no zeros at finite distances)[5].

As noted above, n is large in the semiclassical case. Nevertheless, it is legitimate to retain the $1/2$ beside n in (48.2): the next correction terms in the phase would add to its right-hand side only terms $\sim \lambda/L$, small compared with unity (see the remark at the end of § 46)[6].

To normalize the wave function it is enough to integrate $|\psi|^2$ over $b \le x \le a$, since $\psi(x)$ decays exponentially outside it. The argument of the cosine in (48.1) varies rapidly, so the square of the cosine may be replaced, to sufficient accuracy, by its mean value $1/2$. Then

$$\int |\psi|^2 dx \approx \frac{C^2}{2}\int_b^a \frac{dx}{p(x)} = \frac{\pi C^2}{2m\omega} = 1,$$

where $\omega = 2\pi/T$ is the frequency of the classical periodic motion. Thus the normalized semiclassical function is

$$\psi = 2\sqrt{\frac{m\omega}{\pi p}}\cos\left(\frac{1}{\hbar}\int_b^x p\,dx - \frac{\pi}{4}\right). \tag{48.3}$$

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[5] Strictly, the zeros should be counted using the exact form of the wave function near the turning points. Such an analysis confirms the stated result.

[6] In some cases the exact expression for the energy levels $E(n)$ obtained from the exact Schrödinger equation retains its form as $n \to \infty$; examples are the levels in a Coulomb field and those of the harmonic oscillator. Naturally, in these cases the quantization rule (48.2), valid at large n, gives an expression for $E(n)$ that agrees with the exact one.

The frequency ω is a function of energy and is generally different for different levels.

Relation (48.2) also has another interpretation. The integral $\oint p\,dx$ is the area enclosed by the closed classical phase trajectory (the curve in the p, x plane, the particle's phase space). Dividing this area into cells of area $2\pi\hbar$ gives n cells. But n is the number of quantum states whose energies do not exceed the specified value corresponding to the phase trajectory. Thus, in the semiclassical case, each quantum state corresponds to a phase-space cell of area $2\pi\hbar$. Equivalently, the number of states belonging to a phase-space volume element $\Delta p \Delta x$ is

$$\frac{\Delta p \Delta x}{2\pi\hbar}. \tag{48.4}$$

If the wave vector $k = p/\hbar$ is used instead of momentum, this number is $\Delta k \Delta x/2\pi$. As expected, it agrees with the familiar expression for the number of normal modes of a wave field (see II, § 52).

The general distribution of levels in the energy spectrum can be inferred from (48.2). Let ΔE be the separation of two adjacent levels, whose quantum numbers differ by one. Since ΔE is small at large n compared with the level energy itself, (48.2) gives

$$\Delta E \oint \frac{\partial p}{\partial E}\,dx = 2\pi\hbar.$$

But $\partial E/\partial p = v$, and hence

$$\oint \frac{\partial p}{\partial E}\,dx = \oint \frac{dx}{v} = T.$$

Therefore

$$\Delta E = \frac{2\pi\hbar}{T} = \hbar\omega. \tag{48.5}$$

Thus adjacent levels are separated by $\hbar\omega$. For a sequence of neighboring levels (whose differences in n are small compared with n itself), the corresponding frequencies ω may be treated as approximately equal. Hence, within every small interval of the semiclassical part of the spectrum, the levels are equidistant, with common spacing $\hbar\omega$. This could also have been anticipated, since in the semiclassical case the frequencies corresponding to transitions between different energy levels must be integral multiples of the classical frequency ω. It is useful to follow what becomes, in the classical limit, of the matrix elements of a physical quantity f. We start from the fact that the mean value of f in a quantum state must tend simply to its classical value, provided that the limiting state itself describes motion along a definite trajectory. Such a state is represented by a wave packet (see § 6), formed by superposing stationary states with nearby energies. Its wave function is

$$\psi = \sum_n a_n \psi_n,$$

where the a_n differ appreciably from zero only in an interval Δn such that $1 \ll \Delta n \ll n$; the n are large, as required for the stationary states to be semiclassical. By definition, the mean value is

$$\bar{f} = \int \psi^* f \psi\,dx = \sum_n \sum_m a_m^* a_n f_{mn} e^{i\omega_{mn}t},$$

or, replacing the sums over n, m by sums over n and $s = m - n$,

$$\bar{f} = \sum_n \sum_s a^*_{n+s} a_n f_{n+s,n} e^{i\omega_s t},$$

where $\omega_{mn} = \omega s$ by (48.5).

Matrix elements f_{mn} calculated with semiclassical wave functions fall rapidly as $m - n$ increases, while varying slowly with n itself for fixed $m - n$. We may therefore write approximately

$$\bar{f} = \sum_n \sum_s a^*_n a_n f_s e^{i\omega st} = \sum_n |a_n|^2 \sum_s f_s e^{i\omega st},$$

where $f_s = f_{n+s,n}$ and n denotes a mean quantum number in Δn. Since $\sum_n |a_n|^2 = 1$,

$$\bar{f} = \sum_s f_s e^{i\omega st}.$$

This sum is an ordinary Fourier series. Since $\bar{f}$ must tend to the classical quantity $f(t)$, it follows that the matrix elements f_{mn} tend to the components f_{m-n} in the Fourier-series expansion of the classical function $f(t)$.

Similarly, matrix elements for transitions between continuous-spectrum states tend to the components in the Fourier-integral expansion of $f(t)$. The wave functions of the stationary states must here be normalized to a δ-function of energy divided by $\hbar$.

All these results extend directly to systems with many degrees of freedom executing finite motion for which the mechanical (classical) problem permits complete separation of variables in the Hamilton–Jacobi method (so-called conditionally periodic motion; see I, § 52). After separation, the problem for each degree of freedom is one-dimensional, and the corresponding quantization conditions are

$$\oint p_i \, dq_i = 2\pi\hbar(n_i + \gamma_i), \tag{48.6}$$

where the integral is over a period of the generalized coordinate q_i, and γ_i is a number of order unity depending on the boundary conditions for that degree of freedom[7].

For general arbitrary (not conditionally periodic) multidimensional motion, formulating semiclassical quantization conditions requires deeper arguments[8]. The concept of phase-space "cells," however, applies in the same form in the semiclassical approximation. This is clear from its connection, noted above, with the number of normal modes of a wave field in a prescribed spatial volume. In the general case of a system with s degrees of freedom, a phase-space volume element contains

$$\Delta N = \frac{\Delta q_1 \cdots \Delta q_s \, \Delta p_1 \cdots \Delta p_s}{(2\pi\hbar)^s} \tag{48.7}$$

[7] For motion in a centrally symmetric field,

$$\oint p_r \, dr = 2\pi\hbar(n_r + 1/2), \qquad \oint p_\theta \, d\theta = 2\pi\hbar(l - m + 1/2), \qquad \oint p_\varphi \, d\varphi = 2\pi\hbar m$$

(where $n_r = n - l - 1$ is the radial quantum number). The last equality simply reflects that p_φ is the z component of angular momentum and equals $\hbar m$.

[8] See *J., B. Keller* // Ann. Phys. 1958. V. 4. P. 180.

quantum states[9].

Problem

1. Determine approximately the number of discrete energy levels of a particle moving in a field $U(r)$ that satisfies the semiclassical condition.

Solution. The number of states "belonging" to the phase-space volume corresponding to momenta $0 \le p \le p_{\max}$ and to particle coordinates in the volume element dV is

$$\frac{(4\pi/3)p_{\max}^3 dV}{(2\pi\hbar)^3}.$$

At a given r, the particle's classical momentum can satisfy $E = (p^2/2m) + U(r) \le 0$. Substitution of $p_{\max} = \sqrt{-2mU(r)}$ gives the total number of discrete-spectrum states

$$\frac{\sqrt{2}m^{3/2}}{3\pi^2\hbar^3}\int (-U)^{3/2}\, dV,$$

where the integration is over the region in which $U < 0$. This integral diverges (the number of levels is infinite) if U decreases at infinity as r^{-s} with $s < 2$, in agreement with § 18.

2. The same question for a semiclassical centrally symmetric field $U(r)$ (V. L. Pokrovskii).

Solution. In a centrally symmetric field, the number of states does not equal the number of energy levels, because the latter are degenerate with respect to directions of angular momentum. The desired number can be found by noting that the number of levels for a specified angular momentum M equals the number of (nondegenerate) levels for one-dimensional motion in the effective potential $U_{\text{eff}} = U(r) + M^2/(2mr^2)$. At given r and $E \le 0$, the greatest possible p_r is $p_{r\max} = \sqrt{-2mU_{\text{eff}}}$. Hence the number of states (that is, levels) is

$$\frac{1}{2\pi\hbar}\oint dr\, dp_r = \frac{\sqrt{2m}}{\pi\hbar}\int \sqrt{-U - \frac{M^2}{2mr^2}}\, dr.$$

The total number of discrete levels is obtained by integrating this over $dM/\hbar$ (which replaces summation over l in the semiclassical case):

$$\frac{\sqrt{2m}}{4\hbar^2}\int (-U)r\, dr.$$

[9] In particular, for one particle $d^3p/(2\pi\hbar)^3$ is the number of states belonging to a momentum interval d^3p per unit volume of space. This explains the agreement of the two interpretations of the plane-wave normalization (15.8) noted in the footnote on p. 68.

§ 49. Semiclassical motion in a centrally symmetric field

In a centrally symmetric field, the particle's wave function separates, as we know, into angular and radial parts. We first consider the former.

The dependence of the angular wave function on the angle φ (fixed by the quantum number m) is so simple that no question of finding approximate formulas for it arises. Its dependence on the polar angle θ, however, is semiclassical by the general rule when the corresponding quantum number l is large (a more precise statement of this condition is given below).

We restrict the derivation of a semiclassical angular function to the case most important in applications, namely states with zero magnetic quantum number ($m = 0$)[10]. Apart from a constant factor, this function is the Legendre polynomial $P_l(\cos\theta)$ (see (28.8)) and obeys

$$\frac{d^2P_l}{d\theta^2} + \cot\theta\frac{dP_l}{d\theta} + l(l+1)P_l = 0. \tag{49.1}$$

The substitution

$$P_l = \frac{\chi}{\sqrt{\sin\theta}} \tag{49.2}$$

reduces it to the equation

$$\chi'' + \left[\left(l+\frac{1}{2}\right)^2 + \frac{1}{4\sin^2\theta}\right]\chi = 0, \tag{49.3}$$

which contains no first derivative and has the form of a one-dimensional Schrödinger equation.

In (49.3), the role of the "de Broglie wavelength" is played by $\lambda \sim 1/l$. The requirement that $d\lambda/dx$ be small (condition (46.6)) gives

$$\theta l \gg 1, \qquad (\pi-\theta)l \gg 1, \tag{49.4}$$

which are the semiclassical conditions for the angular part of the wave function. For large l they hold through nearly the entire range of θ, failing only for angles very close to zero or to π.

When (49.4) holds, the second term in the square brackets in (49.3) may be neglected against the first:

$$\chi'' + \left(l+\frac{1}{2}\right)^2\chi = 0.$$

The solution is

$$\chi = \sqrt{\sin\theta}\,P_l(\cos\theta) = A\sin\left[\left(l+\frac{1}{2}\right)\theta + \alpha\right] \tag{49.5}$$

(A and α are constants).

[10]The opposite case, $m = l$, must in the limit correspond to motion along a classical orbit lying in the equatorial plane $\theta = \pi/2$. Indeed,

$$P_l^l(\cos\theta) = \text{const}\cdot\sin^l\theta;$$

as $l \to \infty$, this function (and hence also $|\psi|^2$) tends to zero at every $\theta \neq \pi/2$.

For $\theta \ll 1$, we may put $\cot\theta \approx 1/\theta$ in (49.1); also replacing $l(l+1)$ approximately by $(l+1/2)^2$, we obtain

$$\frac{d^2P_l}{d\theta^2} + \frac{1}{\theta}\frac{dP_l}{d\theta} + \left(l+\frac{1}{2}\right)^2 P_l = 0,$$

whose solution is a Bessel function of order zero:

$$P_l(\cos\theta) = J_0\left[\left(l+\frac{1}{2}\right)\theta\right], \qquad \theta \ll 1. \tag{49.6}$$

The constant factor has been set equal to unity because $P_l = 1$ at $\theta = 0$. Approximation (49.6) is valid for all $\theta \ll 1$. In particular, it applies in the region $1/l \ll \theta \ll 1$, where it must match (49.5), valid for every $\theta \gg 1/l$. When $\theta l \gg 1$, the Bessel function may be replaced by its large-argument asymptotic form, giving

$$P_l \approx \sqrt{\frac{2}{\pi l\theta}} \sin\left[\left(l+\frac{1}{2}\right)\theta + \frac{\pi}{4}\right]$$

(the $1/2$ may be neglected against l in the coefficient). Comparison with (49.5) gives $A = \sqrt{2/\pi l}$ and $\alpha = \pi/4$. We therefore obtain the following final expression for $P_l(\cos\theta)$, applicable in the semiclassical case[11]:

$$P_l(\cos\theta) = \sqrt{\frac{2}{\pi l \sin\theta}} \sin\left[\left(l+\frac{1}{2}\right)\theta + \frac{\pi}{4}\right]. \tag{49.7}$$

The normalized spherical function Y_{l0} is consequently (cf. (28.8))

$$Y_{l0} = \frac{1}{\pi\sqrt{\sin\theta}} \sin\left[\left(l+\frac{1}{2}\right)\theta + \frac{\pi}{4}\right]. \tag{49.8}$$

We now turn to the radial part of the wave function. It was shown in § 32 that $\chi(r) = rR(r)$ obeys an equation identical to the one-dimensional Schrödinger equation with potential energy

$$U_l(r) = U(r) + \frac{\hbar^2}{2mr^2} l(l+1).$$

We may therefore use the results of the preceding sections, taking the potential energy there to mean $U_l(r)$.

The case $l = 0$ is simplest. There is no centrifugal energy, and if $U(r)$ satisfies the necessary condition (46.6), the radial wave function is semiclassical throughout space. At $r = 0$ we must have $\chi = 0$, so the semiclassical function $\chi(r)$ is determined in accordance with (47.6).

If $l \neq 0$, the centrifugal energy must also satisfy (46.6). In the region of small r where the centrifugal energy has the order of the total energy, $\lambda = \hbar/p \sim r/l$, and (46.6) gives $l \gg 1$. Thus, when l is not large, the centrifugal energy violates the semiclassical

[11] Notice that it is precisely the replacement of $l(l+1)$ by $(l+1/2)^2$ that has produced an expression multiplied by $(-1)^l$ when θ is replaced by $\pi - \theta$, as required for $P_l(\cos\theta)$.

condition in the region of small r. It is readily verified that the correct phase of the semiclassical wave function $\chi(r)$ is obtained by using the formulas for one-dimensional motion with $l(l+1)$ in $U_l(r)$ replaced by $(l+1/2)^2$[12]:

$$U_l(r) = U(r) + \frac{\hbar^2}{2mr^2}\left(l + \frac{1}{2}\right)^2 . \tag{49.9}$$

The applicability of the semiclassical approximation to a Coulomb field $U = \pm\alpha/r$ requires separate consideration. Of the entire region of motion, the most significant part corresponds to distances at which $|U| \sim |E|$, that is, $r \sim \alpha/|E|$. Semiclassical motion in this region requires the reduced wavelength $\lambda\!\!\!^{-} \sim \hbar/\sqrt{2m|E|}$ to be small compared with the size $\alpha/|E|$ of the region. This gives

$$|E| \ll \frac{m\alpha^2}{\hbar^2}, \tag{49.10}$$

so the absolute value of the energy must be small compared with the particle's energy in the first Bohr orbit. Condition (49.10) may also be written

$$\frac{\hbar v}{\alpha} \gg 1, \tag{49.11}$$

where $v \sim \sqrt{|E|/m}$ is the particle's speed. Notice that this is the reverse of condition (45.7) for applicability of perturbation theory to the Coulomb field.

The region of small distances ($|U(r)| \gg E$) is of no interest in a repulsive Coulomb field, since the semiclassical wave functions decay exponentially when $U > E$. In an attractive field and for small l, however, the particle can enter a region where $|U| \gg |E|$, raising the question of the range of validity of the semiclassical approximation there. We use the general condition (46.7), setting

$$F = -\frac{dU}{dr} = \frac{\alpha}{r^2}, \qquad p \approx \sqrt{2m|U|} \sim \sqrt{\frac{m\alpha}{r}}.$$

It follows that the range of validity is restricted to distances

$$r \gg \frac{\hbar^2}{m\alpha}, \tag{49.12}$$

that is, distances large compared with the "radius" of the first Bohr orbit.

Problem

Determine the behavior of the wave function near the origin if, as $r \to 0$, the field tends to infinity as $\pm\alpha/r^s$, with $s > 2$.

Solution. For sufficiently small r, the wavelength is

$$\lambda\!\!\!^{-} \sim \frac{\hbar}{\sqrt{m|U|}} \sim \frac{\hbar r^{s/2}}{\sqrt{m\alpha}},$$

[12] Thus, in the simplest case of free motion ($U = 0$), the phase calculated from (48.1) with U_l from (49.9) agrees at large r, as it should, with the phase of (33.12).

and hence

$$\frac{d\lambda\!\!\!^{-}}{dr} \sim \frac{\hbar r^{s/2-1}}{\sqrt{m\alpha}} \ll 1;$$

thus the semiclassical condition is satisfied. In an attractive field, $U \to -\infty$ as $r \to 0$. The region near the origin is then classically accessible, and the radial wave function $\chi \sim 1/\sqrt{p}$, whence

$$\psi \sim r^{s/4-1}.$$

In a repulsive field the region of small r is classically inaccessible. The wave function then tends exponentially to zero as $r \to 0$. Omitting the factor multiplying the exponential, we have

$$\psi \sim \exp\left[-\frac{2\sqrt{m\alpha}}{(s-2)\hbar} r^{-(s/2-1)}\right].$$

§ 50. Transmission through a potential barrier

Consider a particle moving in a field of the type shown in Fig. 13, with a *potential barrier*: a region in which the potential energy $U(x)$ exceeds the particle's total energy E. In classical mechanics such a barrier is impenetrable. In quantum mechanics, however, a particle has a nonzero probability of passing "through the barrier"; this is also called the *tunnel effect*.[13] If $U(x)$ satisfies the semiclassical conditions, the transmission coefficient of the barrier can be found in a general form. These conditions imply in particular that the barrier must be wide, so its transmission coefficient is small in the semiclassical case.

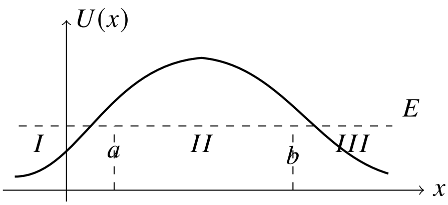

Fig. 13

To avoid interrupting the subsequent calculation, first solve the following problem. Let the semiclassical wave function to the right of the turning point $x = b$, where $U(x) < E$, have the form of a traveling wave:

$$\psi = \frac{C}{\sqrt{p}} \exp\left(\frac{i}{\hbar}\int_b^x p\,dx + \frac{i\pi}{4}\right). \tag{50.1}$$

We must find the wave function of the same state in the region $x < b$. Use the same continuation in the complex x plane as in § 47.

Put

$$E - U(x) \approx F_0(x-b), \qquad F_0 > 0,$$

[13] Examples of this kind were already considered in Problems 2 and 4 of § 25.

and write (50.1) as

$$\psi(x) = \frac{C}{(2mF_0)^{1/4}} \frac{1}{(x-b)^{1/4}} \exp\left\{\frac{i}{\hbar}(2mF_0)^{1/2} \int_b^x \sqrt{x-b}\, dx + \frac{i\pi}{4}\right\}.$$

Continue it from right to left along a semicircle in the upper half-plane:

$$x - b = \rho e^{i\varphi}, \qquad i\int_b^x \sqrt{x-b}\, dx = \frac{2}{3}\rho^{3/2}\left(-\sin\frac{3\varphi}{2} + i\cos\frac{3\varphi}{2}\right),$$

where φ varies from 0 to π. During this continuation the modulus of $\psi(x)$ first decreases and then increases; at the end it becomes

$$\psi(x) = \frac{C}{(2mF_0)^{1/4}} \frac{1}{(b-x)^{1/4} e^{i\pi/4}} \exp\left\{\frac{1}{\hbar}\int_x^b (2mF_0)^{1/2}\sqrt{b-x}\, dx + \frac{i\pi}{4}\right\}.$$

We thus obtain the connection rule[14]

$$\frac{C}{\sqrt{p}} \exp\left\{\frac{i}{\hbar}\int_b^x p\, dx + \frac{i\pi}{4}\right\}\bigg|_{x>b} \longrightarrow \frac{C}{\sqrt{|p|}} \exp\left\{\frac{1}{\hbar}\left|\int_b^x p\, dx\right|\right\}\bigg|_{x<b}. \tag{50.2}$$

This rule presupposes a particular wave in the classically allowed region, a wave traveling to the right, and must be applied specifically in passing from that region to the classically inaccessible region.

Return now to the transmission coefficient of the potential barrier. Let the particle be incident on the barrier from region I, moving from left to right. Behind the barrier, in region III, there is only the transmitted wave moving to the right. Write its wave function as

$$\psi = \sqrt{\frac{D}{v}} \exp\left(\frac{i}{\hbar}\int_b^x p\, dx + \frac{i\pi}{4}\right), \tag{50.3}$$

where $v = p/m$ is the particle velocity and D is the current density in the wave. Rule (50.2) gives the wave function within the barrier, in region II:

$$\psi = \sqrt{\frac{D}{|v|}} \exp\left(\frac{1}{\hbar}\left|\int_x^b p\, dx\right|\right) = \sqrt{\frac{D}{|v|}} \exp\left(\frac{1}{\hbar}\left|\int_a^b p\, dx\right| - \frac{1}{\hbar}\left|\int_a^x p\, dx\right|\right). \tag{50.4}$$

Finally, application of (47.5) gives, in region I before the barrier,

$$\psi = 2\sqrt{\frac{D}{v}} \exp\left(\frac{1}{\hbar}\int_a^b |p|\, dx\right) \cos\left(\frac{1}{\hbar}\int_x^a p\, dx - \frac{\pi}{4}\right).$$

If in this function we put

$$D = \exp\left(-\frac{2}{\hbar}\int_a^b |p|\, dx\right), \tag{50.5}$$

[14] If the continuation from right to left is instead made through the lower half-plane, $\psi(x)$ first increases and then decreases in modulus, becoming exponentially small on the negative half-axis ($\varphi \to -\pi$). Retaining it against the exponentially large function (50.2) would be illegitimate. Over the part of the path where $\psi(x)$ is exponentially large, the inaccuracy of the semiclassical approximation loses an exponentially small addition which, as $\varphi \to -\pi$, could have become an exponentially large term; that term is therefore lost as well.

it takes the form

$$\psi = \frac{2}{\sqrt{v}} \cos\left(\frac{1}{\hbar}\int_a^x p\,dx + \frac{\pi}{4}\right)$$
$$= \frac{1}{\sqrt{v}} \exp\left(\frac{i}{\hbar}\int_a^x p\,dx + \frac{i\pi}{4}\right) + \frac{1}{\sqrt{v}} \exp\left(-\frac{i}{\hbar}\int_a^x p\,dx - \frac{i\pi}{4}\right).$$

The first term, which reduces as $x \to -\infty$ to the plane wave $\psi \sim e^{ipx/\hbar}$, describes the incident wave; the second describes the reflected wave. The chosen normalization gives unit current density in the incident wave. Hence D, the current density in the transmitted wave, is the required barrier transmission coefficient. Formula (50.5) applies only when its exponent is large and D itself is small.[15]

So far $U(x)$ has been assumed semiclassical throughout the barrier, except in the immediate neighborhoods of the turning points. In practice one often encounters barriers whose potential-energy curve is so steep on one side that the semiclassical approximation fails. The principal exponential factor in D remains the same as in (50.5), but its prefactor, unity in (50.5), changes. To calculate it, one must in principle find the exact wave function in the non-semiclassical region and use its matching to determine the semiclassical wave function inside the barrier.

Problems

Problem

1. Find the transmission coefficient of the barrier shown in Fig. 14: $U(x) = 0$ for $x < 0$, and $U(x) = U_0 - Fx$ for $x > 0$. Calculate only the exponential factor.

Solution. A direct calculation gives

$$D \sim \exp\left[-\frac{4\sqrt{2m}}{3\hbar F}(U_0 - E)^{3/2}\right].$$

Problem

2. Find the probability that a particle of zero angular momentum escapes from a spherically symmetric potential well: $U(r) = -U_0$ for $r < r_0$, and $U(r) = \alpha/r$ for $r > r_0$ (Fig. 15).[16]

Solution. The spherically symmetric problem reduces to a one-dimensional one, so the preceding formulas apply. We have

$$w \sim \exp\left[-\frac{2}{\hbar}\int_{r_0}^{\alpha/E} \sqrt{2m\left(\frac{\alpha}{r} - E\right)}\,dr\right].$$

[15] The exponential smallness of D is also why the incident and reflected amplitudes in region I have come out equal: the semiclassical approximation loses the exponentially small difference between them.

[16] This problem was first considered by G. F. Gamow (1928), and by R. W. Gurney and E. U. Condon (1929), in connection with the theory of radioactive α decay.

Evaluation of the integral gives

$$w \sim \exp\left\{-\frac{2\alpha}{\hbar}\sqrt{\frac{2m}{E}}\left[\arccos\sqrt{\frac{Er_0}{\alpha}}-\sqrt{\frac{Er_0}{\alpha}\left(1-\frac{Er_0}{\alpha}\right)}\right]\right\}.$$

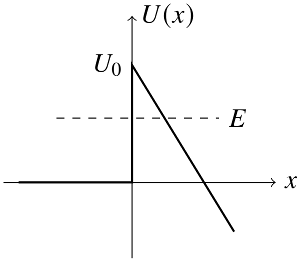

Fig. 14

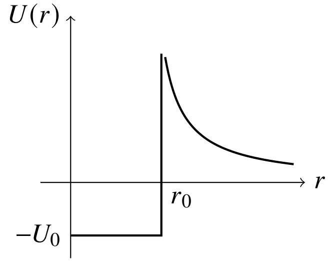

Fig. 15

In the limit $r_0 \to 0$, this becomes

$$w \sim \exp\left(-\frac{\pi\alpha}{\hbar}\sqrt{\frac{2m}{E}}\right) = \exp\left(-\frac{2\pi\alpha}{\hbar v}\right).$$

These formulas apply when the exponent is large, $\alpha/\hbar v \gg 1$. As it should, this condition coincides with the semiclassical condition (49.11) for motion in a Coulomb field.

Problem

3. The field $U(x)$ consists of two symmetric potential wells (I and II in Fig. 16) separated by a barrier. If the barrier were impenetrable, there would be equal energy levels for motion confined to either well. Passage through the barrier splits each such level into two nearby levels corresponding to states in which the particle moves in both wells. Find the splitting, assuming that $U(x)$ is semiclassical.

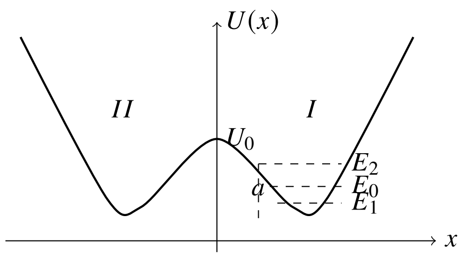

Fig. 16

Solution. Neglecting transmission through the barrier, construct an approximate solution using the semiclassical wave function $\psi_0(x)$ for motion with energy E_0 in one well, say I. It decays exponentially on both sides of that well and is normalized so that its squared integral over well I is unity. When the small tunneling probability is included, E_0 splits into E_1 and E_2. The proper zeroth-order wave functions are the symmetric and antisymmetric combinations of $\psi_0(x)$ and $\psi_0(-x)$:

$$\psi_1(x) = \frac{1}{\sqrt{2}}[\psi_0(x) + \psi_0(-x)], \qquad \psi_2(x) = \frac{1}{\sqrt{2}}[\psi_0(x) - \psi_0(-x)]. \tag{1}$$

In well I, $\psi_0(-x)$ is negligibly small compared with $\psi_0(x)$, while the reverse holds in well II. Thus $\psi_0(x)\psi_0(-x)$ is negligible everywhere, and the functions (1) are normalized so that their squared integrals over wells I and II equal unity.

Write the Schrödinger equations

$$\psi_0'' + \frac{2m}{\hbar^2}(E_0 - U)\psi_0 = 0, \qquad \psi_1'' + \frac{2m}{\hbar^2}(E_1 - U)\psi_1 = 0.$$

Multiply the first by ψ_1, the second by ψ_0, subtract term by term, and integrate from 0 to ∞. At $x = 0$, $\psi_1 = \sqrt{2}\psi_0$ and $\psi_1' = 0$, while

$$\int_0^\infty \psi_0\psi_1\, dx \approx \frac{1}{\sqrt{2}}\int_0^\infty \psi_0^2\, dx = \frac{1}{\sqrt{2}}.$$

We obtain

$$E_1 - E_0 = -\frac{\hbar^2}{m}\psi_0(0)\psi_0'(0).$$

The same calculation for $E_2 - E_0$ gives the opposite sign, and therefore

$$E_2 - E_1 = \frac{2\hbar^2}{m}\psi_0(0)\psi_0'(0).$$

Formula (47.1), with the coefficient C from (48.3), gives

$$\psi_0(0) = \sqrt{\frac{\omega}{2\pi v_0}}\exp\left(-\frac{1}{\hbar}\int_0^a |p|\, dx\right), \qquad \psi_0'(0) = \frac{mv_0}{\hbar}\psi_0(0),$$

where $v_0 = \sqrt{2(U_0 - E_0)/m}$. Hence

$$E_2 - E_1 = \frac{\omega\hbar}{\pi}\exp\left(-\frac{1}{\hbar}\int_{-a}^a |p|\, dx\right),$$

where a is the turning point for energy E_0 (see Fig. 16).

Problem

4. Find the exact transmission coefficient D, without assuming it small, for the parabolic potential barrier $U(x) = -kx^2/2$ (E. C. Kemble, 1935).[17]

[17] The solution also applies to transmission sufficiently near the top of any barrier whose $U(x)$ is quadratic near its maximum.

SOLUTION. For any k and E, the motion is semiclassical at sufficiently large $|x|$, where

$$p = \sqrt{2m\left(E + \frac{1}{2}kx^2\right)} \approx x\sqrt{mk} + E\sqrt{\frac{m}{k}}\frac{1}{x},$$

and the asymptotic solution of the Schrödinger equation is

$$\psi = \mathrm{const} \cdot \xi^{\pm i\varepsilon - 1/2} \exp(\pm i\xi^2/2),$$

where

$$\xi = x\left(\frac{mk}{\hbar^2}\right)^{1/4}, \qquad \varepsilon = \frac{E}{\hbar}\sqrt{\frac{m}{k}}.$$

We require the solution which, as $x \longrightarrow +\infty$, contains only the transmitted wave moving from left to right. Put

$$\psi = B\xi^{i\varepsilon - 1/2} \exp(i\xi^2/2) \qquad \text{as } x \longrightarrow \infty, \tag{1}$$

$$\psi = (-\xi)^{-i\varepsilon - 1/2} \exp(-i\xi^2/2) + A(-\xi)^{i\varepsilon - 1/2} \exp(i\xi^2/2) \qquad \text{as } x \longrightarrow -\infty. \tag{2}$$

The first term in (2) is the incident wave and the second is the reflected wave; propagation is in the direction of increasing phase. The relation between A and B follows because the asymptotic expression for ψ is valid throughout the sufficiently remote region of the complex ξ plane. Continue (1) along a large semicircle of radius ρ in the upper half-plane:

$$\xi = \rho e^{i\varphi}, \qquad i\xi^2 = \rho^2(-\sin 2\varphi + i\cos 2\varphi),$$

where φ varies from 0 to π. The continuation turns (1) into the second term of (2), with

$$A = B(e^{i\pi})^{i\varepsilon - 1/2} = -iBe^{-\pi\varepsilon}; \tag{3}$$

on the part of the path $\pi/2 < \varphi < \pi$, where $|\exp(i\xi^2/2)|$ is exponentially large, an exponentially small quantity that should produce the first term of (2) is lost.[18]

With the incident-wave normalization used in (2), conservation of particle number gives

$$|A|^2 + |B|^2 = 1. \tag{4}$$

Equations (3) and (4) yield

$$D = |B|^2 = 1/(1 + e^{-2\pi\varepsilon}).$$

This is valid for every E. If E is negative and large in magnitude, $D \approx e^{-2\pi|\varepsilon|}$, in agreement with (50.5). For $E > 0$,

$$R = 1 - D = 1/(1 + e^{2\pi\varepsilon})$$

is the above-barrier reflection coefficient.

[18] Continuation through the lower half-plane would be unsuitable for determining A because on the part $-\pi < \varphi < -\pi/2$ adjacent to its left end, where ψ is given by (2), the term containing $\exp(i\xi^2/2)$ is exponentially small compared with the term containing $\exp(-i\xi^2/2)$.

§ 51. Calculation of semiclassical matrix elements

Direct calculation of matrix elements of a physical quantity f using semiclassical wave functions presents serious difficulties. We assume that the energies of the states connected by the matrix element are not close, so the latter does not reduce to a Fourier component of f (§ 48). The difficulty arises because the exponential wave functions, with large imaginary exponents, make the integrand rapidly oscillating. Consider one-dimensional motion in $U(x)$ and, for simplicity, let the operator of f be just a function of x. Let ψ_1, ψ_2 be real wave functions for energies E_1, E_2, with $E_2 > E_1$ (Fig. 17). We must calculate

$$f_{12} = \int_{-\infty}^{+\infty} \psi_1 f \psi_2 \, dx. \tag{51.1}$$

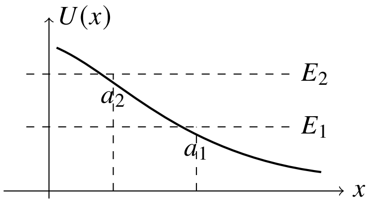

Fig. 17

By (47.5), on the two sides of $x = a_1$ and sufficiently far from it,

$$\psi_1 = \begin{cases} \dfrac{C_1}{\sqrt{|p_1|}} \exp\left(-\dfrac{1}{\hbar} \displaystyle\int_x^{a_1} |p_1| \, dx \right), & x < a_1, \\ \dfrac{2C_1}{\sqrt{p_1}} \cos\left(\dfrac{1}{\hbar} \displaystyle\int_{a_1}^{x} p_1 \, dx - \dfrac{\pi}{4} \right), & x > a_1, \end{cases} \tag{51.2}$$

and similarly for ψ_2, with 1 replaced by 2.

Substitution of these asymptotic expressions in (51.1) would give a wrong result. As shown below, the integral is exponentially small although its integrand is not. A relatively small change in the integrand may therefore change the order of the integral. This difficulty can be avoided as follows.

Write $\psi_2 = \psi_2^+ + \psi_2^-$ by resolving the cosine for $x > a_2$ into two exponentials. According to (50.2),

$$\psi_2^+ = \begin{cases} \dfrac{-iC_2}{2\sqrt{|p_2|}} \exp\left(\dfrac{1}{\hbar} \displaystyle\int_x^{a_2} |p_2| \, dx \right), & x < a_2, \\ \dfrac{C_2}{2\sqrt{p_2}} \exp\left(\dfrac{i}{\hbar} \displaystyle\int_{a_2}^{x} p_2 \, dx - \dfrac{i\pi}{4} \right), & x > a_2, \end{cases} \tag{51.3}$$

and $\psi_2^- = (\psi_2^+)^*$.

Accordingly, (51.1) is the sum of two complex-conjugate integrals, $f_{12} = f_{12}^+ + f_{12}^-$. The integral

$$f_{12}^+ = \int_{-\infty}^{+\infty} \psi_1 f \psi_2^+ \, dx$$

converges: although ψ_2^+ grows exponentially for $x < a_2$, ψ_1 decreases still faster for $x < a_1$, since $|p_1| > |p_2|$ throughout $x < a_2$.

Regard x as complex and displace the integration path into the upper half-plane. A positive imaginary increment of x produces a growing term in ψ_1 for $x > a_1$, but ψ_2^+ decreases faster because $p_2 > p_1$ there. The integrand therefore decreases.

The displaced path no longer crosses a_1, a_2, where the semiclassical approximation fails. Along the entire path we may use their asymptotic expressions in the upper half-plane:

$$\begin{aligned} \psi_1 &= \frac{C_1}{2[2m(U-E_1)]^{1/4}} \exp\left(\frac{1}{\hbar}\int_{a_1}^{x} \sqrt{2m(U-E_1)}\,dx\right), \\ \psi_2^+ &= \frac{-iC_2}{2[2m(U-E_2)]^{1/4}} \exp\left(-\frac{1}{\hbar}\int_{a_2}^{x} \sqrt{2m(U-E_2)}\,dx\right), \end{aligned} \tag{51.4}$$

where the roots are chosen positive on the real axis for $x < a_2$. Thus

$$\begin{aligned} f_{12}^+ = \frac{-iC_1C_2}{4\sqrt{2m}} \int \exp\biggl[\frac{1}{\hbar}\int_{a_1}^{x} &\sqrt{2m(U-E_1)}\,dx \\ - \frac{1}{\hbar}\int_{a_2}^{x} \sqrt{2m(U-E_2)}\,dx\biggr] &\frac{f(x)\,dx}{[(U-E_1)(U-E_2)]^{1/4}}. \end{aligned} \tag{51.5}$$

We seek to move the contour so as to reduce the exponential factor. Its exponent has an extremum only where $U(x) = \infty$; for $E_1 \neq E_2$ its derivative vanishes nowhere else. The only restriction on moving the contour upward is therefore the need to pass around the singular points of $U(x)$, which, by the general theory of linear differential equations, are also the singular points of $\psi(x)$.

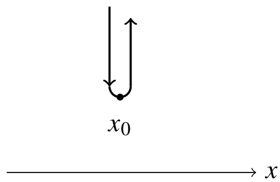

Fig. 18

The contour depends on the form of $U(x)$. If U has only one singular point x_0 in the upper half-plane, the path may be chosen as in Fig. 18. The immediate neighborhood of the singularity gives the principal contribution, so $f_{12} = 2\,\mathrm{Re}\, f_{12}^+$ is mainly proportional to

$$f_{12} \sim \exp\left\{-\frac{1}{\hbar}\,\mathrm{Im}\left[\int^{x_0} \sqrt{2m(E_2-U)}\,dx - \int^{x_0} \sqrt{2m(E_1-U)}\,dx\right]\right\}. \tag{51.6}$$

(L. D. Landau, 1932).[19] The lower limits may be any points in classically accessible

[19]Replacing the wave functions by their asymptotic expressions in deriving (51.5), (51.6) is legitimate because the order of the integral over the Fig. 18 contour is fixed by the order of its integrand; a relatively small change of the latter has no material effect.

regions and do not affect the imaginary parts. If $U(x)$ has several upper-half-plane singularities, x_0 in (51.6) is the one giving the exponent of smallest absolute magnitude.[20]

When E_1, E_2 are close, (51.6) simplifies and the matrix element reduces, by § 48, to a time Fourier component of the classical $f[x(t)]$. Put $E_{2,1} = E \pm \hbar\omega_{21}/2$ and expand in $\hbar\omega_{21}$:

$$f_{12} \sim \exp\left(-\omega_{21}\,\mathrm{Im}\int^{x_0}\frac{dx}{v}\right) = \exp(-\omega_{21}\,\mathrm{Im}\,\tau). \tag{51.6a}$$

Here

$$\tau = \int^{x_0}\frac{dx}{v}, \qquad v(x) = \sqrt{2(E-U(x))/m},$$

may be viewed as the *complex time* needed to reach x_0 in the complex x plane, and $v(x)$ as the corresponding "complex velocity." Formula (51.6a) indeed approximates the Fourier component when $\omega_{21}\,\mathrm{Im}\,\tau \gg 1$.

For motion in a central field the calculation is similar, but $U(r)$ is now the effective potential energy, the sum of potential and centrifugal terms, and differs for different l. For later use denote the two effective potentials generally by $U_1(r), U_2(r)$. The exponent in the integrand of (51.5) then has extrema not only at infinities of U_1 or U_2, but also at points satisfying

$$U_1(r) + U_2(r) = E_1 + E_2. \tag{51.7}$$

Thus in

$$f_{12} \sim \exp\left\{-\frac{1}{\hbar}\,\mathrm{Im}\left[\int^{r_0}\sqrt{2m(E_2-U_2)}\,dr - \int^{r_0}\sqrt{2m(E_1-U_1)}\,dr\right]\right\} \tag{51.8}$$

the competing r_0 include both singular points and the roots of (51.7).

The central-field case also differs because the radial integral runs from 0, not $-\infty$, to $+\infty$:

$$f_{12} = \int_0^\infty \chi_1 f \chi_2\,dr.$$

If the integrand is even, the range may formally be extended to the whole axis and nothing changes. This can occur when U_1, U_2 are even. Then χ_1, χ_2 are even or odd (see § 21),[21] and if $f(r)$ too is even or odd, their product may be even. If the integrand is not even, as necessarily happens if $U(r)$ is not even, the contour cannot be moved away from its initial point $r = 0$, and $r_0 = 0$ must also be included among the competing values in (51.8).

Problems

[20] We assume that $f(x)$ itself has no singularities. Estimate (51.6) also assumes a prefactor of normal order. It may be anomalously small because of the particular problem. The simplest example is $f(x) = \text{const}$: the matrix element vanishes by orthogonality, which (51.6) does not reveal.

[21] For even $U(r)$, the radial wave function $R(r)$ is even (odd) for even (odd) l, as follows from $R \sim r^l$ at small r.

Problem

1. Calculate the semiclassical matrix elements, retaining only the exponential factor, in the field $U = U_0 e^{-\alpha x}$.

Solution. The only infinity of $U(x)$ is at $x \to -\infty$, so put $x_0 = -\infty$ in (51.6), extending the integrals to $+\infty$. Each separately diverges at $-\infty$; first integrate from $-x$ to $+\infty$ and then let $x \to \infty$. The result is

$$f_{12} \sim \exp\left[-\frac{\pi m}{\alpha\hbar}(v_2 - v_1)\right],$$

where $v_1 = \sqrt{2E_1/m}$ and $v_2 = \sqrt{2E_2/m}$ are the asymptotic free velocities as $x \to \infty$.

Problem

2. Do the same in the Coulomb field $U = \alpha/r$ for transitions between states with $l = 0$.

Solution. The only singular point of $U(r)$ is $r = 0$. The corresponding integral was evaluated in Problem 2 of § 50. Formula (51.8) gives

$$f_{12} \sim \exp\left[\frac{\pi\alpha}{\hbar}\left(\frac{1}{v_2} - \frac{1}{v_1}\right)\right].$$

Problem

3. Do the same for an anharmonic oscillator with

$$U(x) = \frac{m\omega^2}{2}x^2 + \beta x^4$$

under the condition

$$\hbar\omega \ll E_1, E_2 \ll \frac{m^2\omega^4}{\beta}. \tag{1}$$

Solution. Extension of the argument to finite motion shows that (51.6) still applies. Choose $x_0 \to \pm\infty$; both points contribute at the same order. Thus

$$f_{12} \sim \exp\left\{-\frac{1}{\hbar}\left[\int_{a_2}^{-\infty}\sqrt{2m(U - E_2)}\,dx - \int_{a_1}^{-\infty}\sqrt{2m(U - E_1)}\,dx\right]\right\}.$$

Under condition (1), the main contribution comes from

$$\sqrt{\frac{E_1}{m\omega^2}}, \sqrt{\frac{E_2}{m\omega^2}} \ll |x| \ll \sqrt{\frac{m\omega^2}{\beta}}, \tag{2}$$

where

$$\frac{m\omega^2}{2}x^2 \gg E_1, E_2, \beta x^4.$$

Expanding the exponent in $E_{1,2}/U$, canceling the zeroth-order terms, and neglecting βx^4, we obtain

$$f_{12} \sim \exp\left(-\frac{E_2}{\hbar\omega}\int\frac{d|x|}{|x|} + \frac{E_1}{\hbar\omega}\int\frac{d|x|}{|x|}\right).$$

Cut off the logarithmic integrals at the boundaries of (2): above at $x \sim \sqrt{m\omega^2/\beta}$ and below at $x \sim a_{2,1} \sim \sqrt{E_{2,1}/(m\omega^2)}$. Then

$$f_{12} \sim \exp\left(-\frac{E_2}{2\hbar\omega}\ln\frac{m^2\omega^4}{\beta E_2} + \frac{E_1}{2\hbar\omega}\ln\frac{m^2\omega^4}{\beta E_1}\right).$$

Introducing the state numbers

$$n_1 \approx E_1/\hbar\omega, \qquad n_2 \approx E_2/\hbar\omega,$$

write the result as

$$f_{12} \sim \frac{n_2^{n_2/2}}{n_1^{n_1/2}}\left(\frac{\beta\hbar}{m\omega^3}\right)^{(n_2-n_1)/2}.$$

Since large x are essential to the solution, this result applies when $f(x)$ does not grow too rapidly at infinity. If $f(x)$ is a polynomial, its degree must be small compared with $n_2 - n_1$.

CHAPTER IX

Spin

§ 54. Spin

In both classical and quantum mechanics, conservation of angular momentum follows from the isotropy of space with respect to a closed system. This already reveals the connection between angular momentum and rotational symmetry. In quantum mechanics, however, that connection is especially deep: it becomes essentially the principal content of the concept of angular momentum. Moreover, the classical definition of a particle's angular momentum as the product $\mathbf{r} \times \mathbf{p}$ loses its direct meaning here, because the radius vector and momentum cannot be measured simultaneously.

We saw in § 28 that specifying l and m determines the angular dependence of a particle's wave function, and hence all its symmetry properties under rotations. In its most general form, the statement of these properties amounts to giving the transformation law of wave functions when the coordinate system is rotated.

The wave function ψ_{LM} of a system of particles, with specified angular momentum L and projection M, remains unchanged[1] only when the coordinate system is rotated about the z axis. Any rotation that changes the direction of the z axis makes the projection of the angular momentum on that axis cease to have a definite value. Thus, in the new coordinate axes, the wave function generally becomes a superposition (linear combination) of the $2L + 1$ functions corresponding to the possible values of M for the given L. One may say that rotations of the coordinate system transform these $2L + 1$ functions ψ_{LM} into one another.[2] The transformation law—the superposition coefficients as functions of the rotation angles of the coordinate axes—is completely fixed by the value of L. Angular momentum thereby acquires the meaning of a quantum number that classifies the states of a system by their transformation properties under rotations of the coordinate system. This aspect of angular momentum in quantum mechanics is particularly important because it is not directly tied to an explicit angular dependence of the wave functions: their law of transformation into one another can be stated independently, without reference to such a dependence.

Consider a composite particle (an atomic nucleus, for example) that is at rest as a whole and is in a definite internal state. Besides a definite internal energy, it has angular momentum of definite magnitude L, associated with the motion of the particles within it; this angular momentum may have $2L + 1$ different spatial orientations. In other words, when treating the motion of a composite particle as a whole, we must assign to it, in addition to its coordinates, one discrete variable: the projection of its internal angular momentum on some selected direction in space.

[1] Apart from an inessential phase factor.

[2] In mathematical terminology, these functions realize the so-called *irreducible representations* of the rotation group. The number of functions transformed into one another is called the *dimension of the representation*; it is understood that no choice of other linear combinations of the functions can reduce this number.

With the meaning of angular momentum understood as above, however, its origin becomes immaterial. We are led naturally to the idea of an "intrinsic" angular momentum that must be assigned to a particle irrespective of whether it is "composite" or "elementary."

Thus quantum mechanics requires an elementary particle to possess an "intrinsic" angular momentum unrelated to its motion through space. This is a specifically quantum property of elementary particles (it disappears in the limit $\hbar \to 0$) and therefore admits no classical interpretation in principle.[3]

The intrinsic angular momentum of a particle is called its *spin*, in distinction to the angular momentum associated with its motion in space, which is called *orbital angular momentum*.[4] This may refer either to an elementary particle or to a particle which, though composite, behaves as an elementary one in the range of phenomena under consideration (an atomic nucleus, for example). We denote a particle's spin, measured like orbital angular momentum in units of $\hbar$, by s. For a particle with spin, a wave-function description of the state must determine not only the probabilities of its various positions in space but also those of the possible orientations of its spin. Thus the wave function must depend not only on three continuous variables, the particle's coordinates, but also on a discrete *spin variable*. This variable gives the spin projection on a selected direction in space (the z axis) and takes a finite set of discrete values, denoted below by σ.

Let $\psi(x, y, z; \sigma)$ be such a wave function. It is, in essence, a set of several coordinate functions, one for each value of σ; these functions will be called the *spin components* of the wave function. The integral

$$\int |\psi(x, y, z; \sigma)|^2 \, dV$$

gives the probability that the particle has a specified value of σ. The probability that it is in the volume element dV, with arbitrary σ, is

$$dV \sum_\sigma |\psi(x, y, z; \sigma)|^2.$$

When applied to the wave function, the quantum-mechanical spin operator acts on the spin variable σ. In other words, it transforms the components of the wave function into one another in a certain way. The form of this operator will be established below. From quite general considerations, however, it is already easy to see that $\hat{s}_x$, $\hat{s}_y$, and $\hat{s}_z$ obey the same commutation relations as the orbital-angular-momentum operators.

The angular-momentum operator essentially coincides with the operator of an infinitesimal rotation. In deriving the orbital-angular-momentum operator in § 26, we considered the result of applying a rotation to a function of the coordinates. Such a derivation is meaningless for spin angular momentum, since its operator acts on the spin variable rather than on the coordinates. To obtain the required commutation relations,

[3] In particular, it would be entirely meaningless to picture the "intrinsic" angular momentum of an elementary particle as arising from rotation "about its own axis."

[4] The physical idea that the electron possesses intrinsic angular momentum was proposed by Uhlenbeck and Goudsmit (G. Uhlenbeck, S. Goudsmit, 1925). Spin was introduced into quantum mechanics by Pauli (W. Pauli, 1927).

we must therefore consider an infinitesimal rotation in general, as a rotation of the coordinate system. Carry out successive infinitesimal rotations about the x and y axes, and then about the same axes in the reverse order. Direct calculation readily shows that the difference between the two resulting operations is equivalent to an infinitesimal rotation about the z axis, through an angle equal to the product of the angles about x and y. We shall not give this simple calculation here. It again yields the usual commutation relations between the component operators of angular momentum, which must therefore hold also for the spin operators:

$$\{\hat{s}_x, \hat{s}_y\} = i\hat{s}_z \tag{54.1}$$

with all their physical consequences.

The commutation relations (54.1) determine the possible magnitudes and components of spin. The entire derivation in § 27, formulas (27.7)–(27.9), used only the commutation relations and applies here without change, with s substituted for L. Formulas (27.7) show that the eigenvalues of the spin projection form a sequence differing by unity. We can no longer assert, however, that these values themselves must be integers, as they were for the orbital projection L_z. The derivation at the beginning of § 27 does not apply, since it uses (26.14), the expression for $\hat{l}_z$ specific to orbital angular momentum.

The sequence of eigenvalues s_z is bounded above and below by values equal in magnitude and opposite in sign; denote them by $\pm s$. The difference $2s$ between the largest and smallest values of s_z must be an integer or zero. Hence s may have the values $0, 1/2, 1, 3/2, \ldots$

Thus the eigenvalues of the square of the spin are

$$s^2 = s(s+1), \tag{54.2}$$

where s may be either an integer, including zero, or a half-integer. For a given s, the spin component s_z may take the $2s + 1$ values $s, s-1, \ldots, -s$. Correspondingly, the wave function of a particle of spin s has $2s + 1$ components.[5] Experiment shows that most elementary particles—electrons, positrons, protons, neutrons, μ mesons, and all hyperons (Λ, Σ, Ξ)—have spin 1/2. There are also elementary particles, the π mesons and K mesons, whose spin is zero.

The total angular momentum of a particle is the sum of its orbital angular momentum $\mathbf{l}$ and spin $\mathbf{s}$. Their operators act on functions of entirely different variables and therefore commute with each other.

The eigenvalues of the total angular momentum

$$\mathbf{j} = \mathbf{l} + \mathbf{s} \tag{54.3}$$

are determined by the same "vector-model" rule as the sum of the orbital angular momenta of two different particles (§ 31). For specified l and s, the total angular momentum may have the values $l + s, l + s - 1, \ldots, |l - s|$. Thus an electron of spin 1/2 with nonzero

[5] Since s is a fixed number for each kind of particle, the spin angular momentum $\hbar s$ tends to zero in the classical limit $\hbar \to 0$. This reasoning does not apply to orbital angular momentum, because l may take arbitrary values. The classical limit requires $\hbar$ to tend to zero and l simultaneously to infinity, while $\hbar l$ remains finite.

orbital angular momentum l may have total angular momentum $j = l \pm 1/2$; for $l = 0$, of course, j has only the single value $j = 1/2$.

The operator $\mathbf{J}$ of the total angular momentum of a system of particles is the sum of their individual $\mathbf{j}$ operators, so its values are again determined by the vector-model rules. It may be written

$$\mathbf{J} = \sum_a \mathbf{j}_a = \mathbf{L} + \mathbf{S}, \qquad \mathbf{L} = \sum_a \mathbf{l}_a, \quad \mathbf{S} = \sum_a \mathbf{s}_a, \tag{54.4}$$

where $\mathbf{S}$ may be called the total spin and $\mathbf{L}$ the total orbital angular momentum of the system.

If the total spin of a system is half-integral (or integral), the same is true of its total angular momentum, since orbital angular momentum is always integral. In particular, a system of an even number of identical particles necessarily has integral total spin, and therefore integral total angular momentum as well.

The total-angular-momentum operators $\mathbf{j}$ of a particle (or $\mathbf{J}$ of a system) obey the same commutation rules as the orbital- or spin-angular-momentum operators, since these are the general commutation rules for any angular momentum. Formulas (27.13) for the matrix elements of angular momentum, which follow from these rules, likewise apply to any angular momentum, provided the matrix elements are defined with respect to the eigenfunctions of that same angular momentum. With the appropriate changes of notation, formulas (29.7)–(29.10) for matrix elements of arbitrary vector quantities also remain valid.

Problem

A particle of spin $1/2$ is in a state with the definite value $s_z = 1/2$. Find the probabilities of the possible values of the spin projection on an axis z' inclined at angle θ to the z axis.

Solution.

The mean spin vector $\bar{\mathbf{s}}$ is plainly directed along the z axis and has magnitude $1/2$. Its projection on the z' axis gives the mean spin in that direction as $\bar{s}_{z'} = (1/2)\cos\theta$. On the other hand, $\bar{s}_{z'} = (1/2)(w_+ - w_-)$, where $w_\pm$ are the probabilities of the values $s_{z'} = \pm 1/2$. Together with $w_+ + w_- = 1$, this gives

$$w_+ = \cos^2(\theta/2), \qquad w_- = \sin^2(\theta/2).$$

§ 55. The spin operator

In the remainder of this chapter we shall not be concerned with the coordinate dependence of wave functions. When discussing, for example, the behavior of $\psi(x, y, z; \sigma)$ under a rotation of the coordinate system, one may suppose the particle to be at the origin. Its coordinates then remain unchanged by the rotation, and the results characterize specifically the dependence of ψ on the spin variable σ.

The variable σ differs from ordinary variables (coordinates) in being discrete. The most general linear operator acting on functions of a discrete variable σ may be written

$$(\hat{f}\psi)(\sigma) = \sum_{\sigma'} f_{\sigma\sigma'}\psi(\sigma'), \tag{55.1}$$

where $f_{\sigma\sigma'}$ are constants. The parentheses around $\hat{f}\psi$ emphasize that the following spin argument belongs not to the original function ψ, but to the function produced by $\hat{f}$. It is readily seen that the $f_{\sigma\sigma'}$ coincide with the matrix elements of the operator, defined by the usual rule (11.5).[6] Integration over the coordinates in (11.5) is now replaced by summation over the discrete variable, so the matrix element is defined by

$$f_{\sigma_2\sigma_1} = \sum_{\sigma} \psi^*_{\sigma_2}(\sigma)[\hat{f}\psi_{\sigma_1}(\sigma)]. \tag{55.2}$$

Here $\psi_{\sigma_1}(\sigma)$ and $\psi_{\sigma_2}(\sigma)$ are eigenfunctions of $\hat{s}_z$ with eigenvalues $s_z = \sigma_1$ and $s_z = \sigma_2$. Each function describes a state in which the particle has a definite s_z; only one component of the wave function is nonzero.[7]

$$\psi_{\sigma_1}(\sigma) = \delta_{\sigma\sigma_1}, \qquad \psi_{\sigma_2}(\sigma) = \delta_{\sigma\sigma_2}. \tag{55.3}$$

According to (55.1),

$$(\hat{f}\psi_{\sigma_1})(\sigma) = \sum_{\sigma'} f_{\sigma\sigma'}\psi_{\sigma_1}(\sigma') = \sum_{\sigma'} f_{\sigma\sigma'}\delta_{\sigma'\sigma_1} = f_{\sigma\sigma_1}.$$

Substitution of this result and $\psi_{\sigma_2}(\sigma)$ in (55.2) makes that equation an identity, proving the assertion.

Operators acting on functions of σ can thus be represented by matrices of order $2s + 1$. This applies in particular to the spin operator itself, whose action on a wave function is, by (55.1),

$$(\hat{\mathbf{s}}\psi)(\sigma) = \sum_{\sigma'} \mathbf{s}_{\sigma\sigma'}\psi(\sigma'). \tag{55.4}$$

As stated above (at the end of § 54), the matrices $\hat{s}_x$, $\hat{s}_y$, $\hat{s}_z$ coincide with the matrices L_x, L_y, L_z obtained in § 27, with L, M replaced by s, σ:

$$\begin{aligned}
(s_x)_{\sigma,\sigma-1} &= (s_x)_{\sigma-1,\sigma} = \frac{1}{2}\sqrt{(s+\sigma)(s-\sigma+1)},\\
(s_y)_{\sigma,\sigma-1} &= -(s_y)_{\sigma-1,\sigma} = -\frac{i}{2}\sqrt{(s+\sigma)(s-\sigma+1)},\\
(s_z)_{\sigma\sigma} &= \sigma.
\end{aligned} \tag{55.5}$$

This defines the spin operator.

[6] Notice that the indices of the matrix elements on the right-hand side of (55.1) occur in an order which is, in a certain sense, the reverse of the usual order in (11.11).

[7] More precisely, one should write $\psi_{\sigma_1}(\sigma) = \psi(x, y, z)\,\delta_{\sigma_1\sigma}$; coordinate factors irrelevant here have been omitted in (55.3).

We emphasize once again the need to distinguish the specified eigenvalue s_z (σ_1 or σ_2) from the independent variable σ. This is the reason for the difference between the notation in (11.11) and (55.1).

In the most important case of spin 1/2 ($s = 1/2$, $\sigma = \pm 1/2$), these are matrices of order two. They are written

$$\hat{\mathbf{s}} = \frac{1}{2}\hat{\boldsymbol{\sigma}}, \tag{55.6}$$

where[8]

$$\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{55.7}$$

The matrices (55.7) are called the *Pauli matrices*. The matrix $\hat{s}_z = \frac{1}{2}\hat{\sigma}_z$ is diagonal, as it must be when defined with respect to the eigenfunctions of s_z itself.[9]

We note several special properties of the Pauli matrices. Direct multiplication of (55.7) gives

$$\hat{\sigma}_y\hat{\sigma}_z = i\hat{\sigma}_x, \qquad \hat{\sigma}_z\hat{\sigma}_x = i\hat{\sigma}_y, \qquad \hat{\sigma}_x\hat{\sigma}_y = i\hat{\sigma}_z. \tag{55.8}$$

Combining these equalities with the general commutation rules (54.1), we find

$$\hat{\sigma}_i\hat{\sigma}_k + \hat{\sigma}_k\hat{\sigma}_i = 0 \quad (i \neq k), \tag{55.9}$$

so the Pauli matrices anticommute. These relations readily give the useful formulas

$$\hat{\boldsymbol{\sigma}}^{,2} = 3, \qquad (\hat{\boldsymbol{\sigma}}\mathbf{a})(\hat{\boldsymbol{\sigma}}\mathbf{b}) = \mathbf{ab} + i\hat{\boldsymbol{\sigma}}(\mathbf{a}\times\mathbf{b}), \tag{55.10}$$

where $\mathbf{a}$ and $\mathbf{b}$ are arbitrary vectors.[10] It follows from these relations that any scalar polynomial expression in the matrices $\hat{\sigma}_i$ reduces to terms independent of $\hat{\boldsymbol{\sigma}}$ and terms of first degree in $\hat{\boldsymbol{\sigma}}$. Hence any scalar function whatsoever of the operator $\hat{\boldsymbol{\sigma}}$ reduces to a linear function (see Problem 1). Finally, the traces (sums of diagonal elements) of the Pauli matrices and their products are

$$\operatorname{Tr}\hat{\sigma}_i = 0, \qquad \operatorname{Tr}(\hat{\sigma}_i\hat{\sigma}_k) = 2\delta_{ik}. \tag{55.11}$$

The following sections of this chapter give a detailed study of the spin properties of wave functions, including their behavior under arbitrary rotations of the coordinate system. Here we immediately note one important property: their behavior under rotations about the z axis.

Perform an infinitesimal rotation through $\delta\varphi$ about the z axis. The operator of this rotation is expressed through the angular-momentum operator, here the spin, as $1 + i\delta\varphi\hat{s}_z$. The functions $\psi(\sigma)$ therefore become $\psi(\sigma) + \delta\psi(\sigma)$, where

$$\delta\psi(\sigma) = i\delta\varphi\hat{s}_z\psi(\sigma) = i\sigma\psi(\sigma)\delta\varphi.$$

[8]When the matrices are written as in (55.7), their rows and columns are numbered by the values of σ, the row number corresponding to the first matrix-element index and the column number to the second. Here these numbers are $+1/2, -1/2$. According to (55.4), the operator acts by multiplying row σ of the matrix by the wave-function components arranged as the column $\psi = \begin{pmatrix} \psi(1/2) \\ \psi(-1/2) \end{pmatrix}$.

[9]Using the same letter for the spin projection and the Pauli matrices causes no ambiguity: the Pauli matrices carry a hat.

[10]Terms on the right-hand sides of (55.8)–(55.10) that do not depend on $\hat{\boldsymbol{\sigma}}$ are, of course, to be understood as constants multiplied by the unit matrix of order two.

Writing this as $d\psi/d\varphi = i\sigma\psi(\sigma)$ and integrating, we find that a finite rotation through φ transforms $\psi(\sigma)$ into

$$\psi'(\sigma) = e^{i\sigma\varphi}\psi(\sigma). \tag{55.12}$$

In particular, a rotation through 2π multiplies the functions by $e^{2\pi i\sigma}$, the same factor for every σ, equal to $(-1)^{2s}$ (the integer 2σ always has the same parity as $2s$). Thus a complete rotation of the coordinate system about the z axis restores the original wave functions of a particle with integral spin, whereas those of a particle with half-integral spin change sign.

Problem

1. Reduce an arbitrary function of the scalar $a + \mathbf{b}\hat{\boldsymbol{\sigma}}$, which is linear in the Pauli matrices, to a linear function.

Solution.

To determine the coefficients in

$$f(a + \mathbf{b}\hat{\boldsymbol{\sigma}}) = A + B\mathbf{b}\hat{\boldsymbol{\sigma}},$$

choose the z axis along $\mathbf{b}$. The eigenvalues of $a+\mathbf{b}\hat{\boldsymbol{\sigma}}$ are then $a \pm b$, and the corresponding eigenvalues of $f(a + \mathbf{b}\hat{\boldsymbol{\sigma}})$ are $f(a \pm b)$. Hence

$$A = \frac{1}{2}[f(a+b) + f(a-b)], \qquad B = \frac{1}{2b}[f(a+b) - f(a-b)].$$

2. Determine the values of the scalar product $\mathbf{s}_1\mathbf{s}_2$ of the spins (1/2) of two particles in states where their resultant spin $\mathbf{S} = \mathbf{s}_1 + \mathbf{s}_2$ has a definite value, 0 or 1.

Solution.

The general formula (31.3), valid for the addition of any two angular momenta, gives

$$\mathbf{s}_1\mathbf{s}_2 = 1/4 \quad \text{for } S = 1, \qquad \mathbf{s}_1\mathbf{s}_2 = -3/4 \quad \text{for } S = 0.$$

3. Which powers of the operator s_z for arbitrary spin s are independent?

Solution.

The operator

$$(\hat{s}_z - s)(\hat{s}_z - s + 1)\cdots(\hat{s}_z + s),$$

formed from the differences between $\hat{s}_z$ and all its possible eigenvalues, gives zero when applied to any wave function and is therefore itself zero. It follows that $(\hat{s}_z)^{2s+1}$ can be expressed in terms of lower powers of $\hat{s}_z$, so only its powers from 1 through $2s$ are independent.

§ 56. Spinors

For zero spin the wave function has only one component, $\psi(0)$. Application of the spin operator gives zero: $\hat{\mathbf{s}}\psi = 0$. Since $\mathbf{s}$ is connected with the operator of infinitesimal rotations, this means that the wave function of a spin-zero particle is unchanged by rotations of the coordinate system; it is therefore a scalar.

The wave function of a spin-1/2 particle has two components, $\psi(1/2)$ and $\psi(-1/2)$. For convenience in later generalizations, distinguish them by the upper indices 1 and 2, respectively. The two-component quantity

$$\psi = \begin{pmatrix} \psi^1 \\ \psi^2 \end{pmatrix} = \begin{pmatrix} \psi(1/2) \\ \psi(-1/2) \end{pmatrix} \tag{56.1}$$

is called a *spinor*.

Under an arbitrary rotation of the coordinate system, the spinor components undergo the linear transformation

$$\psi^{1'} = a\psi^1 + b\psi^2, \qquad \psi^{2'} = c\psi^1 + d\psi^2. \tag{56.2}$$

This can be written

$$\psi^{\lambda'} = (\hat{U}\psi)^\lambda, \qquad \hat{U} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \tag{56.3}$$

where $\hat{U}$ is the transformation matrix.[11] In general, the elements of this matrix are complex functions of the angles through which the coordinate axes are rotated. Relations among them follow directly from the physical requirements imposed on a spinor as a particle's wave function.

Consider the bilinear form

$$\psi^1\varphi^2 - \psi^2\varphi^1, \tag{56.4}$$

where ψ and φ are two spinors. A direct calculation gives

$$\psi^{1'}\varphi^{2'} - \psi^{2'}\varphi^{1'} = (ad - bc)(\psi^1\varphi^2 - \psi^2\varphi^1).$$

Thus (56.4) transforms into itself under a rotation of the coordinate system. If there is only one function transforming into itself, it can be regarded as corresponding to zero spin and must consequently be a scalar, remaining unchanged under every rotation. We therefore obtain

$$ad - bc = 1; \tag{56.5}$$

the determinant of the transformation matrix is unity.[12]

Further relations arise from the requirement that

$$\psi^1\psi^{1*} + \psi^2\psi^{2*} \tag{56.6}$$

be a scalar. This expression determines the probability of finding the particle at a given point in space. A transformation that leaves invariant the sum of the squared moduli of the transformed quantities is unitary, so $\hat{U}^+ = \hat{U}^{-1}$ (see § 12). Subject to (56.5), the inverse matrix is

$$\hat{U}^{-1} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

[11]The notation $\hat{U}\psi$ means multiplication of the rows of $\hat{U}$ by the column ψ.

[12]Such a transformation of two quantities is called *binary*.

Equating it to the adjoint matrix

$$\hat{U}^{+} = \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix},$$

we find

$$a = d^*, \qquad b = -c^*. \tag{56.7}$$

By (56.5) and (56.7), the four complex quantities a, b, c, d actually contain only three independent real parameters, corresponding to the three angles that specify a rotation of a three-dimensional coordinate system.

Comparison of the scalar expressions (56.4) and (56.6) shows that ψ^{1*}, ψ^{2*} must transform like $\psi^2, -\psi^1$. It is readily verified from (56.5) and (56.7) that they do.[13]

Spinor algebra can be cast in a form analogous to tensor algebra. Alongside the contravariant components ψ^1, ψ^2, bearing upper indices, introduce the *covariant* components, bearing lower indices, by the definition

$$\psi_1 = \psi^2, \qquad \psi_2 = -\psi^1. \tag{56.8}$$

The invariant combination (56.4) of two spinors then takes the form of the scalar product

$$\psi_\lambda \varphi^\lambda = \psi_1 \varphi^1 + \psi_2 \varphi^2. \tag{56.9}$$

Here and below, as in tensor algebra, summation is understood over any twice repeated (dummy) index. The following rule must be kept in mind in spinor algebra. We have $\psi^\lambda \varphi_\lambda = \psi^1 \varphi_1 + \psi^2 \varphi_2 = -\psi^2 \varphi^1 - \psi^1 \varphi^2$, and therefore

$$\psi^\lambda \varphi_\lambda = -\psi_\lambda \varphi^\lambda. \tag{56.10}$$

It follows at once that the scalar product of any spinor with itself is zero:

$$\psi_\lambda \psi^\lambda = 0. \tag{56.11}$$

As stated above, ψ_1, ψ_2 transform like ψ^{1*}, ψ^{2*}; hence

$$\psi'_\lambda = (\hat{U}^* \psi)_\lambda. \tag{56.12}$$

The product $\hat{U}^* \psi$ may also be written $\psi \hat{U}^*$ with the transposed matrix $\hat{U}^*$. Since $\hat{U}$ is unitary, $\hat{U}^* = \widetilde{\hat{U}}^{-1}$, so $\widetilde{\psi}' = (\widetilde{\psi} \hat{U}^{-1})$, or[14]

$$\psi'_\lambda = \psi_\mu (U^{-1})^\mu{}_\lambda. \tag{56.13}$$

[13] This property is closely connected with time-reversal symmetry. Time reversal corresponds (see § 18) to replacing the wave function by its complex conjugate, but it also reverses the signs of angular-momentum projections. Functions complex conjugate to the components $\psi^1 = \psi(1/2)$ and $\psi^2 = \psi(-1/2)$ must therefore be equivalent in their properties to the components corresponding to spin projections $-1/2$ and $1/2$, respectively.

[14] Notation of the form $\psi \hat{U}$, with ψ to the left of $\hat{U}$, means multiplication of the row of components (ψ_1, ψ_2) by the columns of $\hat{U}$.

Just as tensors are introduced from vectors in ordinary tensor algebra, one can introduce *spinors of higher rank*. A second-rank spinor, for example, is a four-component quantity $\psi^{\lambda\mu}$ whose components transform like the products $\psi^{\lambda}\varphi^{\mu}$ of components of two first-rank spinors. Along with the contravariant components $\psi^{\lambda\mu}$, one may consider the covariant components $\psi_{\lambda\mu}$ and mixed components $\psi_{\lambda}{}^{\mu}$, transforming respectively like $\psi_{\lambda}\varphi_{\mu}$ and $\psi_{\lambda}\varphi^{\mu}$. Spinors of any rank are defined similarly. Passage from contravariant to covariant spinor components and back can be written

$$\psi_{\lambda} = g_{\lambda\mu}\psi^{\mu}, \qquad \psi^{\lambda} = g^{\mu\lambda}\psi_{\mu}, \tag{56.14}$$

where

$$g_{11} = g_{22} = 0, \qquad g_{12} = -g_{21} = 1 \tag{56.15}$$

is the *metric spinor* in a two-dimensional vector space.[15] Similarly, for example,

$$\psi_{\lambda}{}^{\mu} = g_{\lambda\nu}\psi^{\nu\mu}, \qquad \psi_{\lambda\mu} = g_{\lambda\nu}g_{\mu\rho}\psi^{\nu\rho},$$

so that $\psi_1{}^2 = -\psi^{11}$, $\psi_2{}^1 = -\psi^{21}$, $\psi_{11} = \psi^{22}$, and so forth.

The $g_{\lambda\mu}$ themselves form the antisymmetric unit spinor of second rank. It is readily verified that their components remain unchanged under coordinate transformations and that

$$g_{\lambda\nu}g^{\mu\nu} = \delta^{\mu}_{\lambda}, \tag{56.16}$$

where $\delta^1_1 = \delta^2_2 = 1$ and $\delta^2_1 = \delta^1_2 = 0$.

As in ordinary tensor algebra, spinor algebra has two fundamental operations: multiplication and simplification (or contraction) over a pair of indices. The product of two spinors is a spinor of higher rank. Thus the second- and third-rank spinors $\psi_{\lambda\mu}$ and $\varphi^{\nu\rho\sigma}$ form the fifth-rank spinor $\psi_{\lambda\mu}\varphi^{\nu\rho\sigma}$. Simplification over a pair of indices, that is, summation of components over equal values of one covariant and one contravariant index, lowers the rank by two. For example, simplifying $\psi_{\lambda\mu}{}^{\nu\rho\sigma}$ over μ and ν gives the third-rank spinor $\psi_{\lambda\mu}{}^{\mu\rho\sigma}$; simplifying $\varphi_{\lambda}{}^{\mu}$ gives the scalar $\varphi_{\lambda}{}^{\lambda}$. A rule analogous to (56.10) applies: interchanging the upper and lower positions of the indices over which simplification is performed changes the sign, so $\varphi_{\lambda}{}^{\lambda} = -\varphi^{\lambda}{}_{\lambda}$. In particular, if a spinor is symmetric in any two of its indices, simplification over those indices gives zero. Thus a symmetric second-rank spinor $\varphi_{\lambda}{}^{\mu}$ has $\varphi_{\lambda}{}^{\lambda} = 0$.

A *symmetric spinor* of rank n is one symmetric in all its indices. A symmetric spinor can be constructed from an asymmetric one by symmetrization, that is, by summing the components obtained from all possible permutations of the indices. It follows from the foregoing that the components of a symmetric spinor cannot be contracted to form a spinor of lower rank.

A spinor antisymmetric in all its indices, on the other hand, can only have rank two. Indeed, because every index takes only two values, with three or more indices at least two must have equal values, and the spinor components then vanish identically. Every antisymmetric second-rank spinor reduces to a scalar multiplied by the unit spinor $g_{\lambda\mu}$. We record the following consequence:

$$g_{\lambda\mu}\psi_{\nu} + g_{\mu\nu}\psi_{\lambda} + g_{\nu\lambda}\psi_{\mu} = 0, \tag{56.17}$$

[15] The matrix (56.15) coincides with $i\hat{\sigma}_y$.

where ψ_λ is an arbitrary spinor. This rule follows simply because the expression on the left is, as can readily be verified, an antisymmetric third-rank spinor.

The spinor formed by multiplying $\psi_{\lambda\mu}$ by itself and contracting one pair of indices is antisymmetric in the other pair; indeed,

$$\psi_{\lambda\nu}\psi_\mu{}^\nu = -\psi_\lambda{}^\nu\psi_{\mu\nu}.$$

It must therefore reduce to $g_{\lambda\mu}$ multiplied by a scalar. Fixing the latter so that contraction over the second pair gives the correct result, we find

$$\psi_{\lambda\nu}\psi_\mu{}^\nu = -\frac{1}{2}\psi_{\rho\sigma}\psi^{\rho\sigma}g_{\lambda\mu}. \tag{56.18}$$

The components of the spinor $\varphi^{\lambda\mu\cdots}$ complex conjugate to $\psi_{\lambda\mu\dots}$ transform as components of a contravariant spinor, and conversely. The sum of the squared moduli of the components of any spinor is therefore invariant.

§ 57. Wave functions of particles with arbitrary spin

Having developed the formal algebra of spinors of arbitrary rank, we can turn to the task at hand: studying the properties of wave functions of particles with arbitrary spin.

It is convenient to approach this question by considering a collection of n particles of spin $1/2$. The largest possible value of the z component of the system's total spin is $n/2$; it is obtained when every particle has $s_z = 1/2$ (all spins point in the same direction, along the z axis). In this case the total spin S of the system must also equal $n/2$. All components of the system's wave function $\psi(\sigma_1, \sigma_2, \dots, \sigma_n)$ then vanish except one, $\psi(1/2, 1/2, \dots, 1/2)$. If the wave function is written as a product of n spinors $\psi^\lambda\varphi^\mu\dots$, each referring to one particle, only the component with $\lambda, \mu, \dots = 1$ is nonzero in every factor. Thus only the product $\psi^1\varphi^1\dots$ is nonzero. The collection of all these products, however, is an nth-rank spinor symmetric in all its indices. After a transformation of the coordinate system (so that the spins no longer point along z), it becomes a general nth-rank spinor, but remains symmetric.

The spin properties of wave functions are essentially their properties under rotations of the coordinate system. They are therefore identical for a particle of spin s and for a system of $n = 2s$ spin-$1/2$ particles oriented so that the system's total spin is s. It follows that the wave function of a particle of spin s is a symmetric spinor of rank $n = 2s$.

The number of independent components of a symmetric spinor of rank $2s$ is, as required, $2s + 1$. Indeed, the only distinct components are those whose indices contain $2s$ ones and no twos, $2s - 1$ ones and one two, and so on, down to no ones and $2s$ twos.

Mathematically, symmetric spinors classify the possible transformation types of quantities under rotations of the coordinate system. If $2s + 1$ distinct quantities transform linearly among themselves (and their number cannot be reduced by any choice of linear combinations), their transformation law is equivalent to that of the components of a symmetric spinor of rank $2s$. Any set of any number of functions which transform linearly among themselves under rotations can be reduced, by a suitable linear transformation, to one or more symmetric spinors[16].

[16] In other words, symmetric spinors realize irreducible representations of the rotation group (see § 98).

Thus an arbitrary nth-rank spinor $\psi^{\lambda\mu\nu\cdots}$ can be reduced to symmetric spinors of ranks $n, n-2, n-4, \ldots$. This reduction can be carried out as follows. Symmetrizing $\psi^{\lambda\mu\nu\cdots}$ over all indices gives a symmetric spinor of the same rank n. Next, contracting the original spinor $\psi^{\lambda\mu\nu\cdots}$ over different pairs of indices gives spinors of rank $n-2$, of the form $\psi^{\lambda}{}_{\lambda}{}^{\nu\cdots}$; symmetrizing these in turn produces symmetric spinors of rank $n-2$. Symmetrizing the spinors obtained by contracting $\psi^{\lambda\mu\cdots}{}_{\lambda\mu}$ over two pairs of indices produces symmetric spinors of rank $n-4$, and so forth.

It remains to relate the components of the symmetric spinor of rank $2s$ to the $2s+1$ functions $\psi(\sigma)$, where $\sigma = s, s-1, \ldots, -s$. The component

$$\psi^{\overbrace{11\ldots1}^{s+\sigma}\overbrace{22\ldots2}^{s-\sigma}},$$

in which 1 occurs $s+\sigma$ times and 2 occurs $s-\sigma$ times, corresponds to a spin projection σ on the z axis. To see this, again replace one particle of spin s by a system of $n = 2s$ spin-1/2 particles. The component written above then corresponds to the product

$$\underbrace{\psi^1\varphi^1\ldots}_{s+\sigma}\underbrace{\chi^2\rho^2\ldots}_{s-\sigma}.$$

This product describes a state in which $s+\sigma$ particles have spin projection $+1/2$ and $s-\sigma$ particles have projection $-1/2$; the total projection is therefore $\frac{1}{2}(s+\sigma) - \frac{1}{2}(s-\sigma) = \sigma$. Finally, choose the proportionality factor between this spinor component and $\psi(\sigma)$ so that

$$\sum_{\sigma=-s}^{+s} |\psi(\sigma)|^2 = \sum_{\lambda,\mu,\ldots=1}^{2} |\psi^{\lambda\mu\cdots}|^2 \tag{57.1}$$

(this sum is a scalar, as it must be, since it determines the probability of finding the particle at a given point). In the sum on the right, components with $s+\sigma$ indices equal to 1 occur

$$\frac{(2s)!}{(s+\sigma)!(s-\sigma)!}$$

times. Hence the correspondence is

$$\psi(\sigma) = \left[\frac{(2s)!}{(s+\sigma)!(s-\sigma)!}\right]^{1/2} \psi^{\overbrace{11\ldots1}^{s+\sigma}\overbrace{22\ldots2}^{s-\sigma}}. \tag{57.2}$$

Relation (57.2) ensures not only condition (57.1), but also, as is readily verified, the more general condition

$$\psi^{\lambda\mu\cdots}\varphi_{\lambda\mu\ldots} = \sum_{\sigma}(-1)^{s-\sigma}\psi(\sigma)\varphi(-\sigma), \tag{57.3}$$

where $\psi^{\lambda\mu\cdots}$ and $\varphi_{\lambda\mu\ldots}$ are two different spinors of the same rank, while $\psi(\sigma)$ and $\varphi(\sigma)$ are the functions associated with them by (57.2). The factor $(-1)^{s-\sigma}$ appears because raising all indices changes the sign once for every index equal to 2.

Equations (55.5) specify the action of the spin operator on the wave functions $\psi(\sigma)$. Its action on a wave function written as a spinor of rank $2s$ is also easily found. For spin $1/2$, the functions $\psi(+1/2)$ and $\psi(-1/2)$ are the spinor components ψ^1 and ψ^2. According to (55.6) and (55.7), the spin operators act on them as

$$\begin{aligned} (\hat{s}_x\psi)^1 &= \tfrac{1}{2}\psi^2, \quad (\hat{s}_y\psi)^1 = -\tfrac{i}{2}\psi^2, \quad (\hat{s}_z\psi)^1 = \tfrac{1}{2}\psi^1, \\ (\hat{s}_x\psi)^2 &= \tfrac{1}{2}\psi^1, \quad (\hat{s}_y\psi)^2 = \tfrac{i}{2}\psi^1, \quad\;\; (\hat{s}_z\psi)^2 = -\tfrac{1}{2}\psi^2. \end{aligned} \tag{57.4}$$

To pass to arbitrary spin, once more consider a system of $2s$ spin-$1/2$ particles and write its wave function as a product of $2s$ spinors. The spin operator of the system is the sum of the spin operators of its particles; each acts only on the corresponding spinor, according to (57.4). Returning then to arbitrary symmetric spinors, and hence to the wave functions of a particle of spin s, gives

$$\begin{aligned} (\hat{s}_x\psi)^{\overbrace{11\ldots}^{s+\sigma}\overbrace{22\ldots}^{s-\sigma}} &= \frac{s+\sigma}{2}\psi^{\overbrace{11\ldots}^{s+\sigma-1}\overbrace{22\ldots}^{s-\sigma+1}} + \frac{s-\sigma}{2}\psi^{\overbrace{11\ldots}^{s+\sigma+1}\overbrace{22\ldots}^{s-\sigma-1}}, \\ (\hat{s}_y\psi)^{\overbrace{11\ldots}^{s+\sigma}\overbrace{22\ldots}^{s-\sigma}} &= -i\frac{s+\sigma}{2}\psi^{\overbrace{11\ldots}^{s+\sigma-1}\overbrace{22\ldots}^{s-\sigma+1}} + i\frac{s-\sigma}{2}\psi^{\overbrace{11\ldots}^{s+\sigma+1}\overbrace{22\ldots}^{s-\sigma-1}}, \\ (\hat{s}_z\psi)^{\overbrace{11\ldots}^{s+\sigma}\overbrace{22\ldots}^{s-\sigma}} &= \sigma\psi^{\overbrace{11\ldots}^{s+\sigma}\overbrace{22\ldots}^{s-\sigma}}. \end{aligned} \tag{57.5}$$

So far spinors have been discussed as wave functions of the intrinsic angular momentum of elementary particles. Formally, however, there is no distinction between the spin of a single particle and the total angular momentum of any system considered as a whole, with its internal structure disregarded. The transformation properties of spinors therefore apply equally to the behavior under spatial rotations of the wave functions ψ_{jm} of any particle (or system of particles) with total angular momentum j, irrespective of whether it is orbital or spin angular momentum. A definite correspondence must consequently exist between the transformation laws of the eigenfunctions ψ_{jm} under rotations and those of the components of a symmetric spinor of rank $2j$.

In establishing this correspondence, two aspects of the dependence of wave functions on the angular-momentum projection m (for fixed j) must be clearly distinguished. A wave function may be regarded as a probability amplitude for different values of m, or as an eigenfunction belonging to a specified value of m.

Both aspects occurred at the beginning of § 55, where the eigenfunction $\delta_{\sigma\sigma_0}$ of s_z, corresponding to $s_z = \sigma_0$, was considered. Their mathematical difference is particularly clear for a particle of spin $s = 1/2$. As a function of σ, its spin function is a contravariant spinor of first rank and should therefore be written $\delta^{\lambda}{}_{\sigma_0}$. With respect to σ_0 it is consequently a covariant spinor.

This fact is evidently general: the eigenfunctions ψ_{jm} can be put in correspondence with the components of a covariant symmetric spinor of rank $2j$ by formulas analogous to (57.2)[17]:

$$\psi_{jm} = \left[\frac{(2j)!}{(j+m)!(j-m)!}\right]^{1/2} \psi_{\underbrace{11\ldots}_{j+m}\underbrace{22\ldots}_{j-m}}. \tag{57.6}$$

The eigenfunctions of integer angular momentum j are the spherical functions Y_{jm}. The case $j = 1$ is especially important. The three spherical functions Y_{1m} are

$$Y_{10} = i\sqrt{\frac{3}{4\pi}}\cos\theta = i\sqrt{\frac{3}{4\pi}}n_z, \qquad Y_{1,\pm1} = \mp i\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\varphi} = \mp i\sqrt{\frac{3}{8\pi}}(n_x \pm in_y),$$

where **n** is the unit vector along the radius vector. Their transformation properties are therefore equivalent to those of the components of a vector **a**, with the correspondence

$$\psi_{10} = ia_z, \qquad \psi_{11} = -\frac{i}{\sqrt{2}}(a_x + ia_y), \qquad \psi_{1,-1} = \frac{i}{\sqrt{2}}(a_x - ia_y). \tag{57.7}$$

Comparison with (57.6) shows that the components of a symmetric second-rank spinor correspond to the components of a vector according to

$$\psi_{11} = -\frac{i}{\sqrt{2}}(a_x + ia_y), \qquad \psi_{12} = \psi_{21} = \frac{i}{\sqrt{2}}a_z, \qquad \psi_{22} = \frac{i}{\sqrt{2}}(a_x - ia_y), \tag{57.8}$$

or

$$\psi^{11} = \frac{i}{\sqrt{2}}(a_x - ia_y), \qquad \psi^{12} = \psi^{21} = -\frac{i}{\sqrt{2}}a_z, \qquad \psi^{22} = -\frac{i}{\sqrt{2}}(a_x + ia_y). \tag{57.9}$$

Conversely,

$$a_z = i\sqrt{2}\psi^{12}, \qquad a_x = \frac{i}{\sqrt{2}}(\psi^{22} - \psi^{11}), \qquad a_y = \frac{1}{\sqrt{2}}(\psi^{11} + \psi^{22}). \tag{57.10}$$

With these definitions it is readily checked that

$$\psi^{\lambda\mu}\varphi_{\lambda\mu} = \mathbf{ab}, \tag{57.11}$$

where **a** and **b** correspond to the symmetric spinors $\psi^{\lambda\mu}$ and $\varphi_{\lambda\mu}$. One can likewise verify the correspondence[18]

$$\psi^{\lambda}{}_{h}o\varphi^{\rho\mu} - \varphi^{\lambda}{}_{h}o\psi^{\rho\mu} \longleftrightarrow i\sqrt{2}(\mathbf{a}\times\mathbf{b}). \tag{57.12}$$

Equations (57.10) can be written compactly with the Pauli matrices:

$$a_i = \frac{i}{\sqrt{2}}(\hat{\sigma}_i)^{\lambda}{}_{m}u\psi^{\mu}{}_{l}ambda. \tag{57.13}$$

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[17]The same result can be reached in a slightly different way. Expand the wave function ψ of a particle in a state of angular momentum j in the eigenfunctions ψ_{jm}: $\psi = \sum_m a_m\psi_{jm}$. The coefficients a_m are probability amplitudes for the different values of m. In this sense they correspond to the "components" $\psi(m)$ of the spin wave function, which determines their transformation law. On the other hand, the value of ψ at a given point cannot depend on the choice of coordinates, so $\sum a_m\psi_{jm}$ must be a scalar. Comparison with the scalar (57.3) shows that a_m must transform as $(-1)^{j-m}\psi_{j,-m}$.

[18]The mixed components of a symmetric spinor may be written as $\psi^{\lambda}{}_{m}u$, without distinguishing $\psi^{\lambda\mu}$ and $\psi^{\mu\lambda}$.

The matrix indices on σ are placed above and below to match the positions of the spinor indices on $\psi^{\mu}{}_{\lambda}$. The origin of this formula is easily understood in the special case where the second-rank spinor $\psi^{\mu}{}_{\lambda}$ is a product of a first-rank spinor ψ^{μ} and its complex conjugate ψ^{*}_{λ}. Then

$$\frac{1}{2}\psi^{*}_{\lambda}\hat{\boldsymbol{\sigma}}^{\lambda}{}_{m}u\psi^{\mu}$$

is the mean spin (for a particle with wave function ψ^{μ}), so its vector character is evident.

The correspondence (57.8), or (57.9), is a special case of a general rule: to every symmetric spinor of even rank $2j$, with integer j, there corresponds a symmetric tensor of half that rank, j, which vanishes upon contraction over any pair of indices (such a tensor will be called irreducible). This already follows from the equality of the numbers of independent components of the spinor and tensor, both $2j + 1$, as a direct count shows[19]. The correspondence between their components can be found from (57.8)–(57.10) by regarding the spinor as a product of several second-rank spinors and the tensor as a product of vectors.

Problem

1. Rewrite the definition (57.4) of the spin-1/2 operator in terms of the spinor components of the vector **s**.

Solution.

Using (57.9), which relates the vector **s** to the spinor $s^{\lambda\mu}$, definition (57.4) becomes

$$s^{\lambda\mu}\psi^{\nu} = \frac{i}{2\sqrt{2}}(\psi^{\lambda}g^{\mu\nu} + \psi^{\mu}g^{\lambda\nu}).$$

2. Write formulas defining the action of the spin operator on the vector wave function of a spin-1 particle.

Solution.

The components of the vector function $\boldsymbol{\psi}$ are related to those of the spinor $\psi^{\lambda\mu}$ by (57.9), while the last of (57.5) gives

$$\hat{s}_z\psi_+ = -\psi_+, \qquad \hat{s}_z\psi_- = \psi_-, \qquad \hat{s}_z\psi_z = 0,$$

where $\psi_{\pm} = \psi_x \pm i\psi_y$, or

$$\hat{s}_z\psi_x = -i\psi_y, \qquad \hat{s}_z\psi_y = i\psi_x, \qquad \hat{s}_z\psi_z = 0.$$

The remaining formulas follow by cyclic permutation of x, y, z. Together they are

$$\hat{s}_i\psi_k = -ie_{ik\ell}\psi_\ell.$$

The complex vector $\boldsymbol{\psi}$ can be represented as $\boldsymbol{\psi} = e^{i\alpha}(\mathbf{u} + i\mathbf{v})$, where **u** and **v** are real vectors which, by a suitable choice of the common phase α, can be made mutually perpendicular. They determine a plane with the property that the spin projection on its normal can take only the values ± 1.

[19] In other words, the $2j + 1$ components of an irreducible tensor of rank j (integer j), the $2j + 1$ spherical functions Y_{jm}, and the $2j + 1$ components of a symmetric spinor of rank $2j$ all realize the same irreducible representation of the rotation group.

§ 58. The finite-rotation operator

We return to the transformation of spinors and show how its coefficients can be expressed explicitly in terms of the angles through which the coordinate axes are rotated.

By the definition of the angular-momentum operator (here, the spin operator), $1 + i\delta\varphi\,\mathbf{n}\hat{\mathbf{s}}$ is the operator for a rotation through $\delta\varphi$ about the direction specified by the unit vector $\mathbf{n}$. For the wave function of a spin-1/2 particle, that is, a first-rank spinor, one must set $\hat{\mathbf{s}} = \hat{\boldsymbol{\sigma}}/2$. The operator for a finite rotation through φ about the same direction is therefore

$$\hat{U}_{\mathbf{n}} = \exp(i\varphi\,\mathbf{n}\hat{\boldsymbol{\sigma}}/2) \tag{58.1}$$

(cf. (15.13)). Like any function of the Pauli matrices (see problem 1, § 55), it reduces to an expression linear in those matrices:

$$\hat{U}_{\mathbf{n}} = \cos(\varphi/2) + i\mathbf{n}\hat{\boldsymbol{\sigma}}\sin(\varphi/2). \tag{58.2}$$

Thus, for a rotation about the z axis,

$$\hat{U}_z(\varphi) = \cos\frac{\varphi}{2} + i\hat{\sigma}_z\sin\frac{\varphi}{2} = \begin{pmatrix} e^{i\varphi/2} & 0 \\ 0 & e^{-i\varphi/2} \end{pmatrix}. \tag{58.3}$$

The spinor components consequently transform as

$$\psi^{1'} = \psi^1 e^{i\varphi/2}, \qquad \psi^{2'} = \psi^2 e^{-i\varphi/2}.$$

In particular, a rotation through 2π reverses the signs of the spinor components; spinors of every odd rank therefore have the same property (cf. the end of § 55).

Similarly, the matrices for rotations through φ about the x and y axes are

$$\hat{U}_x(\varphi) = \begin{pmatrix} \cos(\varphi/2) & i\sin(\varphi/2) \\ i\sin(\varphi/2) & \cos(\varphi/2) \end{pmatrix}, \qquad \hat{U}_y(\varphi) = \begin{pmatrix} \cos(\varphi/2) & \sin(\varphi/2) \\ -\sin(\varphi/2) & \cos(\varphi/2) \end{pmatrix}. \tag{58.4}$$

For the special case of a rotation through π about the y axis,

$$\psi^{1'} = \psi^2, \qquad \psi^{2'} = -\psi^1, \qquad \text{i.e.} \qquad \psi^{\lambda'} = \psi_\lambda. \tag{58.5}$$

The transformation matrix for an arbitrary rotation can now readily be written in terms of the Euler angles that specify it.

A rotation of the axes specified by Euler angles α, β, γ is performed in three steps: 1) a rotation through α $(0 \leqslant \alpha \leqslant 2\pi)$ about the z axis; 2) a rotation through β $(0 \leqslant \beta \leqslant \pi)$ about the new position of the y axis (ON in Fig. 20, the so-called *line of nodes*); 3) a rotation through γ $(0 \leqslant \gamma \leqslant 2\pi)$ about the resulting final position z' of the z axis[20].

Clearly, α and β coincide with the spherical angles φ and θ of the new z' axis relative to the xyz axes: $\alpha = \varphi, \beta = \theta$.

[20]The systems xyz and $x'y'z'$ are, as always, right-handed, and the positive sense of the angles is the direction of a right-handed screw advancing along the positive direction of the rotation axis.

The definition of Euler angles used here (customary in quantum-mechanical applications) differs from that in § 35 (see vol. I) in that the second rotation is made about the y axis rather than the x axis. The angles α, β, γ are related to the angles φ, θ, ψ of vol. I (not to be confused with the spherical angles φ, θ) by $\psi = \gamma - \pi/2$.

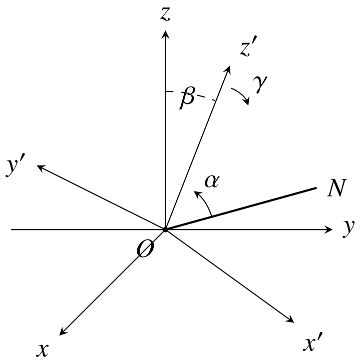

Figure IX.1. Definition of the Euler angles.

For this sequence of rotations the matrix of the complete transformation is the product of three matrices (58.3), (58.4):

$$\hat{U}(\alpha,\beta,\gamma) = \hat{U}_z(\gamma)\hat{U}_y(\beta)\hat{U}_z(\alpha).$$

Direct multiplication gives

$$\hat{U}(\alpha,\beta,\gamma) = \begin{pmatrix} \cos(\beta/2)e^{i(\alpha+\gamma)/2} & \sin(\beta/2)e^{-i(\alpha-\gamma)/2} \\ -\sin(\beta/2)e^{i(\alpha-\gamma)/2} & \cos(\beta/2)e^{-i(\alpha+\gamma)/2} \end{pmatrix}. \tag{58.6}$$

By definition, higher-rank spinors transform as products of first-rank spinor components. In physical applications, however, it is generally not the transformation laws of the spinors themselves that are of principal interest, but those of the corresponding wave functions ψ_{jm}.

Let ψ_{jm}, with $m = j, j-1, \ldots, -j$, describe in the coordinate system xyz a state with a definite angular momentum j, and let $\psi_{jm'}$ describe the same state relative to the axes $x'y'z'$. In the first case m is the value of J_z, while in the second m' is that of $J_{z'}$. The two sets of functions are related linearly; write

$$\psi_{jm} = \sum_{m'} D^{(j)}_{m'm}\psi_{jm'}. \tag{58.7}$$

The coefficients $D^{(j)}_{m'm}$ form, in the indices m', m, a matrix of order $2j+1$, the *finite-rotation matrix* $\hat{D}^{(j)}$. Its elements are functions of the angles α, β, γ through which the system $x'y'z'$ is rotated relative to xyz.

The finite-rotation matrix can be constructed using the spinor representation of the functions ψ_{jm}.

For $j = 1/2$, the two functions $\psi_{1/2,m}$, $m = \pm 1/2$, form a covariant first-rank spinor. According to (56.13), its transformation from $x'y'z'$ to xyz is effected by the matrix $\hat{U}$ of (58.6), so that $\hat{D}^{(1/2)} = \hat{U}$[21]. Write its elements as

$$D^{(1/2)}_{m'm}(\alpha,\beta,\gamma) = e^{im'\gamma}d^{(1/2)}_{m'm}(\beta)e^{im\alpha},$$

[21] Notice that the matrix indices in (58.7) are ordered precisely as required for multiplication of the columns of $\hat{D}^{(j)}$ by the functions $\psi_{jm'}$ arranged in a row. Symbolically, (58.7) would be written $\psi_{jm} = (\boldsymbol{\psi}_j\hat{D}^{(j)})_m$, in accordance with (56.13).

where

$$\begin{array}{c|cc} d^{(1/2)}_{m'm}(\beta) & \frac{1}{2} & -\frac{1}{2} \\ \hline \frac{1}{2} & \cos(\beta/2) & \sin(\beta/2) \\ -\frac{1}{2} & -\sin(\beta/2) & \cos(\beta/2) \end{array}. \tag{58.8}$$

For arbitrary j, the functions ψ_{jm} are related by (57.6) to the components of a symmetric covariant spinor of rank $2j$. Its transformation matrix is a product of $2j$ matrices $\hat{D}^{(1/2)}$, each acting on one spinor index. Multiplication and return to the functions ψ_{jm} gives

$$D^{(j)}_{m'm}(\alpha,\beta,\gamma) = e^{im'\gamma} d^{(j)}_{m'm}(\beta) e^{im\alpha}, \tag{58.9}$$

where[22]

$$\begin{aligned} d^{(j)}_{m'm}(\beta) = &\left[\frac{(j+m')!(j-m')!}{(j+m)!(j-m)!}\right]^{1/2} \left(\cos\frac{\beta}{2}\right)^{m'+m} \\ &\times \left(\sin\frac{\beta}{2}\right)^{m'-m} P^{(m'-m,m'+m)}_{j-m'}(\cos\beta), \end{aligned} \tag{58.10}$$

and

$$\begin{aligned} P^{(a,b)}_n(\cos\beta) = &\frac{(-1)^n}{2^n n!}(1-\cos\beta)^{-a}(1+\cos\beta)^{-b} \\ &\times \left(\frac{d}{d\cos\beta}\right)^n \left[(1-\cos\beta)^{a+n}(1+\cos\beta)^{b+n}\right] \end{aligned} \tag{58.11}$$

are the so-called Jacobi polynomials[23]. Note that

$$d^{(j)}_{m'm}(\beta) = 0 \quad \text{if } |m| > j \text{ or } |m'| > j. \tag{58.12}$$

The functions $d^{(j)}_{m'm}$ have several symmetry properties. They could be read from (58.11) and (58.12), but follow more simply from their definition as coefficients of a rotational transformation.

As a rotational-transformation matrix, $\hat{D}^{(j)}$ is unitary. The inverse of the rotation (α,β,γ) is the rotation $(-\gamma,-\beta,-\alpha)$; since $\hat{d}$ is real, it follows that

$$d^{(j)}_{m'm}(-\beta) = d^{(j)}_{mm'}(\beta). \tag{58.13}$$

Further,

$$d^{(j)}_{m'm}(\beta) = d^{(j)}_{-m,-m'}(\beta), \tag{58.14}$$

and

$$d^{(j)}_{m'm}(\pi) = (-1)^{j+m}\delta_{m',-m}, \qquad d^{(j)}_{m'm}(-\pi) = (-1)^{j-m}\delta_{m',-m}, \qquad d^{(j)}_{m'm}(0) = \delta_{m'm}. \tag{58.15}$$

[22] Details of the calculation may be found in A. R. Edmonds, *Angular Momentum in Quantum Mechanics*, Princeton, 1957 (see also the Russian translation of an article by the same author in the collection *Deformation of Atomic Nuclei*, Moscow: IL, 1958). The definition of $D^{(j)}_{m'm}$ in (58.9), (58.10) differs from Edmonds's book by interchange of α and γ (which is more natural in the present approach), and from the article also by reversal of the signs of all angles.

[23] For the relation of these polynomials to the hypergeometric series, see § e, formula (e.11).

For $j = 1/2$ these relations are immediate from (58.8), and their extension to arbitrary j follows directly from the construction of the transformation matrix described above.

Represent a rotation through $\pi - \beta$ as successive rotations through π and $-\beta$:

$$d^{(j)}_{m'm}(\pi - \beta) = \sum_{m''} d^{(j)}_{m'm''}(\pi) d^{(j)}_{m''m}(-\beta) = (-1)^{j-m'} d^{(j)}_{-m',m}(-\beta),$$

or, using (58.13),

$$d^{(j)}_{m'm}(\pi - \beta) = (-1)^{j-m'} d^{(j)}_{m,-m'}(\beta). \tag{58.16}$$

The result of two rotations about the same axis is independent of their order. Thus the same result must follow if the rotations $-\beta$ and π are made in the reverse order. Doing this and comparing with (58.16) gives

$$d^{(j)}_{m'm}(\beta) = (-1)^{m'-m} d^{(j)}_{-m',-m}(\beta). \tag{58.17}$$

Equations (58.17), (58.14), and (58.13) imply

$$d^{(j)}_{m'm}(\beta) = (-1)^{m'-m} d^{(j)}_{mm'}(\beta) = (-1)^{m'-m} d^{(j)}_{m'm}(-\beta). \tag{58.18}$$

Various symmetry properties of the full functions $D^{(j)}_{m'm}$ follow from (58.13)–(58.18). In particular, their complex conjugates are

$$D^{(j)*}_{m'm}(\alpha, \beta, \gamma) = D^{(j)}_{m'm}(-\alpha, \beta, -\gamma) = (-1)^{m'-m} D^{(j)}_{-m',-m}(\alpha, \beta, \gamma). \tag{58.19}$$

Mathematically, the matrices $\hat{D}^{(j)}$ realize unitary irreducible representations of the rotation group of dimension $2j+1$ (see § 98 below). The orthogonality and normalization relation follows at once:

$$\int D^{(j_1)*}_{m_1'm_1}(\alpha, \beta, \gamma) D^{(j_2)}_{m_2'm_2}(\alpha, \beta, \gamma) \frac{d\omega}{8\pi^2} = \frac{1}{2j_1 + 1} \delta_{j_1 j_2} \delta_{m_1 m_2} \delta_{m_1' m_2'}, \tag{58.20}$$

where $d\omega = \sin\beta \, d\alpha \, d\beta \, d\gamma$. Orthogonality in the indices m and m' is ensured by the factor $\exp\{i(m\alpha + m'\gamma)\}$. Orthogonality in j is associated with the functions $d^{(j)}_{m'm}$, for which

$$\int_0^\pi d^{(j_1)}_{m'm}(\beta) d^{(j_2)}_{m'm}(\beta) \sin\beta \, d\beta = \frac{2}{2j_1 + 1} \delta_{j_1 j_2}. \tag{58.21}$$

For reference, we finally give the functions $d^{(j)}_{m'm}$ for several special parameter values. For $j = 1$,

$$\begin{array}{c|ccc} d^{(1)}_{m'm}(\beta) & 1 & 0 & -1 \\ \hline 1 & \frac{1}{2}(1+\cos\beta) & \frac{1}{\sqrt{2}}\sin\beta & \frac{1}{2}(1-\cos\beta) \\ 0 & -\frac{1}{\sqrt{2}}\sin\beta & \cos\beta & \frac{1}{\sqrt{2}}\sin\beta \\ -1 & \frac{1}{2}(1-\cos\beta) & -\frac{1}{\sqrt{2}}\sin\beta & \frac{1}{2}(1+\cos\beta) \end{array} \tag{58.22}$$

For integer $j = l$ and $m' = 0$, equations (58.10), (58.11) give

$$d^{(l)}_{0m}(\beta) = \left[\frac{(l-m)!}{(l+m)!}\right]^{1/2} P_l^m(\cos\beta). \tag{58.23}$$

The origin of this formula is readily traced from the initial definition (58.7). Refer the functions $\psi_{jm'}$ on the right of (58.7) to the z' axis, on which, for $j = l$,

$$\psi_{lm'}(\mathbf{n}_{z'}) = i^l \sqrt{\frac{2l+1}{4\pi}} \delta_{m'0}. \tag{58.24}$$

The function ψ_{jm} on the left is then the spherical function $Y_{lm}(\beta, \alpha)$ of the spherical angles $\varphi = \alpha$, $\theta = \beta$ of the z' direction. Substitution of (58.24) into (58.7) gives

$$Y_{lm}(\beta, \alpha) = i^l \sqrt{\frac{2l+1}{4\pi}} D^{(l)}_{0m}(\alpha, \beta, \gamma), \tag{58.25}$$

which is equivalent to (58.23). Finally, when one of the indices m, m' has its largest possible value,

$$d^{(j)}_{jm}(\beta) = (-1)^{j-m} d^{(j)}_{-j,m}(\beta) = \left[\frac{(2j)!}{(j+m)!(j-m)!}\right]^{1/2} \left(\cos\frac{\beta}{2}\right)^{j+m} \left(\sin\frac{\beta}{2}\right)^{j-m}. \tag{58.26}$$

§ 59. Partial polarization of particles

For any given spinor ψ^λ, the wave function of a spin-1/2 particle, the direction of the z axis can always be chosen so that one of its components, say ψ^2, vanishes. This is already clear because a direction in space is specified by two quantities (angles): the number of parameters at our disposal is exactly equal to the number of quantities, the real and imaginary parts of the complex ψ^2, that are to be made zero.

Physically, this means that if a spin-1/2 particle—for definiteness, an electron—is in a state described by a certain spin wave function, there is a direction in space along which its spin projection has the definite value $\sigma = 1/2$. The electron in such a state may be said to be *fully polarized*.

There are, however, electron states that may be called *partially polarized*. They are described not by wave functions but only by density matrices; that is, they are mixed states with respect to spin (see § 14).

The spin (or *polarization*) density matrix of the electron is a second-rank spinor $\rho^\lambda{}_m u$, normalized by

$$\rho^\lambda{}_l ambda = \rho^1{}_1 + \rho^2{}_2 = 1 \tag{59.1}$$

and satisfying the "Hermiticity" condition

$$(\rho^\lambda{}_m u)^* = \rho^\mu{}_l ambda. \tag{59.2}$$

For a pure, that is fully polarized, electron spin state, the spinor $\rho^\lambda{}_m u$ reduces to a product of wave-function components:

$$\rho^\lambda{}_m u = \psi^\lambda \psi^*_\mu. \tag{59.3}$$

The diagonal components of the density matrix give the probabilities of the values $+1/2$ and $-1/2$ of the electron's spin projection on the z axis. The mean value of this projection is therefore

$$\bar{s}_z = \frac{1}{2}(\rho^1{}_1 - \rho^2{}_2),$$

or, on using (59.1),

$$\rho^1{}_1 = \frac{1}{2} + \bar{s}_z, \qquad \rho^2{}_2 = \frac{1}{2} - \bar{s}_z. \tag{59.4}$$

In a pure state, the mean values of $s_\pm = s_x \pm i s_y$ are calculated as

$$\bar{s}_\pm = \psi^*_\lambda \hat{s}_\pm \psi^\lambda.$$

Since, by (55.6) and (55.7), the operators $\hat{s}_\pm$ have the matrices

$$\hat{s}_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \hat{s}_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

we find

$$\bar{s}_+ = \psi^*_1 \psi^2, \qquad \bar{s}_- = \psi^*_2 \psi^1.$$

Accordingly, in a mixed state,

$$\rho^1{}_2 = \bar{s}_-, \qquad \rho^2{}_1 = \bar{s}_+. \tag{59.5}$$

With the Pauli matrices, (59.4) and (59.5) can be combined as

$$\rho^\lambda{}_m u = \frac{1}{2}(\delta^\lambda{}_m u + 2\hat{\boldsymbol{\sigma}}^\lambda{}_m u \bar{\mathbf{s}}). \tag{59.6}$$

Thus every component of the electron's polarization density matrix is expressed in terms of the mean components of its spin vector. In other words, the real vector $\bar{\mathbf{s}}$ completely determines the polarization properties of a spin-$1/2$ particle. In the limiting case of full polarization, one component of this vector, with a suitable choice of axes, equals $1/2$, while the other two vanish. In the opposite, unpolarized case, all three components vanish. For arbitrary partial polarization and an arbitrary choice of coordinate system, in general $0 \leqslant p \leqslant 1$, where

$$p = 2(\bar{s}_x^2 + \bar{s}_y^2 + \bar{s}_z^2)^{1/2}$$

may be called the electron's *degree of polarization*. For a particle of arbitrary spin s, the density matrix is the rank-$4s$ spinor $\rho^{\lambda\mu\ldots}{}_{\rho\sigma\ldots}$, symmetric in its first $2s$ and its last $2s$ indices, and satisfying

$$\rho^{\lambda\mu\ldots}{}_{\lambda\mu\ldots} = 1, \tag{59.7}$$

$$(\rho^{\lambda\mu\ldots}{}_{\rho\sigma\ldots})^* = \rho^{\rho\sigma\ldots}{}_{\lambda\mu\ldots}. \tag{59.8}$$

To count the independent density-matrix components, note that among the possible sets of values of $\lambda, \mu, \ldots$ (or of $\rho, \sigma, \ldots$) only $2s+1$ are essentially different. Since the components of $\rho^{\lambda\mu\ldots}{}_{\rho\sigma\ldots}$ are also related by the single condition (59.7), their number is $(2s+1)^2 - 1 = 4s(s+1)$. Although these components are complex, (59.8) ensures that this does not increase the total number of independent quantities characterizing a partially polarized state; that number is therefore $4s(s+1)$.[24] For comparison, a fully polarized state is described by only $4s$ quantities: the $2s+1$ complex components of $\psi^{\lambda\mu\cdots}$, subject to one normalization condition and containing one common phase irrelevant to the description of the state.

Like any spinor of rank $4s$, the spinor $\rho^{\lambda\mu\cdots}{}_{\rho\sigma\ldots}$ is equivalent to a set of irreducible tensors of ranks $4s, 4s-2, \ldots, 0$. Here there is exactly one tensor of each rank. Indeed, by the symmetry properties of $\rho^{\lambda\mu\cdots}{}_{\rho\sigma\ldots}$, each contraction can be performed in only one way: over any one of the indices $\lambda, \mu, \ldots$ and any one of $\rho, \sigma, \ldots$. Moreover, the scalar (rank-zero tensor) is absent altogether, since condition (59.7) reduces it to unity.

§ 60. Time reversal and Kramers' theorem

In quantum mechanics, symmetry of motion under reversal of the sign of time means the following: if ψ is the wave function of some stationary state of a system, then the "time-reversed" wave function, denoted by ψ^{rev}, also describes a possible state with the same energy. It was stated at the end of § 18 that ψ^{rev} coincides with the complex-conjugate function ψ^*. In this simple form, the statement applies to wave functions when particle spin is disregarded. With spin present, it requires qualification.

Represent the wave function of a spin-s particle by the contravariant spinor $\psi^{\lambda\mu\cdots}$ of rank $2s$. On passing to the complex-conjugate functions $\psi^{\lambda\mu\cdots *}$, however, we obtain quantities that transform as components of a covariant spinor. Time reversal therefore replaces $\psi^{\lambda\mu\cdots}$ by a new wave function whose covariant components are defined by

$$\psi^{\text{rev}}_{\lambda\mu\ldots} = \psi^*_{\lambda\mu\ldots}. \tag{60.1}$$

For a specified set of index values $\lambda, \mu, \ldots$, the covariant and contravariant spinor components correspond to angular-momentum projections of opposite sign. In terms of the functions $\psi_{s\sigma}$, time reversal thus replaces $\psi_{s\sigma}$ by $\psi_{s,-\sigma}$, as it must, since reversing time reverses the direction of angular momentum. The precise correspondence following from (60.1) is

$$\psi^{\text{rev}}_{s,-\sigma} = \psi^*_{s\sigma}(-1)^{s-\sigma}. \tag{60.2}$$

In other words, the replacement $\psi_{s\sigma} \longrightarrow \psi^*_{s\sigma}$ required by time reversal means the replacement[25]

$$\psi_{s\sigma} \longrightarrow \psi_{s,-\sigma}(-1)^{s-\sigma}. \tag{60.3}$$

Applying this operation twice gives

$$\psi_{s\sigma} \longrightarrow \psi_{s,-\sigma}(-1)^{s-\sigma} \longrightarrow \psi_{s\sigma}(-1)^{s-\sigma}(-1)^{s+\sigma} = \psi_{s\sigma}(-1)^{2s}.$$

Thus two successive time reversals restore the original wave function only for integral spin; for half-integral spin they change its sign.

[24] Specifying these quantities is equivalent to specifying the mean values of the components of **s** and all their powers and products of degrees $2, 3, \ldots, 2s$ which cannot be reduced to still lower degrees (see Problem 3 of § 55).

[25] Notice the agreement of the complex-conjugation rule for a spherical function, (28.9), with the general rule (60.3).

Consider an arbitrary system of interacting particles. When relativistic interactions are included, the orbital and spin angular momenta of such a system are not in general conserved separately; only the total angular momentum J is conserved. In the absence of an external field, each energy level of the system is $(2j + 1)$-fold degenerate. An external field generally removes this degeneracy. The question is whether the degeneracy can be removed completely, leaving the system with only nondegenerate levels. This question is closely connected with time-reversal symmetry. Classical electrodynamics is invariant under reversal of the sign of time if the electric field is left unchanged and the sign of the magnetic field is reversed.[26] This fundamental property of motion must persist in quantum mechanics. Time-reversal symmetry therefore holds not only for a closed system, but also in any external electric field, provided no magnetic field is present.

The wave functions of the system are spinors $\psi^{\lambda\mu\cdots}$ of rank n equal to twice the sum of the spins of all particles, $n = 2\sum s_a$; this sum need not coincide with the total spin S of the system. From the foregoing, in an arbitrary electric field a wave function and its time reverse must correspond to states of equal energy. A necessary condition for a level to be nondegenerate is that these states be identical, so their wave functions must agree apart from a constant factor. Both must, of course, be expressed as spinors of the same kind, covariant or contravariant.

Write $\psi^{\text{rev}}_{\lambda\mu\ldots} = C\psi_{\lambda\mu\ldots}$, or, by (60.1),

$$\psi^*_{\lambda\mu\ldots} = C\psi_{\lambda\mu\ldots}, \tag{60.4}$$

where C is constant. Complex conjugation of the two sides gives

$$\psi^{\lambda\mu\cdots} = C^*\psi^{\lambda\mu\ldots *}.$$

Lower the indices on the left and correspondingly raise those on the right. This means multiplying both sides by $g_{\alpha\lambda}g_{\beta\mu}\ldots$ and summing over $\lambda, \mu, \ldots$; on the right one must use

$$g_{\alpha\lambda}g_{\beta\mu}\ldots = (-1)^n g_{\lambda\alpha}g_{\mu\beta}\ldots.$$

The result is

$$\psi_{\lambda\mu\ldots} = C^*(-1)^n\psi^*_{\lambda\mu\ldots}.$$

Substituting $\psi^*_{\lambda\mu\ldots}$ from (60.4), we find

$$\psi_{\lambda\mu\ldots} = (-1)^n CC^*\psi_{\lambda\mu\ldots}.$$

This equality must hold identically, so $(-1)^n CC^* = 1$. But $|C|^2$ is necessarily positive; the condition is therefore possible only for even n, that is, for an integral value of $\sum s_a$. For odd n, or half-integral $\sum s_a$,[27] condition (60.4) cannot be satisfied.

We thus reach the result that an electric field can remove the degeneracy completely only in a system with an integral sum of particle spins. In a system with a half-integral spin sum, every level in an arbitrary electric field must be doubly degenerate; the

[26] See II, § 17; see also the remark at the end of § 111.

[27] When $\sum s_a$ is integral (half-integral), all possible values of the total spin S of the system are likewise integral (half-integral).

two distinct states of equal energy correspond to complex-conjugate spinors.[28] (H. A. Kramers, 1930).

We add one mathematical remark. A relation of the form (60.4), with real C, is mathematically the condition that some set of real quantities can be put in correspondence with the spinor components; it may be called a "reality" condition for the spinor.[29] The impossibility of (60.4) for odd n means that no real quantity can correspond to a spinor of odd rank. For even n, by contrast, (60.4) can hold, and C can be real. In particular, a real vector can be put in correspondence with a symmetric second-rank spinor when (60.4) holds with $C = 1$:

$$\psi^{\lambda\mu *} = \psi_{\lambda\mu}$$

(as is easily verified from (57.8), (57.9)). In general, (60.4) with $C = 1$ is the "reality" condition for a symmetric spinor of any even rank.

[28] If the electric field has high (cubic) symmetry, fourfold degeneracy may also occur (see § 99 and its problem).

[29] It is meaningless to speak of a spinor as literally real, since complex-conjugate spinors have different transformation laws.

CHAPTER X

Identity of Particles

§ 61. The indistinguishability principle for identical particles

In classical mechanics, identical particles (electrons, for example) retain their "individuality" despite the identity of their physical properties. One can imagine numbering the particles belonging to a given physical system at some instant and subsequently following each particle along its trajectory; the particles can then be identified at every later instant.

The situation is entirely different in quantum mechanics. As has been noted repeatedly, the uncertainty principle deprives the electron trajectory of all meaning. Even if an electron's position is known exactly at the present instant, its coordinates have no definite values at all at the next instant. Localizing and numbering the electrons at one instant therefore does nothing to permit their later identification. If an electron is localized at some point at a later instant, it is impossible to say which electron has arrived there. Thus quantum mechanics provides no possibility in principle of following each identical particle separately and thereby distinguishing them. Identical particles may be said to lose their "individuality" completely. The identity of their physical properties has a profound consequence: their complete indistinguishability.

This *indistinguishability principle* for identical particles plays a fundamental part in the quantum theory of systems composed of such particles. Begin with a system of only two particles. Because the particles are identical, states obtained from one another merely by interchanging the two particles must be physically equivalent. Hence such an interchange can alter the wave function only by an inessential phase factor. Let $\psi(\xi_1, \xi_2)$ be the system's wave function, where ξ_1 and ξ_2 denote conventionally the three coordinates and the spin projection of each particle. Then

$$\psi(\xi_1, \xi_2) = e^{i\alpha}\psi(\xi_2, \xi_1),$$

where α is a real constant. A second interchange returns the system to its initial state, while multiplying ψ by $e^{2i\alpha}$. Thus $e^{2i\alpha} = 1$, or $e^{i\alpha} = \pm 1$, and consequently $\psi(\xi_1, \xi_2) = \pm\psi(\xi_2, \xi_1)$.

There are therefore only two possibilities: the wave function is either symmetric (unchanged by interchange) or antisymmetric (its sign is reversed). Clearly, the wave functions of all states of the same system must have the same symmetry. Otherwise a state formed by superposing states of different symmetries would have a wave function that was neither symmetric nor antisymmetric.

This result extends directly to a system containing any number of identical particles. If one pair is described, say, by symmetric wave functions, then every other pair of the same particles must have the same property. The wave function must therefore either remain wholly unchanged when any pair is interchanged (and hence under any permutation whatever), or reverse sign when each pair is interchanged. These are called, respectively, a *symmetric* and an *antisymmetric* wave function. Which kind occurs depends on the species of particle. Particles described by antisymmetric functions obey *Fermi–Dirac statistics* and are called *fermions*; particles described by symmetric functions obey *Bose–Einstein statistics* and are called *bosons*[1]. Relativistic quantum mechanics makes it possible to show (see IV, § 25) that the statistics obeyed by particles is uniquely connected with their spin: particles of half-integer spin are fermions, while those of integer spin are bosons.

The statistics of a composite particle is determined by the parity of the number of elementary fermions composing it. Indeed, interchanging two identical composite particles is equivalent to simultaneously interchanging several pairs of identical elementary particles. Interchanging bosons leaves the wave function unchanged, while interchanging fermions reverses its sign. Composite particles containing an odd number of elementary fermions therefore obey Fermi statistics, and those containing an even number obey Bose statistics. This agrees, of course, with the general rule just stated: a composite particle has integer or half-integer spin according as the number of half-integer-spin particles composing it is even or odd.

Thus atomic nuclei of odd atomic weight (consisting of an odd number of protons and neutrons) obey Fermi statistics, and those of even weight obey Bose statistics. For atoms, which contain electrons as well as nuclei, the statistics is evidently determined by whether the sum of the atomic weight and atomic number is even or odd.

Consider a system of N identical particles whose mutual interaction may be neglected. Let $\psi_1, \psi_2, \ldots$ be the wave functions of the different stationary states available to each particle separately. A state of the whole system can be specified by listing the states occupied by the individual particles. We must determine how the wave function ψ of the complete system is to be constructed from $\psi_1, \psi_2, \ldots$.

Let $p_1, p_2, \ldots, p_N$ be the labels of the states occupied by the individual particles; some labels may coincide. For bosons the wave function $\psi(\xi_1, \xi_2, \ldots, \xi_N)$ is the sum of products of the form

$$\psi_{p_1}(\xi_1)\psi_{p_2}(\xi_2)\cdots\psi_{p_N}(\xi_N) \tag{61.1}$$

over all permutations of the distinct indices $p_1, p_2, \ldots$; this sum plainly has the required symmetry. For two particles in different states ($p_1 \neq p_2$), for example,

$$\psi(\xi_1, \xi_2) = \frac{1}{\sqrt{2}}\left[\psi_{p_1}(\xi_1)\psi_{p_2}(\xi_2) + \psi_{p_1}(\xi_2)\psi_{p_2}(\xi_1)\right]. \tag{61.2}$$

The factor $1/\sqrt{2}$ provides normalization (the functions $\psi_1, \psi_2, \ldots$ are mutually orthogonal and normalized). In the general case of N particles the normalized wave function is

$$\psi_{N_1 N_2 \ldots} = \left(\frac{N_1! N_2! \cdots}{N!}\right)^{1/2} \sum \psi_{p_1}(\xi_1)\psi_{p_2}(\xi_2)\cdots\psi_{p_N}(\xi_N), \tag{61.3}$$

where the sum is over all permutations of distinct arrangements of the indices $p_1, p_2, \ldots, p_N$, and N_i is the number of these indices whose value is i (so $\sum N_i = N$).

[1]This terminology comes from the statistics describing an ideal gas of particles with, respectively, antisymmetric or symmetric wave functions. In fact the distinction here is not merely between two statistics, but essentially between two mechanics. Fermi statistics was proposed by E. Fermi for electrons in 1926, and its connection with quantum mechanics was established by Dirac (1926). Bose statistics was proposed by S. Bose for light quanta and generalized by Einstein (1924).

On integrating $|\psi|^2$ over $d\xi_1 d\xi_2 \cdots d\xi_N$[2], all terms vanish except the squared moduli of the individual terms in the sum. Since the number of terms in (61.3) is $N!/(N_1!N_2!\cdots)$, this gives the normalization factor shown there.

For fermions, ψ is the antisymmetric combination of the products (61.1). For two particles,

$$\psi(\xi_1, \xi_2) = \frac{1}{\sqrt{2}}\left[\psi_{p_1}(\xi_1)\psi_{p_2}(\xi_2) - \psi_{p_1}(\xi_2)\psi_{p_2}(\xi_1)\right]. \tag{61.4}$$

For N particles the system's wave function is the determinant

$$\psi_{N_1 N_2 \ldots} = \frac{1}{\sqrt{N!}}\begin{vmatrix} \psi_{p_1}(\xi_1) & \psi_{p_1}(\xi_2) & \cdots & \psi_{p_1}(\xi_N) \\ \psi_{p_2}(\xi_1) & \psi_{p_2}(\xi_2) & \cdots & \psi_{p_2}(\xi_N) \\ \cdots & \cdots & \cdots & \cdots \\ \psi_{p_N}(\xi_1) & \psi_{p_N}(\xi_2) & \cdots & \psi_{p_N}(\xi_N) \end{vmatrix}. \tag{61.5}$$

Interchanging two particles interchanges two columns of the determinant and therefore reverses its sign.

Equation (61.5) has an important consequence. If two of the labels $p_1, p_2, \ldots$ coincide, two rows of the determinant are identical and the determinant vanishes identically. It is nonzero only when all the labels are different. Thus two or more identical fermions cannot simultaneously occupy the same state. This is the *Pauli principle* (W. Pauli, 1925).

§ 62. Exchange interaction

The fact that the Schrödinger equation does not take account of particle spin does not diminish the value of either the equation or the results obtained from it. The reason is that the electric interaction of particles is independent of their spins[3].

Mathematically, the Hamiltonian of a system of electrically interacting particles (in the absence of a magnetic field) contains no spin operators and therefore does not act on the spin variables when applied to the wave function. The Schrödinger equation is thus satisfied by each component of the wave function separately. In other words, the system's wave function can be written as a product

$$\psi(\xi_1, \xi_2) = \chi(\sigma_1, \sigma_2, \ldots)\varphi(\mathbf{r}_1, \mathbf{r}_2, \ldots),$$

where φ depends only on the particle coordinates and χ only on their spins. We shall call the former the *coordinate* or *orbital* wave function, and the latter the *spin* wave function.

The Schrödinger equation essentially determines only the coordinate function φ, leaving χ arbitrary. Whenever the spins themselves are not of interest, the Schrödinger equation may therefore be used with the coordinate function alone as the wave function, as was done in the preceding chapters.

[2]Integration over $d\xi$ is understood here (and in ß 64, 65) to mean integration over the coordinates together with summation over σ.

[3]This statement holds only in the nonrelativistic approximation. When relativistic effects are included, the interaction of charged particles depends on spin.

Despite this independence of the electric interaction from spin, however, the energy of a system has a distinctive dependence on its total spin, arising ultimately from the indistinguishability principle for identical particles.

Consider a system consisting of two identical particles. Solving the Schrödinger equation gives a series of energy levels, each associated with a definite symmetric or antisymmetric coordinate wave function $\varphi(\mathbf{r}_1, \mathbf{r}_2)$.

Indeed, because the particles are identical, the system's Hamiltonian, and hence its Schrödinger equation, is invariant under their interchange. If the energy levels are nondegenerate, interchanging $\mathbf{r}_1$ and $\mathbf{r}_2$ can change $\varphi(\mathbf{r}_1, \mathbf{r}_2)$ only by a constant factor; performing the interchange again shows that this factor can only be ± 1[4].

Suppose first that the particles have zero spin. There is then no spin factor, and the wave function is simply $\varphi(\mathbf{r}_1, \mathbf{r}_2)$, which must be symmetric because zero-spin particles obey Bose statistics. Consequently, not all the energy levels obtained by formally solving the Schrödinger equation can actually occur: those associated with antisymmetric φ are impossible for this system.

Interchanging two identical particles is equivalent to inversion of the coordinate system, whose origin is chosen at the midpoint of the line joining them. Under inversion, on the other hand, φ is multiplied by $(-1)^l$, where l is the orbital angular momentum of their relative motion (see § 30). It follows that a system of two identical zero-spin particles can have only even orbital angular momentum.

Next let the system contain two spin-1/2 particles, say electrons. Its complete wave function, the product of $\varphi(\mathbf{r}_1, \mathbf{r}_2)$ and $\chi(\sigma_1, \sigma_2)$, must be antisymmetric under interchange of the two particles. A symmetric coordinate function must therefore be accompanied by an antisymmetric spin function, and conversely. Write the spin function as a second-rank spinor $\chi^{\lambda\mu}$, with each index referring to the spin of one electron. A function symmetric in the two spins corresponds to a symmetric spinor, $\chi^{\lambda\mu} = \chi^{\mu\lambda}$, and an antisymmetric function to an antisymmetric spinor, $\chi^{\lambda\mu} = -\chi^{\mu\lambda}$. A symmetric second-rank spinor describes a system of total spin one, whereas an antisymmetric spinor reduces to a scalar and corresponds to zero spin.

We thus obtain the following result. Energy levels associated with symmetric solutions $\varphi(\mathbf{r}_1, \mathbf{r}_2)$ of the Schrödinger equation can actually occur when the system's total spin is zero, that is, when the two electron spins are "antiparallel" and add to zero. Energy values associated with antisymmetric $\varphi(\mathbf{r}_1, \mathbf{r}_2)$ require total spin one, so that the two electron spins are "parallel."

In other words, the possible energies of an electron system depend on its total spin. This permits one to speak of a distinctive interaction of the particles that produces this dependence. It is called the *exchange interaction*. It is a purely quantum effect and disappears completely, like spin itself, in the limiting transition to classical mechanics.

For the two-electron system just considered, each energy level corresponds to one definite value of total spin, either 0 or 1. As will be seen below (§ 63), this one-to-one correspondence between spin and energy levels remains valid for systems containing any number of electrons. It does not, however, hold for systems of particles whose spin

[4]If degeneracy is present, linear combinations of the functions belonging to the level can always be chosen to satisfy the same condition.

exceeds 1/2.

Consider two particles of arbitrary spin s. Their spin wave function is a spinor of rank $4s$,

$$\chi^{\overbrace{\lambda\mu\ldots}^{2s}\overbrace{\rho\sigma\ldots}^{2s}},$$

half of whose indices, $2s$, refer to one particle, while the other half refer to the second. The spinor is symmetric in the indices within each group. Interchanging the particles interchanges every index $\lambda, \mu, \ldots$ in the first group with the indices $\rho, \sigma, \ldots$ in the second. To obtain the spin function of a state with total spin S, contract this spinor over $2s - S$ pairs of indices (each pair containing one index from each group) and symmetrize over the remaining indices. The result is a symmetric spinor of rank $2S$.

Contraction over a pair of spinor indices, however, means forming a combination antisymmetric in those indices. On interchanging the particles, the spin wave function is therefore multiplied by $(-1)^{2s-S}$.

The complete wave function of the two-particle system must, under their interchange, be multiplied by $(-1)^{2s}$, which is $+1$ for integer s and -1 for half-integer s. Hence the symmetry of the coordinate wave function under interchange is determined by the factor $(-1)^S$, which depends only on S. Thus the coordinate wave function of two identical particles is symmetric for even total spin and antisymmetric for odd total spin.

Recalling the connection between interchange and inversion, we also conclude that for even (odd) spin S the system can have only even (odd) orbital angular momentum.

Here too there is a dependence between the possible energies and the total spin, but it is not one-to-one. Levels associated with symmetric (antisymmetric) coordinate wave functions can occur for every even (odd) value of S.

Let us count all the states of two particles with even and odd values of S. The quantity S takes $2s + 1$ values, $2s, 2s - 1, \ldots, 0$. For each S there are $2S + 1$ states distinguished by the z component of spin, making $(2s + 1)^2$ states altogether. Let s be an integer. Among the $2s + 1$ values of S, $s + 1$ are even and s are odd. The total number of even-S states is

$$\sum_{S=0,2,\ldots,2s} (2S + 1) = (2s + 1)(s + 1);$$

the remaining $s(2s + 1)$ states have odd S. Similarly, for half-integer s there are $s(2s + 1)$ states with even S and $(s + 1)(2s + 1)$ with odd S.

Problem

1. Determine the exchange splitting of the energy levels of a two-electron system, treating the electron interaction as a perturbation.

Solution. Let the particles, when their interaction is neglected, occupy states with orbital wave functions $\varphi_1(\mathbf{r}_1)$ and $\varphi_2(\mathbf{r}_2)$. States of the system with total spin $S = 0$ and $S = 1$ correspond respectively to the symmetrized and antisymmetrized products

$$\varphi = \frac{1}{\sqrt{2}}[\varphi_1(\mathbf{r}_1)\varphi_2(\mathbf{r}_2) \pm \varphi_1(\mathbf{r}_2)\varphi_2(\mathbf{r}_1)].$$

The mean values of the particle-interaction operator $U(\mathbf{r}_2 - \mathbf{r}_1)$ in these states are $A \pm J$, where

$$A = \iint U|\varphi_1(\mathbf{r}_1)|^2|\varphi_2(\mathbf{r}_2)|^2\, dV_1 dV_2,$$

$$J = \iint U\varphi_1(\mathbf{r}_1)\varphi_1^*(\mathbf{r}_2)\varphi_2(\mathbf{r}_2)\varphi_2^*(\mathbf{r}_1)\, dV_1 dV_2.$$

The integral J is called the *exchange integral*. Dropping the additive constant A, which is not of exchange character, gives the level shifts $\Delta E_0 = J$, $\Delta E_1 = -J$, where the subscript is S. These quantities can be represented as eigenvalues of the spin *exchange operator*[5]

$$\widehat{V}_{\text{ex}} = -\frac{1}{2}J(1 + 4\hat{\mathbf{s}}_1\hat{\mathbf{s}}_2) \tag{1}$$

(for the eigenvalues of $\mathbf{s}_1\mathbf{s}_2$, see problem 2 of § 55).

If, for example, the electrons belong to different atoms, the exchange integral decreases exponentially as the distance R between the atoms increases. Its integrand shows that the integral is determined by the "overlap" of the wave functions $\varphi_1(\mathbf{r}_1)$ and $\varphi_2(\mathbf{r}_2)$. Using the asymptotic decrease of discrete-spectrum wave functions (cf. (21.6)), one finds

$$J \sim \exp[-(\varkappa_1 + \varkappa_2)R], \qquad \varkappa_1 = \frac{1}{\hbar}\sqrt{2m|E_1|}, \qquad \varkappa_2 = \frac{1}{\hbar}\sqrt{2m|E_2|},$$

where E_1, E_2 are the electron energy levels in the two atoms.

2. Do the same for a system of three electrons.

Solution. Using formula (1) of problem 1, write the operator for the pairwise exchange interaction of three electrons as

$$\widehat{V}_{\text{ex}} = -\sum J_{ab}(1/2 + 2\hat{\mathbf{s}}_a\hat{\mathbf{s}}_b), \tag{1}$$

where the sum is over the particle pairs 12, 13, and 23. From (55.6), the matrix elements of $\hat{\mathbf{s}}_a\hat{\mathbf{s}}_b$ between states with different pairs σ_a, σ_b are

$$\langle 1/2, 1/2|\mathbf{s}_a\mathbf{s}_b|1/2, 1/2\rangle = 1/4, \quad \langle 1/2, -1/2|\mathbf{s}_a\mathbf{s}_b|1/2, -1/2\rangle = -1/4,$$

$$\langle 1/2, -1/2|\mathbf{s}_a\mathbf{s}_b| - 1/2, 1/2\rangle = 1/2.$$

First determine the energy for the largest possible projection of the total spin, $M_S = \sigma_1 + \sigma_2 + \sigma_3 = 3/2$; this also determines the energy of the state with $S = 3/2$. The corresponding diagonal matrix element of (1) is

$$\Delta E_{3/2} = -(J_{12} + J_{13} + J_{23}).$$

Now pass to states with $M_S = 1/2$. This value can occur in three ways, according to which one of $\sigma_1, \sigma_2, \sigma_3$ equals $-1/2$ (the other two being $1/2$). A cubic secular equation would therefore result. The calculation can be simplified at once by observing that one root must be the energy already found for $S = 3/2$; hence the secular equation must be

[5] This operator was introduced by Dirac.

divisible by $\Delta E - \Delta E_{3/2}$. In this case this observation avoids calculation of the constant term of the cubic[6].

The leading terms of the equation are

$$(\Delta E)^3 + (J_{12} + J_{13} + J_{23})(\Delta E)^2 \\ + [J_{12}J_{13} + J_{12}J_{23} + J_{13}J_{23} - (J_{12}^2 + J_{13}^2 + J_{23}^2)]\Delta E + \ldots = 0,$$

and division by $\Delta E + J_{12} + J_{13} + J_{23}$ gives the two energy levels belonging to states of spin $S = 1/2$:

$$\Delta E_{1/2} = \pm[J_{12}^2 + J_{13}^2 + J_{23}^2 - J_{12}J_{13} - J_{12}J_{23} - J_{13}J_{23}]^{1/2}.$$

There are thus three energy levels in all, in agreement with the count made in problem 1 of § 63.

3. From which states can a ^{8}Be nucleus decay into two α particles?

SOLUTION. Since an α particle has no spin, a system of two α particles can have only even orbital angular momentum (equal to its total angular momentum), and its states have even parity. The stated decay is therefore possible only from even-parity states of ^{8}Be with even total angular momentum.

[6]This device is especially useful in analogous calculations for systems with larger numbers of particles.